

XCPC Template Library

Morning Flower
and Evening Oath

Tongji University

双码风算法模板与数学速查

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状态说明

本稿包含首批已进行本地验证的模块与完整来源覆盖表的入口。尚未完成 kuangbin、WIDA 全部条目的双码风迁移，尚未完成近三年中国赛站逐题覆盖审计，最大流、平衡树、双连通分量、可持久化等部分接口已有在线 AC 记录，具体题号、源码摘要及验证范围见 `verification/oj.json`。其余接口须分别核验，不能由同文件中某算法通过推断全部通过。本地对拍通过不代表对所有输入的绝对正确性证明；适用范围、数值上限与退化约定属于接口的一部分。

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使用与验证

英文署名已于 QOJ 队伍账号 `ucup-team7502` 核对。两套风格依据训练提交 2932575、2930805、2931279 提取，不擅自将共享队号的提交归属到个人。

代码采用 C++20。以赛时快速抄写为先，一行一条语句，函数体展开。无状态的简单算法使用独立函数，有状态算法使用轻量 `struct`。动态版默认需要 `bits/stdc++.h` 和 `cassert`；传统版各头文件列明依赖。抄写单个函数或结构体时需同时抄写其依赖结构。同名的不同风格实现通常用于不同提交，不建议混在同一编译单元中。点号除特别注明外为 1-based，字符串和多项式为 0-based；所有区间约定应以条目说明为准。

测试源码位于 `tests/`，固定种子便于复现。验证包含独立暴力参考、边界案例、静态与动态实现、AddressSanitizer 和 UndefinedBehaviorSanitizer。`verification/` 保存实际运行记录与源码摘要；OJ 状态只在获得可查提交结果后登记。最大流记录为 <https://www.luogu.com.cn/record/297501480> 与 <https://www.luogu.com.cn/record/297502364>。

原始来源目录共登记 713 项，见 `docs/coverage.csv`。它包含不同版本、例题及公式，不应将其数目视为独立算法数，也不能将“已登记”视为“已完成”。

第一部分

动态容器风格

第一章 Tarjan 与缩点

1.1 Tarjan 强连通分量与缩点

TarjanSCC(n), 点号 1..n; add(u,v) 加有向边, run() 从头计算。bel[u] 为 1..cnt 的强连通分量编号, 跨分量边总从大编号指向小编号, 可按 cnt..1 做拓扑 DP。dag() 返回缩点邻接表, 保留重边; 若 DP 需要对不同邻点去重, 调用者应排序去重。支持自环、重边、不连通图和重复 run。时间及图存储 $O(n+m)$, 递归 DFS 另需 $O(n)$ 调用栈; 本地验证应配置足够栈内存。注意已有 SCC 模板的编号方向与本模板相反。

```
1 struct TarjanSCC
2 {
3     int n, timer = 0, cnt = 0;
4     vector<vector<int>> g;
5     vector<int> dfn, low, bel, st;
6
7     TarjanSCC(int n) : n(n), g(n + 1)
8     {
9     }
10
11     void add(int u, int v)
12     {
13         assert(1 <= u && u <= n && 1 <= v && v <= n);
14         g[u].push_back(v);
15     }
16
17     void dfs(int u)
18     {
19         dfn[u] = low[u] = ++timer;
20         st.push_back(u);
21         for (int v : g[u])
22         {
23             if (!dfn[v])
24             {
25                 dfs(v);
26                 low[u] = min(low[u], low[v]);
27             }
28             else if (!bel[v])
29                 low[u] = min(low[u], dfn[v]);
30         }
31         if (low[u] == dfn[u])
32         {
33             ++cnt;
34             while (true)
35             {
36                 int v = st.back();
37                 st.pop_back();
38                 bel[v] = cnt;
39                 if (v == u)
40                     break;
41             }
42         }
43     }
```

```
44
45 // IDs 1..cnt are in reverse topological order. Recursive DFS needs O(n) stack.
46 void run()
47 {
48     timer = 0;
49     cnt = 0;
50     dfn.assign(n + 1, 0);
51     low = dfn;
52     bel = dfn;
53     st.clear();
54     for (int u = 1; u <= n; u++)
55     {
56         if (!dfn[u])
57             dfs(u);
58     }
59 }
60
61 // Retains parallel edges. Use after run().
62 vector<vector<int>> dag() const
63 {
64     vector<vector<int>> result(cnt + 1);
65     for (int u = 1; u <= n; u++)
66     {
67         for (int v : g[u])
68         {
69             if (bel[u] != bel[v])
70                 result[bel[u]].push_back(bel[v]);
71         }
72     }
73     return result;
74 }
75 };
```

第二章 双连通分量与圆方树

2.1 Tarjan 点双、边双与圆方树

Biconnected(n), 点号 1..n; add(u,v) 返回原边的 0-based 编号, run() 从头计算。blocks 为点双的顶点列表: 桥的两端构成一个块, 孤立点独立成块; 同一个割点可出现在多个块。cut[u] 标记割点, bridge[id] 标记桥; bel[u] 是 1..cnt 的边双编号。block_forest() 返回圆方森林, 原点 1..n, blocks[i] 的方点编号 n+i+1; 孤立点也连接自己的单点方块。bridge_forest() 返回边双缩点森林, 邻接项为 (相邻分量, 原桥编号)。重边按边号处理; 自环保留原边编号但不参与 DFS, 不是桥, 不额外产生点双。全程 $O(n+m)$, 递归 DFS, 分量栈不替代递归调用。两种森林必须在 run 后获取。

```
1 struct Biconnected
2 {
3     int n, timer = 0, cnt = 0;
4     vector<vector<pair<int, int>>> g;
5     vector<pair<int, int>> edges;
6     vector<vector<int>> blocks;
7     vector<int> dfn, low, cut, bridge, bel, st;
8
9     Biconnected(int n) : n(n), g(n + 1)
10    {
11    }
12
13    int add(int u, int v)
14    {
15        assert(1 <= u && u <= n && 1 <= v && v <= n);
16        int id = edges.size();
17        edges.push_back({u, v});
18        if (u != v)
19        {
20            g[u].push_back({v, id});
21            g[v].push_back({u, id});
22        }
23        return id;
24    }
25
26    void dfs(int u, int parent_edge)
27    {
28        dfn[u] = low[u] = ++timer;
29        st.push_back(u);
30        int children = 0;
31        for (auto [v, id] : g[u])
32        {
33            if (id == parent_edge)
34                continue;
35            if (!dfn[v])
36            {
37                children++;
38                dfs(v, id);
39                low[u] = min(low[u], low[v]);
40                if (low[v] > dfn[u])
41                    bridge[id] = 1;
```

```

42         if (low[v] >= dfn[u])
43         {
44             if (parent_edge != -1 || children > 1)
45                 cut[u] = 1;
46             blocks.push_back({u});
47             while (true)
48             {
49                 int x = st.back();
50                 st.pop_back();
51                 blocks.back().push_back(x);
52                 if (x == v)
53                     break;
54             }
55         }
56     }
57     else
58         low[u] = min(low[u], dfn[v]);
59 }
60 if (parent_edge == -1 && children == 0)
61     blocks.push_back({u});
62 }
63
64 void paint(int u)
65 {
66     bel[u] = cnt;
67     for (auto [v, id] : g[u])
68     {
69         if (!bridge[id] && !bel[v])
70             paint(v);
71     }
72 }
73
74 void run()
75 {
76     timer = 0;
77     cnt = 0;
78     dfn.assign(n + 1, 0);
79     low = dfn;
80     cut = dfn;
81     bel = dfn;
82     bridge.assign(edges.size(), 0);
83     blocks.clear();
84     st.clear();
85     for (int u = 1; u <= n; u++)
86     {
87         if (!dfn[u])
88         {
89             dfs(u, -1);
90             st.pop_back();
91         }
92     }
93     for (int u = 1; u <= n; u++)
94     {
95         if (bel[u])
96             continue;
97         ++cnt;

```



```

98         paint(u);
99     }
100 }
101
102 // Original vertices 1..n; block i has node n+i+1. Includes isolated vertices.
103 vector<vector<int>> block_forest() const
104 {
105     vector<vector<int>> tree(n + blocks.size() + 1);
106     for (int i = 0; i < (int)blocks.size(); i++)
107     {
108         int b = n + i + 1;
109         for (int u : blocks[i])
110         {
111             tree[u].push_back(b);
112             tree[b].push_back(u);
113         }
114     }
115     return tree;
116 }
117
118 // Edge-connected component IDs 1..cnt; forest edges carry original edge IDs.
119 vector<vector<pair<int, int>>> bridge_forest() const
120 {
121     vector<vector<pair<int, int>>> tree(cnt + 1);
122     for (int id = 0; id < (int)edges.size(); id++)
123     {
124         if (!bridge[id])
125             continue;
126         auto [u, v] = edges[id];
127         u = bel[u];
128         v = bel[v];
129         tree[u].push_back({v, id});
130         tree[v].push_back({u, id});
131     }
132     return tree;
133 }
134 };

```

第三章 奇环顶点判定

3.1 属于简单奇环的全部顶点

先对 Biconnected 调用 run, 再调用 odd_cycle_vertices(graph), 返回 answer[1..n], 值为 1 表示该点属于某个长度至少为 3 的简单奇环, answer[0]=0。允许重边、断开图及空图; 自环不计入这里的奇环。某点双非二分当且仅当其中存在奇环, 此时该点双每个顶点都属于某个奇环; 答案为所有非二分点双的顶点并集。通过圆方森林把每条非自环边分到唯一点双: 两端中森林深度较大者的父方点即该块, 同深时二者父方点相同。随后只对块内边染色, 避免在每个块中重复扫描割点的全部邻边。两次 DFS 均递归, 分解后额外 $O(n+m)$ 时间和空间。可重排 blocks, 不能改变分解内容。六点以内所有简单图枚举奇环对拍, 另有重边/自环、长链、星形和大奇环测试; POJ2942 补图驱动本地验证, 在线待补。依赖 biconnected.hpp, 传统版函数为 Odd_Cycle_Vertices。

```
1 // After run(). Simple odd cycles have length >= 3; self-loops are ignored.
2 inline vector<int> odd_cycle_vertices(const Biconnected &graph)
3 {
4     int n = graph.n;
5     auto forest = graph.block_forest();
6     vector<int> parent(forest.size(), depth(forest.size()));
7     auto root_tree = [&](auto &&self, int u, int p) -> void
8     {
9         parent[u] = p;
10        for (int v : forest[u])
11            if (v != p)
12            {
13                depth[v] = depth[u] + 1;
14                self(self, v, u);
15            }
16    };
17    for (int u = 1; u <= n; u++)
18        if (!depth[u])
19        {
20            depth[u] = 1;
21            root_tree(root_tree, u, 0);
22        }
```

```

23     vector<vector<pair<int, int>>> edges(graph.blocks.size());
24     for (auto [u, v] : graph.edges)
25         if (u != v)
26         {
27             int block = parent[depth[u] >= depth[v] ? u : v] - n - 1;
28             assert(0 <= block && block < (int)edges.size());
29             edges[block].push_back({u, v});
30         }
31     vector<vector<int>> g(n + 1);
32     vector<int> color(n + 1), answer(n + 1);
33     for (int block = 0; block < (int)edges.size(); block++)
34     {
35         for (int u : graph.blocks[block])
36         {
37             g[u].clear();
38             color[u] = 0;
39         }
40         for (auto [u, v] : edges[block])
41         {
42             g[u].push_back(v);
43             g[v].push_back(u);
44         }
45         bool bipartite = true;
46         auto paint = [&](auto &&self, int u) -> void
47         {
48             for (int v : g[u])
49                 if (!color[v])
50                 {
51                     color[v] = -color[u];
52                     self(self, v);
53                 }
54             else if (color[v] == color[u])
55                 bipartite = false;
56         };
57         int root = graph.blocks[block][0];
58         color[root] = 1;
59         paint(paint, root);
60         if (!bipartite)
61             for (int u : graph.blocks[block])
62                 answer[u] = 1;
63     }
64     return answer;
65 }

```

第四章 删点连通性

4.1 由点双分解计算删点连通块数

在 `Biconnected.run()` 后调用 `removal_components(graph)`, 返回 (原连通块数 `before`, 删点结果 `after`)。 `after[u]` 为独立删除原顶点 `u` 后剩余连通块数, 点号 `1..n`; `after[0]=0` 不参与计算。传统版 `Removal_Components(graph)` 自动推导 `Biconnected_Graph<N>` 的 `N`。 `before` 等于边双数量减去桥数量; `after[u]` 等于 `before-1` 加包含 `u` 的非单点点双数量。单点块必须跳过, 因此孤立点删去后会少一个连通块。空图返回 `(0,{0})`。不修改原分解, $O(n+m)$ 时间、 $O(n)$ 额外空间。若只需要割点、桥和删点增量, 直接用更短的 `Lowlink/Low_Link`; 已经需要点双分解时可用本函数, 避免再次 DFS。

```
1 // Call after run(). Each vertex is deleted independently from the original graph.
2 inline pair<int, vector<int>> removal_components(const Biconnected &graph)
3 {
4     int before = graph.cnt - count(graph.bridge.begin(), graph.bridge.end(), 1);
5     vector<int> after(graph.n + 1, before - 1);
6     after[0] = 0;
7     for (const auto &block : graph.blocks)
8         if (block.size() > 1)
9             for (int u : block)
10                 after[u]++;
11     return {before, after};
12 }
```

第五章 边双缩点与边定向

5.1 边双缩点的大小与内部边数

在 `Biconnected.run()` 后构造 `EdgeCompression(graph)`, 传统版 `Edge_Compression(graph)` 自动推导 `Biconnected_Graph<N>` 的 `N`。 `n` 为边双数量; `size[c]` 为第 `c` 块的原顶点数, `inside[c]` 为块内逻辑无向边数, 分量号 `1..n`, `0` 号槽忽略。 `edges` 每项为 {块 `u`, 块 `v`, 原边 `id`} , 只含跨块的桥, 每条桥出现一次, 按原边号递增; 是无向森林, 不承诺端点为父子方向。 重边逐条计数, 自环在所属块内部计一次, 不能用邻接表度数直接计边而漏掉这些约定。 空图返回 `n=0`、长度 `1` 的零计数数组和空边表。 时间 $O(N+M)$, 输出空间 $O(N+M)$, N/M 指原图规模。

```
1 struct EdgeCompression
2 {
3     int n;
4     vector<int> size, inside;
5     vector<array<int, 3>> edges;
6
7     EdgeCompression(const Biconnected &graph) : n(graph.cnt), size(n + 1), inside(n + 1)
8     {
9         for (int u = 1; u <= graph.n; u++)
10             size[graph.bel[u]]++;
11         for (int id = 0; id < (int)graph.edges.size(); id++)
12         {
13             auto [u, v] = graph.edges[id];
14             u = graph.bel[u];
15             v = graph.bel[v];
16             if (u == v)
17                 inside[u]++;
18             else
19                 edges.push_back({u, v, id});
20         }
21     }
22 };
```

5.2 无向边定向与强连通定向

`orient_edges(graph)` 接受 Lowlink 对象，传统版 `Orient_Edges` 接受 `Low_Link`。返回长度等于逻辑边数的方向数组，第 `id` 项为该边定向后的 {起点, 终点}，顶点 1-based、边 `id` 为 0-based。每条无向边仅处理一次：DFS 树边向下，非树边指向祖先，自环保留 {u,u}，重边不合并。支持非连通图；每个边双块内定向后强连通，桥只给出某个方向，不保证跨桥互相可达。若要求整个非空图强连通，需另外 `run/Run` 检查 `components==1` 且无桥；不满足则不存在这样的方向。该函数只读取邻接表与边数，不依赖已有 DFS 状态，不修改原图，调用时无须先 `run`。复杂度 $O(N+M)$ 时间、 $O(N+M)$ 额外空间，DFS 保持递归。

```

1 // Tree edges go down; other edges go to ancestors. Handles each logical edge once.
2 inline vector<pair<int, int>> orient_edges(const Lowlink &graph)
3 {
4     vector<pair<int, int>> direction(graph.edges);
5     vector<int> seen(graph.n + 1);
6     auto dfs = [&](auto &&self, int u) -> void
7     {
8         seen[u] = 1;
9         for (auto [v, id] : graph.g[u])
10         {
11             if (direction[id].first)
12                 continue;
13             direction[id] = {u, v};
14             if (!seen[v])
15                 self(self, v);
16         }
17     };
18     for (int u = 1; u <= graph.n; u++)
19         if (!seen[u])
20             dfs(dfs, u);
21     return direction;
22 }
```

第六章 桥树最少加边

6.1 连通无向图最少加边方案

先对非空连通无向图调用 `Biconnected.run()`，再调用 `bridge_augmentation(graph)`。返回原图点号构成的新增边列表，不修改原图。允许新增重边，不产生自环；不适用于非连通图。已无桥时返回空列表。桥树有 L 个叶子时最少加 $(L+1)/2$ 条边（整数除法）。递归 DFS 顺序收集叶子，奇数时复制首叶，前后半段配对。每条桥一侧的叶子是循环连续区间，半周配对必跨过该割，故所有桥被消除；每个叶子至少需要一个新边端点，得到同样的下界。分解后本函数 $O(n+m)$ 时间与空间。依赖 `biconnected.hpp`。P2860 双风格完整程序本地验证；新增方案以逐边删除连通性独立验证，尚无本库在线 AC。

```
1 // Connected, nonempty graph after run(); new parallel edges are allowed.
2 inline vector<pair<int, int>> bridge_augmentation(const Biconnected &graph)
3 {
4     assert(graph.n > 0 && graph.cnt > 0);
5     int bridges = count(graph.bridge.begin(), graph.bridge.end(), 1);
6     assert(graph.cnt - bridges == 1);
7     if (graph.cnt == 1)
8         return {};
9     auto tree = graph.bridge_forest();
10    vector<int> representative(graph.cnt + 1);
11    for (int u = 1; u <= graph.n; u++)
12        representative[graph.bel[u]] = u;
13    vector<int> leaves;
14    auto dfs = [&](auto &&self, int u, int parent) -> void
15    {
16        if (tree[u].size() == 1)
17            leaves.push_back(u);
18        for (auto [v, id] : tree[u])
19            if (v != parent)
20                self(self, v, u);
21    };
22    dfs(dfs, 1, 0);
23    if (leaves.size() % 2)
24        leaves.push_back(leaves[0]);
25    int half = leaves.size() / 2;
26    vector<pair<int, int>> added;
27    for (int i = 0; i < half; i++)
28        added.push_back({representative[leaves[i]], representative[leaves[i + half]]});
29    return added;
30 }
```

第七章 数据结构

7.1 并查集

`init(n)`、`find(x)`、`merge(x,y)`，点号 $1..n$ ；`siz[find(x)]` 为集合大小。路径压缩与按大小合并，均摊 $O(\alpha(n))$ 。

```
1 struct DSU
2 {
3     vector<int> fa, siz;
4
5     DSU(int n = 0)
6     {
7         init(n);
8     }
9
10    void init(int n)
11    {
12        fa.resize(n + 1);
13        iota(fa.begin(), fa.end(), 0);
14        siz.assign(n + 1, 1);
15    }
16
17    int find(int x)
18    {
19        while (x != fa[x])
20        {
21            fa[x] = fa[fa[x]];
22            x = fa[x];
23        }
24        return x;
25    }
26
27    bool merge(int x, int y)
28    {
29        x = find(x);
30        y = find(y);
31        if (x == y)
32            return false;
33        if (siz[x] < siz[y])
34            swap(x, y);
35        fa[y] = x;
36        siz[x] += siz[y];
37        return true;
38    }
39 };
```


7.2 可撤销并查集

构造后 `snapshot()` 保存历史长度，`rollback(t)` 回退。禁止路径压缩；`find/merge` 为 $O(\log n)$ ，回退每次成功合并 $O(1)$ 。

```

1 struct RollbackDSU
2 {
3     vector<int> fa, siz;
4     vector<pair<int, int>> his;
5
6     RollbackDSU(int n) : fa(n + 1), siz(n + 1, 1)
7     {
8         iota(fa.begin(), fa.end(), 0);
9     }
10
11     int find(int x) const
12     {
13         while (x != fa[x])
14             x = fa[x];
15         return x;
16     }
17
18     int snapshot() const
19     {
20         return (int)his.size();
21     }
22
23     bool merge(int x, int y)
24     {
25         x = find(x);
26         y = find(y);
27         if (x == y)
28             return false;
29         if (siz[x] < siz[y])
30             swap(x, y);
31         his.push_back({y, siz[x]});
32         fa[y] = x;
33         siz[x] += siz[y];
34         return true;
35     }
36
37     void rollback(int t)
38     {
39         assert(0 <= t && t <= snapshot());
40         while (snapshot() > t)
41         {
42             auto [y, s] = his.back();
43             his.pop_back();
44             siz[fa[y]] = s;
45             fa[y] = y;
46         }
47     }
48 };

```

7.3 树状数组与第 k 小

$\text{add}(x,v)$ 、 $\text{sum}(x)$ 、 $\text{query}(l,r)$ ，下标 $1..n$ ，闭区间； $\text{kth}(k)$ 要求所有频率非负且 $k>0$ ，超过总和返回 $n+1$ 。单次 $O(\log n)$ 。

```

1 template <class T = long long> struct Fenwick
2 {
3     int n;
4     vector<T> a;
5
6     Fenwick(int n = 0) : n(n), a(n + 1)
7     {
8     }
9
10    void add(int x, T v)
11    {
12        assert(x >= 1);
13        for (; x <= n; x += x & -x)
14            a[x] += v;
15    }
16
17    T sum(int x) const
18    {
19        T ans = 0;
20        for (; x; x -= x & -x)
21            ans += a[x];
22        return ans;
23    }
24
25    T query(int l, int r) const
26    {
27        return sum(r) - sum(l - 1);
28    }
29
30    // Nonnegative frequencies; returns n+1 if k exceeds total.
31    int kth(T k) const
32    {
33        assert(k > 0);
34        int x = 0, b = 1;
35        while (b <= n / 2)
36            b *= 2;
37        for (; b; b >>= 1)
38            if (x + b <= n && a[x + b] < k)
39            {
40                k -= a[x + b];
41                x += b;
42            }
43        return x + 1;
44    }
45 };

```

7.4 区间加与区间和

初值全零； $\text{add}(l,r,v)$ 、 $\text{query}(l,r)$ ， $1 \leq l \leq r \leq n$ 。维护 sum 与加法 tag ；所有中间和与长度乘积须在 long long 范围内。单次 $O(\log n)$ 。

```

1 struct LazySeg
2 {
3     int n;
4     vector<long long> sum, tag;
5
6     LazySeg(int n) : n(n), sum(4 * n + 4), tag(4 * n + 4)
7     {
8         assert(n > 0);
9     }
10
11     void apply(int p, int l, int r, long long v)
12     {
13         sum[p] += (r - l + 1) * v;
14         tag[p] += v;
15     }
16
17     void push(int p, int l, int r)
18     {
19         int m = (l + r) / 2;
20         apply(p * 2, l, m, tag[p]);
21         apply(p * 2 + 1, m + 1, r, tag[p]);
22         tag[p] = 0;
23     }
24
25     void add(int l, int r, long long v)
26     {
27         add(1, 1, n, l, r, v);
28     }
29
30     void add(int p, int l, int r, int ql, int qr, long long v)
31     {
32         if (ql <= l && r <= qr)
33         {
34             apply(p, l, r, v);
35             return;
36         }
37         push(p, l, r);
38         int m = (l + r) / 2;
39         if (ql <= m)
40             add(p * 2, l, m, ql, qr, v);
41         if (qr > m)
42             add(p * 2 + 1, m + 1, r, ql, qr, v);
43         sum[p] = sum[p * 2] + sum[p * 2 + 1];
44     }
45
46     long long query(int l, int r)
47     {
48         return query(1, 1, n, l, r);
49     }
50
51     long long query(int p, int l, int r, int ql, int qr)
52     {
53         if (ql <= l && r <= qr)
54             return sum[p];
55         push(p, l, r);

```

```
56     int m = (l + r) / 2;
57     long long ans = 0;
58     if (ql <= m)
59         ans += query(p * 2, l, m, ql, qr);
60     if (qr > m)
61         ans += query(p * 2 + 1, m + 1, r, ql, qr);
62     return ans;
63 }
64 };
```

7.5 64 位异或线性基

值类型 U 为 `unsigned long long`，完整支持 64 位。默认空基；`init/Init` 清空。`insert/Insert` 返回是否增秩；`contains/Contains` 判断可表示性（允许空子集，所以 0 总可表示）。`query/Query(x=0)` 返回 x 与所张成空间异或后的最大无符号值。`kth/Kth(k,nonempty=true)` 返回第 k 小的不同异或值，1-based，参数 k 为 `__uint128_t`，非法序号返回 `nullopt`。默认只允许非空输入子集，0 仅在 `dependent` 为真时出现；传 `false` 则包括空子集所得的 0。`minimum/Minimum` 同此约定，空输入的非空最小值不存在。`merge/Merge` 将另一基所代表的输入多重集并入；重复元素和自合并可新增异或为 0 的非空方案。`rank` 是维数，`dependent` 表示存在非空零异或子集。插入、成员判断、最大值为 $O(64)$ ；合并 $O(64^2)$ 。`kth` 会在需要时 `rebuild/Rebuild` 规范化缓存，首次有效查询 $O(64^2)$ ，后续 $O(64)$ ；仅新增独立向量使缓存失效。所有成员仅供只读，不要直接改数组或标志。

```

1 struct XorBasis
2 {
3     using U = unsigned long long;
4     using Wide = __uint128_t;
5     U a[64]{};
6     vector<U> p;
7     int rank = 0;
8     bool dependent = false, dirty = false;
9
10    void init()
11    {
12        fill(a, a + 64, 0);
13        p.clear();
14        rank = 0;
15        dependent = false;
16        dirty = false;
17    }
18
19    bool insert(U x)
20    {
21        for (int i = 63; i >= 0; i--)
22            if (x >> i & 1)
23            {
24                if (!a[i])
25                {
26                    a[i] = x;
27                    rank++;
28                    dirty = true;
29                    return true;
30                }
31                x ^= a[i];
32            }
33        dependent = true;
34        return false;
35    }

```

```
37     bool contains(U x) const
38     {
39         for (int i = 63; i >= 0; i--)
40             if (x >> i & 1)
41                 x ^= a[i];
42         return x == 0;
43     }
44
45     U query(U x = 0) const
46     {
47         for (int i = 63; i >= 0; i--)
48             x = max(x, x ^ a[i]);
49         return x;
50     }
51
52     void rebuild()
53     {
54         vector<U> b(a, a + 64);
55         for (int i = 0; i < 64; i++)
56             if (b[i])
57                 for (int j = i + 1; j < 64; j++)
58                     if (b[j] >> i & 1)
59                         b[j] ^= b[i];
60         p.clear();
61         for (U x : b)
62             if (x)
63                 p.push_back(x);
64         dirty = false;
65     }
```

```

67 // k is 1-based; nonempty excludes the empty input subset, not value zero.
68 optional<U> kth(Wide k, bool nonempty = true)
69 {
70     if (!k)
71         return nullopt;
72     Wide index = k;
73     if (!nonempty || dependent)
74         index--;
75     if (index >= (Wide(1) << rank))
76         return nullopt;
77     if (dirty)
78         rebuild();
79     U answer = 0;
80     for (int i = 0; i < rank; i++)
81         if (index >> i & 1)
82             answer ^= p[i];
83     return answer;
84 }
85
86 optional<U> minimum(bool nonempty = true)
87 {
88     return kth(1, nonempty);
89 }
90
91 void merge(const XorBasis &other)
92 {
93     dependent |= other.dependent;
94     for (U x : other.a)
95         if (x)
96             insert(x);
97 }
98 };

```

第八章 带位置线性基

8.1 带位置限制的异或线性基

默认构造为空，init/Init 可清空。insert/Insert(x,at) 插入 unsigned long long 值及非负 int 标签，支持完整 64 位，标签可重复、乱序。contains/Contains(x,lower=0) 判断 x 能否由标签至少为 lower 的输入异或得到；query/Query(lower=0,x=0) 返回 x 与这些输入子集异或后的最大值；rank/Rank(lower=0) 返回对应空间维数。lower 为非负 long long，可大于所有 int 标签，空集仍可产生 0。全部操作 $O(64)$ ，空间 $O(64)$ 。消元时同一主元优先保留标签较大的向量，较旧向量继续消元；按阈值筛选后仍是正确线性基。标签是组合可用范围的界，不是选点方案；数组仅供只读。区间 [l,r] 应按右端点排序询问，扫描并仅插入到 r，再查询 lower=l；接口本身没有上界，插入超过 r 后不能只用 lower 恢复旧区间。若需保留历史，可复制每个前缀的对象，但空间为 $O(64n)$ 。不提供非空零异或判定或第 k 小；普通线性基的 dependent 不能直接移植到所有阈值。

```
1 struct PositionBasis
2 {
3     using U = unsigned long long;
4     U a[64];
5     int pos[64];
6
7     PositionBasis()
8     {
9         init();
10    }
11
12    void init()
13    {
14        fill(a, a + 64, 0);
15        fill(pos, pos + 64, -1);
16    }
```



```

18 void insert(U x, int at)
19 {
20     assert(at >= 0);
21     for (int i = 63; i >= 0; i--)
22         if (x >> i & 1)
23             {
24                 if (!a[i])
25                     {
26                         a[i] = x;
27                         pos[i] = at;
28                         return;
29                     }
30                 if (at > pos[i])
31                     {
32                         swap(a[i], x);
33                         swap(pos[i], at);
34                     }
35                 x ^= a[i];
36             }
37 }
38
39 bool contains(U x, long long lower = 0) const
40 {
41     assert(lower >= 0);
42     for (int i = 63; i >= 0; i--)
43         if ((x >> i & 1) && pos[i] >= lower)
44             x ^= a[i];
45     return x == 0;
46 }
47
48 U query(long long lower = 0, U x = 0) const
49 {
50     assert(lower >= 0);
51     for (int i = 63; i >= 0; i--)
52         if (pos[i] >= lower)
53             x = max(x, x ^ a[i]);
54     return x;
55 }
56
57 int rank(long long lower = 0) const
58 {
59     assert(lower >= 0);
60     int answer = 0;
61     for (int i = 0; i < 64; i++)
62         answer += pos[i] >= lower;
63     return answer;
64 }
65 };

```

第九章 线性空间求交

9.1 异或线性空间求交

`basis_intersection(a,b)` 返回 `span(a)` 与 `span(b)` 的交空间的一组独立生成元，结果类型 `XorBasis`；传统版 `Basis_Intersection` 接受并返回 `Xor_Basis`。支持完整 64 位无符号值，不修改输入，也不要求先 rebuild。使用第一个空间的主元，消去第二个空间的向量，同时记录来自第二个空间的部分；消元为零时，该部分属于交空间。时间 $O(64^2)$ ，额外空间 $O(64)$ 。输出按独立生成元解释：`dependent` 为 `false`，输入的重复次数及非空零异或关系不传播；表示的是子空间，不是两组原始输入元素的集合交。枚举整个交空间应调用 `result.kth(k,false)`，包含零；零空间返回空基，`rank=0`、`contains(0)=true`。

```
1 inline XorBasis basis_intersection(const XorBasis &a, const XorBasis &b)
2 {
3     using U = unsigned long long;
4     U pivot[64], from_b[64]{};
5     copy(a.a, a.a + 64, pivot);
6     XorBasis result;
7     for (U value : b.a)
8     {
9         U x = value, part = value;
10        for (int i = 63; i >= 0 && x; i--)
11            if (x >> i & 1)
12            {
13                if (!pivot[i])
14                {
15                    pivot[i] = x;
16                    from_b[i] = part;
17                    break;
18                }
19                x ^= pivot[i];
20                part ^= from_b[i];
21            }
22        if (!x && part)
23            result.insert(part);
24    }
25    return result;
26 }
```

9.2 Zassenhaus：同时求和空间与交空间

`basis_sum_intersection(a,b)` 返回和空间基、交空间基这一对结果。传统接口名为 `Basis_Sum_Intersection`。和空间由两组基共同张成，一般不是两个空间的集合并；两结果都是独立生成元，其 `dependent` 均为 `false`。将 `a` 的向量 `x` 编成 `(x,x)`，`b` 的向量 `y` 编成 `(y,0)`，用 `__uint128_t` 储存高低各 64 位并按高位到低位消元。高半部非零的行投影构成和空间的基，高半部为零的行取低半部构成交空间的基。必须先提升到 128 位再左移 64；不能用 64 位变量移位 64。时间 $O(128^2)$ ，额外空间 $O(128)$ 。结果满足 $\dim(A+B)+\dim(A \cap B)=\dim(A)+\dim(B)$ ，但维数恒等式只是检查条件，不能单独证明返回向量正确。包含空空间、相同空间和输入别名。

```

1 // Return independent generators of (sum space, intersection space).
2 inline pair<XorBasis, XorBasis> basis_sum_intersection(const XorBasis &a,
3                                                         const XorBasis &b)
4 {
5     using W = __uint128_t;
6     W pivot[128]{};
7     auto insert = [&](W x)
8     {
9         for (int i = 127; i >= 0 && x; i--)
10             if (x >> i & 1)
11             {
12                 if (!pivot[i])
13                 {
14                     pivot[i] = x;
15                     return;
16                 }
17                 x ^= pivot[i];
18             }
19     };
20     for (auto x : a.a)
21         insert((W(x) << 64) | x);
22     for (auto x : b.a)
23         insert(W(x) << 64);
24     XorBasis sum, intersection;
25     for (int i = 0; i < 128; i++)
26         if (pivot[i])
27         {
28             if (i < 64)
29                 intersection.insert((unsigned long long)pivot[i]);
30             else
31                 sum.insert((unsigned long long)(pivot[i] >> 64));
32         }
33     return {sum, intersection};
34 }
```

第十章 线段树常用修改

10.1 线段树：区间乘、加、赋值与模意义求和

`AffineSegTree(a,mod)` 传入非空 0-based 初值数组；查询与修改采用 1-based 非空闭区间。`update(l,r,mul,add)` 将区间元素 x 改为 $mul \cdot x + add$ ；乘法用 $(mul,0)$ ，加法用 $(1,add)$ ，赋值用 $(0,value)$ 。`query(l,r)` 返回区间和。 $1 \leq mod \leq INT_MAX$ ，模数无需为素数，初值和修改参数允许 long long 全范围，入口统一归一化。旧标记 (m,b) 后应用新标记 (a,c) ，复合为 $(m \cdot a, b \cdot a + c)$ ，乘法取模后再相加。建树 $O(n)$ ，单次 $O(\log n)$ ，空间 $O(n)$ 。内部 `build`、`apply` 等函数不作为公共接口；动态版重建新对象，传统版可重复 `Init`。

```
1 struct AffineSegTree
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         ll sum = 0, mul = 1, add = 0;
8     };
9
10    int n, mod;
11    vector<Node> t;
12
13    AffineSegTree(const vector<ll> &a, int mod) : n(a.size()), mod(mod), t(4 * n + 4)
14    {
15        assert(n > 0 && mod > 0);
16        build(1, 1, n, a);
17    }
18
19    ll norm(ll x) const
20    {
21        x %= mod;
22        if (x < 0)
23            x += mod;
24        return x;
25    }
26
27    void pull(int p)
28    {
29        t[p].sum = (t[p * 2].sum + t[p * 2 + 1].sum) % mod;
30    }
31
32    void build(int p, int l, int r, const vector<ll> &a)
33    {
34        t[p] = Node{};
35        if (l == r)
36        {
37            t[p].sum = norm(a[l - 1]);
38            return;
39        }
40        int m = (l + r) / 2;
41        build(p * 2, l, m, a);
42        build(p * 2 + 1, m + 1, r, a);
```

```

43     pull(p);
44 }
45
46 // New transform follows the old pending transform: x -> x*a+b.
47 void apply(int p, int len, ll a, ll b)
48 {
49     t[p].sum = (t[p].sum * a % mod + len * b % mod) % mod;
50     t[p].mul = t[p].mul * a % mod;
51     t[p].add = (t[p].add * a % mod + b) % mod;
52 }
53
54 void push(int p, int l, int r)
55 {
56     if (t[p].mul == 1 && t[p].add == 0)
57         return;
58     int m = (l + r) / 2;
59     apply(p * 2, m - l + 1, t[p].mul, t[p].add);
60     apply(p * 2 + 1, r - m, t[p].mul, t[p].add);
61     t[p].mul = 1;
62     t[p].add = 0;
63 }
64
65 void update(int l, int r, ll a, ll b)
66 {
67     assert(1 <= l && l <= r && r <= n);
68     update(1, 1, n, l, r, norm(a), norm(b));
69 }
70
71 void update(int p, int l, int r, int ql, int qr, ll a, ll b)
72 {
73     if (ql <= l && r <= qr)
74     {
75         apply(p, r - l + 1, a, b);
76         return;
77     }
78     push(p, l, r);
79     int m = (l + r) / 2;
80     if (ql <= m)
81         update(p * 2, l, m, ql, qr, a, b);
82     if (qr > m)
83         update(p * 2 + 1, m + 1, r, ql, qr, a, b);
84     pull(p);
85 }
86
87 ll query(int l, int r)
88 {
89     assert(1 <= l && l <= r && r <= n);
90     return query(1, 1, n, l, r);
91 }
92
93 ll query(int p, int l, int r, int ql, int qr)
94 {
95     if (ql <= l && r <= qr)
96         return t[p].sum;

```

```
97     push(p, l, r);
98     int m = (l + r) / 2;
99     ll answer = 0;
100    if (ql <= m)
101        answer += query(p * 2, l, m, ql, qr);
102    if (qr > m)
103        answer += query(p * 2 + 1, m + 1, r, ql, qr);
104    return answer % mod;
105 }
106 };
```

第十一章 网络流

11.1 最大流与方案

Dinic(n), add(u,v,c) 返回偶数原边编号; flow(s,t,limit) 返回本次新增流; used(id) 查询原边净流, cut(s) 返回残量可达点集。允许重边、自环, 容量非负, $s \neq t$; 重复求流须使用相同源汇。只有求尽最大流后 cut 才是最小割。一般 $O(V^2E)$, 递归增广深度 $O(V)$ 。总流不得超过 long long 上限。

```
1 struct Dinic
2 {
3     using ll = long long;
4
5     struct Edge
6     {
7         int from, to;
8         ll cap, initial;
9     };
10
11     int n;
12     vector<Edge> e;
13     vector<vector<int>> g;
14     vector<int> dep, cur;
15
16     Dinic(int n) : n(n), g(n + 1), dep(n + 1), cur(n + 1)
17     {
18     }
19
20     int add(int u, int v, ll c)
21     {
22         assert(c >= 0);
23         int id = (int)e.size();
24         e.push_back({u, v, c, c});
25         e.push_back({v, u, 0, 0});
26         g[u].push_back(id);
27         g[v].push_back(id + 1);
28         return id;
29     }
30
31     bool bfs(int s, int t)
32     {
33         fill(dep.begin(), dep.end(), -1);
34         queue<int> q;
35         dep[s] = 0;
36         q.push(s);
37         while (!q.empty())
38         {
39             int u = q.front();
40             q.pop();
41             for (int id : g[u])
42                 if (e[id].cap && dep[e[id].to] < 0)
43                 {
44                     dep[e[id].to] = dep[u] + 1;
```

```

45         q.push(e[id].to);
46     }
47 }
48 return dep[t] >= 0;
49 }
50
51 ll dfs(int u, int t, ll f)
52 {
53     if (u == t)
54         return f;
55     for (int &i = cur[u]; i < (int)g[u].size(); i++)
56     {
57         int id = g[u][i], v = e[id].to;
58         if (!e[id].cap || dep[v] != dep[u] + 1)
59             continue;
60         ll d = dfs(v, t, min(f, e[id].cap));
61         if (d)
62         {
63             e[id].cap -= d;
64             e[id ^ 1].cap += d;
65             return d;
66         }
67     }
68     return 0;
69 }
70
71 // Returns additional flow; reuse only with the same source/sink.
72 ll flow(int s, int t, ll limit = LLONG_MAX)
73 {
74     assert(s != t && limit >= 0);
75     ll ans = 0;
76     while (ans < limit && bfs(s, t))
77     {
78         fill(cur.begin(), cur.end(), 0);
79         while (ans < limit)
80         {
81             ll d = dfs(s, t, limit - ans);
82             if (!d)
83                 break;
84             ans += d;
85         }
86     }
87     return ans;
88 }
89
90 ll used(int id) const
91 {
92     return e[id].initial - e[id].cap;
93 }
94
95 vector<int> cut(int s)

```



```

96     {
97         bfs(s, s);
98         vector<int> ans;
99         for (int u = 1; u <= n; u++)
100             if (dep[u] >= 0)
101                 ans.push_back(u);
102         return ans;
103     }
104 };

```

11.2 势能最小费用最大流

$\text{add}(u, v, \text{cap}, \text{cost})$ 返回边编号, $\text{flow}(s, t, \text{limit})$ 返回新增流量及 128 位费用, $\text{used}(id)$ 返回原边流量。允许负费用, 初始残量网络必须没有负环; 检测到负环抛异常。Bellman-Ford 初始化 $O(VE)$, 每次增广 $O(E \log V)$; 增广次数不能简单假设为 $O(V)$ 。另要求 $\sum(\text{cap} \times |\text{cost}|) < 2^{120}$, 防止总费用超过 int128 。

```

1 struct MinCostFlow
2 {
3     using ll = long long;
4     using i128 = __int128_t;
5
6     struct Edge
7     {
8         int from, to;
9         ll cap, cost, initial;
10    };
11
12    int n;
13    vector<Edge> e;
14    vector<vector<int>>> g;
15
16    MinCostFlow(int n) : n(n), g(n + 1)
17    {
18    }
19
20    int add(int u, int v, ll cap, ll cost)
21    {
22        assert(cap >= 0 && cost != LLONG_MIN);
23        int id = (int)e.size();
24        e.push_back({u, v, cap, cost, cap});
25        e.push_back({v, u, 0, -cost, 0});
26        g[u].push_back(id);
27        g[v].push_back(id + 1);
28        return id;
29    }
30
31    ll used(int id) const
32    {
33        return e[id].initial - e[id].cap;
34    }
35
36    // Negative costs allowed; initial residual graph must have no negative cycle.
37    pair<ll, i128> flow(int s, int t, ll limit = LLONG_MAX)
38    {
39        assert(s != t && limit >= 0);

```

```

40  const i128 inf = i128(1) << 120;
41  vector<i128> h(n + 1, 0), d(n + 1);
42  vector<int> pre(n + 1);
43  // Super-source Bellman-Ford: feasible potentials for every vertex.
44  for (int it = 0; it < n; it++)
45  {
46      bool changed = false;
47      for (auto &a : e)
48          if (a.cap && h[a.to] > h[a.from] + a.cost)
49          {
50              h[a.to] = h[a.from] + a.cost;
51              changed = true;
52          }
53      if (!changed)
54          break;
55      if (it == n - 1)
56          throw invalid_argument("negative cycle");
57  }
58  ll f = 0;
59  i128 cost = 0;
60  while (f < limit)
61  {
62      fill(d.begin(), d.end(), inf);
63      d[s] = 0;
64      priority_queue<pair<i128, int>,
65                    vector<pair<i128, int>>,
66                    greater<pair<i128, int>>>
67          q;
68      q.push({0, s});
69      while (!q.empty())
70      {
71          auto [du, u] = q.top();
72          q.pop();
73          if (du != d[u])
74              continue;
75          for (int id : g[u])
76          {
77              auto &a = e[id];
78              i128 nd = du + a.cost + h[u] - h[a.to];
79              if (a.cap && nd < d[a.to])
80              {
81                  d[a.to] = nd;
82                  pre[a.to] = id;
83                  q.push({nd, a.to});
84              }
85          }
86      }
87      if (d[t] == inf)
88          break;
89      for (int u = 1; u <= n; u++)
90          if (d[u] != inf)
91              h[u] += d[u];
92      ll take = limit - f;
93      for (int u = t; u != s; u = e[pre[u]].from)
94          take = min(take, e[pre[u]].cap);
95      for (int u = t; u != s; u = e[pre[u]].from)

```

```

96         {
97             int id = pre[u];
98             e[id].cap -= take;
99             e[id ^ 1].cap += take;
100             cost += i128(take) * e[id].cost;
101         }
102         f += take;
103     }
104     return {f, cost};
105 }
106 };

```

11.3 上下界可行循环流

$\text{add}(u, v, lo, hi)$, $0 \leq lo \leq hi$; $\text{solve}()$ 仅调用一次, 返回可行性, 仅成功后读取 $\text{used}(id)$ 。约束为每点流量守恒; 不是带源汇最大流。所有上下界及其总和须在 long long 范围内。依赖 Dinic。

```

1 struct BoundedCirculation
2 {
3     using ll = long long;
4     int n;
5     Dinic g;
6     vector<ll> balance, lower;
7     vector<int> ids;
8     bool solved = false, feasible = false;
9
10    BoundedCirculation(int n) : n(n), g(n + 2), balance(n + 1)
11    {
12    }
13
14    int add(int u, int v, ll lo, ll hi)
15    {
16        assert(!solved && 0 <= lo && lo <= hi);
17        int id = ids.size();
18        ids.push_back(g.add(u, v, hi - lo));
19        lower.push_back(lo);
20        balance[u] -= lo;
21        balance[v] += lo;
22        return id;
23    }
24
25    bool solve()
26    {
27        assert(!solved);
28        solved = true;
29        ll need = 0;
30        for (int u = 1; u <= n; u++)
31            if (balance[u] > 0)
32            {
33                g.add(n + 1, u, balance[u]);
34                need += balance[u];
35            }
36        else if (balance[u] < 0)

```

```
37         g.add(u, n + 2, -balance[u]);
38     return feasible = (g.flow(n + 1, n + 2) == need);
39 }
40
41 ll used(int id) const
42 {
43     assert(feasible);
44     return lower[id] + g.used(ids[id]);
45 }
46 };
```

第十二章 最大权闭合子图

12.1 最大权闭合子图与选点方案

`maximum_closure(weight, dependencies)` 返回 (最大权值, 选中点列表), `weight` 为 0-based `vector<long long>`, 图中点号 `1..n`。依赖边 `(u,v)` 表示选 `u` 必须选 `v`, 允许环、自环与重边; 允许空选集, 因此答案不小于 0。依赖 `flow.hpp` 中的 `Dinic`。正权连源、负权连汇, 依赖边容量取正权总和加 1; 要求所有正权之和严格小于 `LLONG_MAX`。负权先提升到 `int128` 再取负, 并截断到依赖边容量, 因此 `LLONG_MIN` 也可处理。返回残量图源点可达的原顶点, 权值为正权总和减最小割; 超大负权边不会出现在最优割中, 截断不改变原问题最优值。每次调用重建网络。复杂度沿用 `Dinic`: 一般 $O(V^2E)$, 空间 $O(V+E)$, 增广保持递归。返回一种最优闭合集, 不承诺最大或最小选点个数。

```
1 inline pair<long long, vector<int>>
2 maximum_closure(const vector<long long> &weight,
3                 const vector<pair<int, int>> &dependencies)
4 {
5     using I = __int128_t;
6     int n = weight.size(), s = n + 1, t = n + 2;
7     I positive = 0;
8     for (long long w : weight)
9         if (w > 0)
10             positive += w;
11     assert(positive < LLONG_MAX);
12     long long inf = (long long)positive + 1;
13     Dinic graph(n + 2);
14     for (int u = 1; u <= n; u++)
15     {
16         long long w = weight[u - 1];
17         if (w > 0)
18             graph.add(s, u, w);
19         else if (w < 0)
20             graph.add(u, t, (long long)min(-I(w), I(inf)));
21     }
22     for (auto [u, v] : dependencies)
23     {
24         assert(1 <= u && u <= n && 1 <= v && v <= n);
25         graph.add(u, v, inf);
26     }
27     long long answer = (long long)positive - graph.flow(s, t);
28     vector<int> selected;
29     for (int u : graph.cut(s))
30         if (u <= n)
31             selected.push_back(u);
32     return {answer, selected};
33 }
```

第十三章 图论

13.1 非负权最短路

`add(u,v,w)`、`run(s)`，`dis` 为距离，`pre` 为前驱，`path(t)` 返回点序列。不可达用 `LLONG_MAX`；要求所有有限最短距离严格小于该哨兵， $w \geq 0$ 。 $O((V+E)\log V)$ 。

```
1 struct Dijkstra
2 {
3     using ll = long long;
4     static constexpr ll inf = LLONG_MAX;
5     int n;
6     vector<vector<pair<int, ll>>> g;
7     vector<ll> dis;
8     vector<int> pre;
9
10    Dijkstra(int n) : n(n), g(n + 1)
11    {
12    }
13
14    void add(int u, int v, ll w)
15    {
16        assert(w >= 0);
17        g[u].push_back({v, w});
18    }
19
20    void run(int s)
21    {
22        dis.assign(n + 1, inf);
23        pre.assign(n + 1, -1);
24        dis[s] = 0;
25        priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<pair<ll, int>>> q;
26        q.push({0, s});
27        while (!q.empty())
28        {
29            auto [d, u] = q.top();
30            q.pop();
31            if (d != dis[u])
32                continue;
33            for (auto [v, w] : g[u])
34                if (w < inf - d && d + w < dis[v])
35                {
36                    dis[v] = d + w;
37                    pre[v] = u;
38                    q.push({dis[v], v});
39                }
40        }
41    }
42
43    vector<int> path(int t) const
44    {
```

```

45     if (dis[t] == inf)
46         return {};
47     vector<int> ans;
48     for (; t != -1; t = pre[t])
49         ans.push_back(t);
50     reverse(ans.begin(), ans.end());
51     return ans;
52 }
53 };

```

13.2 强连通分量

add(u,v)、run(), bel[1..n] 为分量编号, cnt 为数量; 分量编号按缩点图拓扑序递增。递归 Kosaraju, 按用户要求保留 DFS 写法; $O(V+E)$, 同时存正反图。

```

1 struct SCC
2 {
3     int n, cnt = 0;
4     vector<vector<int>> g, rg;
5     vector<int> bel;
6
7     SCC(int n) : n(n), g(n + 1), rg(n + 1)
8     {
9     }
10
11     void add(int u, int v)
12     {
13         g[u].push_back(v);
14         rg[v].push_back(u);
15     }
16
17     void dfs1(int u, vector<int> &vis, vector<int> &order)
18     {
19         vis[u] = 1;
20         for (int v : g[u])
21         {
22             if (!vis[v])
23                 dfs1(v, vis, order);
24         }
25         order.push_back(u);
26     }
27
28     void dfs2(int u)
29     {
30         bel[u] = cnt;
31         for (int v : rg[u])
32         {
33             if (!bel[v])
34                 dfs2(v);
35         }
36     }
37
38     // Recursive Kosaraju; component IDs are in topological order, 1..cnt.
39     void run()
40     {

```

```

41     vector<int> vis(n + 1), order;
42     cnt = 0;
43     bel.assign(n + 1, 0);
44     for (int s = 1; s <= n; s++)
45     {
46         if (!vis[s])
47             dfs1(s, vis, order);
48     }
49     reverse(order.begin(), order.end());
50     for (int s : order)
51     {
52         if (!bel[s])
53         {
54             ++cnt;
55             dfs2(s);
56         }
57     }
58 }
59 };

```

13.3 2-SAT 与任意可行方案

add(x,a,y,b) 添加 (x=a) 或 (y=b), 变量 1..n; solve() 返回可满足性, 仅成功时读取 ans。依赖同文件 SCC; $O(V+E)$, 不保证字典序最小。

```

1 struct TwoSAT
2 {
3     int n;
4     SCC g;
5     vector<int> ans;
6
7     TwoSAT(int n) : n(n), g(2 * n)
8     {
9     }
10
11     int id(int x, bool value) const
12     {
13         return 2 * x - 1 + value;
14     }
15
16     void add(int x, bool a, int y, bool b)
17     {
18         g.add(id(x, !a), id(y, b));
19         g.add(id(y, !b), id(x, a));
20     }
21
22     bool solve()
23     {
24         g.run();
25         ans.assign(n + 1, 0);

```



```

26     for (int x = 1; x <= n; x++)
27     {
28         if (g.bel[id(x, 0)] == g.bel[id(x, 1)])
29             return false;
30         ans[x] = g.bel[id(x, 1)] > g.bel[id(x, 0)];
31     }
32     return true;
33 }
34 };

```

13.4 Hopcroft–Karp 最大匹配与最小点覆盖

左右两侧分别编号 $1..n$ 、 $1..m$ ，可有空侧及重边； $\text{add}(u,v)$ 加边， $\text{solve}()$ 返回最大匹配总数， $l[u]$ 与 $r[v]$ 为匹配端点，0 表示未匹配。保留已有合法匹配，可加边后再次 solve ；不支持删边或手动更改匹配数组。BFS 用 $\text{dep}[0]$ 记录最短终点层，递归 DFS 只沿最短层增广，并使用当前弧。 $V=n+m$ ，单次 solve 最坏 $O((V+E)\sqrt{V})$ ，空间 $O(V+E)$ 。 $\text{cover}()$ 在 solve 后、下一次改图前调用，返回 (左侧点列表, 右侧点列表)，为最小点覆盖；其补集是最大独立集。覆盖与匹配方案均可直接输出。

```

1 struct BipartiteMatching
2 {
3     int n, m;
4     vector<vector<int>> g;
5     vector<int> l, r, dep, cur;
6
7     BipartiteMatching(int n, int m)
8         : n(n), m(m), g(n + 1), l(n + 1), r(m + 1), dep(n + 1), cur(n + 1)
9     {
10    }
11
12    void add(int u, int v)
13    {
14        assert(1 <= u && u <= n && 1 <= v && v <= m);
15        g[u].push_back(v);
16    }
17
18    bool bfs()
19    {
20        queue<int> q;
21        fill(dep.begin(), dep.end(), n + 1);
22        fill(cur.begin(), cur.end(), 0);
23        for (int u = 1; u <= n; u++)
24        {
25            if (!l[u])
26            {
27                dep[u] = 0;
28                q.push(u);
29            }
30        }
31        while (!q.empty())
32        {
33            int u = q.front();
34            q.pop();
35            if (dep[u] >= dep[0])
36                continue;

```

```

37     for (int v : g[u])
38     {
39         int next = r[v];
40         if (dep[next] != n + 1)
41             continue;
42         dep[next] = dep[u] + 1;
43         if (next)
44             q.push(next);
45     }
46 }
47 return dep[0] != n + 1;
48 }
49
50 bool dfs(int u)
51 {
52     if (!u)
53         return true;
54     for (int &i = cur[u]; i < (int)g[u].size(); i++)
55     {
56         int v = g[u][i];
57         if (dep[r[v]] == dep[u] + 1 && dfs(r[v]))
58         {
59             l[u] = v;
60             r[v] = u;
61             return true;
62         }
63     }
64     dep[u] = n + 1;
65     return false;
66 }
67
68 // Retains an existing matching; edges may be added between calls.
69 int solve()
70 {
71     int answer = 0;
72     for (int u = 1; u <= n; u++)
73         answer += (l[u] != 0);
74     while (bfs())
75     {
76         for (int u = 1; u <= n; u++)
77         {
78             if (!l[u] && dfs(u))
79                 answer++;
80         }
81     }
82     return answer;
83 }
84
85 // Call after solve(), before adding more edges. Returns left/right vertex IDs.
86 pair<vector<int>, vector<int>> cover() const
87 {
88     vector<int> left, right;
89     for (int u = 1; u <= n; u++)
90     {
91         if (dep[u] == n + 1)
92             left.push_back(u);

```

```

93     }
94     for (int v = 1; v <= m; v++)
95     {
96         if (r[v] && dep[r[v]] != n + 1)
97             right.push_back(v);
98     }
99     return {left, right};
100 }
101 };

```

13.5 割点与桥

无向多重图，点号 1..n；add/Insert(u,v) 返回逻辑无向边的 0-based 编号，自环和重边均可输入。run/Run 后 cut[u] 为割点标记，bridge[id] 为桥标记，components 为原图连通块数。delta[u] 是独立删除顶点 u 及其关联边后增加的连通块数，剩余块数为 components+delta[u]；孤立点 delta=-1，普通非割点为 0，割点为正。根的增量为 DFS 子树数减 1；非根统计满足 low[child]>=dfn[u] 的子树数。n=1 的点删除后剩余 0 块。DFS 按父边号跳过，支持重边、非连通图和重复运行。复杂度 $O(n+m)$ ，递归 DFS；新增边后需重新 run。每个删点结果都针对原图，不表示逐次删除。

```

1 struct Lowlink
2 {
3     int n, timer = 0, components = 0;
4     vector<vector<pair<int, int>>> g;
5     vector<int> dfn, low, cut, bridge, delta;
6     int edges = 0;
7
8     Lowlink(int n) : n(n), g(n + 1)
9     {
10    }
11
12     int add(int u, int v)
13     {
14         g[u].push_back({v, edges});
15         g[v].push_back({u, edges});
16         return edges++;
17    }

```

```

19 void dfs(int u, int pe)
20 {
21     dfn[u] = low[u] = ++timer;
22     int children = 0;
23     for (auto [v, id] : g[u])
24     {
25         if (id == pe)
26             continue;
27         if (!dfn[v])
28         {
29             ++children;
30             dfs(v, id);
31             low[u] = min(low[u], low[v]);
32             if (low[v] > dfn[u])
33                 bridge[id] = 1;
34             if (pe != -1 && low[v] >= dfn[u])
35             {
36                 cut[u] = 1;
37                 delta[u]++;
38             }
39         }
40         else
41             low[u] = min(low[u], dfn[v]);
42     }
43     if (pe == -1 && children > 1)
44         cut[u] = 1;
45     if (pe == -1)
46         delta[u] = children - 1;
47 }
48
49 void run()
50 {
51     timer = 0;
52     components = 0;
53     dfn.assign(n + 1, 0);
54     low = dfn;
55     cut = dfn;
56     delta = dfn;
57     bridge.assign(edges, 0);
58     for (int u = 1; u <= n; u++)
59         if (!dfn[u])
60         {
61             components++;
62             dfs(u, -1);
63         }
64 }
65 };

```

第十四章 字典序最小 2-SAT

14.1 递归染色与回滚求最小赋值

构造 LexTwoSAT(n), 变量号 $1..n$ 。add(x,a,y,b) 加入子句 $(x=a) \text{ OR } (y=b)$, force(x,v) 强制变量值; solve() 成功后 answer[$1..n$] 是按变量编号、0 优先于 1 的字典序最小赋值。失败时 answer 为空, $n=0$ 时可满足, answer[0] 始终仅为占位。每个未确定变量先尝试 0, 递归传播所有蕴含; 冲突时仅撤销本次新增标记, 再试 1。history 是撤销记录, 不替代递归调用栈。可重复求解, 追加子句会清空旧答案; marked/history 仅为内部工作区, 失败时不可当作方案。内部 $\text{id}(x,v)=2(x-1)+v$, 从 0 开始, 可用异或 1 取反; 不要混用另一份 SCC 2-SAT 的 1-based 字面量编号。最坏 $O(n(n+m))$ 时间、 $O(n+m)$ 空间, 递归栈 $O(n)$; 100000 变量测试是特定蕴含链, 不代表一般大规模输入的性能保证。HDU1814 完整程序本地对拍, 在线待补。

```
1 struct LexTwoSAT
2 {
3     int n;
4     vector<vector<int>> g;
5     vector<int> marked, history, answer;
6
7     LexTwoSAT(int n) : n(n), g(2 * n), marked(2 * n)
8     {
9         assert(n >= 0);
10    }
11
12    int id(int x, bool value) const
13    {
14        assert(1 <= x && x <= n);
15        return 2 * (x - 1) + value;
16    }
17
18    // (x == a) OR (y == b). Internal literal IDs are 0-based.
19    void add(int x, bool a, int y, bool b)
20    {
21        int u = id(x, a), v = id(y, b);
22        g[u ^ 1].push_back(v);
23        g[v ^ 1].push_back(u);
24        answer.clear();
25    }
26
27    void force(int x, bool value)
28    {
29        add(x, value, x, value);
30    }
```

```
32 bool paint(int u)
33 {
34     if (marked[u ^ 1])
35         return false;
36     if (marked[u])
37         return true;
38     marked[u] = 1;
39     history.push_back(u);
40     for (int v : g[u])
41         if (!paint(v))
42             return false;
43     return true;
44 }
45
46 bool solve()
47 {
48     fill(marked.begin(), marked.end(), 0);
49     history.clear();
50     answer.clear();
51     for (int x = 1; x <= n; x++)
52     {
53         int u = id(x, false);
54         if (marked[u] || marked[u ^ 1])
55             continue;
56         int checkpoint = history.size();
57         if (!paint(u))
58         {
59             while ((int)history.size() > checkpoint)
60             {
61                 marked[history.back()] = 0;
62                 history.pop_back();
63             }
64             if (!paint(u ^ 1))
65                 return false;
66         }
67     }
68     answer.assign(n + 1, 0);
69     for (int x = 1; x <= n; x++)
70         answer[x] = marked[id(x, true)];
71     return true;
72 }
73 };
```

第十五章 有向欧拉路

15.1 有向图字典序最小欧拉路

构造 `DirectedEuler(n)`, $n \geq 1$, 点号 $1..n$; `add(u,v)` 返回从 0 开始的逻辑边编号, 支持自环与重边。`run()` 返回是否存在欧拉路; 参数 `start` 默认 0, `lexicographic` 默认 `true`。`start=0` 自动选择最小可行起点, 否则固定起点; 孤立点不影响有边部分的连通性。成功后 `vertices` 长度为 $m+1$, `edge_ids` 长度为 m , 第 i 条原边从 `vertices[i]` 指向 `vertices[i+1]`; 失败清空两数组。无边时返回单点路径, 自动起点为 1。可重复 `run`, `add` 会清空旧方案。默认求顶点序列字典序最小方案; 关闭排序后只保证任意方案, 邻接顺序可能保留上次排序。开启排序总时间 $O(n+m \log(m+1))$, 否则 $O(n+m)$, 空间 $O(n+m)$ 。DFS 递归深度可能达到 $m+1$, 不是 n 。度数检查后, 以消费边数判定是否遍历全部非孤立部分, 逆后序形成连续走法。小图全部走法对拍同时验证存在性、字典序及边编号, P7771 双风格完整程序本地验证, 在线待补。

```
1 struct DirectedEuler
2 {
3     int n;
4     vector<vector<pair<int, int>>>> g;
5     vector<pair<int, int>> edges;
6     vector<int> in, out, next, vertices, edge_ids;
7
8     DirectedEuler(int n) : n(n), g(n + 1), in(n + 1), out(n + 1), next(n + 1)
9     {
10         assert(n > 0);
11     }
12
13     int add(int u, int v)
14     {
15         assert(1 <= u && u <= n && 1 <= v && v <= n);
16         int id = edges.size();
17         edges.push_back({u, v});
18         g[u].push_back({v, id});
19         out[u]++;
20         in[v]++;
21         vertices.clear();
22         edge_ids.clear();
23         return id;
24     }
25
26     void dfs(int u, int incoming)
27     {
28         while (next[u] < (int)g[u].size())
29         {
30             auto [v, id] = g[u][next[u]++];
31             dfs(v, id);
32         }
33         vertices.push_back(u);
34         if (incoming != -1)
35             edge_ids.push_back(incoming);
36     }
```

```

38 // start=0 chooses the smallest feasible start. Isolated vertices are ignored.
39 bool run(int start = 0, bool lexicographic = true)
40 {
41     assert(0 <= start && start <= n);
42     vertices.clear();
43     edge_ids.clear();
44     int source = 0, sink = 0;
45     for (int u = 1; u <= n; u++)
46     {
47         int delta = out[u] - in[u];
48         if (delta == 1 && !source)
49             source = u;
50         else if (delta == -1 && !sink)
51             sink = u;
52         else if (delta != 0)
53             return false;
54     }
55     if (bool(source) != bool(sink))
56         return false;
57     if (source)
58     {
59         if (start && start != source)
60             return false;
61         start = source;
62     }
63     if (!start)
64     {
65         start = 1;
66         for (int u = 1; u <= n; u++)
67             if (out[u])
68             {
69                 start = u;
70                 break;
71             }
72     }
73     if (lexicographic)
74         for (int u = 1; u <= n; u++)
75             sort(g[u].begin(), g[u].end());
76     fill(next.begin(), next.end(), 0);
77     dfs(start, -1);
78     if (edge_ids.size() != edges.size())
79     {
80         vertices.clear();
81         edge_ids.clear();
82         return false;
83     }
84     reverse(vertices.begin(), vertices.end());
85     reverse(edge_ids.begin(), edge_ids.end());
86     return true;
87 }
88 };

```


第十六章 字典序单词链

16.1 字典序最小单词链

`word_chain(words)` 返回可空的编号序列 (optional), 有解时为原输入的 0-based 单词编号顺序, 每个单词恰用一次; 相邻单词首尾字母相接, 单词序列字典序最小。无解返回 `nullopt`, 空输入返回有值的空序列。单词必须非空且仅含小写 a-z, 重复词视为不同编号。先对原编号按单词排序, 按此顺序加边, 再调用 `DirectedEuler.run(0,false)`; 不可让底层重新按终点排序。欧拉回路从最小出现首字母开始, 非回路起点由度数差确定。O(m log m) 次字符串比较及 O(L+m) 其他工作, L 为总字符数; 空间 O(m), 递归栈 O(m)。依赖 `directed_euler.hpp`; 传统版调用 `Word_Chain`, 内部使用 `Directed_Euler<26>`。全排列对拍、重复词/前缀词、20 万相同词及 POJ2337 完整输出本地验证, 在线待补。

```
1 // Lowercase nonempty words. Returns original 0-based IDs; duplicates stay distinct.
2 inline optional<vector<int>> word_chain(const vector<string> &words)
3 {
4     int n = words.size();
5     vector<int> order(n);
6     iota(order.begin(), order.end(), 0);
7     for (const auto &word : words)
8     {
9         assert(!word.empty());
10        for (char c : word)
11            assert('a' <= c && c <= 'z');
12    }
13    sort(order.begin(),
14         order.end(),
15         [&](int a, int b)
16         {
17             if (words[a] != words[b])
18                 return words[a] < words[b];
19             return a < b;
20         });
21    DirectedEuler graph(26);
22    for (int id : order)
23        graph.add(words[id].front() - 'a' + 1, words[id].back() - 'a' + 1);
24    // Preserve word order in each adjacency list; do not sort by destination.
25    if (!graph.run(0, false))
26        return nullopt;
27    vector<int> result;
28    for (int id : graph.edge_ids)
29        result.push_back(order[id]);
30    return result;
31 }
```

第十七章 无向欧拉路

17.1 无向图欧拉路与边方向

构造 `UndirectedEuler(n)`, $n \geq 1$, 点号 $1..n$; `add(u,v)` 返回从 0 开始的逻辑边号, 自环和重边均保留。`run()` 返回是否存在欧拉路, 参数 `start` 默认 0 (自动), `lexicographic` 默认 `true`。自动选最小可行起点; 指定起点须满足奇度条件且位于有边部分。成功得到 $m+1$ 个 `vertices` 和 m 个 `edge_ids`, 失败清空两数组。第 i 条边方向是 `vertices[i]` 到 `vertices[i+1]`, 与 `edges[edge_ids[i]]` 比较可判断是否反向, 自环任选方向。`used` 按逻辑边号标记, 使两条半边共同只消费一次; 自环贡献度数 2。孤立点不影响有边部分; 无边图返回指定点或点 1 的单点序列。默认求顶点序列字典序最小方案; 关闭排序只保证任意方案, 可能保留上次排序。可重复 `run`, 加边清空旧方案。排序开启时 $O(n+m \log(m+1))$, 否则 $O(n+m)$, 空间 $O(n+m)$, DFS 递归深度可能达到 $m+1$ 。全小图走法、原边号与方向独立验证, P2731 和 SGU101 完整程序本地验证; 在线待补。

```
1 struct UndirectedEuler
2 {
3     int n;
4     vector<vector<pair<int, int>>> g;
5     vector<pair<int, int>> edges;
6     vector<int> next, used, vertices, edge_ids;
7
8     UndirectedEuler(int n) : n(n), g(n + 1), next(n + 1)
9     {
10         assert(n > 0);
11     }
12
13     int add(int u, int v)
14     {
15         assert(1 <= u && u <= n && 1 <= v && v <= n);
16         int id = edges.size();
17         edges.push_back({u, v});
18         g[u].push_back({v, id});
19         g[v].push_back({u, id});
20         vertices.clear();
21         edge_ids.clear();
22         return id;
23     }
24
25     void dfs(int u, int incoming)
26     {
27         while (next[u] < (int)g[u].size())
28         {
29             auto [v, id] = g[u][next[u]++];
30             if (used[id])
31                 continue;
32             used[id] = 1;
33             dfs(v, id);
34         }
35         vertices.push_back(u);
36         if (incoming != -1)
37             edge_ids.push_back(incoming);
38     }
```

```

40 // start=0 chooses the smallest feasible start. Isolated vertices are ignored.
41 bool run(int start = 0, bool lexicographic = true)
42 {
43     assert(0 <= start && start <= n);
44     vertices.clear();
45     edge_ids.clear();
46     vector<int> odd;
47     for (int u = 1; u <= n; u++)
48         if (g[u].size() % 2)
49             odd.push_back(u);
50     if (!odd.empty() && odd.size() != 2)
51         return false;
52     if (!odd.empty())
53     {
54         if (start && start != odd[0] && start != odd[1])
55             return false;
56         if (!start)
57             start = odd[0];
58     }
59     if (!start)
60     {
61         start = 1;
62         for (int u = 1; u <= n; u++)
63             if (!g[u].empty())
64             {
65                 start = u;
66                 break;
67             }
68     }
69     if (lexicographic)
70         for (int u = 1; u <= n; u++)
71             sort(g[u].begin(), g[u].end());
72     fill(next.begin(), next.end(), 0);
73     used.assign(edges.size(), 0);
74     dfs(start, -1);
75     if (edge_ids.size() != edges.size())
76     {
77         vertices.clear();
78         edge_ids.clear();
79         return false;
80     }
81     reverse(vertices.begin(), vertices.end());
82     reverse(edge_ids.begin(), edge_ids.end());
83     return true;
84 }
85 };

```

第十八章 混合图欧拉定向

18.1 欧拉回路与指定端点定向

`mixed_euler_orientation(n, edges, start, finish)` 返回可空的定向边序列。每条输入为 $\{u, v, \text{type}\}$, $\text{type}=1$ 固定 u 到 v , $\text{type}=0$ 可反转; 点号 $1..n$, $n \geq 1$ 。省略两个端点或均传 0 时请求任意起点的欧拉回路; 两个正端点请求指定起终点的欧拉路径, 相等时为指定起点回路。无解返回 `nullopt`, 有解返回与原边顺序对应的端点对; 无边图可返回有值空序列。忽略无关孤立点, 但正端点须在有边部分。先任意定向可反转边, 扣除目标出入度差; 每条反转将两端差值分别改变 -2 和 $+2$, 奇数差无解, 否则用容量 1 的反转边流调整。检查底层无向连通性后, 满流即得到所需方向。最大流图 $n+2$ 点、至多 $m+n$ 条正向边, 总时间可按 Dinic 的 $O((n+2)^2(m+n))$ 上界计, 空间 $O(n+m)$ 。依赖 `flow.hpp`; 返回边交给 `DirectedEuler` 可生成连续走法, 指定端点时保持指定起点。定向不保证字典序最小。

```
1 // Edge = {u,v,type}: type 1 is fixed u->v; type 0 may be reversed.
2 // start=finish=0 requests a circuit; positive endpoints request a specified trail.
3 inline optional<vector<pair<int, int>>> mixed_euler_orientation(
4     int n, const vector<array<int, 3>> &edges, int start = 0, int finish = 0)
5 {
6     assert(n > 0 && 0 <= start && start <= n && 0 <= finish && finish <= n);
7     assert(bool(start) == bool(finish));
8     vector<int> parent(n + 1), id;
9     vector<long long> degree(n + 1), balance(n + 1);
10    iota(parent.begin(), parent.end(), 0);
11    auto find = [&](auto &&self, int u) -> int
12    {
13        if (parent[u] != u)
14            parent[u] = self(self, parent[u]);
15        return parent[u];
16    };
17    Dinic flow(n + 2);
18    vector<pair<int, int>> direction;
19    for (auto [u, v, type] : edges)
20    {
21        assert(1 <= u && u <= n && 1 <= v && v <= n);
22        assert(type == 0 || type == 1);
23        parent[find(find, u)] = find(find, v);
24        degree[u]++;
25        degree[v]++;
26        balance[u]++;
27        balance[v]--;
28        id.push_back(type == 0 ? flow.add(u, v, 1) : -1);
29        direction.push_back({u, v});
30    }
```

```

31     if (!edges.empty())
32     {
33         int root = find(find, edges[0][0]);
34         for (int u = 1; u <= n; u++)
35             if (degree[u] && find(find, u) != root)
36                 return nullopt;
37         if (start && (!degree[start] || !degree[finish]))
38             return nullopt;
39     }
40     if (start)
41     {
42         balance[start]--;
43         balance[finish]++;
44     }
45     int source = n + 1, sink = n + 2;
46     long long need = 0;
47     for (int u = 1; u <= n; u++)
48     {
49         if (balance[u] % 2)
50             return nullopt;
51         if (balance[u] > 0)
52         {
53             flow.add(source, u, balance[u] / 2);
54             need += balance[u] / 2;
55         }
56         else if (balance[u] < 0)
57             flow.add(u, sink, -balance[u] / 2);
58     }
59     if (flow.flow(source, sink) != need)
60         return nullopt;
61     for (int i = 0; i < (int)edges.size(); i++)
62         if (id[i] != -1 && flow.used(id[i]))
63             swap(direction[i].first, direction[i].second);
64     return direction;
65 }

```

18.2 任意起终点的混合欧拉路径

`mixed_euler_trail(n, edges)` 返回任意可行欧拉路径的定向边序列，格式同上一节。忽略边原方向统计度数，自环计 2；若奇度点为 0 个，请求回路；若恰有 2 个，尝试这两个端点的两种顺序；否则无解。最多两次定向求解。返回方向可交给 `DirectedEuler` 自动选择起点，恢复顶点与原边编号序列。枚举所有可反转边方向、全部固定端点及自由端点对拍；逐边检查固定边不变和连续走法，另有 10 万点长流量路径测试。POJ1637 双驱动本地验证，仅检查回路存在性；完整方向和开放路径依靠本地证据，在线待补。

```

1 // Any endpoints: zero odd degrees requests a circuit; two odd degrees tries both
2 // orders.
3 inline optional<vector<pair<int, int>>>
4 mixed_euler_trail(int n, const vector<array<int, 3>> &edges)
5 {
6     assert(n > 0);
7     vector<long long> degree(n + 1);
8     vector<int> odd;
9     for (auto [u, v, type] : edges)
10     {
11         assert(1 <= u && u <= n && 1 <= v && v <= n);
12         degree[u]++;
13         degree[v]++;
14     }
15     for (int u = 1; u <= n; u++)
16         if (degree[u] % 2)
17             odd.push_back(u);
18     if (odd.empty())
19         return mixed_euler_orientation(n, edges);
20     if (odd.size() != 2)
21         return nullopt;
22     auto answer = mixed_euler_orientation(n, edges, odd[0], odd[1]);
23     if (answer)
24         return answer;
25     return mixed_euler_orientation(n, edges, odd[1], odd[0]);
26 }
```

第十九章 图上异或行走

19.1 无向图最大异或行走

XorWalk(n) 建图，点号 1..n，边权为 unsigned long long，支持完整 64 位。add(u,v,w) 加无向边，允许重边、自环和零权；init(n) 清空图。build(root=1) 用递归 DFS 预处理 root 的连通块，query(u,v) 返回该块内 u 到 v 的最大边权异或和；任一点未被本次 build 覆盖则返回 nullopt，即使这两点在另一连通块相连。加入新边后必须重新 build；可更换根重复构建。空图可初始化，但不能 build。这里的行走允许重复点、重复边；不是最大异或简单路径。u=v 也可能通过绕环得到正值。DFS 距离 dist 给出一条树路径的异或值，每条已访问边形成 dist[u] xor dist[v] xor w 的闭合行走值，加入线性基；两点答案为 cycles.query(dist[u] xor dist[v])。build 后 vis、dist 可只读；cycles 表示环异或空间，不保留具体环或行走方案，不能将其非空子集标志解释为行走非空性。建图 O(n+m)，build 为 O(n+64m)，每次 query 为 O(64)，动态空间 O(n+m+64)。

```
1 struct XorWalk
2 {
3     using U = unsigned long long;
4     int n;
5     bool built = false;
6     vector<vector<pair<int, U>>> g;
7     vector<int> vis;
8     vector<U> dist;
9     XorBasis cycles;
10
11     XorWalk(int n)
12     {
13         init(n);
14     }
15
16     void init(int size)
17     {
18         assert(size >= 0);
19         n = size;
20         g.clear();
21         g.resize(size_t(n) + 1);
22         vis.assign(size_t(n) + 1, 0);
23         dist.assign(size_t(n) + 1, 0);
24         cycles.init();
25         built = false;
26     }
```

```

28 void add(int u, int v, U w)
29 {
30     assert(1 <= u && u <= n && 1 <= v && v <= n);
31     g[u].push_back({v, w});
32     g[v].push_back({u, w});
33     built = false;
34 }
35
36 void dfs(int u)
37 {
38     vis[u] = 1;
39     for (auto [v, w] : g[u])
40     {
41         if (!vis[v])
42         {
43             dist[v] = dist[u] ^ w;
44             dfs(v);
45         }
46         else
47             cycles.insert(dist[u] ^ dist[v] ^ w);
48     }
49 }
50
51 void build(int root = 1)
52 {
53     assert(1 <= root && root <= n);
54     fill(vis.begin(), vis.end(), 0);
55     fill(dist.begin(), dist.end(), 0);
56     cycles.init();
57     dfs(root);
58     built = true;
59 }
60
61 optional<U> query(int u, int v) const
62 {
63     assert(built && 1 <= u && u <= n && 1 <= v && v <= n);
64     if (!vis[u] || !vis[v])
65         return nullopt;
66     return cycles.query(dist[u] ^ dist[v]);
67 }
68 };

```


第二十章 树上算法

20.1 重链剖分与 LCA

add(u,v) 建无向树; build(root) 后读取 fa、dep、siz、son、top、dfn、rk。子树对应 [dfn[u],dfn[u]+siz[u]-1]。lca(u,v) 为 $O(\log n)$, path(u,v,work,edge) 分解路径; work(l,r) 仅适用于可交换聚合, 边权模式排除 LCA 位置。输入必须是连通无环树, 建树 $O(n)$, 两遍递归 DFS; 支持改变根后重复 build, dfn 会重新编号。

```
1 struct HLD
2 {
3     int n, timer = 0;
4     vector<vector<int>> g;
5     vector<int> fa, dep, siz, son, top, dfn, rk;
6
7     HLD(int n)
8         : n(n),
9           g(n + 1),
10          fa(n + 1),
11          dep(n + 1),
12          siz(n + 1),
13          son(n + 1),
14          top(n + 1),
15          dfn(n + 1),
16          rk(n + 1)
17     {
18     }
19
20     void add(int u, int v)
21     {
22         g[u].push_back(v);
23         g[v].push_back(u);
24     }
25
26     void dfs1(int u, int p)
27     {
28         fa[u] = p;
29         siz[u] = 1;
30         son[u] = 0;
31         for (int v : g[u])
32         {
33             if (v == p)
34                 continue;
35             dep[v] = dep[u] + 1;
36             dfs1(v, u);
37             siz[u] += siz[v];
38             if (!son[u] || siz[v] > siz[son[u]])
39                 son[u] = v;
40         }
41     }
42
43     void dfs2(int u, int t)
44     {
```

```

45     top[u] = t;
46     dfn[u] = ++timer;
47     rk[timer] = u;
48     if (son[u])
49         dfs2(son[u], t);
50     for (int v : g[u])
51     {
52         if (v != fa[u] && v != son[u])
53             dfs2(v, v);
54     }
55 }
56
57 // Connected tree only. Recursive DFS; subtree interval is [dfn, dfn+siz-1].
58 void build(int root = 1)
59 {
60     assert(1 <= root && root <= n);
61     timer = 0;
62     dep[root] = 0;
63     dfs1(root, root);
64     dfs2(root, root);
65     assert(timer == n);
66 }
67
68 int lca(int u, int v) const
69 {
70     while (top[u] != top[v])
71     {
72         if (dep[top[u]] < dep[top[v]])
73             swap(u, v);
74         u = fa[top[u]];
75     }
76     return dep[u] < dep[v] ? u : v;
77 }
78
79 // Commutative operations only; edge=true excludes the LCA's position.
80 template <class F> void path(int u, int v, F work, bool edge = false) const
81 {
82     while (top[u] != top[v])
83     {
84         if (dep[top[u]] < dep[top[v]])
85             swap(u, v);
86         work(dfn[top[u]], dfn[u]);
87         u = fa[top[u]];
88     }
89     if (dep[u] > dep[v])
90         swap(u, v);
91     if (dfn[u] + edge <= dfn[v])
92         work(dfn[u] + edge, dfn[v]);
93 }
94 };

```

第二十一章 离线最近公共祖先

21.1 离线 Tarjan LCA

构造 `OfflineLCA(n)`, `add(u,v)` 加无向树边, `add_query(u,v)` 登记查询并返回从 0 开始的编号, `run(root=1)` 后读取 `answer[id]`。输入必须是非空树, 点号 $1..n$; 允许同点、重复查询及零个查询。`run` 可重复调用或换根, 不清空查询; 追加查询后需重新 `run`, 旧答案在重算前不可按新根使用。DFS 与并查集查找均递归。并查集按大小合并并压缩路径, `ancestor[find(u)]` 保存已合并部分对应的当前树祖先; 不能直接拿并查集代表当 LCA。每个子树完成后合并到父点并更新 `ancestor`; 查询另一端已完成时, 从其集合读取祖先。总时间 $O((n+q) \alpha(n))$, 空间 $O(n+q)$ 。全部点对和根的父链对拍、50 万点链及 P3379、POJ1470 完整驱动本地验证, 在线待补。

```
1 struct OfflineLCA
2 {
3     int n;
4     vector<vector<int>>> g;
5     vector<vector<pair<int, int>>>> queries;
6     vector<int> parent, size, ancestor, done, answer;
7
8     OfflineLCA(int n)
9         : n(n),
10          g(n + 1),
11          queries(n + 1),
12          parent(n + 1),
13          size(n + 1),
14          ancestor(n + 1),
15          done(n + 1)
16     {
17     }
18
19     void add(int u, int v)
20     {
21         assert(1 <= u && u <= n && 1 <= v && v <= n);
22         g[u].push_back(v);
23         g[v].push_back(u);
24     }
25
26     int add_query(int u, int v)
27     {
28         assert(1 <= u && u <= n && 1 <= v && v <= n);
29         int id = answer.size();
30         answer.push_back(0);
31         queries[u].push_back({v, id});
32         queries[v].push_back({u, id});
33         return id;
34     }
```

```

36     int find(int u)
37     {
38         if (parent[u] != u)
39             parent[u] = find(parent[u]);
40         return parent[u];
41     }
42
43     void merge(int u, int v)
44     {
45         u = find(u);
46         v = find(v);
47         if (size[u] < size[v])
48             swap(u, v);
49         parent[v] = u;
50         size[u] += size[v];
51     }
52
53     void dfs(int u, int p)
54     {
55         for (int v : g[u])
56             if (v != p)
57             {
58                 dfs(v, u);
59                 merge(u, v);
60                 ancestor[find(u)] = u;
61             }
62         done[u] = 1;
63         for (auto [v, id] : queries[u])
64             if (done[v])
65                 answer[id] = ancestor[find(v)];
66     }
67
68     // Input must be a nonempty tree. Query IDs are 0-based.
69     void run(int root = 1)
70     {
71         assert(1 <= root && root <= n);
72         iota(parent.begin(), parent.end(), 0);
73         iota(ancestor.begin(), ancestor.end(), 0);
74         fill(size.begin(), size.end(), 1);
75         fill(done.begin(), done.end(), 0);
76         fill(answer.begin(), answer.end(), 0);
77         dfs(root, 0);
78     }
79 };

```

第二十二章 在线最近公共祖先

22.1 欧拉序与 ST 表 LCA

构造 EulerLCA(n), add(u,v) 加无向树边, build(root=1) 预处理; lca(u,v) 返回最近公共祖先, distance(u,v) 返回无权路径边数。输入必须为非空树, 点号 1..n; 允许同点查询和重复 build 换根。只可在 build 后查询, 加边会清除 ready 标志。递归 DFS 在进入顶点及从每个儿子返回时记录顶点, euler 长度为 2n-1, first[u] 是从 0 开始的首次出现位置; depth[root]=0。两点首次出现位置间深度最小的顶点就是 LCA, 用 ST 表保存该顶点。预处理 $O(n \log n)$ 时间和空间, 每次查询 $O(1)$ 。ST 内部下标从 0 开始; first、depth、euler 可直接读取。50 万点链、全部小树根与点对以及 P3379、POJ1330 双风格完整程序本地验证; 在线待补。

```
1 struct EulerLCA
2 {
3     int n;
4     bool ready = false;
5     vector<vector<int>> g, st;
6     vector<int> first, depth, euler, lg;
7
8     EulerLCA(int n) : n(n), g(n + 1), first(n + 1), depth(n + 1)
9     {
10
11
12     void add(int u, int v)
13     {
14         assert(1 <= u && u <= n && 1 <= v && v <= n);
15         g[u].push_back(v);
16         g[v].push_back(u);
17         ready = false;
18     }
19
20     int shallower(int u, int v) const
21     {
22         return depth[u] <= depth[v] ? u : v;
23     }
```

```

25 void dfs(int u, int parent)
26 {
27     first[u] = euler.size();
28     euler.push_back(u);
29     for (int v : g[u])
30         if (v != parent)
31             {
32                 depth[v] = depth[u] + 1;
33                 dfs(v, u);
34                 euler.push_back(u);
35             }
36 }
37
38 // Input must be a nonempty tree. Rebuilding may change the root.
39 void build(int root = 1)
40 {
41     assert(1 <= root && root <= n);
42     euler.clear();
43     depth[root] = 0;
44     dfs(root, 0);
45     int m = euler.size();
46     lg.assign(m + 1, 0);
47     for (int i = 2; i <= m; i++)
48         lg[i] = lg[i / 2] + 1;
49     st.assign(lg[m] + 1, {});
50     st[0] = euler;
51     for (int k = 1; k <= lg[m]; k++)
52     {
53         st[k].resize(m - (1 << k) + 1);
54         for (int i = 0; i + (1 << k) <= m; i++)
55             st[k][i] = shallower(st[k - 1][i], st[k - 1][i + (1 << (k - 1))]);
56     }
57     ready = true;
58 }
59
60 int lca(int u, int v) const
61 {
62     assert(ready && 1 <= u && u <= n && 1 <= v && v <= n);
63     int l = first[u], r = first[v];
64     if (l > r)
65         swap(l, r);
66     int k = lg[r - l + 1];
67     return shallower(st[k][l], st[k][r - (1 << k) + 1]);
68 }
69
70 int distance(int u, int v) const
71 {
72     int z = lca(u, v);
73     return depth[u] + depth[v] - 2 * depth[z];
74 }
75 };

```

第二十三章 倍增祖先与路径最值

23.1 倍增 LCA、祖先与最大边权

构造 `LiftingLCA(n)`, `add(u,v,w=1)` 加边, `build(root=1)` 预处理非空无向树, 点号 `1..n`。 `jump(u,k)` 返回 `k` 级祖先, `k` 非负, 超过根返回 `0`; `lca(u,v)` 求最近公共祖先; `distance` 返回路径边数, 不是权值和; `max_edge` 返回路径最大边权, 同点返回 `nullopt`, 其他情况返回有值的 `optional<long long>`。支持负权和 `LLONG_MIN`, `LOW` 只是最大值合并的单位元, 不表示空路径。DFS 递归, 根深度为 `0`; `up[k][u]` 与 `mx[k][u]` 是 `2` 的 `k` 次方长度向上段的祖先与最大边权, `0` 号节点为哨兵。可重复 `build` 换根; 修改图后必须重建。预处理 $O(n \log n)$ 时间与空间, 各查询 $O(\log n)$ 。不需要边权最值时, 可删去 `mx` 和 `max_edge` 相关部分。全小树点对/根、极值权重及 50 万点负权链本地验证; P3379、POJ1330 驱动只验证 LCA, 最大边权仍是本地证据。

```
1 struct LiftingLCA
2 {
3     using ll = long long;
4     static constexpr ll LOW = numeric_limits<ll>::lowest();
5     int n, levels;
6     bool ready = false;
7     vector<vector<pair<int, ll>>> g;
8     vector<vector<int>> up;
9     vector<vector<ll>> mx;
10    vector<int> depth;
11
12    LiftingLCA(int n) : n(n), g(n + 1), depth(n + 1)
13    {
14        assert(n > 0);
15        levels = 1;
16        while ((1LL << levels) <= n)
17            levels++;
18        up.assign(levels, vector<int>(n + 1));
19        mx.assign(levels, vector<ll>(n + 1, LOW));
20    }
21
22    void add(int u, int v, ll w = 1)
23    {
24        assert(1 <= u && u <= n && 1 <= v && v <= n);
25        g[u].push_back({v, w});
26        g[v].push_back({u, w});
27        ready = false;
28    }
```

```

30 void dfs(int u, int parent, ll weight)
31 {
32     up[0][u] = parent;
33     mx[0][u] = weight;
34     for (int k = 1; k < levels; k++)
35     {
36         int p = up[k - 1][u];
37         up[k][u] = up[k - 1][p];
38         mx[k][u] = max(mx[k - 1][u], mx[k - 1][p]);
39     }
40     for (auto [v, w] : g[u])
41         if (v != parent)
42         {
43             depth[v] = depth[u] + 1;
44             dfs(v, u, w);
45         }
46     }
47
48     // Nonempty tree only. Node 0 means no ancestor.
49     void build(int root = 1)
50     {
51         assert(1 <= root && root <= n);
52         depth[root] = 0;
53         dfs(root, 0, LOW);
54         ready = true;
55     }
56
57     int jump(int u, long long steps) const
58     {
59         assert(ready && 1 <= u && u <= n && steps >= 0);
60         if (steps > depth[u])
61             return 0;
62         for (int k = 0; k < levels; k++)
63             if (steps >> k & 1)
64                 u = up[k][u];
65         return u;
66     }

```



```

68  int lca(int u, int v) const
69  {
70      assert(ready && 1 <= u && u <= n && 1 <= v && v <= n);
71      if (depth[u] < depth[v])
72          swap(u, v);
73      u = jump(u, depth[u] - depth[v]);
74      if (u == v)
75          return u;
76      for (int k = levels - 1; k >= 0; k--)
77          if (up[k][u] != up[k][v])
78              {
79                  u = up[k][u];
80                  v = up[k][v];
81              }
82      return up[0][u];
83  }
84
85  int distance(int u, int v) const
86  {
87      return depth[u] + depth[v] - 2 * depth[lca(u, v)];
88  }
89
90  optional<ll> max_edge(int u, int v) const
91  {
92      assert(ready && 1 <= u && u <= n && 1 <= v && v <= n);
93      if (u == v)
94          return nullopt;
95      if (depth[u] < depth[v])
96          swap(u, v);
97      ll answer = LOW;
98      int delta = depth[u] - depth[v];
99      for (int k = 0; k < levels; k++)
100          if (delta >> k & 1)
101              {
102                  answer = max(answer, mx[k][u]);
103                  u = up[k][u];
104              }
105      if (u == v)
106          return answer;
107      for (int k = levels - 1; k >= 0; k--)
108          if (up[k][u] != up[k][v])
109              {
110                  answer = max(answer, max(mx[k][u], mx[k][v]));
111                  u = up[k][u];
112                  v = up[k][v];
113              }
114      return max(answer, max(mx[0][u], mx[0][v]));
115  }
116  };

```

第二十四章 树上路径交

24.1 两条树路径的交集

`path_intersection(a,b,c,d,depth,lca)` 返回路径 `a-b` 与 `c-d` 的公共路径，字段 `u`、`v` 是交集端点，`vertices` 是公共顶点数；空交返回 `{0,0,0}`，单点交的 `u=v`。点号从 1 开始，`depth` 数组与 `lca` 回调必须来自同一棵非空树、同一个根，深度按边数计算，不能传带权根距离。支持数组或 `vector` 深度表，回调接收两个点并返回 LCA。将 `c`、`d` 分别投影到 `a-b` 得 `p`、`q`；投影是 `lca(a,b)`、`lca(a,x)`、`lca(b,x)` 中最深者。若 `p≠q`，`c-d` 必经过 `p-q`；若 `p=q`，检验 `p` 是否在 `c-d` 上即可区分单点交与空交。不依赖顶点编号大小。常数 LCA 调用、 $O(1)$ 额外空间；使用 EulerLCA 时为 $O(1)$ ，HLD/倍增时为 $O(\log n)$ 。公共边数为 $\max(0, \text{vertices}-1)$ 。全小树根与四端点、随机换号树用路径点集独立验证；P3398 完整程序只检查是否相交，端点及数量由本地验证覆盖。

```
1 struct PathIntersection
2 {
3     int u, v, vertices;
4 };
5
6 // Same nonempty tree and root for depth[] and lca(). Empty result is {0,0,0}.
7 template <class Depth, class Lca>
8 PathIntersection
9 path_intersection(int a, int b, int c, int d, const Depth &depth, Lca lca)
10 {
11     int z = lca(a, b);
12     auto project = [&](int x)
13     {
14         int p = z;
15         for (int v : {lca(a, x), lca(b, x)})
16             if (depth[v] > depth[p])
17                 p = v;
18         return p;
19     };
20     auto distance = [&](int u, int v)
21     {
22         return 1LL * depth[u] + depth[v] - 2LL * depth[lca(u, v)];
23     };
24     int p = project(c), q = project(d);
25     if (p == q && distance(c, p) + distance(p, d) != distance(c, d))
26         return {0, 0, 0};
27     return {p, q, int(distance(p, q) + 1)};
28 }
```

第二十五章 点分治

25.1 点分治：树上距离点对计数

CentroidPairs(n), 点号 $1..n$; add(u,v,w) 建树, 允许森林但禁止环、重边、自环, 边权非负; 所有简单路径长度须能用 long long 表示。build() 预处理, 可重复调用; 改变边后须重建。count_leq(k) 统计距离 $\leq k$ 的无序不同顶点对, count_exact(k) 统计距离恰为 k 的点对; 负 k 返回 0, 零距离只统计不同顶点, 跨连通块不配对。parent[c] 是点分树父节点, 根为 0。all[c] 为该层分治块到重心的距离, 含重心距离 0; part[c] 为去掉重心后各子块的距离。预处理 $O(n \log^2 n)$, 空间 $O(n \log n)$, 每次计数 $O(n \log n)$; 固定树上多次查询可以复用排序结果。DFS 递归, 距离和比较采用减法避免溢出。不是动态点分树更新模板。

```
1 struct CentroidPairs
2 {
3     using ll = long long;
4     int n;
5     vector<vector<pair<int, ll>>> g;
6     vector<int> siz, removed, parent;
7     vector<vector<ll>> all;
8     vector<vector<vector<ll>>> part;
9
10    CentroidPairs(int n)
11        : n(n),
12          g(n + 1),
13          siz(n + 1),
14          removed(n + 1),
15          parent(n + 1),
16          all(n + 1),
17          part(n + 1)
18    {
19    }
20
21    void add(int u, int v, ll w)
22    {
23        assert(1 <= u && u <= n && 1 <= v && v <= n && u != v && w >= 0);
24        g[u].push_back({v, w});
25        g[v].push_back({u, w});
26    }
27
28    void size_dfs(int u, int p)
29    {
30        siz[u] = 1;
31        for (auto [v, w] : g[u])
32        {
33            if (v == p || removed[v])
34                continue;
35            size_dfs(v, u);
36            siz[u] += siz[v];
37        }
38    }
39
40    int centroid(int u, int p, int total)
41    {
```

```

42     for (auto [v, w] : g[u])
43     {
44         if (v != p && !removed[v] && siz[v] > total / 2)
45             return centroid(v, u, total);
46     }
47     return u;
48 }
49
50 void collect(int u, int p, ll d, vector<ll> &out)
51 {
52     out.push_back(d);
53     for (auto [v, w] : g[u])
54     {
55         if (v == p || removed[v])
56             continue;
57         assert(d <= numeric_limits<ll>::max() - w);
58         collect(v, u, d + w, out);
59     }
60 }
61
62 void decompose(int entry, int p)
63 {
64     size_dfs(entry, 0);
65     int c = centroid(entry, 0, siz[entry]);
66     removed[c] = 1;
67     parent[c] = p;
68     all[c] = {0};
69     for (auto [v, w] : g[c])
70     {
71         if (removed[v])
72             continue;
73         part[c].push_back({});
74         auto &d = part[c].back();
75         collect(v, c, w, d);
76         sort(d.begin(), d.end());
77         all[c].insert(all[c].end(), d.begin(), d.end());
78     }
79     sort(all[c].begin(), all[c].end());
80     for (auto [v, w] : g[c])
81     {
82         if (!removed[v])
83             decompose(v, c);
84     }
85 }
86
87 // Input must be a forest. Rebuild after changing edges.
88 void build()
89 {
90     fill(removed.begin(), removed.end(), 0);
91     for (int u = 1; u <= n; u++)
92     {
93         all[u].clear();
94         part[u].clear();
95     }
96     for (int u = 1; u <= n; u++)
97     {

```

```

98         if (!removed[u])
99             decompose(u, 0);
100     }
101 }
102
103 ll pairs(const vector<ll> &d, ll k) const
104 {
105     int l = 0, r = (int)d.size() - 1;
106     ll answer = 0;
107     while (l < r)
108     {
109         if (d[l] <= k - d[r])
110         {
111             answer += r - l;
112             l++;
113         }
114         else
115             r--;
116     }
117     return answer;
118 }
119
120 // Unordered pairs of distinct connected vertices. Nonnegative edge weights.
121 ll count_leq(ll k) const
122 {
123     if (k < 0)
124         return 0;
125     ll answer = 0;
126     for (int c = 1; c <= n; c++)
127     {
128         answer += pairs(all[c], k);
129         for (const auto &d : part[c])
130             answer -= pairs(d, k);
131     }
132     return answer;
133 }
134
135 ll count_exact(ll k) const
136 {
137     if (k < 0)
138         return 0;
139     return count_leq(k) - (k ? count_leq(k - 1) : 0);
140 }
141 };

```

第二十六章 树上启发式合并

26.1 树上启发式合并：子树众数颜色和

SubtreeColors(n), 点号 1..n, add(u,v) 建无向树; 要求图连通且无环, n=0 可空运行。run(colors,root=1) 传入长度 n+1 的 int 颜色数组, colors[0] 忽略。answer[u] 为子树内所有最高频颜色的原始编号之和, max_freq[u] 为该频次; 支持负颜色与稀疏编号, 内部离散化, 和用 long long。重复 run 可更换颜色或根。重儿子信息保留, 轻儿子通过 DFS 序区间加入; 清空整棵临时子树后再重置最大值, 不尝试逐次删除时维护众数。O(n log n) 时间, O(n) 空间, 全部 DFS 递归。insert 是内部增量统计, 可按题目修改统计量, 但相应清空规则也必须调整。

```
1 struct SubtreeColors
2 {
3     int n, best = 0;
4     long long sum = 0;
5     vector<vector<int>> g;
6     vector<int> siz, heavy, tin, tout, order, color, value, freq;
7     vector<int> max_freq;
8     vector<long long> answer;
9
10    SubtreeColors(int n) : n(n), g(n + 1)
11    {
12    }
13
14    void add(int u, int v)
15    {
16        assert(1 <= u && u <= n && 1 <= v && v <= n && u != v);
17        g[u].push_back(v);
18        g[v].push_back(u);
19    }
20
21    void prepare(int u, int p)
22    {
23        siz[u] = 1;
24        tin[u] = order.size();
25        order.push_back(u);
26        for (int v : g[u])
27        {
28            if (v == p)
29                continue;
30            prepare(v, u);
31            siz[u] += siz[v];
32            if (siz[v] > siz[heavy[u]])
33                heavy[u] = v;
34        }
35        tout[u] = (int)order.size() - 1;
36    }
37
38    void insert(int u)
39    {
40        int c = color[u];
41        freq[c]++;
42        if (freq[c] > best)
```

```

43     {
44         best = freq[c];
45         sum = value[c];
46     }
47     else if (freq[c] == best)
48         sum += value[c];
49 }
50
51 void solve(int u, int p, bool keep)
52 {
53     for (int v : g[u])
54     {
55         if (v != p && v != heavy[u])
56             solve(v, u, false);
57     }
58     if (heavy[u])
59         solve(heavy[u], u, true);
60     for (int v : g[u])
61     {
62         if (v == p || v == heavy[u])
63             continue;
64         for (int i = tin[v]; i <= tout[v]; i++)
65             insert(order[i]);
66     }
67     insert(u);
68     max_freq[u] = best;
69     answer[u] = sum;
70     if (!keep)
71     {
72         for (int i = tin[u]; i <= tout[u]; i++)
73             freq[color[order[i]]]--;
74         best = 0;
75         sum = 0;
76     }
77 }
78
79 // A connected tree, colors[1..n]. Sum all colors tied for maximum frequency.
80 void run(const vector<int> &colors, int root = 1)
81 {
82     assert((int)colors.size() == n + 1);
83     siz.assign(n + 1, 0);
84     heavy = siz;
85     tin = siz;
86     tout = siz;
87     color = siz;
88     max_freq = siz;
89     answer.assign(n + 1, 0);
90     order.clear();
91     value.assign(colors.begin() + 1, colors.end());
92     sort(value.begin(), value.end());
93     value.erase(unique(value.begin(), value.end()), value.end());
94     freq.assign(value.size(), 0);
95     best = 0;
96     sum = 0;

```

```
97     if (!n)
98         return;
99     assert(1 <= root && root <= n);
100    for (int u = 1; u <= n; u++)
101        color[u] =
102            lower_bound(value.begin(), value.end(), colors[u]) - value.begin();
103    prepare(root, 0);
104    solve(root, 0, false);
105 }
106 };
```


第二十七章 虚树

27.1 虚树：关键点压缩与路径信息

`VirtualTree(n)`, $n \geq 1$, 点号 $1..n$; `add(u,v,w=1)` 建非负权连通树, 无环和重边。 `prepare(root=1)` 递归预处理, 可换根重建; 根到任一点距离须在 `long long` 范围内。 `build(keys)` 自动去重并补齐必要 LCA, 返回 `vertices` 和 `edges`; 全部编号沿用原树, `vertices` 按 DFS 序排列, 非空时首项即虚树根。 `edges` 为 `(from,to,distance,minimum)`, 分别是祖先到后代的虚边、压缩路径长度和路径最小边权。空关键点返回空结果, 单点返回单节点; 原树根不会强制加入, 需要时由调用者加入 `keys`。边按后代 DFS 序排列, 可逆序做自底向上的 DP。不分配或清空 n 个查询邻接表; 每次仅返回 $O(k)$ 结果。预处理 $O(n \log n)$ 时间空间, 单次 $O(k \log k + k \log n)$ 。 `lca`、`ancestor` 也可直接使用, 要求有效点号且已经 `prepare`。构造用的祖先链栈属于虚树算法, 原树 DFS 保持递归。

```
1 struct VirtualTree
2 {
3     using ll = long long;
4
5     struct Edge
6     {
7         int from, to;
8         ll distance, minimum;
9     };
10
11     struct Result
12     {
13         vector<int> vertices;
14         vector<Edge> edges;
15     };
16
17     int n, log = 1, timer = 0;
18     vector<vector<pair<int, ll>>> g;
19     vector<vector<int>> up;
20     vector<vector<ll>> mn;
21     vector<int> depth, tin, tout;
22     vector<ll> dist;
23
24     VirtualTree(int n) : n(n), g(n + 1)
25     {
26     }
27
28     void add(int u, int v, ll w = 1)
29     {
30         assert(1 <= u && u <= n && 1 <= v && v <= n && u != v && w >= 0);
31         g[u].push_back({v, w});
32         g[v].push_back({u, w});
33     }
34
35     void dfs(int u, int p)
36     {
37         tin[u] = ++timer;
38         up[0][u] = p;
39         for (int j = 1; j < log; j++)
```

```

40     {
41         up[j][u] = up[j - 1][up[j - 1][u]];
42         mn[j][u] = min(mn[j - 1][u], mn[j - 1][up[j - 1][u]]);
43     }
44     for (auto [v, w] : g[u])
45     {
46         if (v == p)
47             continue;
48         assert(dist[u] <= numeric_limits<ll>::max() - w);
49         depth[v] = depth[u] + 1;
50         dist[v] = dist[u] + w;
51         mn[0][v] = w;
52         dfs(v, u);
53     }
54     tout[u] = timer;
55 }
56
57 void prepare(int root = 1)
58 {
59     assert(1 <= root && root <= n);
60     log = 1;
61     while ((1LL << log) <= n)
62         log++;
63     up.assign(log, vector<int>(n + 1));
64     mn.assign(log, vector<ll>(n + 1, numeric_limits<ll>::max()));
65     depth.assign(n + 1, 0);
66     tin = depth;
67     tout = depth;
68     dist.assign(n + 1, 0);
69     timer = 0;
70     dfs(root, root);
71 }
72
73 bool ancestor(int u, int v) const
74 {
75     return tin[u] <= tin[v] && tout[v] <= tout[u];
76 }
77
78 int lca(int u, int v) const
79 {
80     if (ancestor(u, v))
81         return u;
82     if (ancestor(v, u))
83         return v;
84     for (int j = log - 1; j >= 0; j--)
85     {
86         if (!ancestor(up[j][u], v))
87             u = up[j][u];
88     }
89     return up[0][u];
90 }
91
92 Edge path(int u, int v) const
93 {
94     assert(u != v && ancestor(u, v));
95     ll minimum = numeric_limits<ll>::max();

```

```

96     int x = v, steps = depth[v] - depth[u];
97     for (int j = 0; j < log; j++)
98     {
99         if (steps >> j & 1)
100         {
101             minimum = min(minimum, mn[j][x]);
102             x = up[j][x];
103         }
104     }
105     return {u, v, dist[v] - dist[u], minimum};
106 }
107
108 // Uses original vertex IDs. Empty input gives an empty result.
109 Result build(vector<int> keys) const
110 {
111     if (keys.empty())
112         return {};
113     for (int u : keys)
114         assert(1 <= u && u <= n);
115     auto cmp = [&](int u, int v)
116     {
117         return tin[u] < tin[v];
118     };
119     sort(keys.begin(), keys.end(), cmp);
120     keys.erase(unique(keys.begin(), keys.end()), keys.end());
121     int k = keys.size();
122     for (int i = 1; i < k; i++)
123         keys.push_back(lca(keys[i - 1], keys[i]));
124     sort(keys.begin(), keys.end(), cmp);
125     keys.erase(unique(keys.begin(), keys.end()), keys.end());
126     Result result{keys, {}};
127     vector<int> st;
128     for (int u : keys)
129     {
130         while (!st.empty() && !ancestor(st.back(), u))
131             st.pop_back();
132         if (!st.empty())
133             result.edges.push_back(path(st.back(), u));
134         st.push_back(u);
135     }
136     return result;
137 }
138 };

```

第二十八章 字符串

28.1 前缀函数

`prefix_function(s)` 返回每个前缀的最长真 border 长度，0-based；空串返回空数组。O(n)，无状态，直接抄写函数。

```
1 inline vector<int> prefix_function(const string &s)
2 {
3     int n = (int)s.size();
4     vector<int> p(n);
5     for (int i = 1; i < n; i++)
6     {
7         int j = p[i - 1];
8         while (j && s[i] != s[j])
9             j = p[j - 1];
10        p[i] = j + (s[i] == s[j]);
11    }
12    return p;
13 }
```

28.2 KMP 匹配

`kmp_match(s,t)` 返回非空模式串 `t` 在 `s` 中的所有起点 (0-based)，允许重叠。需要同章前缀函数。O(|s|+|t|)。

```
1 inline vector<int> kmp_match(const string &s, const string &t)
2 {
3     assert(!t.empty());
4     auto p = prefix_function(t);
5     vector<int> ans;
6     for (int i = 0, j = 0; i < (int)s.size(); i++)
7     {
8         while (j && s[i] != t[j])
9             j = p[j - 1];
10        j += (s[i] == t[j]);
11        if (j == (int)t.size())
12        {
13            ans.push_back(i - j + 1);
14            j = p[j - 1];
15        }
16    }
17    return ans;
18 }
```

28.3 Z 函数

`z_function(s)` 返回后缀 `s[i..]` 与原串的 LCP，约定 `z[0]=n`，空串返回空数组。O(n)。

```

1 inline vector<int> z_function(const string &s)
2 {
3     int n = (int)s.size();
4     vector<int> a(n);
5     if (n)
6         a[0] = n;
7     for (int i = 1, l = 0, r = 0; i < n; i++)
8     {
9         if (i < r)
10            a[i] = min(r - i, a[i - l]);
11        while (i + a[i] < n && s[a[i]] == s[i + a[i]])
12            ++a[i];
13        if (i + a[i] > r)
14        {
15            l = i;
16            r = i + a[i];
17        }
18    }
19    return a;
20 }
```

28.4 Manacher 回文半径

`manacher(s)` 返回 `odd` 与 `even`；`odd[i]` 的半径包含中心，`even[i]` 的中心在 `i` 前。字符串及下标为 0-based，空串合法，O(n)。

```

1 // odd[i] includes the center; even[i] is centered before i.
2 inline pair<vector<int>, vector<int>> manacher(const string &s)
3 {
4     int n = (int)s.size();
5     vector<int> odd(n), even(n);
6     for (int i = 0, l = 0, r = -1; i < n; i++)
7     {
8         int k = i > r ? 1 : min(odd[l + r - i], r - i + 1);
9         while (i - k >= 0 && i + k < n && s[i - k] == s[i + k])
10            ++k;
11        odd[i] = k--;
12        if (i + k > r)
13        {
14            l = i - k;
15            r = i + k;
16        }
17    }
18    for (int i = 0, l = 0, r = -1; i < n; i++)
19    {
20        int k = i > r ? 0 : min(even[l + r - i + 1], r - i + 1);
21        while (i - k - 1 >= 0 && i + k < n && s[i - k - 1] == s[i + k])
22            ++k;
```

```

23     even[i] = k--;
24     if (i + k > r)
25     {
26         l = i - k - 1;
27         r = i + k;
28     }
29 }
30 return {odd, even};
31 }

```

28.5 最小循环表示

`minimum_rotation(s)` 返回字典序最小循环表示的起点 (0-based), 空串返回 0; 按无符号字节比较, 与 `string` 字典序一致。O(n)。

```

1 inline int minimum_rotation(const string &s)
2 {
3     int n = (int)s.size(), i = 0, j = 1, k = 0;
4     if (!n)
5         return 0;
6     while (i < n && j < n && k < n)
7     {
8         unsigned char a = s[(i + k) % n], b = s[(j + k) % n];
9         if (a == b)
10            {
11                ++k;
12                continue;
13            }
14         if (a > b)
15            {
16                i += k + 1;
17                if (i == j)
18                    ++i;
19            }
20         else
21            {
22                j += k + 1;
23                if (i == j)
24                    ++j;
25            }
26         k = 0;
27     }
28     return min(i, j);
29 }

```

28.6 AC 自动机出现次数

字母表 `a..z`, `add(s)` 返回终止状态, 要求模式非空; 全部插入后 `build()`, 只允许一次构建。`count(text)` 返回每个状态对应模式的出现次数, 可重复查询; 重复模式共享状态。构建 $O(26N)$, 查询 $O(|text|+N)$ 。

```

1 struct AhoCorasick
2 {
3     struct Node
4     {

```

```

5      array<int, 26> go{};
6      int fail = 0;
7  };
8
9  vector<Node> a{Node{}};
10 vector<int> order;
11 bool built = false;
12
13 int add(const string &s)
14 {
15     assert(!built && !s.empty());
16     int u = 0;
17     for (char ch : s)
18     {
19         int c = ch - 'a';
20         assert(0 <= c && c < 26);
21         if (!a[u].go[c])
22         {
23             int v = (int)a.size();
24             a[u].go[c] = v;
25             a.push_back(Node{});
26         }
27         u = a[u].go[c];
28     }
29     return u;
30 }
31
32 void build()
33 {
34     assert(!built);
35     built = true;
36     queue<int> q;
37     for (int v : a[0].go)
38         if (v)
39             q.push(v);
40     while (!q.empty())
41     {
42         int u = q.front();
43         q.pop();
44         order.push_back(u);
45         for (int c = 0; c < 26; c++)
46         {
47             int v = a[u].go[c];
48             if (v)
49             {
50                 a[v].fail = a[a[u].fail].go[c];
51                 q.push(v);
52             }
53             else
54                 a[u].go[c] = a[a[u].fail].go[c];
55         }
56     }
57 }
58
59 vector<long long> count(const string &s) const
60 {

```

```
61     assert(built);
62     vector<long long> ans(a.size());
63     int u = 0;
64     for (char ch : s)
65     {
66         u = a[u].go[ch - 'a'];
67         ++ans[u];
68     }
69     for (int i = (int)order.size() - 1; i >= 0; i--)
70         ans[a[order[i]].fail] += ans[order[i]];
71     return ans;
72 }
73 };
```


28.7 倍增后缀数组与 LCP

构造 `SuffixArray(s)` 支持任意字节串，按 `unsigned char` 的 0..255 数值比较，包含内嵌零字节；也可构造 `SuffixArray(a,alphabet)`，整数序列要求 $0 \leq a[i] < \text{alphabet}$ 、 $1 \leq \text{alphabet} < \text{INT_MAX}$ 。`init(a,alphabet)` 可重复重建，传统版为 `Init`。长度须小于 `INT_MAX`。`sa[k]` 为第 k 个非空后缀起点，`rk[i]` 为排名，全部 0-based；`lcp[k]` 为 `sa[k-1]` 与 `sa[k]` 的最长公共前缀，`lcp[0]=0`。空序列合法，三数组均为空。无需手动追加哨兵，也不输出空后缀。每轮按第二关键字已有顺序、第一关键字稳定计数排序，倍增 $O(n \log n + \text{alphabet})$ ，空间 $O(n + \text{alphabet})$ ；字节串 `alphabet=256`。大整数先自行保序离散化，避免按最大原值开桶。Kasai 求 `lcp` 为 $O(n)$ 。不同起点 x,y 的 LCP 为 `lcp[min(rk[x],rk[y])+1..max(rk[x],rk[y])]` 的最小值；本结构不内建 RMQ，需要频繁查询时另接静态 RMQ。同一起点的 LCP 为 `n-x`。覆盖倍增构造，不替代 DC3/SA-IS。

```

1 struct SuffixArray
2 {
3     vector<int> sa, rk, lcp;
4
5     SuffixArray(const string &s)
6     {
7         vector<int> a;
8         a.reserve(s.size());
9         for (unsigned char c : s)
10             a.push_back(c);
11         init(a, 256);
12     }
13
14     SuffixArray(const vector<int> &s, int alphabet)
15     {
16         init(s, alphabet);
17     }
18
19     void init(const vector<int> &s, int alphabet)
20     {
21         assert(s.size() < INT_MAX);
22         assert(1 <= alphabet && alphabet < INT_MAX);
23         int n = (int)s.size();
24         sa.resize(n);
25         rk.resize(n);
26         lcp.assign(n, 0);
27         if (!n)
28             return;
29         vector<int> order(n), old(n), cnt(max(n + 1, alphabet + 1));
30         int m = alphabet;
31         for (int i = 0; i < n; i++)
32         {
33             assert(0 <= s[i] && s[i] < alphabet);
34             rk[i] = s[i] + 1;
35             cnt[rk[i]]++;
36         }
37         for (int i = 1; i <= m; i++)
38             cnt[i] += cnt[i - 1];
39         for (int i = n - 1; i >= 0; i--)
40             sa[--cnt[rk[i]]] = i;

```

```

41     for (int k = 1; k < n; k *= 2)
42     {
43         int p = 0;
44         for (int i = n - k; i < n; i++)
45             order[p++] = i;
46         for (int i : sa)
47             if (i >= k)
48                 order[p++] = i - k;
49         fill(cnt.begin(), cnt.begin() + m + 1, 0);
50         for (int x : rk)
51             cnt[x]++;
52         for (int i = 1; i <= m; i++)
53             cnt[i] += cnt[i - 1];
54         for (int i = n - 1; i >= 0; i--)
55             sa[--cnt[rk[order[i]]]] = order[i];
56         old.swap(rk);
57         auto second = [&](int i)
58         {
59             return i < n - k ? old[i + k] : 0;
60         };
61         m = 1;
62         rk[sa[0]] = m;
63         for (int i = 1; i < n; i++)
64         {
65             int a = sa[i - 1], b = sa[i];
66             if (old[a] != old[b] || second(a) != second(b))
67                 m++;
68             rk[b] = m;
69         }
70         if (m == n)
71             break;
72     }
73     for (int i = 0; i < n; i++)
74         rk[sa[i]] = i;
75     for (int i = 0, k = 0; i < n; i++)
76     {
77         if (!rk[i])
78         {
79             k = 0;
80             continue;
81         }
82         int j = sa[rk[i] - 1];
83         while (k < n - i && k < n - j && s[i + k] == s[j + k])
84             ++k;
85         lcp[rk[i]] = k;
86         if (k)
87             --k;
88     }
89 }
90 };

```

28.8 后缀自动机

`extend(c)` 接收整数 0..25; `distinct()` 返回不同非空子串数, `counts()` 返回 `endpos` 次数且不修改结构, 可重复调用。状态 `a`, 初始状态 0, `link` 根为 -1; 空间 $O(26n)$, 构建 $O(26n)$ 。

```

1 struct SuffixAutomaton
2 {
3     struct Node
4     {
5         array<int, 26> go{};
6         int link = -1, len = 0;
7         long long occ = 0;
8     };
9
10    vector<Node> a{Node{}};
11    int last = 0;
12
13    void extend(int c)
14    {
15        assert(0 <= c && c < 26);
16        int cur = (int)a.size();
17        a.push_back(Node{});
18        a[cur].len = a[last].len + 1;
19        a[cur].occ = 1;
20        int p = last;
21        while (p != -1 && !a[p].go[c])
22        {
23            a[p].go[c] = cur;
24            p = a[p].link;
25        }
26        if (p == -1)
27            a[cur].link = 0;
28        else
29        {
30            int q = a[p].go[c];
31            if (a[p].len + 1 == a[q].len)
32                a[cur].link = q;
33            else
34            {
35                int clone = (int)a.size();
36                Node copy = a[q];
37                a.push_back(copy);
38                a[clone].len = a[p].len + 1;
39                a[clone].occ = 0;
40                while (p != -1 && a[p].go[c] == q)
41                {
42                    a[p].go[c] = clone;
43                    p = a[p].link;
44                }
45                a[q].link = a[cur].link = clone;
46            }
47        }
48        last = cur;
49    }
50
51    vector<long long> counts() const

```

```
52 {
53     vector<int> bucket(a[last].len + 1), order(a.size());
54     vector<long long> ans(a.size());
55     for (auto &v : a)
56         ++bucket[v.len];
57     partial_sum(bucket.begin(), bucket.end(), bucket.begin());
58     for (int i = 0; i < (int)a.size(); i++)
59     {
60         order[--bucket[a[i].len]] = i;
61         ans[i] = a[i].occ;
62     }
63     for (int i = (int)order.size() - 1; i > 0; i--)
64         ans[a[order[i]].link] += ans[order[i]];
65     return ans;
66 }
67
68 long long distinct() const
69 {
70     long long ans = 0;
71     for (int i = 1; i < (int)a.size(); i++)
72         ans += a[i].len - a[a[i].link].len;
73     return ans;
74 }
75 };
```

第二十九章 后缀查询与重复子串

29.1 后缀 LCP 与子串比较

先构造 SuffixArray, 再构造 SuffixLCP(sa); 传统版对应 Suffix_Array 和 Suffix_LCP。init(sa) / Init(sa) 可重新建表。复制所需排名与 LCP 数据, 不保存原字符串或源对象引用, 源对象可销毁或重建。query(x,y) 返回 s[x..n) 与 s[y..n) 的最长公共前缀, $0 \leq x, y \leq n$, 允许空后缀; 同起点返回 n-x。compare(l1,r1,l2,r2) 比较两个 0-based 半开子串, 允许空串, 返回 -1/0/1; 传统接口为 Query/Compare。若公共前缀覆盖较短子串, 按长度比较, 否则原后缀排名给出首个不同字符的顺序。字节排序及整数字母表约束继承 SuffixArray。预处理稀疏表 $O(n \log n)$ 时间、空间, 每次查询 $O(1)$, 无需哈希和原串。n、rk、lg、st 可读, 禁止调用方破坏。

```
1 struct SuffixLCP
2 {
3     int n;
4     vector<int> rk, lg;
5     vector<vector<int>>> st;
6
7     SuffixLCP(const SuffixArray &suffix)
8     {
9         init(suffix);
10    }
11
12    void init(const SuffixArray &suffix)
13    {
14        n = suffix.sa.size();
15        rk = suffix.rk;
16        lg.assign(n + 1, 0);
17        for (int i = 2; i <= n; i++)
18            lg[i] = lg[i / 2] + 1;
19        st.assign(lg[n] + 1, {});
20        st[0] = suffix.lcp;
21        for (int k = 1; k <= lg[n]; k++)
22        {
23            int len = 1 << k;
24            st[k].resize(n - len + 1);
25            for (int i = 0; i <= n - len; i++)
26                st[k][i] = min(st[k - 1][i], st[k - 1][i + len / 2]);
27        }
28    }
```

```

30  int query(int x, int y) const
31  {
32      assert(0 <= x && x <= n && 0 <= y && y <= n);
33      if (x == y || x == n || y == n)
34          return n - max(x, y);
35      int a = rk[x], b = rk[y];
36      if (a > b)
37          swap(a, b);
38      int k = lg[b - a];
39      return min(st[k][a + 1], st[k][b - (1 << k) + 1]);
40  }
41
42  // Compare [l1,r1) and [l2,r2); return -1, 0 or 1.
43  int compare(int l1, int r1, int l2, int r2) const
44  {
45      assert(0 <= l1 && l1 <= r1 && r1 <= n);
46      assert(0 <= l2 && l2 <= r2 && r2 <= n);
47      int a = r1 - l1, b = r2 - l2;
48      if (query(l1, l2) >= min(a, b))
49          return (a > b) - (a < b);
50      return rk[l1] < rk[l2] ? -1 : 1;
51  }
52 };

```

29.2 两个前缀的最长公共后缀

`prefix_lcs(reversed,x,y)` 返回原序列 `s[0..x)` 与 `s[0..y)` 的最长公共后缀，端点满足 $0 \leq x, y \leq n$ 。reversed 必须在原序列的反转序列上构造 SuffixLCP，不能传正向对象；转换为 `reversed.query(n-x,n-y)`，单次 $O(1)$ 。传统版 Prefix_Lcs 接受 Suffix_LCP。只需前向 LCP 时不必构造反向对象；该接口中的 LCS 是最长公共后缀，不是最长公共子序列。对于两个任意子串，以右端点查询后还需截断到两个子串长度的最小值。

```

1  // reversed must be built on the reversed original sequence.
2  inline int prefix_lcs(const SuffixLCP &reversed, int x, int y)
3  {
4      assert(0 <= x && x <= reversed.n && 0 <= y && y <= reversed.n);
5      return reversed.query(reversed.n - x, reversed.n - y);
6  }

```

29.3 重复子串 AA 的起止位置计数

`square_counts(forward,reversed)` 返回 `(start,finish)` 两个长度 `n` 的 `long long` 数组, `start[i]/finish[i]` 为开始/结束于位置 `i` 的非空 AA 型子串个数; 结束位置是最后一个字符下标, 均 0-based。两对象必须分别对应同一原序列及其反序列, 长度相同, 允许空序列。传统接口为 `Square_Counts`。枚举半长 `len`, 并以其整数倍 `i` 为锚点; 向左匹配截断为 `len-1`, 向右截断为 `len`, 由 LCP/LCS 得出合法起点连续区间, 差分累计。左侧不能放宽到 `len`, 否则相邻锚点会重复计数。预处理查询器后, 此函数 $O(n \log n)$ 时间、 $O(n)$ 额外空间。不同出现位置、不同半长分别计数; 不要求 `A` 为最小周期。P1117 的 AABB 拆分数为所有分界点 `finish[i]*start[i+1]` 之和, 累加类型需按题目范围选择, 函数本身只返回起止计数。

```

1 // Count nonempty squares AA by their starting and ending positions.
2 inline pair<vector<long long>, vector<long long>>
3 square_counts(const SuffixLCP &forward, const SuffixLCP &reversed)
4 {
5     int n = forward.n;
6     assert(reversed.n == n);
7     vector<long long> start(n + 1), finish(n + 1);
8     for (int len = 1; len <= n / 2; len++)
9         for (int i = 0; i < n - len; i += len)
10            {
11                int j = i + len;
12                int left = min(len - 1, prefix_lcs(reversed, i, j));
13                int right = min(len, forward.query(i, j));
14                if (left + right < len)
15                    continue;
16                int l = i - left, r = i + right - len;
17                start[l]++;
18                start[r + 1]--;
19                finish[l + 2 * len - 1]++;
20                finish[r + 2 * len]--;
21            }
22     for (int i = 1; i < n; i++)
23     {
24         start[i] += start[i - 1];
25         finish[i] += finish[i - 1];
26     }
27     start.resize(n);
28     finish.resize(n);
29     return {start, finish};
30 }
```

第三十章 回文树

30.1 回文树 PAM: 回文后缀与出现次数

`add(ch)` 追加小写英文字母, 返回最长回文后缀节点, 亦保存于 `last`。节点 0 是长度 0 根, 1 是长度 -1 根, 实际回文节点从 2 开始; `distinct()` 为不同非空回文串数。`t[u].len` 为长度, `fail` 为最长真回文后缀链接, `suffixes` 为该节点的非空回文后缀数量; 追加后的 `t[last].suffixes` 就是本位置结尾的回文串数。`total` 为当前全部回文子串的出现次数之和。`end` 为首次出现时的 0-based 右端点, 可恢复原串子段; `hits` 仅统计作为最长达后缀的次数, 不能当出现次数。`occurrences()` 返回按节点编号的完整出现次数副本, 根置零, $O(n)$; 重复调用以及查询后继续追加均安全。`init()` 清空重建。固定 26 字符集下建树 $O(n)$, 空间 $O(26n)$ 。

```
1 struct PalindromicTree
2 {
3     struct Node
4     {
5         int next[26]{};
6         int len = 0, fail = 0, suffixes = 0, end = -1;
7         long long hits = 0;
8     };
9
10    vector<Node> t;
11    vector<int> s;
12    int last = 0;
13    long long total = 0;
14
15    PalindromicTree()
16    {
17        init();
18    }
19
20    void init()
21    {
22        t.assign(2, Node{});
23        t[0].fail = 1;
24        t[1].len = -1;
25        s.assign(1, -1);
26        last = 0;
27        total = 0;
28    }
29
30    int get_fail(int u) const
31    {
32        int pos = s.size() - 1;
33        while (s[pos - t[u].len - 1] != s[pos])
34            u = t[u].fail;
35        return u;
36    }
37
38    // Append a lowercase letter; return the longest palindromic suffix node.
39    int add(char ch)
40    {
41        int c = ch - 'a';
```



```

42     assert(0 <= c && c < 26);
43     s.push_back(c);
44     int u = get_fail(last);
45     if (!t[u].next[c])
46     {
47         int v = t.size();
48         t.push_back(Node{});
49         t[v].len = t[u].len + 2;
50         t[v].fail = t[get_fail(t[u].fail)].next[c];
51         t[v].suffixes = t[t[v].fail].suffixes + 1;
52         t[v].end = (int)s.size() - 2;
53         t[u].next[c] = v;
54     }
55     last = t[u].next[c];
56     t[last].hits++;
57     total += t[last].suffixes;
58     return last;
59 }
60
61 int distinct() const
62 {
63     return (int)t.size() - 2;
64 }
65
66 // Copy-and-propagate: repeated calls and later appends remain valid.
67 vector<long long> occurrences() const
68 {
69     vector<long long> count(t.size());
70     for (int u = 2; u < (int)t.size(); u++)
71         count[u] = t[u].hits;
72     for (int u = (int)t.size() - 1; u >= 2; u--)
73         count[t[u].fail] += count[u];
74     count[0] = 0;
75     count[1] = 0;
76     return count;
77 }
78 };

```

第三十一章 数论

31.1 模运算、素性、CRT 与类欧几里德

mul/power 支持无符号 64 位正模数，乘积用 uint128。prime 使用确定性 64 位底数。inverse(a,m) 无逆元返回 -1。crt(r,m,b,n) 合并 $x \equiv r \pmod{m}$ 与 $x \equiv b \pmod{n}$ ，要求 $m,n > 0$ 。从 $r=0$ 、 $m=1$ 开始逐项合并；成功后 r 为最小非负解、 m 为最小正周期。余数允许完整 long long 范围。无解返回 false；有解但新周期超过 LLONG_MAX 抛 overflow_error。两种失败都会先将 r 归一化， m 不变；失败后不能继续当作完整解使用。递归 exgcd 和乘法中间量用 int128，单次 $O(\log \min(m,n))$ ，抄写 CRT 时只需类型别名、exgcd 和 crt。floor_sum 为 $i=0..n-1$ 的 $\text{floor}((ai+b)/m)$ 之和，支持负 a 、 b ，返回 int128；约定 $n \leq 10^9$ 、 $m \leq 10^9$ ， a 、 b 为有符号 64 位。

```
1 struct NumberTheory
2 {
3     using ll = long long;
4     using ull = unsigned long long;
5     using i128 = __int128_t;
6     using u128 = __uint128_t;
7
8     static ull mul(ull a, ull b, ull m)
9     {
10         return u128(a) * b % m;
11     }
12
13     static ull power(ull a, ull b, ull m)
14     {
15         assert(m);
16         ull r = 1 % m;
17         a %= m;
18         for (; b >>= 1, a = mul(a, a, m))
19             if (b & 1)
20                 r = mul(r, a, m);
21         return r;
22     }
23
24     static bool prime(ull n)
25     {
26         if (n < 2)
27             return false;
28         for (ull p : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37})
29             if (n % p == 0)
30                 return n == p;
31         ull d = n - 1;
32         int s = 0;
33         while (!(d & 1))
34         {
35             d >>= 1;
36             ++s;
37         }
38         for (ull a :
39             {2ULL, 325ULL, 9375ULL, 28178ULL, 450775ULL, 9780504ULL, 1795265022ULL})
40         {
41             if (a % n == 0)
```

```

42         continue;
43         ull x = power(a, d, n);
44         if (x == 1 || x == n - 1)
45             continue;
46         bool ok = false;
47         for (int j = 1; j < s; j++)
48         {
49             x = mul(x, x, n);
50             if (x == n - 1)
51             {
52                 ok = true;
53                 break;
54             }
55         }
56         if (!ok)
57             return false;
58     }
59     return true;
60 }
61
62 static i128 exgcd(i128 a, i128 b, i128 &x, i128 &y)
63 {
64     if (!b)
65     {
66         x = 1;
67         y = 0;
68         return a;
69     }
70     i128 d = exgcd(b, a % b, y, x);
71     y -= a / b * x;
72     return d;
73 }
74
75 static ll inverse(ll a, ll m)
76 {
77     assert(m > 0);
78     a %= m;
79     if (a < 0)
80         a += m;
81     i128 x, y;
82     if (exgcd(a, m, x, y) != 1)
83         return -1;
84     return (x % m + m) % m;
85 }
86
87 // x = r (mod m); false means inconsistent. Throws if lcm exceeds int64.
88 static bool crt(ll &r, ll &m, ll b, ll n)
89 {
90     assert(m > 0 && n > 0);
91     r %= m;
92     if (r < 0)
93         r += m;
94     b %= n;
95     if (b < 0)
96         b += n;
97     i128 x, y;

```

```

98     ll g = (ll)exgcd(m, n, x, y);
99     i128 diff = i128(b) - r;
100     if (diff % g)
101         return false;
102     i128 q = n / g, k = (diff / g * x % q + q) % q, mod = i128(m) * q;
103     if (mod > LLONG_MAX)
104         throw overflow_error("CRT modulus");
105     r = (r + i128(m) * k) % mod;
106     m = (ll)mod;
107     return true;
108 }
109
110 // sum_{i=0}^{n-1} floor((a*i+b)/m), signed a,b; 0<=n<=1e9, 1<=m<=1e9.
111 static i128 floor_sum(ll n, ll m, ll a, ll b)
112 {
113     assert(n >= 0 && m > 0);
114     i128 ans = 0;
115     auto norm = [&](ll &v)
116     {
117         ll q = v / m;
118         if (v % m < 0)
119             --q;
120         v = (ll)(i128(v) - i128(q) * m);
121         return q;
122     };
123     ans += i128(norm(a)) * n * (n - 1) / 2;
124     ans += i128(norm(b)) * n;
125     while (true)
126     {
127         ans += i128(n) * (n - 1) / 2 * (a / m);
128         a %= m;
129         ans += i128(n) * (b / m);
130         b %= m;
131         i128 y = i128(a) * n + b;
132         if (y < m)
133             break;
134         n = (ll)(y / m);
135         b = (ll)(y % m);
136         swap(a, m);
137     }
138     return ans;
139 }
140 };

```

31.2 Pollard-Rho 因数分解

`factor(n)` 返回升序质因子（含重数）， $n \geq 1$ ；依赖 `NumberTheory`。返回因子经过整除和素性检查；Las Vegas 随机运行时间，固定默认种子方便复现，可由调用者更换。无确定性最坏时间保证。

```

1 struct PollardRho
2 {
3     using ull = unsigned long long;
4     using u128 = __uint128_t;
5     mt19937_64 rng;
6
7     PollardRho(ull seed = 712367821) : rng(seed)
8     {
9     }
10
11     ull rho(ull n)
12     {
13         if (n % 2 == 0)
14             return 2;
15         for (;;)
16         {
17             ull c = rng() % (n - 1) + 1, x = rng() % n, y = x, d = 1;
18             auto f = [&](ull v)
19             {
20                 return (NumberTheory::mul(v, v, n) + u128(c)) % n;
21             };
22             // Retry bounded attempts; Las Vegas: only exact divisors returned.
23             for (int i = 0; i < 200000 && d == 1; i++)
24             {
25                 x = f(x);
26                 y = f(f(y));
27                 d = gcd(x > y ? x - y : y - x, n);
28             }
29             if (1 < d && d < n)
30                 return d;
31         }
32     }
33
34     void split(ull n, vector<ull> &a)
35     {
36         if (n == 1)
37             return;
38         if (NumberTheory::prime(n))
39         {
40             a.push_back(n);
41             return;
42         }
43         ull d = rho(n);
44         split(d, a);
45         split(n / d, a);
46     }

```

```

47
48     vector<ull> factor(ull n)
49     {
50         assert(n >= 1);
51         vector<ull> a;
52         split(n, a);
53         sort(a.begin(), a.end());
54         return a;
55     }
56 };

```

31.3 线性筛、欧拉函数与 Möbius 函数

构造参数 $n \geq 0$, prime 为质数表, lp/phi/mu 为 $0..n$ 数组。phi[1]=mu[1]=1, $O(n)$ 时间和空间。

```

1 struct LinearSieve
2 {
3     vector<int> prime, lp, phi, mu;
4
5     LinearSieve(int n) : lp(n + 1), phi(n + 1), mu(n + 1)
6     {
7         if (n)
8             phi[1] = mu[1] = 1;
9         for (int i = 2; i <= n; i++)
10         {
11             if (!lp[i])
12             {
13                 lp[i] = i;
14                 prime.push_back(i);
15                 phi[i] = i - 1;
16                 mu[i] = -1;
17             }
18             for (int p : prime)
19             {
20                 if (p > n / i)
21                     break;
22                 int j = i * p;
23                 lp[j] = p;
24                 if (i % p == 0)
25                 {
26                     phi[j] = phi[i] * p;
27                     mu[j] = 0;
28                     break;
29                 }
30                 phi[j] = phi[i] * (p - 1);
31                 mu[j] = -mu[i];
32             }
33         }
34     }
35 };

```

31.4 静态模整数

模板参数必须为不超过 `INT_MAX` 的素数且 ≥ 2 。加减乘、`pow`、`inv`、除法；求逆要求非零，`pow` 指数非负。
复杂度：加减乘 $O(1)$ ，幂和逆 $O(\log \text{mod})$ 。

```

1 template <int mod> struct ModInt
2 {
3     int v;
4
5     ModInt(long long x = 0) : v((x % mod + mod) % mod)
6     {
7     }
8
9     ModInt operator+(ModInt b) const
10    {
11        return ModInt((long long)v + b.v);
12    }
13
14    ModInt operator-(ModInt b) const
15    {
16        return ModInt((long long)v - b.v);
17    }
18
19    ModInt operator*(ModInt b) const
20    {
21        return ModInt(1LL * v * b.v);
22    }
23
24    ModInt pow(long long e) const
25    {
26        assert(e >= 0);
27        ModInt a = *this, r = 1;
28        for (; e >>= 1, a = a * a)
29            if (e & 1)
30                r = r * a;
31        return r;
32    }
33
34    ModInt inv() const
35    {
36        assert(v);
37        return pow(mod - 2);
38    } // prime modulus
39
40    ModInt operator/(ModInt b) const
41    {
42        return *this * b.inv();
43    }
44 };

```

31.5 阶乘组合数

`Binomial<prime>(n=0)` 预处理 $0..n$ ，要求 `prime` 为素数且 $0 \leq n < \text{prime}$ 。`init(n)` 只扩大当前上限，较小或相同的 n 不改变表；`choose(n,k)` 返回 $C(n,k)$ ，`permute(n,k)` 返回 $n!/(n-k)!$ 。查询前显式预处理到 n ，查询本身不扩表。 $n < 0$ 、 $k < 0$ 或 $k > n$ 返回 0， $C(0,0)=P(0,0)=1$ 。`fac`、`ifac` 为可直接读取的阶乘与逆阶乘表，增长可能使旧引用失效。单次查询 $O(1)$ ；从 `old` 扩到 n 新增 $O(n-\text{old}+\log \text{prime})$ 次域运算，重分配还可能复制旧表。首次预处理 $O(n+\log \text{prime})$ ，空间 $O(n)$ 。超过模数不能使用阶乘除法：小素数用 `Lucas`，小合数用 `exLucas`。

```

1  template <int mod> struct Binomial
2  {
3      using Z = ModInt<mod>;
4      vector<Z> fac{Z(1)}, ifac{Z(1)};
5
6      Binomial(int n = 0)
7      {
8          init(n);
9      }
10
11     void init(int n)
12     {
13         assert(0 <= n && n < mod);
14         int old = (int)fac.size() - 1;
15         if (n <= old)
16             return;
17         fac.resize(n + 1);
18         ifac.resize(n + 1);
19         for (int i = old + 1; i <= n; i++)
20             fac[i] = fac[i - 1] * i;
21         ifac[n] = fac[n].inv();
22         for (int i = n; i > old; i--)
23             ifac[i - 1] = ifac[i] * i;
24     }
25
26     Z choose(int n, int k) const
27     {
28         if (k < 0 || k > n)
29             return 0;
30         assert(n < (int)fac.size());
31         return fac[n] * ifac[k] * ifac[n - k];
32     }
33
34     Z permute(int n, int k) const
35     {
36         if (k < 0 || k > n)
37             return 0;
38         assert(n < (int)fac.size());
39         return fac[n] * ifac[n - k];
40     }
41 };

```


第三十二章 批量模逆元

32.1 任意数组的批量逆元

`batch_inverse(a,mod)` 返回与 `a` 同序的逆元向量，传统接口为 `Batch_Inverse`。要求 `mod` 为正 `long long`，每个输入可为完整 `long long`；只要某一项与 `mod` 不互素，整个结果返回 `nullopt`，不输出部分逆元。空输入返回有效的空向量；模数 1 约定返回全零，与单个逆元接口一致。模数可以是合数。函数按值接收数组并归一化；不再需要原数组时传 `move(a)` 可避免输入副本。依赖 `NumberTheory` 的 `inverse` 和 `exgcd`，传统版对应 `Inverse`、`Exgcd`。仅求一次逆元，时间 $O(n+\log \text{mod})$ ，空间 $O(n)$ ；128 位中间乘法防止溢出。

```
1 inline optional<vector<long long>> batch_inverse(vector<long long> a, long long mod)
2 {
3     assert(mod > 0);
4     int n = a.size();
5     vector<long long> prefix(n + 1, 1 % mod);
6     for (int i = 0; i < n; i++)
7     {
8         a[i] %= mod;
9         if (a[i] < 0)
10             a[i] += mod;
11         prefix[i + 1] = __int128_t(prefix[i]) * a[i] % mod;
12     }
13     long long inverse = NumberTheory::inverse(prefix[n], mod);
14     if (inverse == -1)
15         return nullopt;
16     for (int i = n - 1; i >= 0; i--)
17     {
18         long long value = a[i];
19         a[i] = __int128_t(inverse) * prefix[i] % mod;
20         inverse = __int128_t(inverse) * value % mod;
21     }
22     return a;
23 }
```

第三十三章 混合进制中国剩余定理

33.1 Garner：大周期 CRT 与混合进制

`garner_digits(b,moduli)` 返回按输入顺序排列的混合进制各位；`garner(b,moduli,target)` 返回最小非负 CRT 解对 `target` 的余数。传统接口为 `Garner_Digits`、`Garner`。两个输入向量等长，长度可为 0；模数为正且两两互素，允许模数 1，余数支持完整 `long long` 范围。`target` 为正 `long long`，与输入模数无需互素。输入不互素时抛 `invalid_argument`，即使方程有解也不适用；此时用扩展 CRT。空输入的规范解取 0。算法从不构造模数乘积，中间运算使用 `int128`。 k 个模数耗时 $O(k^2+k \log M)$ ，空间 $O(k)$ ； M 为最大输入模数。依赖 `NumberTheory` 的 `inverse` 和 `exgcd`，传统版对应 `Inverse`、`Exgcd`。

```
1 // Pairwise coprime positive moduli. Digits use the input order.
2 inline vector<long long> garner_digits(const vector<long long> &b,
3                                       const vector<long long> &moduli)
4 {
5     using ll = long long;
6     using i128 = __int128_t;
7     int n = b.size();
8     assert(moduli.size() == b.size());
9     vector<ll> value(n), coefficient(n, 1), digit(n);
10    for (ll m : moduli)
11        assert(m > 0);
12    for (int i = 0; i < n; i++)
13    {
14        ll m = moduli[i];
15        ll inverse = NumberTheory::inverse(coefficient[i], m);
16        if (inverse == -1)
17            throw invalid_argument("Garner moduli are not coprime");
18        i128 delta = (i128(b[i]) - value[i]) % m;
19        if (delta < 0)
20            delta += m;
21        digit[i] = delta * inverse % m;
22        for (int j = i + 1; j < n; j++)
23        {
24            value[j] = (value[j] + i128(coefficient[j]) * digit[i]) % moduli[j];
25            coefficient[j] = i128(coefficient[j]) * m % moduli[j];
26        }
27    }
28    return digit;
29 }
```

规范解对目标数取模

```
31 // Return the least nonnegative CRT solution modulo target, not all solutions.
32 inline long long
33 Garner(const vector<long long> &b, const vector<long long> &moduli, long long target)
34 {
35     assert(target > 0);
36     auto digit = Garner_digits(b, moduli);
37     long long answer = 0;
38     for (int i = (int)digit.size() - 1; i >= 0; i--)
39         answer = (__int128_t(answer) * moduli[i] + digit[i]) % target;
40     return answer;
41 }
```

第三十四章 乘法阶与原根

34.1 乘法阶与原根

构造 `PrimitiveRoot(n)`，要求 $2 \leq n \leq \text{INT_MAX}$ ；`init(n)` 可重建。`order(a)` 返回 a 模 n 的乘法阶， a 不可逆时返回 0；`is_root(a)` 判断该剩余类是否为原根， a 可为完整 `long long`。`minimum()` 返回 $1..n-1$ 内最小原根，无原根返回 -1；`all()` 返回升序的全部原根，无原根返回空向量。`mod`、`phi`、`exists` 只读；`factors` 为 `phi` 的不同素因子，辅助 `factor` 为试除分解。依赖 `NumberTheory` 的 `power`。初始化 $O(\sqrt{n})$ ，求阶 $O(\log^2 n)$ ，原根判定 $O((1+\omega(\phi))\log n)$ 。若最小原根为 G ，顺序枚举最坏检查 G 个候选，不能假定 G 总是常数。枚举全部另需 $O(\phi \log \phi + c \log c)$ 时间、 $O(c)$ 空间， c 为原根数；大模数只求最小原根或阶时，不应调用 `all`。

```
1 struct PrimitiveRoot
2 {
3     int mod, phi;
4     bool exists;
5     vector<int> factors;
6
7     static vector<int> factor(int n)
8     {
9         vector<int> result;
10        for (int p = 2; p <= n / p; p++)
11            if (n % p == 0)
12            {
13                result.push_back(p);
14                while (n % p == 0)
15                    n /= p;
16            }
17        if (n > 1)
18            result.push_back(n);
19        return result;
20    }
21
22    PrimitiveRoot(int n)
23    {
24        init(n);
25    }
```

```
27 void init(int n)
28 {
29     assert(n >= 2);
30     mod = n;
31     phi = n;
32     for (int p : factor(n))
33         phi -= phi / p;
34     int odd = n;
35     if (odd % 2 == 0)
36         odd /= 2;
37     exists = n == 2 || n == 4 || (odd % 2 && factor(odd).size() == 1);
38     factors = factor(phi);
39 }
40
41 int order(long long a) const
42 {
43     a %= mod;
44     if (a < 0)
45         a += mod;
46     if (gcd(a, (long long)mod) != 1)
47         return 0;
48     int result = phi;
49     for (int p : factors)
50         while (result % p == 0 && NumberTheory::power(a, result / p, mod) == 1)
51             result /= p;
52     return result;
53 }
```

```

55     bool is_root(long long a) const
56     {
57         if (!exists)
58             return false;
59         a %= mod;
60         if (a < 0)
61             a += mod;
62         if (gcd(a, (long long)mod) != 1)
63             return false;
64         for (int p : factors)
65             if (NumberTheory::power(a, phi / p, mod) == 1)
66                 return false;
67         return true;
68     }
69
70     int minimum() const
71     {
72         if (!exists)
73             return -1;
74         for (int g = 1; g < mod; g++)
75             if (is_root(g))
76                 return g;
77         return -1;
78     }
79
80     vector<int> all() const
81     {
82         int g = minimum();
83         vector<int> roots;
84         if (g == -1)
85             return roots;
86         long long value = 1;
87         for (int k = 1; k <= phi; k++)
88         {
89             value = value * g % mod;
90             if (gcd(k, phi) == 1)
91                 roots.push_back(value);
92         }
93         sort(roots.begin(), roots.end());
94         return roots;
95     }
96 };

```

第三十五章 Carmichael 函数

35.1 Carmichael: 模乘法群的公共指数

`carmichael(n)` 返回 $\lambda(n)$, 即使所有与 n 互素的 a 都满足 a 的 $\lambda(n)$ 次幂模 n 为 1 的最小正指数; 约定 $\lambda(1)=1$ 。要求 $1 \leq n \leq \text{INT_MAX}$ 。传统接口为 `Carmichael`。试除分解后将各素数幂的指数取 lcm, 特别处理 2 的三次及以上幂。时间 $O(\sqrt{n})$, 额外空间 $O(1)$; 所有中间值不超过有符号 64 位范围, 结果不超过 n 。该指数只适用于可逆底数, 不能据此直接对非互素底数的幂降指数。函数不依赖原根模块。

```
1 inline int carmichael(int n)
2 {
3     assert(n >= 1);
4     long long answer = 1;
5     for (int p = 2; p <= n / p; p++)
6         if (n % p == 0)
7             {
8                 int power = 1;
9                 while (n % p == 0)
10                     {
11                         n /= p;
12                         power *= p;
13                     }
14                 int value = power / p * (p - 1);
15                 if (p == 2 && power >= 8)
16                     value /= 2;
17                 answer = answer / gcd(answer, (long long)value) * value;
18             }
19     if (n > 1)
20         answer = answer / gcd(answer, (long long)(n - 1)) * (n - 1);
21     return answer;
22 }
```

第三十六章 素数模数组组合数

36.1 Lucas：小素数模的大下标组合数

Lucas(p) 预处理素数模数 p, choose(n,k) 返回 $C(n,k)$ 模 p; n、k 为完整 unsigned long long 范围, $k > n$ 返回 0, $C(0,0)=1$ 。init(p) 可更换模数并重新预处理; 不检测素性, 要求调用者保证 p 为素数且 $p \geq 2$ 。预处理时间 $O(p + \log p)$, 动态版空间 $O(p)$; 单次查询 $O(1 + \log_p(n+1))$ 。适合小素数模数, 通常 $p \leq 10^6$; 大素数不能因查询下标小就忽略这里整段 $O(p)$ 的预处理。fac、ifac 下标为 $0..p-1$, 模数和表只读; 中间乘法用 long long。合数模数请使用 exLucas。

```
1 struct Lucas
2 {
3     int mod;
4     vector<int> fac, ifac;
5
6     Lucas(int p)
7     {
8         init(p);
9     }
10
11     void init(int p)
12     {
13         assert(p >= 2);
14         mod = p;
15         fac.resize(p);
16         ifac.resize(p);
17         fac[0] = 1;
18         for (int i = 1; i < p; i++)
19             fac[i] = 1LL * fac[i - 1] * i % p;
20         long long a = fac[p - 1], r = 1;
21         for (int e = p - 2; e; e >>= 1)
22         {
23             if (e & 1)
24                 r = r * a % p;
25             a = a * a % p;
26         }
27         ifac[p - 1] = r;
28         for (int i = p - 1; i; i--)
29             ifac[i - 1] = 1LL * ifac[i] * i % p;
30     }
```



```
32  int choose(unsigned long long n, unsigned long long k) const
33  {
34      if (k > n)
35          return 0;
36      long long answer = 1;
37      while (n || k)
38      {
39          int a = n % mod, b = k % mod;
40          if (b > a)
41              return 0;
42          answer = answer * fac[a] % mod;
43          answer = answer * ifac[b] % mod;
44          answer = answer * ifac[a - b] % mod;
45          n /= mod;
46          k /= mod;
47      }
48      return answer;
49  }
50  };
```

第三十七章 合数模数组合数

37.1 合数模数组合数 (扩展 Lucas)

ExLucas(mod) 预处理固定模数, choose(n,k) 返回 $C(n,k)$ 模 mod; n、k 为 unsigned long long, $k > n$ 返回 0, mod=1 返回 0。适合 $\text{mod} \leq 10^6$ 的合数或素数模数, 不能用于 10^9 级素数的整段预处理。分解 mod 为互素素数幂 $q = p^e$, 预处理每段内不含 p 因子的前缀积; unit(n,f) 去除 $n!$ 中全部 p 因子, valuation 统计指数。分子分母指数之差不少于 e 时该项为零, 否则只对互素单位部分求逆, 最后 CRT 合并。预处理 $O(\sqrt{\text{mod}} + \sum q)$, 空间 $O(\sum(q+1))$; 每次查询 $O(\sum(\log_p(n+1) \cdot \log(n+1) + \log q))$, 求和遍历不同素因子。所有乘法用 long long; 同一对象可重复查询, factors、fac 与辅助函数视为内部数据。

```
1 struct ExLucas
2 {
3     using ull = unsigned long long;
4     using ll = long long;
5
6     struct Factor
7     {
8         int p, q, exponent, start, crt;
9     };
10
11     int mod;
12     vector<int> fac;
13     vector<Factor> factors;
14
15     static ll power(ll a, ull b, int m)
16     {
17         ll result = 1;
18         while (b)
19         {
20             if (b & 1)
21                 result = result * a % m;
22             a = a * a % m;
23             b >>= 1;
24         }
25         return result;
26     }
27
28     ExLucas(int m) : mod(m)
29     {
30         assert(m >= 1);
31         int x = m;
32         for (int p = 2; p <= x / p; p++)
33         {
34             if (x % p)
35                 continue;
36             int q = 1, e = 0;
37             do
38             {
39                 x /= p;
40                 q *= p;
41                 e++;
```

```

42         } while (x % p == 0);
43         add_factor(p, q, e);
44     }
45     if (x > 1)
46         add_factor(x, x, 1);
47 }
48
49 void add_factor(int p, int q, int e)
50 {
51     int start = fac.size();
52     fac.push_back(1);
53     for (int i = 1; i < q; i++)
54         fac.push_back(1ll(fac.back()) * (i % p ? i : 1) % q);
55     fac.push_back(fac.back());
56     int phi = q - q / p;
57     int crt = 1ll(mod / q) * power(mod / q % q, phi - 1, q) % mod;
58     factors.push_back({p, q, e, start, crt});
59 }
60
61 static ull valuation(ull n, int p)
62 {
63     ull result = 0;
64     while (n)
65     {
66         n /= p;
67         result += n;
68     }
69     return result;
70 }
71
72 ll unit(ull n, const Factor &f) const
73 {
74     ll result = 1;
75     while (n)
76     {
77         result = result * power(fac[f.start + f.q], n / f.q, f.q) % f.q;
78         result = result * fac[f.start + n % f.q] % f.q;
79         n /= f.p;
80     }
81     return result;
82 }

```

```
84     int choose(ull n, ull k) const
85     {
86         if (k > n || mod == 1)
87             return 0;
88         ll answer = 0;
89         for (const auto &f : factors)
90         {
91             ull e = valuation(n, f.p) - valuation(k, f.p) - valuation(n - k, f.p);
92             if (e >= ull(f.exponent))
93                 continue;
94             ll denominator = unit(k, f) * unit(n - k, f) % f.q;
95             ll residue =
96                 unit(n, f) * power(denominator, f.q - f.q / f.p - 1, f.q) % f.q;
97             residue = residue * power(f.p, e, f.q) % f.q;
98             answer = (answer + residue * f.crt) % mod;
99         }
100         return answer;
101     }
102 };
```

第三十八章 二次剩余

38.1 素数模平方根 (Cipolla)

`mod_sqrt(n,p)` 返回 `vector<long long>`, 按升序列出全部不同根; 无解返回空向量, 零或模数 2 返回单根。要求 p 为素数, $2 \leq p \leq \text{INT_MAX}$, n 支持完整 `long long` 范围并在入口归一化。先用 Euler 判别; $p \equiv 3 \pmod{4}$ 时直接快速幂, 其余使用 Cipolla 二次扩域。函数内保存固定种子的伪随机生成器; 随机性影响寻找非剩余的尝试次数, 不影响结果, 通常期望 $O(\log p)$ 时间、 $O(1)$ 辅助空间。乘法用 `long long`, 扩域实部的三因子乘积中间先取模; 指数中的 $p+1$ 先转 `long long`, 避免 `INT_MAX` 溢出。此版本不处理合数模数及 64 位大素数, 也不用于浮点开方。

```
1 inline vector<long long> mod_sqrt(long long n, int p)
2 {
3     using ll = long long;
4     assert(p >= 2);
5     n = (n % p + p) % p;
6     if (n == 0 || p == 2)
7         return {n};
8     auto power = [p](ll a, ll b)
9     {
10         ll result = 1;
11         while (b)
12         {
13             if (b & 1)
14                 result = result * a % p;
15             a = a * a % p;
16             b >>= 1;
17         }
18         return result;
19     };
20     if (power(n, (p - 1) / 2) != 1)
21         return {};
22     ll answer;
23     if (p % 4 == 3)
24         answer = power(n, (ll(p) + 1) / 4);
25     else
26     {
27         static mt19937_64 rng(712367);
28         ll a, w;
29         do
30         {
31             a = rng() % p;
32             w = (a * a % p - n + p) % p;
33         } while (power(w, (p - 1) / 2) != p - 1);
34         using P = pair<ll, ll>;
35         auto mul = [p, w](P x, P y)
36         {
37             ll real = (x.first * y.first + x.second * y.second % p * w) % p;
38             ll imag = (x.first * y.second + x.second * y.first) % p;
39             return P{real, imag};
40         };
41         P result{1, 0}, base{a, 1};
```

```
42     ll exponent = (ll(p) + 1) / 2;
43     while (exponent)
44     {
45         if (exponent & 1)
46             result = mul(result, base);
47         base = mul(base, base);
48         exponent >>= 1;
49     }
50     answer = result.first;
51 }
52 ll other = p - answer;
53 if (answer > other)
54     swap(answer, other);
55 return {answer, other};
56 }
```

第三十九章 多项式插值

39.1 拉格朗日插值与连续点优化

先声明 `Lagrange.init(xs,ys,p)` 使用任意横坐标, `xs`、`ys` 为等长 0-based `vector<long long>`; `consecutive(ys,p)` 专用于横坐标 $0..n-1$ 。两者均覆盖旧数据。要求 `p` 为素数, $2 \leq p \leq \text{INT_MAX}$, 点数 $n \leq p$, 任意横坐标取模后两两不同; 不是仅要求原整数不同。`query(k)` 返回次数小于 `n` 的插值多项式在 `k` 处的值, 允许 `k`、`xs`、`ys` 为完整 `long long` 范围, 内部先归一化; 空数据约定零多项式。通用预处理 $O(n^2+n \log p)$, 连续点预处理 $O(n+\log p)$, 每次查询 $O(n)$, 空间 $O(n)$ 。查询利用前缀积与滚动后缀积, 无需求逆, 因此 `k` 与样本点模 `p` 相等时也正确。若求 $1^d + \dots + k^d$, 要求 $0 \leq d \leq p-2$, 先计算 $0..d+1$ 的前缀和作为 `ys`, 再用 `consecutive`; 用快速幂生成这些样本另花 $O(d \log(d+1))$ 。仅处理素数域中的精确插值, 不适用于浮点曲线拟合或重复节点的导数约束。

```
1 struct Lagrange
2 {
3     using ll = long long;
4     int n, mod;
5     vector<ll> x, w;
6
7     ll norm(ll a) const
8     {
9         a %= mod;
10        if (a < 0)
11            a += mod;
12        return a;
13    }
14
15    ll power(ll a, int b) const
16    {
17        ll result = 1;
18        while (b)
19        {
20            if (b & 1)
21                result = result * a % mod;
22            a = a * a % mod;
23            b >>= 1;
24        }
25        return result;
26    }
27
28    void init(const vector<ll> &xs, const vector<ll> &ys, int p)
29    {
30        n = ys.size();
31        mod = p;
32        assert(p >= 2 && n <= p && xs.size() == ys.size());
33        x.resize(n);
34        w.resize(n);
35        for (int i = 0; i < n; i++)
36            x[i] = norm(xs[i]);
37        for (int i = 0; i < n; i++)
38        {
39            ll denominator = 1;
```

```

40     for (int j = 0; j < n; j++)
41         if (i != j)
42             denominator = denominator * norm(x[i] - x[j]) % mod;
43     assert(denominator != 0);
44     w[i] = norm(ys[i]) * power(denominator, mod - 2) % mod;
45 }
46 }
47
48 void consecutive(const vector<ll> &ys, int p)
49 {
50     n = ys.size();
51     mod = p;
52     assert(p >= 2 && n <= p);
53     x.resize(n);
54     w.resize(n);
55     if (!n)
56         return;
57     vector<ll> invfac(n, 1);
58     ll factorial = 1;
59     for (int i = 1; i < n; i++)
60         factorial = factorial * i % mod;
61     invfac[n - 1] = power(factorial, mod - 2);
62     for (int i = n - 1; i > 0; i--)
63         invfac[i - 1] = invfac[i] * i % mod;
64     for (int i = 0; i < n; i++)
65     {
66         x[i] = i;
67         w[i] = norm(ys[i]) * invfac[i] % mod * invfac[n - 1 - i] % mod;
68         if ((n - 1 - i) & 1)
69             w[i] = norm(-w[i]);
70     }
71 }
72
73 ll query(ll k) const
74 {
75     k = norm(k);
76     vector<ll> pre(n + 1, 1);
77     for (int i = 0; i < n; i++)
78         pre[i + 1] = pre[i] * norm(k - x[i]) % mod;
79     ll answer = 0, suffix = 1;
80     for (int i = n - 1; i >= 0; i--)
81     {
82         answer = (answer + w[i] * pre[i] % mod * suffix) % mod;
83         suffix = suffix * norm(k - x[i]) % mod;
84     }
85     return answer;
86 }
87 };

```


第四十章 多项式

40.1 NTT、形式幂级数、FWT 与递推

依赖 number_theory.hpp。模数固定 998244353，乘法结果长度 $\leq 2^{23}$ ；inverse 要求 $a[0] \neq 0$ ，log 要求 $a[0]=1$ ，exp 要求 $a[0]=0$ ，截断长度 $n>0$ 。朴素逐层 NTT 实现的 FPS 为 $O(n \log n)$ 。FWT 的 op 为 $\wedge$ 、 $|$ 、 $\&$ ；BM 返回 $s[n]=\sum c[i-1]s[n-i]$ 的系数；recurrence 用 $O(k^2 \log n)$ 求第 n 项。BM 对有限样本给出的递推不自动保证未知后续项成立。

```
1 struct Polynomial
2 {
3     static constexpr int mod = 998244353;
4     using Z = ModInt<mod>;
5     using Poly = vector<Z>;
6
7     static void ntt(Poly &a, bool invert = false)
8     {
9         int n = (int)a.size();
10        assert(n > 0 && (n & (n - 1)) == 0 && n <= (1 << 23));
11        for (int i = 1, j = 0; i < n; i++)
12        {
13            int bit = n >> 1;
14            for (; j & bit; bit >>= 1)
15                j ^= bit;
16            j ^= bit;
17            if (i < j)
18                swap(a[i], a[j]);
19        }
20        for (int len = 2; len <= n; len *= 2)
21        {
22            Z step = Z(3).pow((mod - 1) / len);
23            if (invert)
24                step = step.inv();
25            for (int i = 0; i < n; i += len)
26            {
27                Z w = 1;
28                for (int j = 0; j < len / 2; j++, w = w * step)
29                {
30                    Z u = a[i + j], v = a[i + j + len / 2] * w;
31                    a[i + j] = u + v;
32                    a[i + j + len / 2] = u - v;
33                }
34            }
35        }
36        if (invert)
37        {
38            Z inv = Z(n).inv();
39            for (auto &x : a)
40                x = x * inv;
41        }
42    }
43 }
```

```

44 static Poly multiply(Poly a, Poly b)
45 {
46     if (a.empty() || b.empty())
47         return {};
48     int size = (int)a.size() + (int)b.size() - 1, n = 1;
49     assert(size <= (1 << 23));
50     while (n < size)
51         n *= 2;
52     a.resize(n);
53     b.resize(n);
54     ntt(a);
55     ntt(b);
56     for (int i = 0; i < n; i++)
57         a[i] = a[i] * b[i];
58     ntt(a, true);
59     a.resize(size);
60     return a;
61 }
62
63 static Poly inverse(const Poly &a, int n)
64 {
65     assert(n > 0 && !a.empty() && a[0].v);
66     Poly b{a[0].inv()};
67     while ((int)b.size() < n)
68     {
69         int m = min(n, 2 * (int)b.size());
70         Poly f(a.begin(), a.begin() + min(m, (int)a.size()));
71         auto t = multiply(f, b);
72         t.resize(m);
73         for (auto &x : t)
74             x = Z(0) - x;
75         t[0] = t[0] + 2;
76         b = multiply(b, t);
77         b.resize(m);
78     }
79     return b;
80 }
81
82 static Poly derivative(const Poly &a)
83 {
84     Poly b;
85     for (int i = 1; i < (int)a.size(); i++)
86         b.push_back(a[i] * i);
87     return b;
88 }
89
90 static Poly integral(const Poly &a)
91 {
92     Poly b(a.size() + 1);
93     vector<Z> inv(a.size() + 1);
94     if (!a.empty())
95         inv[1] = 1;
96     for (int i = 2; i <= (int)a.size(); i++)
97         inv[i] = Z(mod - mod / i) * inv[mod % i];
98     for (int i = 0; i < (int)a.size(); i++)
99         b[i + 1] = a[i] * inv[i + 1];

```

```

100     return b;
101 }
102
103 static Poly log(const Poly &a, int n)
104 {
105     assert(n > 0 && !a.empty() && a[0].v == 1);
106     auto b = multiply(derivative(a), inverse(a, n));
107     b.resize(n - 1);
108     return integral(b);
109 }
110
111 static Poly exp(const Poly &a, int n)
112 {
113     assert(n > 0 && (a.empty() || a[0].v == 0));
114     Poly b{1};
115     while ((int)b.size() < n)
116     {
117         int m = min(n, 2 * (int)b.size());
118         auto t = log(b, m);
119         for (int i = 0; i < m; i++)
120             t[i] = (i < (int)a.size() ? a[i] : Z(0)) - t[i];
121         t[0] = t[0] + 1;
122         b = multiply(b, t);
123         b.resize(m);
124     }
125     return b;
126 }
127
128 static void fwt(Poly &a, char op, bool invert = false)
129 {
130     int n = (int)a.size();
131     assert(n > 0 && (n & (n - 1)) == 0);
132     const Z inv2 = (mod + 1) / 2;
133     for (int len = 1; len < n; len *= 2)
134         for (int i = 0; i < n; i += len * 2)
135             for (int j = 0; j < len; j++)
136             {
137                 Z &u = a[i + j], &v = a[i + j + len], x = u, y = v;
138                 if (op == '^')
139                 {
140                     u = x + y;
141                     v = x - y;
142                     if (invert)
143                     {
144                         u = u * inv2;
145                         v = v * inv2;
146                     }
147                 }
148                 else if (op == '|')
149                     v = invert ? y - x : y + x;
150                 else
151                 {
152                     assert(op == '&');
153                     u = invert ? x - y : x + y;
154                 }
155             }

```

```

156     }
157
158     // c: s[n]=sum c[i-1]*s[n-i], i=1..L.
159     static Poly berlekamp_massey(const Poly &s)
160     {
161         Poly c{1}, b{1};
162         int l = 0, m = 1;
163         Z last = 1;
164         for (int n = 0; n < (int)s.size(); n++)
165         {
166             Z d = s[n];
167             for (int i = 1; i <= l; i++)
168                 d = d + c[i] * s[n - i];
169             if (!d.v)
170             {
171                 ++m;
172                 continue;
173             }
174             auto old = c;
175             Z coef = d / last;
176             if (c.size() < b.size() + m)
177                 c.resize(b.size() + m);
178             for (int j = 0; j < (int)b.size(); j++)
179                 c[j + m] = c[j + m] - coef * b[j];
180             if (2 * l <= n)
181             {
182                 l = n + 1 - l;
183                 b = old;
184                 last = d;
185                 m = 1;
186             }
187             else
188                 ++m;
189         }
190         c.erase(c.begin());
191         for (auto &x : c)
192             x = Z(0) - x;
193         return c;
194     }
195
196     static Z recurrence(Poly init, const Poly &c, unsigned long long n)
197     {
198         int k = (int)c.size();
199         if (n < init.size())
200             return init[n];
201         if (!k)
202             return 0;
203         assert((int)init.size() >= k);
204         auto combine = [&](const Poly &a, const Poly &b)
205         {
206             Poly t(2 * k - 1);
207             for (int i = 0; i < k; i++)
208                 for (int j = 0; j < k; j++)
209                     t[i + j] = t[i + j] + a[i] * b[j];
210             for (int i = 2 * k - 2; i >= k; i--)
211                 for (int j = 1; j <= k; j++)

```

```
212         t[i - j] = t[i - j] + t[i] * c[j - 1];
213     t.resize(k);
214     return t;
215 };
216 Poly a(k), r(k);
217 r[0] = 1;
218 if (k == 1)
219     a[0] = c[0];
220 else
221     a[1] = 1;
222 for (; n; n >= 1, a = combine(a, a))
223     if (n & 1)
224         r = combine(r, a);
225 Z ans = 0;
226 for (int i = 0; i < k; i++)
227     ans = ans + r[i] * init[i];
228 return ans;
229 }
230 };
```

第四十一章 代数与数论进阶

41.1 模高斯消元、行列式、矩阵树定理

素数模数。solve(a,n) 输入增广矩阵，返回 consistent、rank、特解 particular 和零空间基 kernel。determinant(empty)=1。spanning_trees(n,edges) 使用 0-based 无向图，允许重边，忽略自环；返回生成树数模 p。高斯与行列式 $O(n^3)$ ，矩阵幂 $O(n^3 \log e)$ 。

```
1 template <int mod> struct LinearAlgebra
2 {
3     using Z = ModInt<mod>;
4     using Matrix = vector<vector<Z>>>;
5
6     struct Solution
7     {
8         bool consistent;
9         int rank;
10        vector<Z> particular;
11        Matrix kernel;
12    };
13
14    // Augmented m x (n+1), prime modulus. Empty system needs explicit n.
15    static Solution solve(Matrix a, int n)
16    {
17        int m = (int)a.size(), row = 0;
18        vector<int> where(n, -1);
19        for (const auto &v : a)
20            assert((int)v.size() == n + 1);
21        for (int col = 0; col < n && row < m; col++)
22        {
23            int p = row;
24            while (p < m && !a[p][col].v)
25                ++p;
26            if (p == m)
27                continue;
28            swap(a[p], a[row]);
29            Z inv = a[row][col].inv();
30            for (int j = col; j <= n; j++)
31                a[row][j] = a[row][j] * inv;
32            for (int i = 0; i < m; i++)
33                if (i != row && a[i][col].v)
34                {
35                    Z f = a[i][col];
36                    for (int j = col; j <= n; j++)
37                        a[i][j] = a[i][j] - f * a[row][j];
38                }
39            where[col] = row++;
40        }
41        for (int i = row; i < m; i++)
42            if (a[i][n].v)
43                return {false, row, {}, {}};
44        Solution ans{true, row, vector<Z>(n), {}};
```

```

45     for (int j = 0; j < n; j++)
46         if (where[j] != -1)
47             ans.particular[j] = a[where[j]][n];
48     for (int j = 0; j < n; j++)
49         if (where[j] == -1)
50         {
51             vector<Z> v(n);
52             v[j] = 1;
53             for (int k = 0; k < n; k++)
54                 if (where[k] != -1)
55                     v[k] = Z(0) - a[where[k]][j];
56             ans.kernel.push_back(v);
57         }
58     return ans;
59 }
60
61 static Z determinant(Matrix a)
62 {
63     int n = (int)a.size();
64     Z ans = 1;
65     for (int i = 0; i < n; i++)
66     {
67         assert((int)a[i].size() == n);
68         int p = i;
69         while (p < n && !a[p][i].v)
70             ++p;
71         if (p == n)
72             return 0;
73         if (p != i)
74         {
75             swap(a[p], a[i]);
76             ans = Z(0) - ans;
77         }
78         ans = ans * a[i][i];
79         Z inv = a[i][i].inv();
80         for (int j = i + 1; j < n; j++)
81         {
82             Z f = a[j][i] * inv;
83             for (int k = i; k < n; k++)
84                 a[j][k] = a[j][k] - f * a[i][k];
85         }
86     }
87     return ans;
88 }
89
90 static Matrix multiply(const Matrix &a, const Matrix &b)
91 {
92     assert(!a.empty() && !b.empty());
93     int n = a.size(), m = b[0].size(), k = b.size();
94     assert((int)a[0].size() == k);
95     Matrix c(n, vector<Z>(m));
96     for (int i = 0; i < n; i++)
97         for (int t = 0; t < k; t++)
98             for (int j = 0; j < m; j++)
99                 c[i][j] = c[i][j] + a[i][t] * b[t][j];
100     return c;

```

```

101     }
102
103     static Matrix power(Matrix a, unsigned long long e)
104     {
105         int n = a.size();
106         assert(n > 0 && (int)a[0].size() == n);
107         Matrix r(n, vector<Z>(n));
108         for (int i = 0; i < n; i++)
109             r[i][i] = 1;
110         for (; e >>= 1, a = multiply(a, a))
111             if (e & 1)
112                 r = multiply(r, a);
113         return r;
114     }
115
116     // Undirected multigraph, vertices 0..n-1; loops ignored.
117     static Z spanning_trees(int n, const vector<pair<int, int>> &edges)
118     {
119         assert(n > 0);
120         Matrix lap(n, vector<Z>(n));
121         for (auto [u, v] : edges)
122             if (u != v)
123             {
124                 lap[u][u] = lap[u][u] + 1;
125                 lap[v][v] = lap[v][v] + 1;
126                 lap[u][v] = lap[u][v] - 1;
127                 lap[v][u] = lap[v][u] - 1;
128             }
129         lap.pop_back();
130         for (auto &row : lap)
131             row.pop_back();
132         return determinant(lap);
133     }
134 };

```


41.2 杜教筛

构造预筛上限 $B \geq 1$, $\text{mertens}(n)$ 返回 μ 的前缀和, $\text{totient_sum}(n)$ 返回 φ 的前缀和 (int128)。约定 $0 \leq n \leq 10^{11}$, 按整除分块递归并缓存; 大输入需要合理 B , 不能把常数 B 当作性能保证。

```

1 struct DuJiao
2 {
3     using ll = long long;
4     using I = __int128_t;
5     int limit;
6     vector<ll> pmu, pphi;
7     unordered_map<ll, ll> mmu;
8     unordered_map<ll, I> mphi;
9
10    DuJiao(int limit) : limit(limit), pmu(limit + 1), pphi(limit + 1)
11    {
12        assert(limit >= 1);
13        LinearSieve s(limit);
14        for (int i = 1; i <= limit; i++)
15        {
16            pmu[i] = pmu[i - 1] + s.mu[i];
17            pphi[i] = pphi[i - 1] + s.phi[i];
18        }
19    }
20
21    ll mertens(ll n)
22    {
23        if (n <= limit)
24            return pmu[n];
25        if (mmu.count(n))
26            return mmu[n];
27        ll ans = 1;
28        for (ll l = 2, r; l <= n; l = r + 1)
29        {
30            r = n / (n / l);
31            ans -= (r - l + 1) * mertens(n / l);
32        }
33        return mmu[n] = ans;
34    }
35
36    I totient_sum(ll n)
37    {
38        if (n <= limit)
39            return pphi[n];
40        if (mphi.count(n))
41            return mphi[n];
42        I ans = I(n) * (n + 1) / 2;
43        for (ll l = 2, r; l <= n; l = r + 1)
44        {
45            r = n / (n / l);
46            ans -= I(r - l + 1) * totient_sum(n / l);
47        }
48        return mphi[n] = ans;
49    }
50 };

```

41.3 扩展 BSGS

依赖 `number_theory.hpp` 的求逆与快速幂。`solve(a,b,m)` 返回最小 $x \geq 0$ 使 $a^x \equiv b \pmod{m}$ ，无解返回 `-1`； $1 \leq m \leq 10^{12}$ ， a 、 b 可为完整 `long long` 范围，入口归一化，约定 $0^0=1$ 。先剥离 `gcd` 并逐次检查较小指数，再在互质剩余问题上 BSGS。`baby` 表对同余幂保留最小指数 j ，按递增大步 i 检查区间，因此首个命中即最小解。单次期望时间 $O(\sqrt{m} + \log m)$ ，空间 $O(\sqrt{m})$ ，哈希表先 `reserve`；基于 `unordered_map` 的界为期望界。每次调用独立建表；底数 1 及零指数解直接返回。

```

1 struct DiscreteLog
2 {
3     using ll = long long;
4
5     // 1<=m<=1e12; Expected O(sqrt(m)) time, O(sqrt(m)) storage. Smallest x>=0, or -1.
6     static ll solve(ll a, ll b, ll m)
7     {
8         assert(m >= 1 && m <= 1000000000000LL);
9         a = (a % m + m) % m;
10        b = (b % m + m) % m;
11        if (m == 1 || b == 1)
12            return 0;
13        if (a == 1)
14            return -1;
15        ll offset = 0, k = 1;
16        for (ll g; (g = gcd(a, m)) > 1;)
17        {
18            if (b == k)
19                return offset;
20            if (b % g)
21                return -1;
22            b /= g;
23            m /= g;
24            k = (__int128)k * (a / g) % m;
25            ++offset;
26        }
27        ll target = (__int128)b * NumberTheory::inverse(k, m) % m;
28        ll step = sqrtl(m) + 1;
29        unordered_map<ll, ll> baby;
30        baby.reserve(step);
31        ll cur = 1 % m;
32        for (ll j = 0; j < step; j++)
33        {
34            if (!baby.count(cur))
35                baby[cur] = j;
36            cur = (__int128)cur * a % m;
37        }
38        ll inv = NumberTheory::inverse(NumberTheory::power(a, step, m), m);
39        cur = target;

```

```
40     for (ll i = 0; i <= step; i++)
41     {
42         auto it = baby.find(cur);
43         if (it != baby.end())
44             return offset + i * step + it->second;
45         cur = (__int128)cur * inv % m;
46     }
47     return -1;
48 }
49 };
```

第四十二章 计算几何

42.1 整数精确二维几何

坐标绝对值 $\leq 10^{12}$ ，点数 $\leq 10^6$ 。叉积、点积、线段相交、凸包、点在多边形、点在凸包、旋转卡壳直径。凸包逆时针，不保留共线内点，不重复首点。contains 返回 0 外部、1 边界、2 内部。convex_contains/diameter2 输入必须为本模板 strict hull。PolarLess 要求排除零向量。

```
1 struct IntegerGeometry
2 {
3     using ll = long long;
4     using I = __int128_t;
5
6     // Contract: |coordinate|<=1e12. Differences/products widened before arithmetic.
7     struct Point
8     {
9         ll x = 0, y = 0;
10
11         bool operator<(Point b) const
12         {
13             return tie(x, y) < tie(b.x, b.y);
14         }
15
16         bool operator==(Point b) const
17         {
18             return x == b.x && y == b.y;
19         }
20     };
21
22     static I cross(Point a, Point b, Point c)
23     {
24         return (I(b.x) - a.x) * (I(c.y) - a.y) - (I(b.y) - a.y) * (I(c.x) - a.x);
25     }
26
27     static I dot(Point a, Point b, Point c)
28     {
29         return (I(b.x) - a.x) * (I(c.x) - a.x) + (I(b.y) - a.y) * (I(c.y) - a.y);
30     }
31
32     static I dist2(Point a, Point b)
33     {
34         return dot(a, b, b);
35     }
36
37     static int sign(I x)
38     {
39         return (x > 0) - (x < 0);
40     }
41
42     static bool on_segment(Point p, Point a, Point b)
43     {
44         return cross(a, b, p) == 0 && dot(p, a, b) <= 0;
```

```

45     }
46
47     static bool intersect(Point a, Point b, Point c, Point d)
48     {
49         int x = sign(cross(a, b, c)), y = sign(cross(a, b, d));
50         int z = sign(cross(c, d, a)), w = sign(cross(c, d, b));
51         if (!x && on_segment(c, a, b))
52             return true;
53         if (!y && on_segment(d, a, b))
54             return true;
55         if (!z && on_segment(a, c, d))
56             return true;
57         if (!w && on_segment(b, c, d))
58             return true;
59         return x * y < 0 && z * w < 0;
60     }
61
62     static vector<Point> hull(vector<Point> p)
63     {
64         sort(p.begin(), p.end());
65         p.erase(unique(p.begin(), p.end(), p.end()));
66         int n = (int)p.size();
67         if (n < 3)
68             return p;
69         vector<Point> h;
70         for (Point v : p)
71         {
72             while (h.size() > 1 && cross(h[h.size() - 2], h.back(), v) <= 0)
73                 h.pop_back();
74             h.push_back(v);
75         }
76         int low = (int)h.size();
77         for (int i = n - 2; i >= 0; i--)
78         {
79             while ((int)h.size() > low && cross(h[h.size() - 2], h.back(), p[i]) <= 0)
80                 h.pop_back();
81             h.push_back(p[i]);
82         }
83         h.pop_back();
84         return h; // CCW, no repeated endpoint, no collinear interior points.
85     }
86
87     static I area2(const vector<Point> &p)
88     {
89         I ans = 0;
90         int n = (int)p.size();
91         for (int i = 0; i < n; i++)
92             ans += I(p[i].x) * p[(i + 1) % n].y - I(p[i].y) * p[(i + 1) % n].x;
93         return ans;
94     }
95
96     // 0 outside, 1 boundary, 2 inside. Simple polygon; either orientation.
97     static int contains(const vector<Point> &p, Point q)
98     {
99         int winding = 0, n = (int)p.size();
100        for (int i = 0; i < n; i++)

```

```

101     {
102         Point a = p[i], b = p[(i + 1) % n];
103         if (on_segment(q, a, b))
104             return 1;
105         I c = cross(a, b, q);
106         if (a.y <= q.y && b.y > q.y && c > 0)
107             ++winding;
108         if (a.y > q.y && b.y <= q.y && c < 0)
109             --winding;
110     }
111     return winding ? 2 : 0;
112 }
113
114 // Input must be the strict CCW hull above. O(log n).
115 static int convex_contains(const vector<Point> &p, Point q)
116 {
117     int n = (int)p.size();
118     if (n < 3)
119         return contains(p, q);
120     I a = cross(p[0], p[1], q), b = cross(p[0], p[n - 1], q);
121     if (a < 0 || b > 0)
122         return 0;
123     if (!a)
124         return on_segment(q, p[0], p[1]) ? 1 : 0;
125     if (!b)
126         return on_segment(q, p[0], p[n - 1]) ? 1 : 0;
127     int l = 1, r = n - 1;
128     while (l + 1 < r)
129     {
130         int m = (l + r) / 2;
131         if (cross(p[0], p[m], q) >= 0)
132             l = m;
133         else
134             r = m;
135     }
136     I c = cross(p[l], p[r], q);
137     return c < 0 ? 0 : c == 0 ? 1 : 2;
138 }
139
140 static I diameter2(const vector<Point> &p)
141 {
142     int n = (int)p.size();
143     if (n < 2)
144         return 0;
145     if (n == 2)
146         return dist2(p[0], p[1]);
147     I ans = 0;
148     int j = 1;
149     for (int i = 0; i < n; i++)
150     {
151         int k = (i + 1) % n;
152         while (cross(p[i], p[k], p[(j + 1) % n]) > cross(p[i], p[k], p[j]))
153             j = (j + 1) % n;
154         ans = max({ans, dist2(p[i], p[j]), dist2(p[k], p[j])});
155     }
156     return ans;

```

```

157     }
158
159     struct PolarLess
160     {
161         static int half(Point p)
162         {
163             return p.y < 0 || (p.y == 0 && p.x < 0);
164         }
165
166         bool operator()(Point a, Point b) const
167         {
168             if (half(a) != half(b))
169                 return half(a) < half(b);
170             I c = cross({0, 0}, a, b);
171             if (c)
172                 return c > 0;
173             return dist2({0, 0}, a) < dist2({0, 0}, b);
174         }
175     };
176 };

```

42.2 浮点点线圆构造

long double, 显式 eps 默认 10^{-12} 。要求有限输入及适当尺度；接近相切时按容差分类，不能提供任意实数输入的精确拓扑保证。Result 区分 none/one/two/infinite/degenerate。圆圆交点仅在圆心和正半径完全相同时返回 infinite；零半径圆按点处理，不把近同心圆当作重合。圆相关交点容差按局部距离和半径缩放，不使用固定的单位长度下限。零长度线返回 degenerate；投影与点到线段距离将零线段视为点。overlap 返回交叠面积，小圆弓用级数避免三角函数相减消失。坐标缩放和误差预算仍须按题面复核。

```

1 struct RealGeometry
2 {
3     using R = long double;
4     // Tolerance is explicit, not used in sorting. Requires finite, well-scaled input.
5     R eps;
6
7     RealGeometry(R eps = 1e-12L) : eps(eps)
8     {
9         assert(eps > 0);
10    }
11
12    struct Point
13    {
14        R x = 0, y = 0;
15
16        Point operator+(Point b) const
17        {
18            return {x + b.x, y + b.y};
19        }
20
21        Point operator-(Point b) const
22        {
23            return {x - b.x, y - b.y};
24        }
25

```

```

26     Point operator*(R k) const
27     {
28         return {x * k, y * k};
29     }
30
31     Point operator/(R k) const
32     {
33         return {x / k, y / k};
34     }
35 };
36
37 struct Circle
38 {
39     Point o;
40     R r;
41 };
42 enum class Kind
43 {
44     none,
45     one,
46     two,
47     infinite,
48     degenerate
49 };
50
51 struct Result
52 {
53     Kind kind;
54     vector<Point> p;
55 };
56
57 static R dot(Point a, Point b)
58 {
59     return a.x * b.x + a.y * b.y;
60 }
61
62 static R cross(Point a, Point b)
63 {
64     return a.x * b.y - a.y * b.x;
65 }
66
67 static R norm(Point a)
68 {
69     return hypotl(a.x, a.y);
70 }
71
72 static Point perp(Point a)
73 {
74     return {-a.y, a.x};
75 }
76
77 Point projection(Point p, Point a, Point b) const
78 {
79     Point v = b - a;
80     R d = dot(v, v);
81     if (d == 0)

```



```

82         return a;
83     return a + v * (dot(p - a, v) / d);
84 }
85
86 R segment_distance(Point p, Point a, Point b) const
87 {
88     Point v = b - a;
89     R d = dot(v, v);
90     if (d == 0)
91         return norm(p - a);
92     R t = clamp(dot(p - a, v) / d, R(0), R(1));
93     return norm(p - (a + v * t));
94 }
95
96 Result line_intersection(Point a, Point b, Point c, Point d) const
97 {
98     Point u = b - a, v = d - c;
99     R nu = norm(u), nv = norm(v);
100    if (nu == 0 || nv == 0)
101        return {Kind::degenerate, {}};
102    R det = cross(u, v);
103    if (fabsl(det) <= eps * nu * nv)
104        return {fabsl(cross(c - a, u)) <= eps * nu * max(R(1), norm(c - a))
105                ? Kind::infinite
106                : Kind::none,
107                {}};
108    return {Kind::one, {a + u * (cross(c - a, v) / det)}};
109 }
110
111 Result line_circle(Point a, Point b, Circle c) const
112 {
113     assert(c.r >= 0);
114     Point v = b - a;
115     R len = norm(v);
116     if (len == 0)
117         return {Kind::degenerate, {}};
118     Point h = projection(c.o, a, b);
119     R d = norm(h - c.o), tol = eps * max(d, c.r);
120     if (d > c.r + tol)
121         return {Kind::none, {}};
122     if (fabsl(d - c.r) <= tol)
123         return {Kind::one, {h}};
124     R t = sqrtl(max(R(0), (c.r - d) * (c.r + d)));
125     Point w = v * (t / len);
126     return {Kind::two, {h - w, h + w}};
127 }
128
129 Result circles(Circle a, Circle b) const
130 {
131     assert(a.r >= 0 && b.r >= 0);
132     Point v = b.o - a.o;
133     R d = norm(v);
134     R tol = eps * max({d, a.r, b.r});
135     // Infinite intersections require identical positive-radius circles.
136     if (d == 0)
137         {

```

```

138     if (a.r != b.r)
139         return {Kind::none, {}};
140     if (a.r == 0)
141         return {Kind::one, {a.o}};
142     return {Kind::infinite, {}};
143 }
144 if (a.r == 0 && b.r == 0)
145     return {Kind::none, {}};
146 if (a.r == 0 || b.r == 0)
147 {
148     R radius = max(a.r, b.r);
149     if (fabsl(d - radius) > tol)
150         return {Kind::none, {}};
151     return {Kind::one, {a.r == 0 ? a.o : b.o}};
152 }
153 if (d > a.r + b.r + tol || d < fabsl(a.r - b.r) - tol)
154     return {Kind::none, {}};
155 R x = (d * d + (a.r - b.r) * (a.r + b.r)) / (2 * d);
156 Point h = a.o + v * (x / d);
157 // Near-concentric circles must not be mistaken for internal tangency.
158 if (fabsl(x) > a.r + tol)
159     return {Kind::none, {}};
160 if (fabsl(fabsl(x) - a.r) <= tol)
161     return {Kind::one, {h}};
162 R y = sqrtl(max(R(0), (a.r - x) * (a.r + x)));
163 Point w = perp(v) * (y / d);
164 return {Kind::two, {h - w, h + w}};
165 }
166
167 R overlap(Circle a, Circle b) const
168 {
169     assert(a.r >= 0 && b.r >= 0);
170     R d = norm(a.o - b.o), pi = acosl(-1.L);
171     if (d >= a.r + b.r)
172         return 0;
173     if (d <= fabsl(a.r - b.r))
174         return pi * min(a.r, b.r) * min(a.r, b.r);
175     R difference = (a.r - b.r) * (a.r + b.r);
176     R x = acosl(clamp((d * d + difference) / (2 * d * a.r), R(-1), R(1)));
177     R y = acosl(clamp((d * d - difference) / (2 * d * b.r), R(-1), R(1)));
178     auto segment = [](R angle)
179     {
180         if (angle < 1e-3L)
181         {
182             R z = angle * angle;
183             return angle * z * (R(2) / 3 + z * (-R(2) / 15 + z * R(4) / 315));
184         }
185         return angle - sinl(2 * angle) / 2;
186     };
187     return a.r * a.r * segment(x) + b.r * b.r * segment(y);
188 }
189 };

```

第四十三章 优化与可持久化

43.1 Max-plus 矩阵幂

$a[i][j]$ 为转移收益，不存在的边为 neg ； $power(k)$ 求恰好 k 步的最优转移，单位矩阵对角为 0。所有有限中间值绝对值小于 2^{119} 。 $O(n^3 \log k)$ ；2025 成都 B 的转移类型，具体题目的建模仍需调用者完成。

```
1 struct MaxPlusMatrix
2 {
3     using I = __int128_t;
4     static constexpr I neg = -(I(1) << 120);
5     int n;
6     vector<vector<I>> a;
7
8     MaxPlusMatrix(int n, bool identity = false) : n(n), a(n, vector<I>(n, neg))
9     {
10         if (identity)
11             for (int i = 0; i < n; i++)
12                 a[i][i] = 0;
13     }
14
15     MaxPlusMatrix operator*(const MaxPlusMatrix &b) const
16     {
17         assert(n == b.n);
18         MaxPlusMatrix c(n);
19         for (int i = 0; i < n; i++)
20             for (int k = 0; k < n; k++)
21                 if (a[i][k] != neg)
22                     for (int j = 0; j < n; j++)
23                         if (b.a[k][j] != neg)
24                             c.a[i][j] = max(c.a[i][j], a[i][k] + b.a[k][j]);
25         return c;
26     }
27
28     MaxPlusMatrix power(unsigned long long e) const
29     {
30         MaxPlusMatrix b = *this, r(n, true);
31         for (; e >>= 1, b = b * b)
32             if (e & 1)
33                 r = r * b;
34         return r;
35     }
36 };
```

43.2 离散李超树与最优直线编号

构造传入所有可能查询的横坐标， $add(\{k,b,id\})$ 插入直线， $query(x)$ 返回最小值与最小编号，要求 x 在预登记坐标内且至少有一条线。整数求值用 $int128$ ， k 、 b 、 x 为有符号 64 位且中间值可表示。单次 $O(\log n)$ 。

```
1 struct LiChao
2 {
```

```

3  using ll = long long;
4  using I = __int128_t;
5
6  struct Line
7  {
8      ll k = 0, b = 0;
9      int id = -1;
10
11      I value(ll x) const
12      {
13          return I(k) * x + b;
14      }
15  };
16
17  struct Node
18  {
19      Line line;
20      int left = -1, right = -1;
21  };
22
23  vector<ll> xs;
24  vector<Node> a{Node{}};
25
26  LiChao(vector<ll> x) : xs(x)
27  {
28      sort(xs.begin(), xs.end());
29      xs.erase(unique(xs.begin(), xs.end()), xs.end());
30      assert(!xs.empty());
31  }
32
33  bool better(Line u, Line v, ll x) const
34  {
35      return v.id == -1 ||
36             (u.id != -1 && pair{u.value(x), u.id} < pair{v.value(x), v.id});
37  }
38
39  void add(Line line)
40  {
41      assert(line.id >= 0);
42      add(0, 0, (int)xs.size() - 1, line);
43  }
44
45  void add(int p, int l, int r, Line line)
46  {
47      int m = (l + r) / 2;
48      if (better(line, a[p].line, xs[m]))
49          swap(line, a[p].line);
50      if (l == r || line.id == -1)
51          return;
52      bool left = better(line, a[p].line, xs[l]);
53      bool right = better(line, a[p].line, xs[r]);
54      if (!left && !right)
55          return;
56      int next = left ? a[p].left : a[p].right;
57      if (next == -1)
58          {

```

```

59     next = (int)a.size();
60     a.push_back(Node{});
61     if (left)
62         a[p].left = next;
63     else
64         a[p].right = next;
65 }
66 if (left)
67     add(next, l, m, line);
68 else
69     add(next, m + 1, r, line);
70 }
71
72 pair<I, int> query(ll x) const
73 {
74     int idx = lower_bound(xs.begin(), xs.end(), x) - xs.begin();
75     assert(idx < (int)xs.size() && xs[idx] == x);
76     int p = 0, l = 0, r = (int)xs.size() - 1;
77     Line best;
78     while (p != -1)
79     {
80         if (better(a[p].line, best, x))
81             best = a[p].line;
82         if (l == r)
83             break;
84         int m = (l + r) / 2;
85         if (idx <= m)
86         {
87             p = a[p].left;
88             r = m;
89         }
90         else
91         {
92             p = a[p].right;
93             l = m + 1;
94         }
95     }
96     assert(best.id != -1);
97     return {best.value(x), best.id};
98 }
99 };

```

43.3 主席树：静态区间第 k 小

构造传入 0-based 数组， $\text{kth}(l, r, k)$ 的区间使用 1-based 闭区间；要求 $1 \leq k \leq r - l + 1$ 。离散化后每个前缀一个版本，单次 $O(\log n)$ ，总空间 $O(n \log n)$ 。

```

1 struct PersistentKth
2 {
3     struct Node
4     {
5         int l = 0, r = 0, sum = 0;
6     };
7
8     vector<Node> t{Node{}};

```

```

9     vector<int> root{0};
10    vector<long long> vals;
11
12    PersistentKth(const vector<long long> &a) : vals(a)
13    {
14        sort(vals.begin(), vals.end());
15        vals.erase(unique(vals.begin(), vals.end()), vals.end());
16        for (long long x : a)
17        {
18            int p = lower_bound(vals.begin(), vals.end(), x) - vals.begin();
19            root.push_back(add(root.back(), 0, (int)vals.size() - 1, p));
20        }
21    }
22
23    int add(int old, int l, int r, int x)
24    {
25        int p = t.size();
26        Node copy = t[old];
27        t.push_back(copy);
28        ++t[p].sum;
29        if (l < r)
30        {
31            int m = (l + r) / 2;
32            if (x <= m)
33            {
34                int v = add(t[old].l, l, m, x);
35                t[p].l = v;
36            }
37            else
38            {
39                int v = add(t[old].r, m + 1, r, x);
40                t[p].r = v;
41            }
42        }
43        return p;
44    }
45
46    long long kth(int l, int r, int k) const
47    {
48        assert(1 <= l && l <= r && r < (int)root.size() && 1 <= k && k <= r - l + 1);
49        int u = root[l - 1], v = root[r], L = 0, R = (int)vals.size() - 1;
50        while (L < R)
51        {
52            int m = (L + R) / 2, c = t[t[v].l].sum - t[t[u].l].sum;
53            if (k <= c)
54            {
55                u = t[u].l;
56                v = t[v].l;
57                R = m;
58            }
59            else

```

```
60         {
61             k -= c;
62             u = t[u].r;
63             v = t[v].r;
64             L = m + 1;
65         }
66     }
67     return vals[L];
68 }
69 };
```

第四十四章 可持久化数组

44.1 可持久化数组：历史版本分叉与单点修改

PersistentArray(a,changes=0), a 是非空 0-based 初值数组; 第 0 版为初始状态。set(version,pos,value) 基于任意已有版本修改 1-based 位置, 创建并返回新版本编号; copy(version) 以 $O(1)$ 时间复制版本; query(version,pos) 只读且不自动创建版本。values 支持 long long 全范围。每次修改复制根到叶的节点, 旧版本不变; build 为内部函数, 不可覆盖历史节点。建树 $O(n)$, 修改及查询 $O(\log n)$, 初始 $2n-1$ 个节点, 每次修改至多 $\text{ceil}(\log_2(n))+1$ 个; 总空间 $O(n + \text{修改数} * \log n + \text{版本数})$ 。changes 仅用于预留节点池, 低估后可扩容, 所有外部引用应使用下标, 不保存 vector 元素引用。节点下标用 int, 节点总量须可表示。

```
1 struct PersistentArray
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0;
8         ll value = 0;
9     };
10
11     int n;
12     vector<Node> t{Node{}};
13     vector<int> root;
14
15     PersistentArray(const vector<ll> &a, int changes = 0) : n(a.size())
16     {
17         assert(n > 0 && changes >= 0);
18         int levels = 1;
19         for (int x = n - 1; x; x >>= 1)
20             levels++;
21         t.reserve(2ULL * n + 1ULL * changes * levels + 1);
22         root.push_back(build(1, n, a));
23     }
24
25     int build(int l, int r, const vector<ll> &a)
26     {
27         int p = t.size();
28         t.push_back(Node{});
29         if (l == r)
30             t[p].value = a[l - 1];
31         else
32         {
33             int m = (l + r) / 2;
34             int left = build(l, m, a);
35             int right = build(m + 1, r, a);
36             t[p].l = left;
37             t[p].r = right;
38         }
39         return p;
40     }
41 }
```



```

42 int modify(int old, int l, int r, int pos, ll value)
43 {
44     int p = t.size();
45     Node copy = t[old];
46     t.push_back(copy);
47     if (l == r)
48         t[p].value = value;
49     else
50     {
51         int m = (l + r) / 2;
52         if (pos <= m)
53         {
54             int child = modify(t[old].l, l, m, pos, value);
55             t[p].l = child;
56         }
57         else
58         {
59             int child = modify(t[old].r, m + 1, r, pos, value);
60             t[p].r = child;
61         }
62     }
63     return p;
64 }
65
66 // Create a new version from any existing version. Positions are 1-based.
67 int set(int version, int pos, ll value)
68 {
69     assert(0 <= version && version < (int)root.size());
70     assert(1 <= pos && pos <= n);
71     int p = modify(root[version], 1, n, pos, value);
72     root.push_back(p);
73     return (int)root.size() - 1;
74 }
75
76 int copy(int version)
77 {
78     assert(0 <= version && version < (int)root.size());
79     root.push_back(root[version]);
80     return (int)root.size() - 1;
81 }
82
83 ll query(int version, int pos) const
84 {
85     assert(0 <= version && version < (int)root.size());
86     assert(1 <= pos && pos <= n);
87     int p = root[version], l = 1, r = n;
88     while (l < r)
89     {
90         int m = (l + r) / 2;
91         if (pos <= m)
92         {
93             p = t[p].l;
94             r = m;

```

```
95         }
96         else
97         {
98             p = t[p].r;
99             l = m + 1;
100        }
101    }
102    return t[p].value;
103 }
104 };
```

第四十五章 树上路径第 k 小

45.1 树上路径点权第 k 小

TreePathKth(a) 传入非空 0-based 权值数组，节点编号 1..n；add(u,v) 加无向边，要求恰好构成一棵树。build(root=1) 递归 DFS 建立根到各节点的频数版本与倍增表，可换根重建，旧版本自动清空。kth(u,v,k) 返回简单路径上第 k 小的点权，包含两个端点，每个节点计一次； $1 \leq k \leq$ 路径点数。lca(u,v) 返回当前预处理根下的最近公共祖先。使用 $root[u] + root[v] - root[lca] - root[parent(lca)]$ ，编号 0 为全零版本，避免少算或重复算 LCA。支持重复值与完整 long long 权值，内部只统计频数；权值和树结构在查询阶段保持不变。预处理 $O(n \log n)$ ，每次查询 $O(\log n)$ ，空间 $O(n \log n)$ 。递归深度最坏 $O(n)$ ，保持递归写法。

```
1 struct TreePathKth
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, sum = 0;
8     };
9
10    int n, lg;
11    vector<vector<int>> g, up;
12    vector<int> root, dep;
13    vector<ll> value, vals;
14    vector<Node> t;
15
16    TreePathKth(const vector<ll> &a)
17        : n(a.size()), g(n + 1), root(n + 1), dep(n + 1), value(a)
18    {
19        assert(n > 0);
20        lg = 1;
21        while ((1LL << lg) <= n)
22            lg++;
23        up.assign(lg, vector<int>(n + 1));
24    }
25
26    void add(int u, int v)
27    {
28        g[u].push_back(v);
29        g[v].push_back(u);
30    }
31
32    int insert(int old, int l, int r, int x)
33    {
34        int p = t.size();
35        Node copy = t[old];
36        t.push_back(copy);
37        t[p].sum++;
38        if (l == r)
39            return p;
40        int m = (l + r) / 2;
41        if (x <= m)
```

```

42     {
43         int child = insert(t[old].l, l, m, x);
44         t[p].l = child;
45     }
46     else
47     {
48         int child = insert(t[old].r, m + 1, r, x);
49         t[p].r = child;
50     }
51     return p;
52 }
53
54 void dfs(int u, int p)
55 {
56     up[0][u] = p;
57     for (int j = 1; j < lg; j++)
58         up[j][u] = up[j - 1][up[j - 1][u]];
59     int x = lower_bound(vals.begin(), vals.end(), value[u - 1]) - vals.begin();
60     root[u] = insert(root[p], 0, (int)vals.size() - 1, x);
61     for (int v : g[u])
62     {
63         if (v == p)
64             continue;
65         dep[v] = dep[u] + 1;
66         dfs(v, u);
67     }
68 }
69
70 void build(int r = 1)
71 {
72     assert(1 <= r && r <= n);
73     vals = value;
74     sort(vals.begin(), vals.end());
75     vals.erase(unique(vals.begin(), vals.end()), vals.end());
76     t.assign(1, Node{});
77     dep[r] = 0;
78     dfs(r, 0);
79 }
80
81 int lca(int u, int v) const
82 {
83     if (dep[u] < dep[v])
84         swap(u, v);
85     int diff = dep[u] - dep[v];
86     for (int j = 0; j < lg; j++)
87         if (diff >> j & 1)
88             u = up[j][u];
89     if (u == v)
90         return u;
91     for (int j = lg - 1; j >= 0; j--)
92         if (up[j][u] != up[j][v])
93         {
94             u = up[j][u];
95             v = up[j][v];
96         }
97     return up[0][u];

```

```

98     }
99
100     ll kth(int u, int v, int k) const
101     {
102         int z = lca(u, v);
103         assert(1 <= k && k <= dep[u] + dep[v] - 2 * dep[z] + 1);
104         int a = root[u], b = root[v], c = root[z], d = root[up[0][z]];
105         int l = 0, r = (int)vals.size() - 1;
106         while (l < r)
107         {
108             int count = t[t[a].l].sum + t[t[b].l].sum - t[t[c].l].sum - t[t[d].l].sum;
109             int m = (l + r) / 2;
110             if (k <= count)
111             {
112                 a = t[a].l;
113                 b = t[b].l;
114                 c = t[c].l;
115                 d = t[d].l;
116                 r = m;
117             }
118             else
119             {
120                 k -= count;
121                 a = t[a].r;
122                 b = t[b].r;
123                 c = t[c].r;
124                 d = t[d].r;
125                 l = m + 1;
126             }
127         }
128         return vals[l];
129     }
130 };

```

第四十六章 动态区间顺序统计

46.1 动态区间第 k 小 (树状数组套权值线段树)

`DynamicKth(a,future)` 传入非空 0-based 初值, `future` 包含所有将来可能赋入的新值; 内部自动合并初值并离散化。`set(pos,value)` 单点赋值, `kth(l,r,k)` 查询闭区间第 k 小, 下标均为 1-based; $1 \leq k \leq r-l+1$ 。允许重复值与完整 `long long` 权值; 新值必须已在离散化集合内, 操作可按原顺序执行, 不保留历史版本。每个树状数组节点对应一棵可变权值线段树; 删除旧值后插入新值。空节点回收到 `free_head` 链, 重复修改不会按操作次数累计泄漏节点。设 D 为不同值数, $S=n+\text{future.size()}$, 预处理 $O(S \log S + n \log n \log(D+1))$, 单次修改及查询 $O(\log n \log(D+1))$ 。节点数上界 $B=(\text{ceil}(\log_2(D))+1) \cdot \sum \text{lowbit}(i)$, 求和 $i=1..n$; 空间 $O(S+n+B)$ 。`update`、`modify`、`id` 和节点池均为内部接口; 不能手动减去不存在的值。

```
1 struct DynamicKth
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, sum = 0;
8     };
9
10    int n, free_head = 0;
11    vector<int> bit, rank;
12    vector<ll> vals;
13    vector<Node> t{Node{}};
14
15    DynamicKth(const vector<ll> &a, vector<ll> future = {})
16        : n(a.size()), bit(n + 1), rank(n + 1)
17    {
18        assert(n > 0);
19        vals = future;
20        vals.insert(vals.end(), a.begin(), a.end());
21        sort(vals.begin(), vals.end());
22        vals.erase(unique(vals.begin(), vals.end()), vals.end());
23        for (int i = 1; i <= n; i++)
24        {
25            rank[i] = id(a[i - 1]);
26            modify(i, rank[i], 1);
27        }
28    }
29
30    int id(ll value) const
31    {
32        int p = lower_bound(vals.begin(), vals.end(), value) - vals.begin();
33        assert(p < (int)vals.size() && vals[p] == value);
34        return p;
35    }
36
37    int new_node()
38    {
39        if (!free_head)
```

```

40     {
41         t.push_back(Node{});
42         return (int)t.size() - 1;
43     }
44     int p = free_head;
45     free_head = t[p].l;
46     t[p] = Node{};
47     return p;
48 }
49
50 int update(int p, int l, int r, int x, int delta)
51 {
52     if (!p)
53         p = new_node();
54     t[p].sum += delta;
55     if (l < r)
56     {
57         int m = (l + r) / 2;
58         if (x <= m)
59         {
60             int child = update(t[p].l, l, m, x, delta);
61             t[p].l = child;
62         }
63         else
64         {
65             int child = update(t[p].r, m + 1, r, x, delta);
66             t[p].r = child;
67         }
68     }
69     if (!t[p].sum)
70     {
71         t[p].l = free_head;
72         free_head = p;
73         return 0;
74     }
75     return p;
76 }
77
78 void modify(int pos, int x, int delta)
79 {
80     for (int i = pos; i <= n; i += i & -i)
81         bit[i] = update(bit[i], 0, (int)vals.size() - 1, x, delta);
82 }

```

```

84 void set(int pos, ll value)
85 {
86     assert(1 <= pos && pos <= n);
87     int x = id(value);
88     if (rank[pos] == x)
89         return;
90     modify(pos, rank[pos], -1);
91     modify(pos, x, 1);
92     rank[pos] = x;
93 }
94
95 ll kth(int l, int r, int k) const
96 {
97     assert(1 <= l && l <= r && r <= n && 1 <= k && k <= r - l + 1);
98     vector<int> a, b;
99     for (int i = r; i; i -= i & -i)
100         a.push_back(bit[i]);
101     for (int i = l - 1; i; i -= i & -i)
102         b.push_back(bit[i]);
103     int L = 0, R = (int)vals.size() - 1;
104     while (L < R)
105     {
106         int count = 0;
107         for (int p : a)
108             count += t[t[p].l].sum;
109         for (int p : b)
110             count -= t[t[p].l].sum;
111         bool left = k <= count;
112         if (!left)
113             k -= count;
114         for (int &p : a)
115             p = left ? t[p].l : t[p].r;
116         for (int &p : b)
117             p = left ? t[p].l : t[p].r;
118         int m = (L + R) / 2;
119         if (left)
120             R = m;
121         else
122             L = m + 1;
123     }
124     return vals[L];
125 }
126 };

```


第四十七章 区间不同数统计

47.1 静态区间不同数计数（前驱位置主席树）

PersistentDistinct(a) 传入非空 0-based 数组，query(l,r) 返回闭区间内不同值的个数， $1 \leq l \leq r \leq n$ 。支持重复值与完整 long long 值域；建成后不支持改值，查询顺序任意。令 prev[i] 为同值上次出现的位置，未出现为 0；root[i] 在值域 0..n-1 中统计前 i 个 prev。区间内每种值恰有一次 prev[i]<l，因此答案为 root[r]-root[l-1] 中位置 0..l-1 的频数和。每个元素只复制一条节点路径，预处理 $O(n \log n)$ ，查询 $O(\log n)$ ，空间 $O(n \log n)$ 。动态版预留 $n \cdot (\text{ceil}(\log_2(n))+1)+1$ 个节点，避免扩容峰值；root 和节点池为内部数据。

```
1 struct PersistentDistinct
2 {
3     struct Node
4     {
5         int l = 0, r = 0, sum = 0;
6     };
7
8     int n;
9     vector<Node> t{Node{}};
10    vector<int> root;
11
12    PersistentDistinct(const vector<long long> &a) : n(a.size()), root(n + 1)
13    {
14        assert(n > 0);
15        int height = 1;
16        while ((1LL << (height - 1)) < n)
17            height++;
18        t.reserve(size_t(n) * height + 1);
19        vector<long long> vals = a;
20        sort(vals.begin(), vals.end());
21        vals.erase(unique(vals.begin(), vals.end()), vals.end());
22        vector<int> last(vals.size());
23        for (int i = 1; i <= n; i++)
24        {
25            int x = lower_bound(vals.begin(), vals.end(), a[i - 1]) - vals.begin();
26            root[i] = insert(root[i - 1], 0, n - 1, last[x]);
27            last[x] = i;
28        }
29    }
30
31    int insert(int old, int l, int r, int x)
32    {
33        int p = t.size();
34        Node copy = t[old];
35        t.push_back(copy);
36        t[p].sum++;
37        if (l < r)
38        {
39            int m = (l + r) / 2;
40            if (x <= m)
41            {
42                int child = insert(t[old].l, l, m, x);
```

```

43         t[p].l = child;
44     }
45     else
46     {
47         int child = insert(t[old].r, m + 1, r, x);
48         t[p].r = child;
49     }
50 }
51 return p;
52 }
53
54 int query(int l, int r) const
55 {
56     assert(1 <= l && l <= r && r <= n);
57     int u = root[l - 1], v = root[r];
58     int L = 0, R = n - 1, answer = 0;
59     while (L < R)
60     {
61         int m = (L + R) / 2;
62         if (l - 1 <= m)
63         {
64             u = t[u].l;
65             v = t[v].l;
66             R = m;
67         }
68         else
69         {
70             answer += t[t[v].l].sum - t[t[u].l].sum;
71             u = t[u].r;
72             v = t[v].r;
73             L = m + 1;
74         }
75     }
76     return answer + t[v].sum - t[u].sum;
77 }
78 };

```

第四十八章 图论进阶

48.1 矩形二分图最小权匹配

`solve(cost)` 输入 $n \times m$ 完整权值矩阵, $0 \leq n \leq m$; 每行匹配不同列, 返回最小费用及列号 (0-based)。允许负费用, 使用 int128 势能; $O(n^2m)$ 。禁止边不能用未经证明的有限大常数随意替代, 当前接口仅表示完整二分图。

```
1 struct Hungarian
2 {
3     using I = __int128_t;
4
5     // Minimum cost injection from n rows to m columns, n<=m. Zero-based matrix.
6     static pair<I, vector<int>> solve(const vector<vector<long long>> &cost)
7     {
8         int n = cost.size();
9         if (!n)
10             return {0, {}};
11         int m = cost[0].size();
12         assert(n <= m);
13         vector<I> u(n + 1), v(m + 1);
14         vector<int> p(m + 1), way(m + 1);
15         const I inf = I(1) << 120;
16         for (int i = 1; i <= n; i++)
17         {
18             assert((int)cost[i - 1].size() == m);
19             p[0] = i;
20             int j0 = 0;
21             vector<I> dis(m + 1, inf);
22             vector<bool> used(m + 1);
23             do
24             {
25                 used[j0] = true;
26                 int i0 = p[j0], j1 = 0;
27                 I delta = inf;
28                 for (int j = 1; j <= m; j++)
29                     if (!used[j])
30                     {
31                         I cur = I(cost[i0 - 1][j - 1]) - u[i0] - v[j];
32                         if (cur < dis[j])
33                         {
34                             dis[j] = cur;
35                             way[j] = j0;
36                         }
37                         if (dis[j] < delta)
38                         {
39                             delta = dis[j];
40                             j1 = j;
41                         }
42                     }
43                 for (int j = 0; j <= m; j++)
44                     if (used[j])
45                     {
```

```

46         u[p[j]] += delta;
47         v[j] -= delta;
48     }
49     else
50         dis[j] -= delta;
51     j0 = j1;
52 } while (p[j0]);
53 do
54 {
55     int j1 = way[j0];
56     p[j0] = p[j1];
57     j0 = j1;
58 } while (j0);
59 }
60 vector<int> ans(n);
61 for (int j = 1; j <= m; j++)
62     if (p[j])
63         ans[p[j] - 1] = j - 1;
64 return {-v[0], ans};
65 }
66 };

```

48.2 有向最小树形图

`solve(n, root, edges)`, 点号 $0..n-1$, $n \geq 1$; 返回以 `root` 为根向外可达的最小树形图费用, 无解 `nullopt`。允许负权、重边、自环 (自环不选); $O(VE)$ 。当前仅返回费用, 不恢复边方案。所有中间值须远小于 2^{120} 。

```

1 struct Arborescence
2 {
3     using I = __int128_t;
4
5     struct Edge
6     {
7         int u, v;
8         I w;
9     };
10
11     // Directed minimum rooted OUT-arborescence cost, 0-based, or nullopt.
12     static optional<I> solve(int n, int root, vector<Edge> e)
13     {
14         const I inf = I(1) << 120;
15         I ans = 0;
16         while (true)
17         {
18             vector<I> in(n, inf);
19             vector<int> pre(n, -1), id(n, -1), vis(n, -1);
20             for (auto a : e)
21                 if (a.u != a.v && a.w < in[a.v])
22                 {
23                     in[a.v] = a.w;
24                     pre[a.v] = a.u;
25                 }
26             in[root] = 0;
27             for (I x : in)
28                 {

```

```

29         if (x == inf)
30             return nullopt;
31         ans += x;
32     }
33     int cnt = 0;
34     for (int i = 0; i < n; i++)
35     {
36         int u = i;
37         while (vis[u] != i && id[u] == -1 && u != root)
38         {
39             vis[u] = i;
40             u = pre[u];
41         }
42         if (u != root && id[u] == -1)
43         {
44             id[u] = cnt;
45             for (int v = pre[u]; v != u; v = pre[v])
46                 id[v] = cnt;
47             ++cnt;
48         }
49     }
50     if (!cnt)
51         return ans;
52     for (int i = 0; i < n; i++)
53         if (id[i] == -1)
54             id[i] = cnt++;
55     for (auto &a : e)
56     {
57         int v = a.v;
58         a.u = id[a.u];
59         a.v = id[a.v];
60         if (a.u != a.v)
61             a.w -= in[v];
62     }
63     root = id[root];
64     n = cnt;
65 }
66 }
67 };

```

48.3 无向全局最小割

`solve(w)`, $n \geq 2$; w 是对称非负权矩阵, 重边需先累加, 主对角置零。返回割权与一侧顶点编号 (0-based), 允许不连通图。 $O(n^3)$, 总权值小于 2^{120} 。

```

1 struct StoerWagner
2 {
3     using I = __int128_t;
4
5     // Symmetric nonnegative adjacency matrix; returns weight and one cut side.
6     static pair<I, vector<int>> solve(vector<vector<I>> w)
7     {
8         int n = w.size();
9         assert(n >= 2);
10        vector<int> active(n);
11        iota(active.begin(), active.end(), 0);
12        vector<vector<int>> group(n);
13        for (int i = 0; i < n; i++)
14            group[i] = {i};
15        I best = I(1) << 120;
16        vector<int> side;
17        while (active.size() > 1)
18        {
19            vector<I> dis(n);
20            vector<bool> used(n);
21            int prev = -1;
22            for (int step = 0; step < (int)active.size(); step++)
23            {
24                int sel = -1;
25                for (int v : active)
26                    if (!used[v] && (sel == -1 || dis[v] > dis[sel]))
27                        sel = v;
28                if (step + 1 == (int)active.size())
29                {
30                    if (dis[sel] < best)
31                    {
32                        best = dis[sel];
33                        side = group[sel];
34                    }
35                    for (int v : active)
36                        if (v != prev && v != sel)
37                            w[prev][v] = w[v][prev] = w[prev][v] + w[sel][v];
38                    group[prev].insert(
39                        group[prev].end(), group[sel].begin(), group[sel].end());
40                    active.erase(find(active.begin(), active.end(), sel));
41                    break;
42                }

```

```
43         used[sel] = true;
44         prev = sel;
45         for (int v : active)
46             if (!used[v])
47                 dis[v] += w[sel][v];
48     }
49 }
50 return {best, side};
51 }
52 };
```

第四十九章 带权二分图匹配

49.1 二分图最大权匹配（缺边与可选匹配）

WeightedMatching(n,m), 两侧编号 1..n、1..m。add(x,y,w) 加边，重边保留最大权；边权支持完整 long long 范围，内部势能与答案为 int128。solve() 要求全部左点匹配不同右点，无解返回 nullopt；n=m 时即最大权完美匹配。solve(true) 允许点不匹配，只最大化总权，不优先最大化边数。两种模式均从空匹配重算；l[x]、r[y] 为方案，0 表示未匹配，无解时全部清零。缺边用独立于合法边权的 -2^{120} 哨兵，仅用 add 修改；不把缺边当零权边。允许空侧。全匹配 $O(n^2m)$ ，可选模式增加 n 个虚拟零权右点，为 $O(n^2(m+n))$ ；空间 $O(nm+n+m)$ 。按最短松弛量扩展交错树：每次至少访问一个新右点，最多 n 次后增广；无有限松弛量即无解。

```
1 struct WeightedMatching
2 {
3     using I = __int128_t;
4     static constexpr I inf = I(1) << 120;
5     int n, m;
6     vector<vector<I>> w;
7     vector<int> l, r;
8
9     WeightedMatching(int n, int m)
10         : n(n), m(m), w(n + 1, vector<I>(m + 1, -inf)), l(n + 1), r(m + 1)
11     {
12     }
13
14     void add(int x, int y, long long weight)
15     {
16         assert(1 <= x && x <= n && 1 <= y && y <= m);
17         w[x][y] = max(w[x][y], I(weight));
18     }
19
20     optional<I> solve(bool allow_unmatched = false)
21     {
22         fill(l.begin(), l.end(), 0);
23         fill(r.begin(), r.end(), 0);
24         if (!allow_unmatched && n > m)
25             return nullopt;
26         int cols = m + (allow_unmatched ? n : 0);
27         vector<I> u(n + 1), v(cols + 1);
28         vector<int> p(cols + 1), way(cols + 1);
29         for (int i = 1; i <= n; i++)
30         {
31             p[0] = i;
32             int j0 = 0;
33             vector<I> dis(cols + 1, inf);
34             vector<bool> used(cols + 1);
35             do
36             {
37                 used[j0] = true;
38                 int x = p[j0], j1 = 0;
39                 I delta = inf;
40                 for (int j = 1; j <= cols; j++)
41                 {
```



```

42         if (used[j])
43             continue;
44         I weight = j <= m ? w[x][j] : 0;
45         if (weight != -inf)
46         {
47             I cur = -weight - u[x] - v[j];
48             if (cur < dis[j])
49             {
50                 dis[j] = cur;
51                 way[j] = j0;
52             }
53         }
54         if (dis[j] < delta)
55         {
56             delta = dis[j];
57             j1 = j;
58         }
59     }
60     if (delta == inf)
61         return nullopt;
62     for (int j = 0; j <= cols; j++)
63     {
64         if (used[j])
65         {
66             u[p[j]] += delta;
67             v[j] -= delta;
68         }
69         else if (dis[j] != inf)
70             dis[j] -= delta;
71     }
72     j0 = j1;
73 } while (p[j0]);
74 do
75 {
76     int j1 = way[j0];
77     p[j0] = p[j1];
78     j0 = j1;
79 } while (j0);
80 }
81 I answer = 0;
82 for (int j = 1; j <= m; j++)
83 {
84     if (!p[j])
85         continue;
86     l[p[j]] = j;
87     r[j] = p[j];
88     answer += w[p[j]][j];
89 }
90 return answer;
91 }
92 };

```

第五十章 精确几何进阶

50.1 整数最近点对与 Minkowski 和

`closest_pair(points)` 返回最短距离平方，少于两个点为 `nullopt`，重复点返回 0， $O(n \log n)$ ，全程精确整数。
`minkowski` 输入 strict CCW 凸包，返回不重复首点的凸包；非退化 $O(n+m)$ ，退化时 $O((n+m)\log(n+m))$ 。坐标及坐标和须满足整数几何范围。

```
1 struct GeometryExtra
2 {
3     using G = IntegerGeometry;
4     using P = G::Point;
5     using I = G::I;
6
7     // No pair exists for n<2. Exact squared distance,  $O(n \log n)$ .
8     static optional<I> closest_pair(vector<P> p)
9     {
10         int n = p.size();
11         if (n < 2)
12             return nullopt;
13         sort(p.begin(), p.end());
14         for (int i = 1; i < n; i++)
15             if (p[i] == p[i - 1])
16                 return I(0);
17         vector<P> tmp(n);
18         auto by_y = [](P a, P b)
19         {
20             return tie(a.y, a.x) < tie(b.y, b.x);
21         };
22         function<I(int, int)> solve = [&](int l, int r) -> I
23         {
24             if (r - l <= 3)
25             {
26                 I best = I(1) << 120;
27                 for (int i = l; i < r; i++)
28                     for (int j = i + 1; j < r; j++)
29                         best = min(best, G::dist2(p[i], p[j]));
30                 sort(p.begin() + l, p.begin() + r, by_y);
31                 return best;
32             }
33             int m = (l + r) / 2;
34             long long x = p[m].x;
35             I best = min(solve(l, m), solve(m, r));
36             merge(p.begin() + l,
37                 p.begin() + m,
38                 p.begin() + m,
39                 p.begin() + r,
40                 tmp.begin() + l,
41                 by_y);
42             copy(tmp.begin() + l, tmp.begin() + r, p.begin() + l);
43             vector<P> strip;
44             for (int i = l; i < r; i++)
```

```

45         if ((I(p[i].x) - x) * (I(p[i].x) - x) < best)
46         {
47             for (int j = (int)strip.size() - 1; j >= 0; j--)
48             {
49                 I dy = I(p[i].y) - strip[j].y;
50                 if (dy * dy >= best)
51                     break;
52                 best = min(best, G::dist2(p[i], strip[j]));
53             }
54             strip.push_back(p[i]);
55         }
56         return best;
57     };
58     return solve(0, n);
59 }
60
61 // Inputs strict CCW hulls. Coordinate sums must obey IntegerGeometry's bound.
62 static vector<P> minkowski(vector<P> a, vector<P> b)
63 {
64     if (a.empty() || b.empty())
65         return {};
66     auto plus = [](P u, P v)
67     {
68         return P{u.x + v.x, u.y + v.y};
69     };
70     if (a.size() < 3 || b.size() < 3)
71     {
72         vector<P> p;
73         for (P u : a)
74             for (P v : b)
75                 p.push_back(plus(u, v));
76         return G::hull(p);
77     }
78     auto start = [](vector<P> &p)
79     {
80         auto it = min_element(p.begin(),
81                               p.end(),
82                               [](P u, P v)
83                               {
84                                   return tie(u.y, u.x) < tie(v.y, v.x);
85                               });
86         rotate(p.begin(), it, p.end());
87     };
88     start(a);
89     start(b);
90     int n = a.size(), m = b.size(), i = 0, j = 0;
91     vector<P> result;
92     while (i < n || j < m)
93     {
94         result.push_back(plus(a[i % n], b[j % m]));
95         if (i == n)
96         {
97             ++j;
98             continue;
99         }
100         if (j == m)

```

```
101         {
102             ++i;
103             continue;
104         }
105         P u{a[(i + 1) % n].x - a[i].x, a[(i + 1) % n].y - a[i].y};
106         P v{b[(j + 1) % m].x - b[j].x, b[(j + 1) % m].y - b[j].y};
107         I c = G::cross({0, 0}, u, v);
108         if (c >= 0)
109             ++i;
110         if (c <= 0)
111             ++j;
112     }
113     return result;
114 }
115 };
```

50.2 整数三维共线、共面与方向

坐标绝对值 $\leq 10^9$; $\text{orient}(a,b,c,d)$ 为有向六倍四面体体积, 0 表示共面; collinear 判三点共线, on_segment 支持退化线段。int128 精确判定, 不包含三维凸包构造。

```

1 struct IntegerGeometry3D
2 {
3     using I = __int128_t;
4
5     // |coordinates|<=1e9: triple products fit signed int128.
6     struct Point
7     {
8         long long x, y, z;
9     };
10
11     struct Vector
12     {
13         I x, y, z;
14     };
15
16     static Vector diff(Point a, Point b)
17     {
18         return {I(a.x) - b.x, I(a.y) - b.y, I(a.z) - b.z};
19     }
20
21     static Vector cross(Vector a, Vector b)
22     {
23         return {a.y * b.z - a.z * b.y, a.z * b.x - a.x * b.z, a.x * b.y - a.y * b.x};
24     }
25
26     static I dot(Vector a, Vector b)
27     {
28         return a.x * b.x + a.y * b.y + a.z * b.z;
29     }
30
31     static I orient(Point a, Point b, Point c, Point d)
32     {
33         return dot(cross(diff(b, a), diff(c, a)), diff(d, a));
34     }
35
36     static bool collinear(Point a, Point b, Point c)
37     {
38         auto v = cross(diff(b, a), diff(c, a));
39         return !v.x && !v.y && !v.z;
40     }
41
42     static bool on_segment(Point p, Point a, Point b)
43     {
44         return collinear(p, a, b) && dot(diff(a, p), diff(b, p)) <= 0;
45     }
46 };

```

第五十一章 凸包支撑点查询

51.1 支撑点查询恢复下凸包

静态 `build(left,right,oracle)` 返回从左到右的严格下凸包顶点, `Point` 为 $\{x,y\}$ 。有限非空整数点集无需显式给出; 回调 `oracle(a,b)` 必须精确返回使 $a*x+b*y$ 最大的可行点, 允许任意最优解。 `left/right` 必须分别为最小/最大横坐标中纵坐标最低的点; 全体同横坐标时二者必须相同。所有坐标绝对值不超过 10^{18} , 方向差使用 `long long`, 叉积先提升至 `int128`; 回调内部也须保证数值精确且无溢出。对弦 $a \rightarrow b$ 查询方向 $(b.y-a.y, a.x-b.x)$, 严格位于弦下方则递归分治, 否则确认边; 输出时删除共线中间点。若分治找到 K 个严格改善支撑点, 查询次数为 $2K+1$ (同横坐标时为 0), 额外时间、空间和最坏递归深度为 $O(K+1)$ 。 K 包括可能返回的共线非顶点, 不将其等同于最终顶点数。不能直接传浮点近似优化器。二分图闭包组合: 点为 $(|S|, |N(S)|)$, 左点赋权 a 、右点赋权 b , 左 $\rightarrow$ 邻点依赖; 下凸包方向满足 $a \geq 0, b < 0$, 最优闭包的右点恰为邻集。端点为 $(0,0)$ 和 $(\text{左点数}, \text{全部左点的邻点数})$, 有孤立右点时后者不等于全部右点数。该组合已做子集枚举对拍, 尚不代表关联赛题整题通过。

```
1 struct SupportHull
2 {
3     using ll = long long;
4     using I = __int128_t;
5
6     struct Point
7     {
8         ll x, y;
9     };
10
11     static I cross(Point a, Point b, Point c)
12     {
13         return (I(b.x) - a.x) * (I(c.y) - a.y) - (I(b.y) - a.y) * (I(c.x) - a.x);
14     }
```

```

16 // oracle(a,b) maximizes a*x+b*y exactly; |coordinates| <= 1e18.
17 // left/right: lowest points at the minimum/maximum x.
18 template <class Oracle>
19 static vector<Point> build(Point left, Point right, Oracle &&oracle)
20 {
21     assert(left.x <= right.x);
22     vector<Point> hull{left};
23     auto dfs = [&](auto &&self, Point a, Point b) -> void
24     {
25         Point c = oracle(b.y - a.y, a.x - b.x);
26         I turn = cross(a, b, c);
27         assert(turn <= 0);
28         if (turn < 0)
29         {
30             assert(a.x < c.x && c.x < b.x);
31             self(self, a, c);
32             self(self, c, b);
33         }
34         else
35         {
36             while (hull.size() >= 2 &&
37                 cross(hull[hull.size() - 2], hull.back(), b) == 0)
38                 hull.pop_back();
39             hull.push_back(b);
40         }
41     };
42     if (left.x < right.x)
43         dfs(dfs, left, right);
44     else
45         assert(left.y == right.y);
46     return hull;
47 }
48 };

```

第五十二章 动态树

52.1 Link-cut tree: 动态森林与路径异或

构造 LinkCutTree(n), 点号 1..n, 初始权值为零。set(x,value) 修改点权; link(x,y) 仅在不同连通块之间连边并返回是否成功; cut(x,y) 只删除真实存在的边, 否则返回 false。connected 查询连通性; query(x,y) 返回路径点权异或, 要求两点连通, 支持 x=y。make_root/find_root 维护与查询当前表示树根。每次操作均摊 $O(\log n)$, $O(n)$ 空间, splay 下传使用显式栈, 支持长链。权值为 unsigned long long。仅维护路径异或, 不包含虚子树聚合或动态直径。

```
1 struct LinkCutTree
2 {
3     using U = unsigned long long;
4
5     struct Node
6     {
7         int ch[2] {}, fa = 0;
8         U val = 0, sum = 0;
9         bool rev = false;
10    };
11
12    vector<Node> a;
13    vector<int> st;
14
15    LinkCutTree(int n) : a(n + 1), st(n + 1)
16    {
17    }
18
19    bool is_root(int x) const
20    {
21        int f = a[x].fa;
22        return a[f].ch[0] != x && a[f].ch[1] != x;
23    }
24
25    void pull(int x)
26    {
27        a[x].sum = a[a[x].ch[0]].sum ^ a[x].val ^ a[a[x].ch[1]].sum;
28    }
29
30    void reverse(int x)
31    {
32        if (!x)
33            return;
34        swap(a[x].ch[0], a[x].ch[1]);
35        a[x].rev = !a[x].rev;
36    }
37
38    void push(int x)
39    {
40        if (!a[x].rev)
41            return;
42        reverse(a[x].ch[0]);
```



```

43     reverse(a[x].ch[1]);
44     a[x].rev = false;
45 }
46
47 void rotate(int x)
48 {
49     int y = a[x].fa, z = a[y].fa;
50     int k = a[y].ch[1] == x;
51     int w = a[x].ch[k ^ 1];
52     if (!is_root(y))
53         a[z].ch[a[z].ch[1] == y] = x;
54     a[x].fa = z;
55     a[y].ch[k] = w;
56     if (w)
57         a[w].fa = y;
58     a[x].ch[k ^ 1] = y;
59     a[y].fa = x;
60     pull(y);
61     pull(x);
62 }
63
64 void splay(int x)
65 {
66     int top = 0, y = x;
67     st[top++] = y;
68     while (!is_root(y))
69     {
70         y = a[y].fa;
71         st[top++] = y;
72     }
73     while (top)
74         push(st[--top]);
75     while (!is_root(x))
76     {
77         int y = a[x].fa, z = a[y].fa;
78         if (!is_root(y))
79         {
80             bool same = (a[y].ch[1] == x) == (a[z].ch[1] == y);
81             rotate(same ? y : x);
82         }
83         rotate(x);
84     }
85 }
86
87 void access(int x)
88 {
89     int last = 0;
90     for (int y = x; y; y = a[y].fa)
91     {
92         splay(y);
93         // The old right child's fa remains y: it becomes a path-parent.
94         a[y].ch[1] = last;
95         if (last)
96             a[last].fa = y;
97         pull(y);
98         last = y;

```

```

99     }
100     splay(x);
101 }
102
103 void make_root(int x)
104 {
105     access(x);
106     reverse(x);
107 }
108
109 int find_root(int x)
110 {
111     access(x);
112     push(x);
113     while (a[x].ch[0])
114     {
115         x = a[x].ch[0];
116         push(x);
117     }
118     splay(x);
119     return x;
120 }
121
122 bool connected(int x, int y)
123 {
124     return x == y || find_root(x) == find_root(y);
125 }
126
127 bool link(int x, int y)
128 {
129     make_root(x);
130     if (find_root(y) == x)
131         return false;
132     a[x].fa = y;
133     return true;
134 }
135
136 bool cut(int x, int y)
137 {
138     make_root(x);
139     access(y);
140     if (a[y].ch[0] != x || a[x].ch[1])
141         return false;
142     a[y].ch[0] = 0;
143     a[x].fa = 0;
144     pull(y);
145     return true;
146 }
147
148 void set(int x, U value)
149 {
150     access(x);
151     a[x].val = value;
152     pull(x);

```

```
153     }
154
155     // x and y must be connected. This is a path aggregate, not a subtree sum.
156     U query(int x, int y)
157     {
158         make_root(x);
159         access(y);
160         return a[y].sum;
161     }
162 };
```

第五十三章 随机平衡树

53.1 FHQ Treap: 可重集合与顺序统计

`insert(x)` 插入一个副本, `erase(x)` 删除一个副本并返回是否存在; `rank(x)` 为严格小于 `x` 的元素数加一, `x` 不必存在; `kth(k)` 要求 $1 \leq k \leq \text{size}()$ 。 `prev/next` 为严格前驱/后继, 不存在时返回 `nullopt`。支持重复值、`LLONG_MIN` 和 `LLONG_MAX`, 分裂不使用 `x-1`。随机优先级下单次期望 $O(\log n)$, 并非确定性最坏界。节点池不回收已删除节点, 空间按累计插入次数计算; 可在构造时指定随机种子。

```
1 struct OrderedTreap
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, siz = 0;
8         ll val = 0;
9         unsigned long long pri = 0;
10    };
11
12    vector<Node> a{Node{}};
13    mt19937_64 rng;
14    int root = 0;
15
16    OrderedTreap(unsigned long long seed = 712367821) : rng(seed)
17    {
18    }
19
20    void pull(int p)
21    {
22        a[p].siz = a[a[p].l].siz + a[a[p].r].siz + 1;
23    }
24
25    int node(ll x)
26    {
27        int p = a.size();
28        a.push_back({0, 0, 1, x, rng()});
29        return p;
30    }
31
32    // left: val < x, or val <= x when equal is true. No x-1 overflow.
33    void split(int p, ll x, int &l, int &r, bool equal = false)
34    {
35        if (!p)
36        {
37            l = 0;
38            r = 0;
39            return;
40        }
41        if (a[p].val < x || (equal && a[p].val == x))
42        {
43            l = p;
```

```

44         split(a[p].r, x, a[p].r, r, equal);
45     }
46     else
47     {
48         r = p;
49         split(a[p].l, x, l, a[p].l, equal);
50     }
51     pull(p);
52 }
53
54 int merge(int l, int r)
55 {
56     if (!l || !r)
57         return l ? l : r;
58     if (a[l].pri > a[r].pri)
59     {
60         a[l].r = merge(a[l].r, r);
61         pull(l);
62         return l;
63     }
64     a[r].l = merge(l, a[r].l);
65     pull(r);
66     return r;
67 }
68
69 void insert(ll x)
70 {
71     int p = node(x), l, r;
72     split(root, x, l, r);
73     root = merge(merge(l, p), r);
74 }
75
76 bool erase(ll x)
77 {
78     int l, mid, r;
79     split(root, x, l, mid);
80     split(mid, x, mid, r, true);
81     bool found = mid != 0;
82     if (mid)
83         mid = merge(a[mid].l, a[mid].r);
84     root = merge(l, merge(mid, r));
85     return found;
86 }
87
88 int size() const
89 {
90     return a[root].siz;
91 }
92
93 int less(ll x, bool equal = false) const
94 {
95     int p = root, ans = 0;
96     while (p)
97     {
98         if (a[p].val < x || (equal && a[p].val == x))
99             {

```

```

100         ans += a[a[p].l].siz + 1;
101         p = a[p].r;
102     }
103     else
104         p = a[p].l;
105     }
106     return ans;
107 }
108
109 int rank(ll x) const
110 {
111     return less(x) + 1;
112 }
113
114 ll kth(int k) const
115 {
116     assert(1 <= k && k <= size());
117     int p = root;
118     while (true)
119     {
120         int left = a[a[p].l].siz;
121         if (k == left + 1)
122             return a[p].val;
123         if (k <= left)
124             p = a[p].l;
125         else
126         {
127             k -= left + 1;
128             p = a[p].r;
129         }
130     }
131 }
132
133 optional<ll> prev(ll x) const
134 {
135     int k = less(x);
136     if (!k)
137         return nullopt;
138     return kth(k);
139 }
140
141 optional<ll> next(ll x) const
142 {
143     int k = less(x, true);
144     if (k == size())
145         return nullopt;
146     return kth(k + 1);
147 }
148 };

```

53.2 隐式 Treap：序列翻转、加值与区间和

`insert(k,value)` 在前 k 个元素后插入, $0 \leq k \leq \text{size}()$; `erase(l,r)`、`reverse(l,r)`、`add(l,r,value)`、`query(l,r)` 均使用 1-based 非空闭区间。`values()` 返回当前完整序列, $O(n)$, 使用递归中序遍历。其余操作在随机优先级下期望 $O(\log$

n); 翻转与加法懒标记同时维护。所有值、和、标记及 $\text{value} \times \text{长度}$ 中间结果必须落在 `long long` 范围内。删除不回收节点，内存按累计插入数计算。

```

1 struct SequenceTreap
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, siz = 0;
8         ll val = 0, sum = 0, tag = 0;
9         unsigned long long pri = 0;
10        bool rev = false;
11    };
12
13    vector<Node> a{Node{}};
14    mt19937_64 rng;
15    int root = 0;
16
17    SequenceTreap(unsigned long long seed = 712367821) : rng(seed)
18    {
19    }
20
21    int size() const
22    {
23        return a[root].siz;
24    }
25
26    void pull(int p)
27    {
28        a[p].siz = a[a[p].l].siz + a[a[p].r].siz + 1;
29        a[p].sum = a[a[p].l].sum + a[p].val + a[a[p].r].sum;
30    }
31
32    void apply_add(int p, ll x)
33    {
34        if (!p)
35            return;
36        a[p].val += x;
37        a[p].sum += x * a[p].siz;
38        a[p].tag += x;
39    }
40
41    void apply_reverse(int p)
42    {
43        if (!p)
44            return;
45        swap(a[p].l, a[p].r);
46        a[p].rev = !a[p].rev;
47    }
48
49    void push(int p)
50    {
51        if (a[p].rev)
52        {
53            apply_reverse(a[p].l);

```

```

54         apply_reverse(a[p].r);
55         a[p].rev = false;
56     }
57     if (a[p].tag)
58     {
59         apply_add(a[p].l, a[p].tag);
60         apply_add(a[p].r, a[p].tag);
61         a[p].tag = 0;
62     }
63 }
64
65 // Split into the first k elements and the rest.
66 void split(int p, int k, int &l, int &r)
67 {
68     if (!p)
69     {
70         l = 0;
71         r = 0;
72         return;
73     }
74     push(p);
75     int left = a[a[p].l].siz;
76     if (k > left)
77     {
78         l = p;
79         split(a[p].r, k - left - 1, a[p].r, r);
80     }
81     else
82     {
83         r = p;
84         split(a[p].l, k, l, a[p].l);
85     }
86     pull(p);
87 }
88
89 int merge(int l, int r)
90 {
91     if (!l || !r)
92         return l ? l : r;
93     if (a[l].pri > a[r].pri)
94     {
95         push(l);
96         a[l].r = merge(a[l].r, r);
97         pull(l);
98         return l;
99     }
100    push(r);
101    a[r].l = merge(l, a[r].l);
102    pull(r);
103    return r;
104 }
105
106 // Insert after the first k elements, 0 <= k <= size().
107 void insert(int k, ll value)
108 {
109     assert(0 <= k && k <= size());

```



```

110     int p = a.size(), l, r;
111     a.push_back({0, 0, 1, value, value, 0, rng(), false});
112     split(root, k, l, r);
113     root = merge(merge(l, p), r);
114 }
115
116 // All range operations use 1-based inclusive intervals.
117 void erase(int l, int r)
118 {
119     assert(1 <= l && l <= r && r <= size());
120     int x, y, z;
121     split(root, r, y, z);
122     split(y, l - 1, x, y);
123     root = merge(x, z);
124 }
125
126 void reverse(int l, int r)
127 {
128     assert(1 <= l && l <= r && r <= size());
129     int x, y, z;
130     split(root, r, y, z);
131     split(y, l - 1, x, y);
132     apply_reverse(y);
133     root = merge(merge(x, y), z);
134 }
135
136 void add(int l, int r, ll value)
137 {
138     assert(1 <= l && l <= r && r <= size());
139     int x, y, z;
140     split(root, r, y, z);
141     split(y, l - 1, x, y);
142     apply_add(y, value);
143     root = merge(merge(x, y), z);
144 }
145
146 ll query(int l, int r)
147 {
148     assert(1 <= l && l <= r && r <= size());
149     int x, y, z;
150     split(root, r, y, z);
151     split(y, l - 1, x, y);
152     ll ans = a[y].sum;
153     root = merge(merge(x, y), z);
154     return ans;
155 }
156
157 void collect(int p, vector<ll> &out)
158 {
159     if (!p)
160         return;
161     push(p);
162     collect(a[p].l, out);
163     out.push_back(a[p].val);

```

```
164         collect(a[p].r, out);
165     }
166
167     vector<ll> values()
168     {
169         vector<ll> answer;
170         collect(root, answer);
171         return answer;
172     }
173 };
```

第五十四章 伸展树

54.1 Splay: 可重集合与顺序统计

insert/erase/rank/kth/prev/next 与有序 Treap 同义, 严格前驱后继不存在返回 nullopt。重复键计数存储; 失败查询也伸展最后访问节点, 单次均摊 $O(\log n)$ 。不回收节点, 空间按累计新建节点数计算。支持 long long 全范围; 不包含序列 Splay、虚子树信息或矩阵动态 DP。

```
1 struct OrderedSplay
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int ch[2]{}, fa = 0, cnt = 0, siz = 0;
8         ll val = 0;
9     };
10
11     vector<Node> a{Node{}};
12     int root = 0;
13
14     int size() const
15     {
16         return a[root].siz;
17     }
18
19     void pull(int x)
20     {
21         a[x].siz = a[a[x].ch[0]].siz + a[x].cnt + a[a[x].ch[1]].siz;
22     }
23
24     int node(ll value, int parent)
25     {
26         int x = a.size();
27         a.push_back({{0, 0}, parent, 1, 1, value});
28         return x;
29     }
30
31     void rotate(int x)
32     {
33         int y = a[x].fa, z = a[y].fa;
34         int k = a[y].ch[1] == x;
35         int w = a[x].ch[k ^ 1];
36         if (z)
37             a[z].ch[a[z].ch[1] == y] = x;
38         else
39             root = x;
40         a[x].fa = z;
41         a[y].ch[k] = w;
42         if (w)
43             a[w].fa = y;
44         a[x].ch[k ^ 1] = y;
```

```

45     a[y].fa = x;
46     pull(y);
47     pull(x);
48 }
49
50 void splay(int x)
51 {
52     while (a[x].fa)
53     {
54         int y = a[x].fa, z = a[y].fa;
55         if (z)
56         {
57             bool same = (a[y].ch[1] == x) == (a[z].ch[1] == y);
58             rotate(same ? y : x);
59         }
60         rotate(x);
61     }
62 }
63
64 // Splay the last visited node even when the value does not exist.
65 int find(ll value)
66 {
67     int x = root, last = 0;
68     while (x && a[x].val != value)
69     {
70         last = x;
71         x = a[x].ch[value > a[x].val];
72     }
73     if (x)
74         splay(x);
75     else if (last)
76         splay(last);
77     return x;
78 }
79
80 void insert(ll value)
81 {
82     int x = root, parent = 0;
83     while (x && a[x].val != value)
84     {
85         parent = x;
86         x = a[x].ch[value > a[x].val];
87     }
88     if (x)
89     {
90         ++a[x].cnt;
91         pull(x);
92     }
93     else
94     {
95         x = node(value, parent);
96         if (parent)
97             a[parent].ch[value > a[parent].val] = x;
98         else
99             root = x;
100    }

```

```

101     splay(x);
102 }
103
104 bool erase(ll value)
105 {
106     int x = find(value);
107     if (!x)
108         return false;
109     if (a[x].cnt > 1)
110     {
111         --a[x].cnt;
112         pull(x);
113         return true;
114     }
115     int l = a[x].ch[0], r = a[x].ch[1];
116     if (l)
117         a[l].fa = 0;
118     if (r)
119         a[r].fa = 0;
120     if (!l)
121     {
122         root = r;
123         return true;
124     }
125     root = l;
126     x = l;
127     while (a[x].ch[1])
128         x = a[x].ch[1];
129     splay(x);
130     a[x].ch[1] = r;
131     if (r)
132         a[r].fa = x;
133     pull(x);
134     return true;
135 }
136
137 int rank(ll value)
138 {
139     int x = root, last = 0, ans = 1;
140     while (x)
141     {
142         last = x;
143         if (value <= a[x].val)
144             x = a[x].ch[0];
145         else
146         {
147             ans += a[a[x].ch[0]].siz + a[x].cnt;
148             x = a[x].ch[1];
149         }
150     }
151     if (last)
152         splay(last);
153     return ans;
154 }
155
156 ll kth(int k)

```

```

157 {
158     assert(1 <= k && k <= size());
159     int x = root;
160     while (true)
161     {
162         int left = a[a[x].ch[0]].siz;
163         if (k <= left)
164             x = a[x].ch[0];
165         else if (k <= left + a[x].cnt)
166         {
167             splay(x);
168             return a[x].val;
169         }
170         else
171         {
172             k -= left + a[x].cnt;
173             x = a[x].ch[1];
174         }
175     }
176 }
177
178 optional<ll> prev(ll value)
179 {
180     int x = root, last = 0, answer = 0;
181     while (x)
182     {
183         last = x;
184         if (a[x].val < value)
185         {
186             answer = x;
187             x = a[x].ch[1];
188         }
189         else
190             x = a[x].ch[0];
191     }
192     if (last)
193         splay(last);
194     if (!answer)
195         return nullopt;
196     splay(answer);
197     return a[answer].val;
198 }
199
200 optional<ll> next(ll value)
201 {
202     int x = root, last = 0, answer = 0;
203     while (x)
204     {
205         last = x;
206         if (a[x].val > value)
207         {
208             answer = x;
209             x = a[x].ch[0];
210         }
211         else

```

```
212         x = a[x].ch[1];
213     }
214     if (last)
215         splay(last);
216     if (!answer)
217         return nullopt;
218     splay(answer);
219     return a[answer].val;
220 }
221 };
```

第五十五章 状态序列维护

55.1 状态序列 Treap：分组区间 GCD

build(v) 用数值与状态组成的序列替换当前内容。数值为无符号 64 位，状态为 0 或 1。insert(k,value,tag) 在前 k 项后插入。erase(l,r)、query(l,r,tag) 使用 1-based 非空闭区间；set(k,value) 修改值，toggle(k) 翻转状态。query 返回选中状态的 GCD，无对应元素为 nullopt，实际全零则返回 0。回收删除节点，空间按峰值长度计算。建树 $O(n)$ 次 GCD，随机优先级下单次期望 $O(\log n)$ 次 GCD；删除另需 $O(\text{删除长度})$ 回收。一次 GCD 为 $O(\log U)$ 。覆盖 kuangbin 3.7 的序列操作，尚无该原题 OJ 记录。建树后的聚合与删除节点回收均使用递归遍历；笛卡尔树的构造单调栈保留。

```
1 struct GcdSequenceTreap
2 {
3     using U = unsigned long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, siz = 0, tag = 0;
8         int cnt[2]{};
9         U val = 0, g[2]{}, pri = 0;
10    };
11
12    vector<Node> a{Node{}};
13    vector<int> recycled;
14    mt19937_64 rng;
15    int root = 0;
16
17    GcdSequenceTreap(U seed = 712367821) : rng(seed)
18    {
19    }
20
21    int size() const
22    {
23        return a[root].siz;
24    }
25
26    int node(U value, int tag)
27    {
28        assert(tag == 0 || tag == 1);
29        int p;
30        if (recycled.empty())
31        {
32            p = a.size();
33            a.push_back(Node{});
34        }
35        else
36        {
37            p = recycled.back();
38            recycled.pop_back();
39            a[p] = Node{};
40        }
41        a[p].siz = 1;
```



```

42     a[p].val = value;
43     a[p].tag = tag;
44     a[p].cnt[tag] = 1;
45     a[p].g[tag] = value;
46     a[p].pri = rng();
47     return p;
48 }
49
50 void pull(int p)
51 {
52     int l = a[p].l, r = a[p].r;
53     a[p].siz = a[l].siz + a[r].siz + 1;
54     for (int t = 0; t < 2; t++)
55     {
56         a[p].cnt[t] = a[l].cnt[t] + a[r].cnt[t] + (a[p].tag == t);
57         a[p].g[t] = gcd(a[l].g[t], a[r].g[t]);
58         if (a[p].tag == t)
59             a[p].g[t] = gcd(a[p].g[t], a[p].val);
60     }
61 }
62
63 void split(int p, int k, int &l, int &r)
64 {
65     if (!p)
66     {
67         l = 0;
68         r = 0;
69         return;
70     }
71     int left = a[a[p].l].siz;
72     if (k > left)
73     {
74         l = p;
75         split(a[p].r, k - left - 1, a[p].r, r);
76     }
77     else
78     {
79         r = p;
80         split(a[p].l, k, l, a[p].l);
81     }
82     pull(p);
83 }
84
85 int merge(int l, int r)
86 {
87     if (!l || !r)
88         return l ? l : r;
89     if (a[l].pri > a[r].pri)
90     {
91         a[l].r = merge(a[l].r, r);
92         pull(l);
93         return l;
94     }
95     a[r].l = merge(l, a[r].l);
96     pull(r);
97     return r;

```

```

98     }
99
100    void rebuild(int p)
101    {
102        if (!p)
103            return;
104        rebuild(a[p].l);
105        rebuild(a[p].r);
106        pull(p);
107    }
108
109    void release(int p)
110    {
111        if (!p)
112            return;
113        release(a[p].l);
114        release(a[p].r);
115        recycled.push_back(p);
116    }
117
118    // O(n) GCD calls; replaces the current sequence.
119    void build(const vector<pair<U, int>> &v)
120    {
121        a.assign(1, Node{});
122        recycled.clear();
123        root = 0;
124        vector<int> st;
125        for (auto [value, tag] : v)
126        {
127            int p = node(value, tag), last = 0;
128            while (!st.empty() && a[st.back()].pri < a[p].pri)
129            {
130                last = st.back();
131                st.pop_back();
132            }
133            a[p].l = last;
134            if (!st.empty())
135                a[st.back()].r = p;
136            st.push_back(p);
137        }
138        if (st.empty())
139            return;
140        root = st.front();
141        rebuild(root);
142    }
143
144    // Insert after the first k entries. Range operations below are 1-based.
145    void insert(int k, U value, int tag)
146    {
147        assert(0 <= k && k <= size());
148        int p = node(value, tag), l, r;
149        split(root, k, l, r);
150        root = merge(merge(l, p), r);
151    }
152
153    void erase(int l, int r)

```

```

154     {
155         assert(1 <= l && l <= r && r <= size());
156         int x, y, z;
157         split(root, r, y, z);
158         split(y, l - 1, x, y);
159         root = merge(x, z);
160         release(y);
161     }
162
163     void set(int k, U value)
164     {
165         assert(1 <= k && k <= size());
166         int x, y, z;
167         split(root, k, y, z);
168         split(y, k - 1, x, y);
169         a[y].val = value;
170         pull(y);
171         root = merge(merge(x, y), z);
172     }
173
174     void toggle(int k)
175     {
176         assert(1 <= k && k <= size());
177         int x, y, z;
178         split(root, k, y, z);
179         split(y, k - 1, x, y);
180         a[y].tag ^= 1;
181         pull(y);
182         root = merge(merge(x, y), z);
183     }
184
185     optional<U> query(int l, int r, int tag)
186     {
187         assert(1 <= l && l <= r && r <= size());
188         assert(tag == 0 || tag == 1);
189         int x, y, z;
190         split(root, r, y, z);
191         split(y, l - 1, x, y);
192         optional<U> answer;
193         if (a[y].cnt[tag])
194             answer = a[y].g[tag];
195         root = merge(merge(x, y), z);
196         return answer;
197     }
198 };

```

第五十六章 一般图匹配

56.1 一般图最大基数匹配：带花树

Blossom(n), 点号 $0..n-1$; add(u,v) 加无向边, 忽略自环, 允许重边。solve() 从空匹配重新计算, 返回匹配边数; mate[u] 为配对点, 未匹配为 -1。结果支持直接输出方案。缩花维护交错树, 单次增广 $O(n^2+m)$, 总计 $O(n^3+nm)$, 简单图为 $O(n^3)$; 空间 $O(n+m)$ 。重复调用 solve 或加边后重算均可。不处理边权, 不能代替最大权匹配。

```
1 struct Blossom
2 {
3     int n;
4     vector<vector<int>> g;
5     vector<int> mate, pre, base, q;
6     vector<bool> outer, flower, seen;
7
8     Blossom(int n)
9         : n(n), g(n), mate(n, -1), pre(n), base(n), outer(n), flower(n), seen(n)
10    {
11    }
12
13    void add(int u, int v)
14    {
15        assert(0 <= u && u < n && 0 <= v && v < n);
16        if (u == v)
17            return;
18        g[u].push_back(v);
19        g[v].push_back(u);
20    }
21
22    int lca(int u, int v)
23    {
24        fill(seen.begin(), seen.end(), false);
25        while (true)
26        {
27            u = base[u];
28            seen[u] = true;
29            if (mate[u] == -1)
30                break;
31            u = pre[mate[u]];
32        }
33        while (!seen[base[v]])
34            v = pre[mate[base[v]]];
35        return base[v];
36    }
37
38    void mark(int u, int b, int v)
39    {
40        while (base[u] != b)
41        {
42            flower[base[u]] = true;
43            flower[base[mate[u]]] = true;
44            pre[u] = v;
```

```

45         v = mate[u];
46         u = pre[v];
47     }
48 }
49
50 bool augment(int root)
51 {
52     fill(outer.begin(), outer.end(), false);
53     fill(pre.begin(), pre.end(), -1);
54     iota(base.begin(), base.end(), 0);
55     q.assign(1, root);
56     outer[root] = true;
57     for (int head = 0; head < (int)q.size(); head++)
58     {
59         int u = q[head];
60         for (int v : g[u])
61         {
62             if (base[u] == base[v] || mate[u] == v)
63                 continue;
64             if (v == root || (mate[v] != -1 && pre[mate[v]] != -1))
65             {
66                 int b = lca(u, v);
67                 fill(flower.begin(), flower.end(), false);
68                 mark(u, b, v);
69                 mark(v, b, u);
70                 for (int x = 0; x < n; x++)
71                 {
72                     if (!flower[base[x]])
73                         continue;
74                     base[x] = b;
75                     if (!outer[x])
76                     {
77                         outer[x] = true;
78                         q.push_back(x);
79                     }
80                 }
81             }
82             else if (pre[v] == -1)
83             {
84                 pre[v] = u;
85                 if (mate[v] == -1)
86                 {
87                     while (v != -1)
88                     {
89                         int p = pre[v];
90                         int next = mate[p];
91                         mate[v] = p;
92                         mate[p] = v;
93                         v = next;
94                     }
95                     return true;
96                 }
97                 int x = mate[v];
98                 outer[x] = true;
99                 q.push_back(x);
100             }

```

```
101         }
102     }
103     return false;
104 }
105
106 // Rebuild the maximum matching; mate[u] is -1 for an unmatched vertex.
107 int solve()
108 {
109     fill(mate.begin(), mate.end(), -1);
110     int answer = 0;
111     for (int u = 0; u < n; u++)
112     {
113         if (mate[u] == -1 && augment(u))
114             answer++;
115     }
116     return answer;
117 }
118 };
```

第五十七章 线性递推

57.1 Berlekamp–Massey：最短线性递推

`berlekamp_massey(s,p)` 要求 p 为素数, $2 \leq p \leq \text{INT_MAX}$, 输入均在 $[0,p)$; 返回 c , 使 $s[i] = \sum_{j=1..k} c[j-1]s[i-j]$ ($j=1..k, i \geq k$)。空序列或全零序列返回空系数。有限前缀可能对应多种未来, 不能无条件外推; 若已知原序列满足阶数至多 d 的齐次线性递推, 提供至少 $2d$ 项才能恢复可外推的递推。模运算次数 $O(Nk+N \log p)$, 后项来自逐次求逆; 空间 $O(N)$ 。不适用于合数模数或浮点拟合。

```
1 // Prime p; normalized input. s[i] = sum(c[j-1] * s[i-j]), i >= c.size().
2 vector<int> berlekamp_massey(const vector<int> &s, int p = 998244353)
3 {
4     assert(p >= 2);
5     vector<int> c{1}, b{1};
6     int len = 0, shift = 1, last = 1;
7     for (int i = 0; i < (int)s.size(); i++)
8     {
9         assert(0 <= s[i] && s[i] < p);
10        long long d = s[i];
11        for (int j = 1; j <= len; j++)
12            d = (d + 1LL * c[j] * s[i - j]) % p;
13        if (d == 0)
14        {
15            shift++;
16            continue;
17        }
18        long long inv = 1, x = last;
19        for (int e = p - 2; e; e >>= 1)
20        {
21            if (e & 1)
22                inv = inv * x % p;
23            x = x * x % p;
24        }
25        long long factor = d * inv % p;
26        auto old = c;
27        if (c.size() < b.size() + shift)
28            c.resize(b.size() + shift);
29        for (int j = 0; j < (int)b.size(); j++)
30            c[j + shift] = (c[j + shift] - factor * b[j] % p + p) % p;
31        if (2 * len <= i)
32        {
33            len = i + 1 - len;
34            b = old;
35            last = d;
36            shift = 1;
37        }
38    }
```

```
38         else
39             shift++;
40     }
41     c.resize(len + 1);
42     c.erase(c.begin());
43     for (int &x : c)
44         x = x == 0 ? 0 : p - x;
45     return c;
46 }
```


57.2 特征多项式求线性递推第 k 项

`recurrence_nth(init,c,n,p)` 使用 0-based 下标, n 为 unsigned long long。init 与 c 长度均为阶数 k ; $a[i] = \sum_{j=1}^k c[j-1]a[i-j]$ 。 $2 \leq p \leq \text{INT_MAX}$, 数值均在 $[0,p)$, 本函数允许合数模数。空系数约定全零序列。计算 x^n 对特征多项式取余再与初值点乘, $O(k^2 \log(n+1))$ 时间、 $O(k)$ 空间; 不依赖 BM 或其他模板。较大阶数需快速多项式或 Bostan-Mori 版本, 当前函数不覆盖该复杂度需求。

```

1 // init has exactly k entries; c[j-1] multiplies a[n-j]. Modulus need not be prime.
2 int recurrence_nth(const vector<int> &init,
3                   const vector<int> &c,
4                   unsigned long long n,
5                   int p = 998244353)
6 {
7     assert(p >= 2 && init.size() == c.size());
8     int k = c.size();
9     if (!k)
10         return 0;
11     for (int x : init)
12         assert(0 <= x && x < p);
13     for (int x : c)
14         assert(0 <= x && x < p);
15     if (n < (unsigned)k)
16         return init[n];
17     auto multiply = [&](const vector<int> &a, const vector<int> &b)
18     {
19         vector<int> t(2 * k - 1);
20         for (int i = 0; i < k; i++)
21         {
22             for (int j = 0; j < k; j++)
23                 t[i + j] = (t[i + j] + 1LL * a[i] * b[j]) % p;
24         }
25         for (int i = 2 * k - 2; i >= k; i--)
26         {
27             for (int j = 1; j <= k; j++)
28                 t[i - j] = (t[i - j] + 1LL * t[i] * c[j - 1]) % p;
29         }
30         t.resize(k);
31         return t;
32     };
33     vector<int> answer(k), x(k);
34     answer[0] = 1;
35     if (k == 1)
36         x[0] = c[0];
37     else
38         x[1] = 1;
39     while (n)
40     {
41         if (n & 1)

```

```
42         answer = multiply(answer, x);
43         x = multiply(x, x);
44         n >>= 1;
45     }
46     long long value = 0;
47     for (int i = 0; i < k; i++)
48         value = (value + 1LL * answer[i] * init[i]) % p;
49     return value;
50 }
```

第二部分

传统接口风格

第五十八章 Tarjan 与缩点

58.1 Tarjan 强连通分量与缩点

算法约束见动态版本第 2 页。本节代码独立可抄；采用下列实际接口名。

声明 Tarjan_SCC<N>, 先 Init(n), Insert 加边, Run 求分量, Dag 输出缩点图。bel 与 cnt 可直接读取；大对象放在全局或 static。

```
1 template <int N> struct Tarjan_SCC
2 {
3     int n, timer = 0, cnt = 0;
4     vector<int> g[N + 1], st;
5     int dfn[N + 1], low[N + 1], bel[N + 1];
6
7     void Init(int vertices)
8     {
9         assert(0 <= vertices && vertices <= N);
10        n = vertices;
11        for ( int u = 1; u <= n; u++ )
12            g[u].clear();
13    }
14
15    void Insert(int u, int v)
16    {
17        assert(1 <= u && u <= n && 1 <= v && v <= n);
18        g[u].push_back(v);
19    }
20
21    void Dfs(int u)
22    {
23        dfn[u] = low[u] = ++timer;
24        st.push_back(u);
25        for ( int v : g[u] )
26        {
27            if ( !dfn[v] )
28            {
29                Dfs(v);
30                low[u] = min(low[u], low[v]);
31            }
32            else if ( !bel[v] )
33                low[u] = min(low[u], dfn[v]);
34        }
35        if ( low[u] == dfn[u] )
36        {
37            ++cnt;
38            while ( true )
39            {
40                int v = st.back();
41                st.pop_back();
42                bel[v] = cnt;
43                if ( v == u )
44                    break;
```

```

45     }
46 }
47 }
48
49 // IDs 1..cnt are in reverse topological order. Recursive DFS needs O(n) stack.
50 void Run()
51 {
52     timer = 0;
53     cnt = 0;
54     fill(dfn, dfn + n + 1, 0);
55     fill(low, low + n + 1, 0);
56     fill-bel, bel + n + 1, 0);
57     st.clear();
58     for ( int u = 1; u <= n; u++ )
59     {
60         if ( !dfn[u] )
61             Dfs(u);
62     }
63 }
64
65 // Retains parallel edges. Use after Run().
66 vector<vector<int>> Dag() const
67 {
68     vector<vector<int>> result(cnt + 1);
69     for ( int u = 1; u <= n; u++ )
70     {
71         for ( int v : g[u] )
72         {
73             if ( bel[u] != bel[v] )
74                 result[bel[u]].push_back(bel[v]);
75         }
76     }
77     return result;
78 }
79 };

```

第五十九章 双连通分量与圆方树

59.1 Tarjan 点双、边双与圆方树

算法约束见动态版本第 4 页。本节代码独立可抄；采用下列实际接口名。

声明 `Biconnected_Graph<N>`, `Init(n)` 初始化, `Insert` 加边, `Run` 分解, `Block_Forest` 和 `Bridge_Forest` 输出两种森林。 `blocks`、`cut`、`bridge`、`bel`、`cnt` 可直接读取。大对象放在全局或 `static`。

```
1 template <int N> struct Biconnected_Graph
2 {
3     int n, timer = 0, cnt = 0;
4     vector<pair<int, int>> g[N + 1], edges;
5     vector<vector<int>> blocks;
6     vector<int> bridge, st;
7     int dfn[N + 1], low[N + 1], cut[N + 1], bel[N + 1];
8
9     void Init(int vertices)
10    {
11        assert(0 <= vertices && vertices <= N);
12        n = vertices;
13        for ( int u = 1; u <= n; u++ )
14            g[u].clear();
15        edges.clear();
16    }
17
18    int Insert(int u, int v)
19    {
20        assert(1 <= u && u <= n && 1 <= v && v <= n);
21        int id = edges.size();
22        edges.push_back({u, v});
23        if ( u != v )
24        {
25            g[u].push_back({v, id});
26            g[v].push_back({u, id});
27        }
28        return id;
29    }
30
31    void Dfs(int u, int parent_edge)
32    {
33        dfn[u] = low[u] = ++timer;
34        st.push_back(u);
35        int children = 0;
36        for ( auto [v, id] : g[u] )
37        {
38            if ( id == parent_edge )
39                continue;
40            if ( !dfn[v] )
41            {
42                children++;
43                Dfs(v, id);
44                low[u] = min(low[u], low[v]);
```

```

45         if ( low[v] > dfn[u] )
46             bridge[id] = 1;
47         if ( low[v] >= dfn[u] )
48         {
49             if ( parent_edge != -1 || children > 1 )
50                 cut[u] = 1;
51             blocks.push_back({u});
52             while ( true )
53             {
54                 int x = st.back();
55                 st.pop_back();
56                 blocks.back().push_back(x);
57                 if ( x == v )
58                     break;
59             }
60         }
61     }
62     else
63         low[u] = min(low[u], dfn[v]);
64 }
65 if ( parent_edge == -1 && children == 0 )
66     blocks.push_back({u});
67 }
68
69 void Paint(int u)
70 {
71     bel[u] = cnt;
72     for ( auto [v, id] : g[u] )
73     {
74         if ( !bridge[id] && !bel[v] )
75             Paint(v);
76     }
77 }
78
79 void Run()
80 {
81     timer = 0;
82     cnt = 0;
83     fill(dfn, dfn + n + 1, 0);
84     fill(low, low + n + 1, 0);
85     fill(cut, cut + n + 1, 0);
86     fill(bel, bel + n + 1, 0);
87     bridge.assign(edges.size(), 0);
88     blocks.clear();
89     st.clear();
90     for ( int u = 1; u <= n; u++ )
91     {
92         if ( !dfn[u] )
93         {
94             Dfs(u, -1);
95             st.pop_back();
96         }
97     }
98     for ( int u = 1; u <= n; u++ )
99     {
100         if ( bel[u] )

```

```

101         continue;
102         ++cnt;
103         Paint(u);
104     }
105 }
106
107 // Original vertices 1..n; block i has node n+i+1. Includes isolated vertices.
108 vector<vector<int>> Block_Forest() const
109 {
110     vector<vector<int>> tree(n + blocks.size() + 1);
111     for ( int i = 0; i < (int)blocks.size(); i++ )
112     {
113         int b = n + i + 1;
114         for ( int u : blocks[i] )
115         {
116             tree[u].push_back(b);
117             tree[b].push_back(u);
118         }
119     }
120     return tree;
121 }
122
123 // Edge-connected component IDs 1..cnt; forest edges carry original edge IDs.
124 vector<vector<pair<int, int>>> Bridge_Forest() const
125 {
126     vector<vector<pair<int, int>>> tree(cnt + 1);
127     for ( int id = 0; id < (int)edges.size(); id++ )
128     {
129         if ( !bridge[id] )
130             continue;
131         auto [u, v] = edges[id];
132         u = bel[u];
133         v = bel[v];
134         tree[u].push_back({v, id});
135         tree[v].push_back({u, id});
136     }
137     return tree;
138 }
139 };

```


第六十章 奇环顶点判定

60.1 属于简单奇环的全部顶点

算法约束见动态版本第 7 页。本节代码独立可抄；采用下列实际接口名。

```
1 // After Run(). Simple odd cycles have length >= 3; self-loops are ignored.
2 template <int N> vector<int> Odd_Cycle_Vertices(const Biconnected_Graph<N> &graph)
3 {
4     int n = graph.n;
5     auto forest = graph.Block_Forest();
6     vector<int> parent(forest.size()), depth(forest.size());
7     auto root_tree = [&](auto &&self, int u, int p) -> void
8     {
9         parent[u] = p;
10        for ( int v : forest[u] )
11            if ( v != p )
12            {
13                depth[v] = depth[u] + 1;
14                self(self, v, u);
15            }
16    };
17    for ( int u = 1; u <= n; u++ )
18        if ( !depth[u] )
19        {
20            depth[u] = 1;
21            root_tree(root_tree, u, 0);
22        }
```

```

23     vector<vector<pair<int, int>>> edges(graph.blocks.size());
24     for ( auto [u, v] : graph.edges )
25         if ( u != v )
26         {
27             int block = parent[depth[u] >= depth[v] ? u : v] - n - 1;
28             assert(0 <= block && block < (int)edges.size());
29             edges[block].push_back({u, v});
30         }
31     vector<vector<int>> g(n + 1);
32     vector<int> color(n + 1), answer(n + 1);
33     for ( int block = 0; block < (int)edges.size(); block++ )
34     {
35         for ( int u : graph.blocks[block] )
36         {
37             g[u].clear();
38             color[u] = 0;
39         }
40         for ( auto [u, v] : edges[block] )
41         {
42             g[u].push_back(v);
43             g[v].push_back(u);
44         }
45         bool bipartite = true;
46         auto paint = [&](auto &&self, int u) -> void
47         {
48             for ( int v : g[u] )
49                 if ( !color[v] )
50                 {
51                     color[v] = -color[u];
52                     self(self, v);
53                 }
54             else if ( color[v] == color[u] )
55                 bipartite = false;
56         };
57         int root = graph.blocks[block][0];
58         color[root] = 1;
59         paint(paint, root);
60         if ( !bipartite )
61             for ( int u : graph.blocks[block] )
62                 answer[u] = 1;
63     }
64     return answer;
65 }

```

第六十一章 删点连通性

61.1 由点双分解计算删点连通块数

算法约束见动态版本第 9 页。本节代码独立可抄；采用下列实际接口名。

```
1 // Call after Run(). Each vertex is deleted independently from the original graph.
2 template <int N>
3 inline pair<int, vector<int>> Removal_Components(const Biconnected_Graph<N> &graph)
4 {
5     int before = graph.cnt - count(graph.bridge.begin(), graph.bridge.end(), 1);
6     vector<int> after(graph.n + 1, before - 1);
7     after[0] = 0;
8     for ( const auto &block : graph.blocks )
9         if ( block.size() > 1 )
10             for ( int u : block )
11                 after[u]++;
12     return {before, after};
13 }
```

第六十二章 边双缩点与边定向

62.1 边双缩点的大小与内部边数

算法约束见动态版本第 10 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Edge_Compression
2 {
3     int n;
4     vector<int> size, inside;
5     vector<array<int, 3>> edges;
6
7     template <int N>
8     Edge_Compression(const Biconnected_Graph<N> &graph)
9         : n(graph.cnt), size(n + 1), inside(n + 1)
10    {
11        for ( int u = 1; u <= graph.n; u++ )
12            size[graph.bel[u]]++;
13        for ( int id = 0; id < (int)graph.edges.size(); id++ )
14        {
15            auto [u, v] = graph.edges[id];
16            u = graph.bel[u];
17            v = graph.bel[v];
18            if ( u == v )
19                inside[u]++;
20            else
21                edges.push_back({u, v, id});
22        }
23    }
24 };
```

62.2 无向边定向与强连通定向

算法约束见动态版本第 11 页。本节代码独立可抄；采用下列实际接口名。

```
1 // Tree edges go down; other edges go to ancestors. Handles each logical edge once.
2 inline vector<pair<int, int>> Orient_Edges(const Low_Link &graph)
3 {
4     vector<pair<int, int>> direction(graph.edges);
5     vector<int> seen(graph.n + 1);
6     auto dfs = [&](auto &&self, int u) -> void
7     {
8         seen[u] = 1;
9         for ( auto [v, id] : graph.g[u] )
10         {
11             if ( direction[id].first )
12                 continue;
13             direction[id] = {u, v};
14             if ( !seen[v] )
15                 self(self, v);
16         }
17     };
18     for ( int u = 1; u <= graph.n; u++ )
19         if ( !seen[u] )
20             dfs(dfs, u);
21     return direction;
22 }
```

第六十三章 桥树最少加边

63.1 连通无向图最少加边方案

算法约束见动态版本第 12 页。本节代码独立可抄；采用下列实际接口名。

```
1 // Connected, nonempty graph after Run(); new parallel edges are allowed.
2 template <int N>
3 inline vector<pair<int, int>> Bridge_Augmentation(const Biconnected_Graph<N> &graph)
4 {
5     assert(graph.n > 0 && graph.cnt > 0);
6     int bridges = count(graph.bridge.begin(), graph.bridge.end(), 1);
7     assert(graph.cnt - bridges == 1);
8     if ( graph.cnt == 1 )
9         return {};
10    auto tree = graph.Bridge_Forest();
11    vector<int> representative(graph.cnt + 1);
12    for ( int u = 1; u <= graph.n; u++ )
13        representative[graph.bel[u]] = u;
14    vector<int> leaves;
15    auto dfs = [&](auto &&self, int u, int parent) -> void
16    {
17        if ( tree[u].size() == 1 )
18            leaves.push_back(u);
19        for ( auto [v, id] : tree[u] )
20            if ( v != parent )
21                self(self, v, u);
22    };
23    dfs(dfs, 1, 0);
24    if ( leaves.size() % 2 )
25        leaves.push_back(leaves[0]);
26    int half = leaves.size() / 2;
27    vector<pair<int, int>> added;
28    for ( int i = 0; i < half; i++ )
29        added.push_back({representative[leaves[i]], representative[leaves[i + half]]});
30    return added;
31 }
```

第六十四章 数据结构

64.1 并查集

算法约束见动态版本第 13 页。本节代码独立可抄；采用下列实际接口名。

先声明容量模板参数，再调用 Init(n) 初始化；大对象必须置于全局或 static 存储。

```
1 template <int N> struct Disjoint_Set
2 {
3     int n, fa[N + 1], siz[N + 1];
4
5     void Init(int size)
6     {
7         assert(size <= N);
8         n = size;
9         iota(fa, fa + n + 1, 0);
10        fill(siz, siz + n + 1, 1);
11    }
12
13    int Find(int x)
14    {
15        int root = x;
16        while ( root != fa[root] )
17            root = fa[root];
18        while ( x != root )
19        {
20            int next = fa[x];
21            fa[x] = root;
22            x = next;
23        }
24        return root;
25    }
26
27    bool Merge(int x, int y)
28    {
29        x = Find(x);
30        y = Find(y);
31        if ( x == y )
32            return false;
33        if ( siz[x] < siz[y] )
34            swap(x, y);
35        fa[y] = x;
36        siz[x] += siz[y];
37        return true;
38    }
39 };
```

64.2 可撤销并查集

算法约束见动态版本第 14 页。本节代码独立可抄；采用下列实际接口名。

```

1  template <int N> struct Rollback_DSU
2  {
3      int fa[N + 1], siz[N + 1];
4      vector<pair<int, int>> history;
5
6      void Init(int n)
7      {
8          assert(n <= N);
9          history.clear();
10         iota(fa, fa + n + 1, 0);
11         fill(siz, siz + n + 1, 1);
12     }
13
14     int Find(int x) const
15     {
16         while ( fa[x] != x )
17             x = fa[x];
18         return x;
19     }
20
21     int Snapshot() const
22     {
23         return (int)history.size();
24     }
25
26     bool Merge(int x, int y)
27     {
28         x = Find(x);
29         y = Find(y);
30         if ( x == y )
31             return false;
32         if ( siz[x] < siz[y] )
33             swap(x, y);
34         history.push_back({y, siz[x]});
35         fa[y] = x;
36         siz[x] += siz[y];
37         return true;
38     }
39
40     void Rollback(int time)
41     {
42         assert(0 <= time && time <= Snapshot());
43         while ( Snapshot() > time )
44         {
45             auto [y, size] = history.back();
46             history.pop_back();
47             siz[fa[y]] = size;
48             fa[y] = y;
49         }
50     }
51 };

```


64.3 树状数组与第 k 小

算法约束见动态版本第 15 页。本节代码独立可抄；采用下列实际接口名。

先声明容量模板参数，再调用 Init(n) 初始化；大对象必须置于全局或 static 存储。

```

1 template <int N, class T = long long> struct Binary_Indexed_Tree
2 {
3     int n;
4     T tree[N + 1];
5
6     void Init(int size)
7     {
8         assert(size <= N);
9         n = size;
10        fill(tree, tree + n + 1, T(0));
11    }
12
13    void Insert(int x, T value)
14    {
15        assert(x >= 1);
16        for ( ; x <= n; x += x & -x )
17            tree[x] += value;
18    }
19
20    T Sum(int x) const
21    {
22        T answer = 0;
23        for ( ; x; x -= x & -x )
24            answer += tree[x];
25        return answer;
26    }
27
28    T Query(int L, int R) const
29    {
30        return Sum(R) - Sum(L - 1);
31    }
32
33    int Kth(T k) const
34    {
35        assert(k > 0);
36        int now = 0, step = 1;
37        while ( step <= n / 2 )
38            step <<= 1;
39        for ( ; step; step >>= 1 )
40            if ( now + step <= n && tree[now + step] < k )
41            {
42                now += step;
43                k -= tree[now];
44            }
45        return now + 1;
46    }
47 };

```

64.4 区间加与区间和

算法约束见动态版本第 15 页。本节代码独立可抄；采用下列实际接口名。

先声明容量模板参数，再调用 Init(n) 初始化；大对象必须置于全局或 static 存储。

```

1 template <int N> struct Segment_Tree
2 {
3     int n;
4     long long tree[4 * N + 5], tag[4 * N + 5];
5
6     void Init(int size)
7     {
8         assert(size > 0 && size <= N);
9         n = size;
10        fill(tree, tree + 4 * n + 5, 0);
11        fill(tag, tag + 4 * n + 5, 0);
12    }
13
14    void Update(int now, int L, int R, long long value)
15    {
16        tree[now] += (R - L + 1) * value;
17        tag[now] += value;
18    }
19
20    void Push_Up(int now)
21    {
22        tree[now] = tree[now << 1] + tree[now << 1 | 1];
23    }
24
25    void Push_Down(int now, int L, int R)
26    {
27        int mid = (L + R) >> 1;
28        Update(now << 1, L, mid, tag[now]);
29        Update(now << 1 | 1, mid + 1, R, tag[now]);
30        tag[now] = 0;
31    }
32
33    void Insert(int ql, int qr, long long value)
34    {
35        Insert(1, 1, n, ql, qr, value);
36    }
37
38    void Insert(int now, int L, int R, int ql, int qr, long long value)
39    {
40        if (ql <= L && R <= qr)
41            return Update(now, L, R, value);
42        Push_Down(now, L, R);
43        int mid = (L + R) >> 1;
44        if (ql <= mid)
45            Insert(now << 1, L, mid, ql, qr, value);
46        if (qr > mid)
47            Insert(now << 1 | 1, mid + 1, R, ql, qr, value);
48        Push_Up(now);
49    }
50
51    long long Query(int ql, int qr)

```

```
52 {
53     return Query(1, 1, n, ql, qr);
54 }
55
56 long long Query(int now, int L, int R, int ql, int qr)
57 {
58     if ( ql <= L && R <= qr )
59         return tree[now];
60     Push_Down(now, L, R);
61     int mid = (L + R) >> 1;
62     long long answer = 0;
63     if ( ql <= mid )
64         answer += Query(now << 1, L, mid, ql, qr);
65     if ( qr > mid )
66         answer += Query(now << 1 | 1, mid + 1, R, ql, qr);
67     return answer;
68 }
69 };
```

64.5 64 位异或线性基

算法约束见动态版本第 18 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Xor_Basis
2 {
3     using U = unsigned long long;
4     using Wide = __uint128_t;
5     U base[64]{};
6     vector<U> p;
7     int rank = 0;
8     bool dependent = false, dirty = false;
9
10    void Init()
11    {
12        fill(base, base + 64, 0);
13        p.clear();
14        rank = 0;
15        dependent = false;
16        dirty = false;
17    }
18
19    bool Insert(U x)
20    {
21        for ( int i = 63; i >= 0; i-- )
22            if ( x >> i & 1 )
23            {
24                if ( !base[i] )
25                {
26                    base[i] = x;
27                    rank++;
28                    dirty = true;
29                    return true;
30                }
31                x ^= base[i];
32            }
33        dependent = true;
34        return false;
35    }
36
37    bool Contains(U x) const
38    {
39        for ( int i = 63; i >= 0; i-- )
40            if ( x >> i & 1 )
41                x ^= base[i];
42        return x == 0;
43    }
44
45    U Query(U x = 0) const
46    {
47        for ( int i = 63; i >= 0; i-- )
48            x = max(x, x ^ base[i]);
49        return x;
50    }
```

```

52 void Rebuild()
53 {
54     vector<U> b(base, base + 64);
55     for ( int i = 0; i < 64; i++ )
56         if ( b[i] )
57             for ( int j = i + 1; j < 64; j++ )
58                 if ( b[j] >> i & 1 )
59                     b[j] ^= b[i];
60     p.clear();
61     for ( U x : b )
62         if ( x )
63             p.push_back(x);
64     dirty = false;
65 }
66
67 // k is 1-based; nonempty excludes the empty input subset, not value zero.
68 optional<U> Kth(Wide k, bool nonempty = true)
69 {
70     if ( !k )
71         return nullopt;
72     Wide index = k;
73     if ( !nonempty || dependent )
74         index--;
75     if ( index >= (Wide(1) << rank) )
76         return nullopt;
77     if ( dirty )
78         Rebuild();
79     U answer = 0;
80     for ( int i = 0; i < rank; i++ )
81         if ( index >> i & 1 )
82             answer ^= p[i];
83     return answer;
84 }
85
86 optional<U> Minimum(bool nonempty = true)
87 {
88     return Kth(1, nonempty);
89 }
90
91 void Merge(const Xor_Basis &other)
92 {
93     dependent |= other.dependent;
94     for ( U x : other.base )
95         if ( x )
96             Insert(x);
97 }
98 };

```

第六十五章 带位置线性基

65.1 带位置限制的异或线性基

算法约束见动态版本第 21 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Position_Basis
2 {
3     using U = unsigned long long;
4     U base[64];
5     int pos[64];
6
7     Position_Basis()
8     {
9         Init();
10    }
11
12    void Init()
13    {
14        fill(base, base + 64, 0);
15        fill(pos, pos + 64, -1);
16    }
```

```

18 void Insert(U x, int at)
19 {
20     assert(at >= 0);
21     for ( int i = 63; i >= 0; i-- )
22         if ( x >> i & 1 )
23         {
24             if ( !base[i] )
25             {
26                 base[i] = x;
27                 pos[i] = at;
28                 return;
29             }
30             if ( at > pos[i] )
31             {
32                 swap(base[i], x);
33                 swap(pos[i], at);
34             }
35             x ^= base[i];
36         }
37 }
38
39 bool Contains(U x, long long lower = 0) const
40 {
41     assert(lower >= 0);
42     for ( int i = 63; i >= 0; i-- )
43         if ( (x >> i & 1) && pos[i] >= lower )
44             x ^= base[i];
45     return x == 0;
46 }
47
48 U Query(long long lower = 0, U x = 0) const
49 {
50     assert(lower >= 0);
51     for ( int i = 63; i >= 0; i-- )
52         if ( pos[i] >= lower )
53             x = max(x, x ^ base[i]);
54     return x;
55 }
56
57 int Rank(long long lower = 0) const
58 {
59     assert(lower >= 0);
60     int answer = 0;
61     for ( int i = 0; i < 64; i++ )
62         answer += pos[i] >= lower;
63     return answer;
64 }
65 };

```

第六十六章 线性空间求交

66.1 异或线性空间求交

算法约束见动态版本第 23 页。本节代码独立可抄；采用下列实际接口名。

```
1 inline Xor_Basis Basis_Intersection(const Xor_Basis &a, const Xor_Basis &b)
2 {
3     using U = unsigned long long;
4     U pivot[64], from_b[64]{};
5     copy(a.base, a.base + 64, pivot);
6     Xor_Basis result;
7     for ( U value : b.base )
8     {
9         U x = value, part = value;
10        for ( int i = 63; i >= 0 && x; i-- )
11            if ( x >> i & 1 )
12            {
13                if ( !pivot[i] )
14                {
15                    pivot[i] = x;
16                    from_b[i] = part;
17                    break;
18                }
19                x ^= pivot[i];
20                part ^= from_b[i];
21            }
22        if ( !x && part )
23            result.Insert(part);
24    }
25    return result;
26 }
```


66.2 Zassenhaus：同时求和空间与交空间

算法约束见动态版本第 24 页。本节代码独立可抄；采用下列实际接口名。

```

1 // Return independent generators of (sum space, intersection space).
2 inline pair<Xor_Basis, Xor_Basis> Basis_Sum_Intersection(const Xor_Basis &a,
3                                                         const Xor_Basis &b)
4 {
5     using W = __uint128_t;
6     W pivot[128]{};
7     auto insert = [&](W x)
8     {
9         for ( int i = 127; i >= 0 && x; i-- )
10             if ( x >> i & 1 )
11             {
12                 if ( !pivot[i] )
13                 {
14                     pivot[i] = x;
15                     return;
16                 }
17                 x ^= pivot[i];
18             }
19     };
20     for ( auto x : a.base )
21         insert((W(x) << 64) | x);
22     for ( auto x : b.base )
23         insert(W(x) << 64);
24     Xor_Basis sum, intersection;
25     for ( int i = 0; i < 128; i++ )
26         if ( pivot[i] )
27         {
28             if ( i < 64 )
29                 intersection.Insert((unsigned long long)pivot[i]);
30             else
31                 sum.Insert((unsigned long long)(pivot[i] >> 64));
32         }
33     return {sum, intersection};
34 }
```

第六十七章 线段树常用修改

67.1 线段树：区间乘、加、赋值与模意义求和

算法约束见动态版本第 25 页。本节代码独立可抄；采用下列实际接口名。

声明 `Affine_Segment_Tree<N>` 后 `Init(a,mod)`, `N` 为最大数组长度。`Update(l,r,mul,add)` 修改, `Query(l,r)` 求和；支持重复初始化。大对象放在全局或 `static`。

```
1 template <int N> struct Affine_Segment_Tree
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         ll sum = 0, mul = 1, add = 0;
8     };
9
10    int n, mod;
11    Node t[4 * N + 4];
12
13    void Init(const vector<ll> &a, int modulus)
14    {
15        n = a.size();
16        mod = modulus;
17        assert(0 < n && n <= N && mod > 0);
18        Build(1, 1, n, a);
19    }
20
21    ll Norm(ll x) const
22    {
23        x %= mod;
24        if ( x < 0 )
25            x += mod;
26        return x;
27    }
28
29    void Push_Up(int p)
30    {
31        t[p].sum = (t[p * 2].sum + t[p * 2 + 1].sum) % mod;
32    }
33
34    void Build(int p, int l, int r, const vector<ll> &a)
35    {
36        t[p] = Node{};
37        if ( l == r )
38        {
39            t[p].sum = Norm(a[l - 1]);
40            return;
41        }
42        int m = (l + r) / 2;
43        Build(p * 2, l, m, a);
44        Build(p * 2 + 1, m + 1, r, a);
```

```

45     Push_Up(p);
46 }
47
48 // New transform follows the old pending transform: x -> x*a+b.
49 void Apply(int p, int len, ll a, ll b)
50 {
51     t[p].sum = (t[p].sum * a % mod + len * b % mod) % mod;
52     t[p].mul = t[p].mul * a % mod;
53     t[p].add = (t[p].add * a % mod + b) % mod;
54 }
55
56 void Push_Down(int p, int l, int r)
57 {
58     if ( t[p].mul == 1 && t[p].add == 0 )
59         return;
60     int m = (l + r) / 2;
61     Apply(p * 2, m - l + 1, t[p].mul, t[p].add);
62     Apply(p * 2 + 1, r - m, t[p].mul, t[p].add);
63     t[p].mul = 1;
64     t[p].add = 0;
65 }
66
67 void Update(int l, int r, ll a, ll b)
68 {
69     assert(1 <= l && l <= r && r <= n);
70     Update(1, 1, n, l, r, Norm(a), Norm(b));
71 }
72
73 void Update(int p, int l, int r, int ql, int qr, ll a, ll b)
74 {
75     if ( ql <= l && r <= qr )
76     {
77         Apply(p, r - l + 1, a, b);
78         return;
79     }
80     Push_Down(p, l, r);
81     int m = (l + r) / 2;
82     if ( ql <= m )
83         Update(p * 2, l, m, ql, qr, a, b);
84     if ( qr > m )
85         Update(p * 2 + 1, m + 1, r, ql, qr, a, b);
86     Push_Up(p);
87 }
88
89 ll Query(int l, int r)
90 {
91     assert(1 <= l && l <= r && r <= n);
92     return Query(1, 1, n, l, r);
93 }
94
95 ll Query(int p, int l, int r, int ql, int qr)
96 {
97     if ( ql <= l && r <= qr )
98         return t[p].sum;

```

```
99     Push_Down(p, l, r);
100     int m = (l + r) / 2;
101     ll answer = 0;
102     if ( ql <= m )
103         answer += Query(p * 2, l, m, ql, qr);
104     if ( qr > m )
105         answer += Query(p * 2 + 1, m + 1, r, ql, qr);
106     return answer % mod;
107 }
108 };
```

第六十八章 网络流

68.1 最大流与方案

算法约束见动态版本第 28 页。本节代码独立可抄；采用下列实际接口名。

先声明容量模板参数，再调用 Init(n) 初始化；大对象必须置于全局或 static 存储。

```
1 template <int N, int M> struct Network_Flow
2 {
3     using ll = long long;
4
5     struct Edge
6     {
7         int to, next;
8         ll cap, initial;
9     } edge[2 * M];
10
11     int n, tot, head[N + 1], deep[N + 1], cur[N + 1], que[N + 1];
12
13     void Init(int size)
14     {
15         assert(size <= N);
16         n = size;
17         tot = 0;
18         fill(head, head + n + 1, -1);
19     }
20
21     int Add_Edge(int u, int v, ll cap)
22     {
23         assert(cap >= 0 && tot + 2 <= 2 * M);
24         int id = tot;
25         edge[tot] = {v, head[u], cap, cap};
26         head[u] = tot++;
27         edge[tot] = {u, head[v], 0, 0};
28         head[v] = tot++;
29         return id;
30     }
31
32     bool Bfs(int s, int t)
33     {
34         fill(deep, deep + n + 1, -1);
35         int L = 0, R = 0;
36         que[R++] = s;
37         deep[s] = 0;
38         while ( L < R )
39         {
40             int now = que[L++];
41             for ( int i = head[now]; i != -1; i = edge[i].next )
42             {
43                 int to = edge[i].to;
44                 if ( edge[i].cap && deep[to] == -1 )
45                 {
```

```

46         deep[to] = deep[now] + 1;
47         que[R++] = to;
48     }
49 }
50 }
51 return deep[t] != -1;
52 }
53
54 ll Dfs(int now, int t, ll flow)
55 {
56     if ( now == t )
57         return flow;
58     for ( int &i = cur[now]; i != -1; i = edge[i].next )
59     {
60         int to = edge[i].to;
61         if ( !edge[i].cap || deep[to] != deep[now] + 1 )
62             continue;
63         ll delta = Dfs(to, t, min(flow, edge[i].cap));
64         if ( delta )
65         {
66             edge[i].cap -= delta;
67             edge[i ^ 1].cap += delta;
68             return delta;
69         }
70     }
71     return 0;
72 }
73
74 ll Max_Flow(int s, int t, ll limit = LLONG_MAX)
75 {
76     assert(s != t && limit >= 0);
77     ll answer = 0;
78     while ( answer < limit && Bfs(s, t) )
79     {
80         copy(head, head + n + 1, cur);
81         while ( answer < limit )
82         {
83             ll delta = Dfs(s, t, limit - answer);
84             if ( !delta )
85                 break;
86             answer += delta;
87         }
88     }
89     return answer;
90 }
91
92 ll Used(int id) const
93 {
94     return edge[id].initial - edge[id].cap;
95 }
96
97 vector<int> Cut(int s)

```

```

98     {
99         Bfs(s, s);
100         vector<int> answer;
101         for ( int i = 1; i <= n; i++ )
102             if ( deep[i] != -1 )
103                 answer.push_back(i);
104         return answer;
105     }
106 };

```

68.2 势能最小费用最大流

算法约束见动态版本第 30 页。本节代码独立可抄；采用下列实际接口名。
使用 Insert 加边、Flow 求流、Used 查看方案。

```

1  struct Min_Cost_Flow
2  {
3      using ll = long long;
4      using i128 = __int128_t;
5
6      struct Edge
7      {
8          int from, to;
9          ll cap, cost, initial;
10     };
11
12     int n;
13     vector<Edge> e;
14     vector<vector<int>>> g;
15
16     Min_Cost_Flow(int n) : n(n), g(n + 1)
17     {
18     }
19
20     int Insert(int u, int v, ll cap, ll cost)
21     {
22         assert(cap >= 0 && cost != LLONG_MIN);
23         int id = (int)e.size();
24         e.push_back({u, v, cap, cost, cap});
25         e.push_back({v, u, 0, -cost, 0});
26         g[u].push_back(id);
27         g[v].push_back(id + 1);
28         return id;
29     }
30
31     ll Used(int id) const
32     {
33         return e[id].initial - e[id].cap;
34     }
35
36     // Negative costs allowed; initial residual graph must have no negative cycle.
37     pair<ll, i128> Flow(int s, int t, ll limit = LLONG_MAX)
38     {
39         assert(s != t && limit >= 0);
40         const i128 inf = i128(1) << 120;

```

```

41     vector<i128> h(n + 1, 0), d(n + 1);
42     vector<int> pre(n + 1);
43     // Super-source Bellman-Ford: feasible potentials for every vertex.
44     for ( int it = 0; it < n; it++ )
45     {
46         bool changed = false;
47         for ( auto &a : e )
48             if ( a.cap && h[a.to] > h[a.from] + a.cost )
49             {
50                 h[a.to] = h[a.from] + a.cost;
51                 changed = true;
52             }
53         if ( !changed )
54             break;
55         if ( it == n - 1 )
56             throw invalid_argument("negative cycle");
57     }
58     ll f = 0;
59     i128 cost = 0;
60     while ( f < limit )
61     {
62         fill(d.begin(), d.end(), inf);
63         d[s] = 0;
64         priority_queue<pair<i128, int>,
65                       vector<pair<i128, int>>,
66                       greater<pair<i128, int>>>
67             q;
68         q.push({0, s});
69         while ( !q.empty() )
70         {
71             auto [du, u] = q.top();
72             q.pop();
73             if ( du != d[u] )
74                 continue;
75             for ( int id : g[u] )
76             {
77                 auto &a = e[id];
78                 i128 nd = du + a.cost + h[u] - h[a.to];
79                 if ( a.cap && nd < d[a.to] )
80                 {
81                     d[a.to] = nd;
82                     pre[a.to] = id;
83                     q.push({nd, a.to});
84                 }
85             }
86         }
87         if ( d[t] == inf )
88             break;
89         for ( int u = 1; u <= n; u++ )
90             if ( d[u] != inf )
91                 h[u] += d[u];
92         ll take = limit - f;
93         for ( int u = t; u != s; u = e[pre[u]].from )
94             take = min(take, e[pre[u]].cap);
95         for ( int u = t; u != s; u = e[pre[u]].from )
96         {

```



```
97         int id = pre[u];
98         e[id].cap -= take;
99         e[id ^ 1].cap += take;
100        cost += i128(take) * e[id].cost;
101    }
102    f += take;
103 }
104 return {f, cost};
105 }
106 };
```

68.3 上下界可行循环流

算法约束见动态版本第 32 页。本节代码独立可抄；采用下列实际接口名。

```

1  template <int N, int M> struct Bounded_Circulation
2  {
3      int n, tot;
4      Network_Flow<N + 2, M + N> graph;
5      long long balance[N + 1], lower[M];
6      int id[M];
7      bool solved, feasible;
8
9      void Init(int size)
10     {
11         assert(size <= N);
12         n = size;
13         tot = 0;
14         solved = feasible = false;
15         graph.Init(n + 2);
16         fill(balance, balance + n + 1, 0);
17     }
18
19     int Insert(int u, int v, long long L, long long R)
20     {
21         assert(!solved && 0 <= L && L <= R && tot < M);
22         id[tot] = graph.Add_Edge(u, v, R - L);
23         lower[tot] = L;
24         balance[u] -= L;
25         balance[v] += L;
26         return tot++;
27     }
28
29     bool Solve()
30     {
31         assert(!solved);
32         solved = true;
33         long long need = 0;
34         for ( int u = 1; u <= n; u++ )
35         {
36             if ( balance[u] > 0 )
37             {
38                 graph.Add_Edge(n + 1, u, balance[u]);
39                 need += balance[u];
40             }
41             else if ( balance[u] < 0 )
42                 graph.Add_Edge(u, n + 2, -balance[u]);
43         }
44         return feasible = (graph.Max_Flow(n + 1, n + 2) == need);
45     }
46
47     long long Used(int index) const
48     {
49         assert(feasible);
50         return lower[index] + graph.Used(id[index]);
51     }
52 };

```

第六十九章 最大权闭合子图

69.1 最大权闭合子图与选点方案

算法约束见动态版本第 34 页。本节代码独立可抄；采用下列实际接口名。

Maximum_Closure(weight,dependencies,graph) 额外接收 Network_Flow<N,M> 引用，并在每次调用中 Init 重建该网络。N 至少为原点数加 2，M 至少为依赖边数加非零权点数；这是原有向边容量，不是残量边总数。将网络对象放在全局或 static，模板参数可由对象自动推导。

```
1 template <int N, int M>
2 pair<long long, vector<int>> Maximum_Closure(const vector<long long> &weight,
3                                             const vector<pair<int, int>> &dependencies,
4                                             Network_Flow<N, M> &graph)
5 {
6     using I = __int128_t;
7     int n = weight.size(), s = n + 1, t = n + 2;
8     I positive = 0;
9     for ( long long w : weight )
10         if ( w > 0 )
11             positive += w;
12     assert(positive < LLONG_MAX);
13     long long inf = (long long)positive + 1;
14     graph.Init(n + 2);
15     for ( int u = 1; u <= n; u++ )
16     {
17         long long w = weight[u - 1];
18         if ( w > 0 )
19             graph.Add_Edge(s, u, w);
20         else if ( w < 0 )
21             graph.Add_Edge(u, t, (long long)min(-I(w), I(inf)));
22     }
23     for ( auto [u, v] : dependencies )
24     {
25         assert(1 <= u && u <= n && 1 <= v && v <= n);
26         graph.Add_Edge(u, v, inf);
27     }
28     long long answer = (long long)positive - graph.Max_Flow(s, t);
29     vector<int> selected;
30     for ( int u : graph.Cut(s) )
31         if ( u <= n )
32             selected.push_back(u);
33     return {answer, selected};
34 }
```

第七十章 图论

70.1 非负权最短路

算法约束见动态版本第 35 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Shortest_Path
2 {
3     using ll = long long;
4     static constexpr ll inf = LLONG_MAX;
5     int n;
6     vector<vector<pair<int, ll>>> g;
7     vector<ll> dis;
8     vector<int> pre;
9
10    Shortest_Path(int n) : n(n), g(n + 1)
11    {
12    }
13
14    void Insert(int u, int v, ll w)
15    {
16        assert(w >= 0);
17        g[u].push_back({v, w});
18    }
19
20    void Run(int s)
21    {
22        dis.assign(n + 1, inf);
23        pre.assign(n + 1, -1);
24        dis[s] = 0;
25        priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<pair<ll, int>>> q;
26        q.push({0, s});
27        while ( !q.empty() )
28        {
29            auto [d, u] = q.top();
30            q.pop();
31            if ( d != dis[u] )
32                continue;
33            for ( auto [v, w] : g[u] )
34                if ( w < inf - d && d + w < dis[v] )
35                {
36                    dis[v] = d + w;
37                    pre[v] = u;
38                    q.push({dis[v], v});
39                }
40        }
41    }
42
43    vector<int> Path(int t) const
44    {
```

```

45     if ( dis[t] == inf )
46         return {};
47     vector<int> ans;
48     for ( ; t != -1; t = pre[t] )
49         ans.push_back(t);
50     reverse(ans.begin(), ans.end());
51     return ans;
52 }
53 };

```

70.2 强连通分量

算法约束见动态版本第 36 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Strong_Component
2 {
3     int n, cnt = 0;
4     vector<vector<int>> g, rg;
5     vector<int> bel;
6
7     Strong_Component(int n) : n(n), g(n + 1), rg(n + 1)
8     {
9     }
10
11     void Insert(int u, int v)
12     {
13         g[u].push_back(v);
14         rg[v].push_back(u);
15     }
16
17     void Dfs1(int u, vector<int> &vis, vector<int> &order)
18     {
19         vis[u] = 1;
20         for ( int v : g[u] )
21         {
22             if ( !vis[v] )
23                 Dfs1(v, vis, order);
24         }
25         order.push_back(u);
26     }
27
28     void Dfs2(int u)
29     {
30         bel[u] = cnt;
31         for ( int v : rg[u] )
32         {
33             if ( !bel[v] )
34                 Dfs2(v);
35         }
36     }
37
38     // Recursive Kosaraju; component IDs are in topological order, 1..cnt.
39     void Run()
40     {
41         vector<int> vis(n + 1), order;

```

```

42     cnt = 0;
43     bel.assign(n + 1, 0);
44     for ( int s = 1; s <= n; s++ )
45     {
46         if ( !vis[s] )
47             Dfs1(s, vis, order);
48     }
49     reverse(order.begin(), order.end());
50     for ( int s : order )
51     {
52         if ( !bel[s] )
53         {
54             ++cnt;
55             Dfs2(s);
56         }
57     }
58 }
59 };

```

70.3 2-SAT 与任意可行方案

算法约束见动态版本第 37 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Two_SAT
2 {
3     int n;
4     Strong_Component g;
5     vector<int> ans;
6
7     Two_SAT(int n) : n(n), g(2 * n)
8     {
9     }
10
11     int id(int x, bool value) const
12     {
13         return 2 * x - 1 + value;
14     }
15
16     void Insert(int x, bool a, int y, bool b)
17     {
18         g.Insert(id(x, !a), id(y, b));
19         g.Insert(id(y, !b), id(x, a));
20     }
21
22     bool Solve()
23     {
24         g.Run();
25         ans.assign(n + 1, 0);

```

```

26     for ( int x = 1; x <= n; x++ )
27     {
28         if ( g.bel[id(x, 0)] == g.bel[id(x, 1)] )
29             return false;
30         ans[x] = g.bel[id(x, 1)] > g.bel[id(x, 0)];
31     }
32     return true;
33 }
34 };

```

70.4 Hopcroft–Karp 最大匹配与最小点覆盖

算法约束见动态版本第 38 页。本节代码独立可抄；采用下列实际接口名。

声明 `Bipartite_Matching<N,M>` 后 `Init(n,m)`，两侧容量分别为 `N`、`M`。Insert 加边，Solve 求匹配，Cover 返回最小点覆盖；`l`、`r` 保存方案。大对象放在全局或 `static`。

```

1  template <int N, int M> struct Bipartite_Matching
2  {
3      int n, m;
4      vector<int> g[N + 1];
5      int l[N + 1], r[M + 1], dep[N + 1], cur[N + 1], que[N + 1];
6
7      void Init(int left_size, int right_size)
8      {
9          assert(0 <= left_size && left_size <= N && 0 <= right_size && right_size <= M);
10         n = left_size;
11         m = right_size;
12         fill(l, l + n + 1, 0);
13         fill(r, r + m + 1, 0);
14         for ( int u = 1; u <= n; u++ )
15             g[u].clear();
16     }
17
18     void Insert(int u, int v)
19     {
20         assert(1 <= u && u <= n && 1 <= v && v <= m);
21         g[u].push_back(v);
22     }
23
24     bool Bfs()
25     {
26         int head = 0, tail = 0;
27         fill(dep, dep + n + 1, n + 1);
28         fill(cur, cur + n + 1, 0);
29         for ( int u = 1; u <= n; u++ )
30         {
31             if ( !l[u] )
32             {
33                 dep[u] = 0;
34                 que[tail++] = u;
35             }
36         }
37         while ( head < tail )
38         {
39             int u = que[head++];

```

```

40         if ( dep[u] >= dep[0] )
41             continue;
42         for ( int v : g[u] )
43         {
44             int next = r[v];
45             if ( dep[next] != n + 1 )
46                 continue;
47             dep[next] = dep[u] + 1;
48             if ( next )
49                 que[tail++] = next;
50         }
51     }
52     return dep[0] != n + 1;
53 }
54
55 bool Dfs(int u)
56 {
57     if ( !u )
58         return true;
59     for ( int &i = cur[u]; i < (int)g[u].size(); i++ )
60     {
61         int v = g[u][i];
62         if ( dep[r[v]] == dep[u] + 1 && Dfs(r[v]) )
63         {
64             l[u] = v;
65             r[v] = u;
66             return true;
67         }
68     }
69     dep[u] = n + 1;
70     return false;
71 }
72
73 // Retains an existing matching; edges may be added between calls.
74 int Solve()
75 {
76     int answer = 0;
77     for ( int u = 1; u <= n; u++ )
78         answer += (l[u] != 0);
79     while ( Bfs() )
80     {
81         for ( int u = 1; u <= n; u++ )
82         {
83             if ( !l[u] && Dfs(u) )
84                 answer++;
85         }
86     }
87     return answer;
88 }
89
90 // Call after Solve(), before adding more edges. Returns left/right vertex IDs.
91 pair<vector<int>, vector<int>> Cover() const
92 {
93     vector<int> left, right;
94     for ( int u = 1; u <= n; u++ )
95     {

```



```

96         if ( dep[u] == n + 1 )
97             left.push_back(u);
98     }
99     for ( int v = 1; v <= m; v++ )
100     {
101         if ( r[v] && dep[r[v]] != n + 1 )
102             right.push_back(v);
103     }
104     return {left, right};
105 }
106 };

```

70.5 割点与桥

算法约束见动态版本第 40 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Low_Link
2 {
3     int n, timer = 0, components = 0;
4     vector<vector<pair<int, int>>> g;
5     vector<int> dfn, low, cut, bridge, delta;
6     int edges = 0;
7
8     Low_Link(int n) : n(n), g(n + 1)
9     {
10    }
11
12    int Insert(int u, int v)
13    {
14        g[u].push_back({v, edges});
15        g[v].push_back({u, edges});
16        return edges++;
17    }

```

```

19 void Dfs(int u, int pe)
20 {
21     dfn[u] = low[u] = ++timer;
22     int children = 0;
23     for ( auto [v, id] : g[u] )
24     {
25         if ( id == pe )
26             continue;
27         if ( !dfn[v] )
28         {
29             ++children;
30             Dfs(v, id);
31             low[u] = min(low[u], low[v]);
32             if ( low[v] > dfn[u] )
33                 bridge[id] = 1;
34             if ( pe != -1 && low[v] >= dfn[u] )
35             {
36                 cut[u] = 1;
37                 delta[u]++;
38             }
39         }
40         else
41             low[u] = min(low[u], dfn[v]);
42     }
43     if ( pe == -1 && children > 1 )
44         cut[u] = 1;
45     if ( pe == -1 )
46         delta[u] = children - 1;
47 }
48
49 void Run()
50 {
51     timer = 0;
52     components = 0;
53     dfn.assign(n + 1, 0);
54     low = dfn;
55     cut = dfn;
56     delta = dfn;
57     bridge.assign(edges, 0);
58     for ( int u = 1; u <= n; u++ )
59         if ( !dfn[u] )
60         {
61             components++;
62             Dfs(u, -1);
63         }
64 }
65 };

```

第七十一章 字典序最小 2-SAT

71.1 递归染色与回滚求最小赋值

算法约束见动态版本第 42 页。本节代码独立可抄；采用下列实际接口名。

声明 Lex_Two_SAT<N>, Init(n) 清空子句; Add、Force、Solve 与动态版对应, answer[1..n] 为答案。内部 Id 的字面量编号从 0 开始, 大对象放全局或 static。

```
1 template <int N> struct Lex_Two_SAT
2 {
3     int n;
4     vector<int> g[2 * N], history, answer;
5     int marked[2 * N];
6
7     void Init(int variables)
8     {
9         assert(0 <= variables && variables <= N);
10        n = variables;
11        for ( int u = 0; u < 2 * n; u++ )
12            g[u].clear();
13        history.clear();
14        answer.clear();
15    }
16
17    int Id(int x, bool value) const
18    {
19        assert(1 <= x && x <= n);
20        return 2 * (x - 1) + value;
21    }
22
23    // (x == a) OR (y == b). Internal literal IDs are 0-based.
24    void Add(int x, bool a, int y, bool b)
25    {
26        int u = Id(x, a), v = Id(y, b);
27        g[u ^ 1].push_back(v);
28        g[v ^ 1].push_back(u);
29        answer.clear();
30    }
31
32    void Force(int x, bool value)
33    {
34        Add(x, value, x, value);
35    }
```

```

37  bool Paint(int u)
38  {
39      if ( marked[u ^ 1] )
40          return false;
41      if ( marked[u] )
42          return true;
43      marked[u] = 1;
44      history.push_back(u);
45      for ( int v : g[u] )
46          if ( !Paint(v) )
47              return false;
48      return true;
49  }
50
51  bool Solve()
52  {
53      fill(marked, marked + 2 * n, 0);
54      history.clear();
55      answer.clear();
56      for ( int x = 1; x <= n; x++ )
57      {
58          int u = Id(x, false);
59          if ( marked[u] || marked[u ^ 1] )
60              continue;
61          int checkpoint = history.size();
62          if ( !Paint(u) )
63          {
64              while ( (int)history.size() > checkpoint )
65              {
66                  marked[history.back()] = 0;
67                  history.pop_back();
68              }
69              if ( !Paint(u ^ 1) )
70                  return false;
71          }
72      }
73      answer.assign(n + 1, 0);
74      for ( int x = 1; x <= n; x++ )
75          answer[x] = marked[Id(x, true)];
76      return true;
77  }
78  };

```

第七十二章 有向欧拉路

72.1 有向图字典序最小欧拉路

算法约束见动态版本第 44 页。本节代码独立可抄；采用下列实际接口名。

声明 Directed_Euler<N>, Init(n) 清空图, Insert 加边并返回从 0 开始的边编号, Run 求解（参数默认值同动态版）。vertices、edge_ids、edges 可直接读取；N 为最大点数，大对象放全局或 static。

```
1 template <int N> struct Directed_Euler
2 {
3     int n;
4     vector<pair<int, int>> g[N + 1], edges;
5     vector<int> vertices, edge_ids;
6     int in[N + 1], out[N + 1], next[N + 1];
7
8     void Init(int count)
9     {
10         assert(0 < count && count <= N);
11         n = count;
12         for ( int u = 1; u <= n; u++ )
13         {
14             g[u].clear();
15             in[u] = 0;
16             out[u] = 0;
17         }
18         edges.clear();
19         vertices.clear();
20         edge_ids.clear();
21     }
```

```
23  int Insert(int u, int v)
24  {
25      assert(1 <= u && u <= n && 1 <= v && v <= n);
26      int id = edges.size();
27      edges.push_back({u, v});
28      g[u].push_back({v, id});
29      out[u]++;
30      in[v]++;
31      vertices.clear();
32      edge_ids.clear();
33      return id;
34  }
35
36  void Dfs(int u, int incoming)
37  {
38      while ( next[u] < (int)g[u].size() )
39      {
40          auto [v, id] = g[u][next[u]++];
41          Dfs(v, id);
42      }
43      vertices.push_back(u);
44      if ( incoming != -1 )
45          edge_ids.push_back(incoming);
46  }
```

```

48 // start=0 chooses the smallest feasible start. Isolated vertices are ignored.
49 bool Run(int start = 0, bool lexicographic = true)
50 {
51     assert(0 <= start && start <= n);
52     vertices.clear();
53     edge_ids.clear();
54     int source = 0, sink = 0;
55     for ( int u = 1; u <= n; u++ )
56     {
57         int delta = out[u] - in[u];
58         if ( delta == 1 && !source )
59             source = u;
60         else if ( delta == -1 && !sink )
61             sink = u;
62         else if ( delta != 0 )
63             return false;
64     }
65     if ( bool(source) != bool(sink) )
66         return false;
67     if ( source )
68     {
69         if ( start && start != source )
70             return false;
71         start = source;
72     }
73     if ( !start )
74     {
75         start = 1;
76         for ( int u = 1; u <= n; u++ )
77             if ( out[u] )
78             {
79                 start = u;
80                 break;
81             }
82     }
83     if ( lexicographic )
84         for ( int u = 1; u <= n; u++ )
85             sort(g[u].begin(), g[u].end());
86     fill(next, next + n + 1, 0);
87     Dfs(start, -1);
88     if ( edge_ids.size() != edges.size() )
89     {
90         vertices.clear();
91         edge_ids.clear();
92         return false;
93     }
94     reverse(vertices.begin(), vertices.end());
95     reverse(edge_ids.begin(), edge_ids.end());
96     return true;
97 }
98 };

```

第七十三章 字典序单词链

73.1 字典序最小单词链

算法约束见动态版本第 46 页。本节代码独立可抄；采用下列实际接口名。

```
1 // Lowercase nonempty words. Returns original 0-based IDs; duplicates stay distinct.
2 inline optional<vector<int>> Word_Chain(const vector<string> &words)
3 {
4     int n = words.size();
5     vector<int> order(n);
6     iota(order.begin(), order.end(), 0);
7     for ( const auto &word : words )
8     {
9         assert(!word.empty());
10        for ( char c : word )
11            assert('a' <= c && c <= 'z');
12    }
13    sort(order.begin(),
14         order.end(),
15         [&](int a, int b)
16         {
17             if ( words[a] != words[b] )
18                 return words[a] < words[b];
19             return a < b;
20         });
21    Directed_Euler<26> graph;
22    graph.Init(26);
23    for ( int id : order )
24        graph.Insert(words[id].front() - 'a' + 1, words[id].back() - 'a' + 1);
25    // Preserve word order in each adjacency list; do not sort by destination.
26    if ( !graph.Run(0, false) )
27        return nullopt;
28    vector<int> result;
29    for ( int id : graph.edge_ids )
30        result.push_back(order[id]);
31    return result;
32 }
```


第七十四章 无向欧拉路

74.1 无向图欧拉路与边方向

算法约束见动态版本第 47 页。本节代码独立可抄；采用下列实际接口名。

声明 `Undirected_Euler<N>`，`Init(n)` 清空图，`Insert` 返回从 0 开始的逻辑边编号，`Run` 参数与动态版一致。
`vertices`、`edge_ids` 和原边 `edges` 可直接读取；`N` 为最大点数，大对象放全局或 `static`。

```
1 template <int N> struct Undirected_Euler
2 {
3     int n;
4     vector<pair<int, int>> g[N + 1], edges;
5     vector<int> used, vertices, edge_ids;
6     int next[N + 1];
7
8     void Init(int count)
9     {
10         assert(0 < count && count <= N);
11         n = count;
12         for ( int u = 1; u <= n; u++ )
13         {
14             g[u].clear();
15         }
16         edges.clear();
17         vertices.clear();
18         edge_ids.clear();
19     }
```

```
21  int Insert(int u, int v)
22  {
23      assert(1 <= u && u <= n && 1 <= v && v <= n);
24      int id = edges.size();
25      edges.push_back({u, v});
26      g[u].push_back({v, id});
27      g[v].push_back({u, id});
28      vertices.clear();
29      edge_ids.clear();
30      return id;
31  }
32
33  void Dfs(int u, int incoming)
34  {
35      while ( next[u] < (int)g[u].size() )
36      {
37          auto [v, id] = g[u][next[u]++];
38          if ( used[id] )
39              continue;
40          used[id] = 1;
41          Dfs(v, id);
42      }
43      vertices.push_back(u);
44      if ( incoming != -1 )
45          edge_ids.push_back(incoming);
46  }
```

```

48 // start=0 chooses the smallest feasible start. Isolated vertices are ignored.
49 bool Run(int start = 0, bool lexicographic = true)
50 {
51     assert(0 <= start && start <= n);
52     vertices.clear();
53     edge_ids.clear();
54     vector<int> odd;
55     for ( int u = 1; u <= n; u++ )
56         if ( g[u].size() % 2 )
57             odd.push_back(u);
58     if ( !odd.empty() && odd.size() != 2 )
59         return false;
60     if ( !odd.empty() )
61     {
62         if ( start && start != odd[0] && start != odd[1] )
63             return false;
64         if ( !start )
65             start = odd[0];
66     }
67     if ( !start )
68     {
69         start = 1;
70         for ( int u = 1; u <= n; u++ )
71             if ( !g[u].empty() )
72             {
73                 start = u;
74                 break;
75             }
76     }
77     if ( lexicographic )
78         for ( int u = 1; u <= n; u++ )
79             sort(g[u].begin(), g[u].end());
80     fill(next, next + n + 1, 0);
81     used.assign(edges.size(), 0);
82     Dfs(start, -1);
83     if ( edge_ids.size() != edges.size() )
84     {
85         vertices.clear();
86         edge_ids.clear();
87         return false;
88     }
89     reverse(vertices.begin(), vertices.end());
90     reverse(edge_ids.begin(), edge_ids.end());
91     return true;
92 }
93 };

```

第七十五章 混合图欧拉定向

75.1 欧拉回路与指定端点定向

算法约束见动态版本第 49 页。本节代码独立可抄；采用下列实际接口名。

调用时额外传入 `Network_Flow<N,M>` 工作区作为临时最大流图；`N` 至少为 `n+2`，`M` 至少为可反转边数 `+n`。
大流对象放全局或 `static`。

```
1 // Edge = {u,v,type}: type 1 is fixed u->v; type 0 may be reversed.
2 // start=finish=0 requests a circuit; positive endpoints request a specified trail.
3 template <int N, int M>
4 optional<vector<pair<int, int>>>
5 Mixed_Euler_Orientation(int n,
6                          const vector<array<int, 3>> &edges,
7                          Network_Flow<N, M> &flow,
8                          int start = 0,
9                          int finish = 0)
10 {
11     assert(n > 0 && 0 <= start && start <= n && 0 <= finish && finish <= n);
12     assert(bool(start) == bool(finish));
13     vector<int> parent(n + 1), id;
14     vector<long long> degree(n + 1), balance(n + 1);
15     iota(parent.begin(), parent.end(), 0);
16     auto find = [&](auto &&self, int u) -> int
17     {
18         if ( parent[u] != u )
19             parent[u] = self(self, parent[u]);
20         return parent[u];
21     };
22     flow.Init(n + 2);
23     vector<pair<int, int>> direction;
24     for ( auto [u, v, type] : edges )
25     {
26         assert(1 <= u && u <= n && 1 <= v && v <= n);
27         assert(type == 0 || type == 1);
28         parent[find(find, u)] = find(find, v);
29         degree[u]++;
30         degree[v]++;
31         balance[u]++;
32         balance[v]--;
33         id.push_back(type == 0 ? flow.Add_Edge(u, v, 1) : -1);
34         direction.push_back({u, v});
35     }
```

```

36     if ( !edges.empty() )
37     {
38         int root = find(find, edges[0][0]);
39         for ( int u = 1; u <= n; u++ )
40             if ( degree[u] && find(find, u) != root )
41                 return nullopt;
42         if ( start && (!degree[start] || !degree[finish]) )
43             return nullopt;
44     }
45     if ( start )
46     {
47         balance[start]--;
48         balance[finish]++;
49     }
50     int source = n + 1, sink = n + 2;
51     long long need = 0;
52     for ( int u = 1; u <= n; u++ )
53     {
54         if ( balance[u] % 2 )
55             return nullopt;
56         if ( balance[u] > 0 )
57         {
58             flow.Add_Edge(source, u, balance[u] / 2);
59             need += balance[u] / 2;
60         }
61         else if ( balance[u] < 0 )
62             flow.Add_Edge(u, sink, -balance[u] / 2);
63     }
64     if ( flow.Max_Flow(source, sink) != need )
65         return nullopt;
66     for ( int i = 0; i < (int)edges.size(); i++ )
67         if ( id[i] != -1 && flow.Used(id[i]) )
68             swap(direction[i].first, direction[i].second);
69     return direction;
70 }

```

75.2 任意起终点的混合欧拉路径

算法约束见动态版本第 51 页。本节代码独立可抄；采用下列实际接口名。

调用时额外传入 `Network_Flow<N,M>` 工作区作为临时最大流图；`N` 至少为 `n+2`，`M` 至少为可反转边数 `+n`。
大流对象放全局或 `static`。

```

1 // Any endpoints: zero odd degrees requests a circuit; two odd degrees tries both
2 // orders.
3 template <int N, int M>
4 optional<vector<pair<int, int>>>
5 Mixed_Euler_Trail(int n, const vector<array<int, 3>> &edges, Network_Flow<N, M> &flow)
6 {
7     assert(n > 0);
8     vector<long long> degree(n + 1);
9     vector<int> odd;
10    for ( auto [u, v, type] : edges )
11    {
12        assert(1 <= u && u <= n && 1 <= v && v <= n);
13        degree[u]++;
14        degree[v]++;
15    }
16    for ( int u = 1; u <= n; u++ )
17        if ( degree[u] % 2 )
18            odd.push_back(u);
19    if ( odd.empty() )
20        return Mixed_Euler_Orientation(n, edges, flow);
21    if ( odd.size() != 2 )
22        return nullopt;
23    auto answer = Mixed_Euler_Orientation(n, edges, flow, odd[0], odd[1]);
24    if ( answer )
25        return answer;
26    return Mixed_Euler_Orientation(n, edges, flow, odd[1], odd[0]);
27 }
```

第七十六章 图上异或行走

76.1 无向图最大异或行走

算法约束见动态版本第 52 页。本节代码独立可抄；采用下列实际接口名。

声明 `Xor_Walk<N>`，`Init(n)` 清空，`Insert` 加边，`Build(root)` 预处理指定连通块，`Query(u,v)` 返回可选最大值。N 为最大点号，大对象置于全局或 `static`；空间 $O(N+m+64)$ 。DFS 保持递归。

```
1 template <int N> struct Xor_Walk
2 {
3     using U = unsigned long long;
4     int n = 0;
5     bool built = false;
6     vector<pair<int, U>> g[N + 1];
7     int vis[N + 1];
8     U dist[N + 1];
9     Xor_Basis cycles;
10
11     void Init(int size)
12     {
13         assert(0 <= size && size <= N);
14         for ( int u = 1; u <= n; u++ )
15             g[u] = vector<pair<int, U>>();
16         n = size;
17         cycles.Init();
18         built = false;
19     }
```

```

21 void Insert(int u, int v, U w)
22 {
23     assert(1 <= u && u <= n && 1 <= v && v <= n);
24     g[u].push_back({v, w});
25     g[v].push_back({u, w});
26     built = false;
27 }
28
29 void Dfs(int u)
30 {
31     vis[u] = 1;
32     for ( auto [v, w] : g[u] )
33     {
34         if ( !vis[v] )
35         {
36             dist[v] = dist[u] ^ w;
37             Dfs(v);
38         }
39         else
40             cycles.Insert(dist[u] ^ dist[v] ^ w);
41     }
42 }
43
44 void Build(int root = 1)
45 {
46     assert(1 <= root && root <= n);
47     fill(vis, vis + n + 1, 0);
48     fill(dist, dist + n + 1, 0);
49     cycles.Init();
50     Dfs(root);
51     built = true;
52 }
53
54 optional<U> Query(int u, int v) const
55 {
56     assert(built && 1 <= u && u <= n && 1 <= v && v <= n);
57     if ( !vis[u] || !vis[v] )
58         return nullopt;
59     return cycles.Query(dist[u] ^ dist[v]);
60 }
61 };

```


第七十七章 树上算法

77.1 重链剖分与 LCA

算法约束见动态版本第 54 页。本节代码独立可抄；采用下列实际接口名。

声明 Heavy_Light-Decomposition<N> 后 Init(n), Insert 加边, Build(root) 剖分; Lca 与 Path 对应动态接口。大对象放在全局或 static。

```
1 template <int N> struct Heavy_Light-Decomposition
2 {
3     int n, timer = 0;
4     vector<int> g[N + 1];
5     int fa[N + 1], dep[N + 1], siz[N + 1], son[N + 1];
6     int top[N + 1], dfn[N + 1], rk[N + 1];
7
8     void Init(int vertices)
9     {
10         assert(0 < vertices && vertices <= N);
11         n = vertices;
12         for ( int u = 1; u <= n; u++ )
13             g[u].clear();
14     }
15
16     void Insert(int u, int v)
17     {
18         g[u].push_back(v);
19         g[v].push_back(u);
20     }
21
22     void Dfs1(int u, int p)
23     {
24         fa[u] = p;
25         siz[u] = 1;
26         son[u] = 0;
27         for ( int v : g[u] )
28         {
29             if ( v == p )
30                 continue;
31             dep[v] = dep[u] + 1;
32             Dfs1(v, u);
33             siz[u] += siz[v];
34             if ( !son[u] || siz[v] > siz[son[u]] )
35                 son[u] = v;
36         }
37     }
38
39     void Dfs2(int u, int t)
40     {
41         top[u] = t;
42         dfn[u] = ++timer;
43         rk[timer] = u;
44         if ( son[u] )
```

```

45     Dfs2(son[u], t);
46     for ( int v : g[u] )
47     {
48         if ( v != fa[u] && v != son[u] )
49             Dfs2(v, v);
50     }
51 }
52
53 // Connected tree only. Recursive DFS; subtree interval is [dfn, dfn+siz-1].
54 void Build(int root = 1)
55 {
56     assert(1 <= root && root <= n);
57     timer = 0;
58     dep[root] = 0;
59     Dfs1(root, root);
60     Dfs2(root, root);
61     assert(timer == n);
62 }
63
64 int Lca(int u, int v) const
65 {
66     while ( top[u] != top[v] )
67     {
68         if ( dep[top[u]] < dep[top[v]] )
69             swap(u, v);
70         u = fa[top[u]];
71     }
72     return dep[u] < dep[v] ? u : v;
73 }
74
75 // Commutative operations only; edge=true excludes the LCA's position.
76 template <class F> void Path(int u, int v, F work, bool edge = false) const
77 {
78     while ( top[u] != top[v] )
79     {
80         if ( dep[top[u]] < dep[top[v]] )
81             swap(u, v);
82         work(dfn[top[u]], dfn[u]);
83         u = fa[top[u]];
84     }
85     if ( dep[u] > dep[v] )
86         swap(u, v);
87     if ( dfn[u] + edge <= dfn[v] )
88         work(dfn[u] + edge, dfn[v]);
89 }
90 };

```

第七十八章 离线最近公共祖先

78.1 离线 Tarjan LCA

算法约束见动态版本第 56 页。本节代码独立可抄；采用下列实际接口名。

声明 Offline_LCA<N>, Init(n) 清空树与查询；Insert 加无向树边，Add_Query 返回从 0 开始的查询编号，Run(root) 统一计算，answer[id] 为答案。大对象放全局或 static。

```
1 template <int N> struct Offline_LCA
2 {
3     int n;
4     vector<int> g[N + 1], answer;
5     vector<pair<int, int>> queries[N + 1];
6     int parent[N + 1], size[N + 1], ancestor[N + 1], done[N + 1];
7
8     void Init(int vertices)
9     {
10         assert(0 <= vertices && vertices <= N);
11         n = vertices;
12         for ( int u = 1; u <= n; u++ )
13         {
14             g[u].clear();
15             queries[u].clear();
16         }
17         answer.clear();
18     }
19
20     void Insert(int u, int v)
21     {
22         assert(1 <= u && u <= n && 1 <= v && v <= n);
23         g[u].push_back(v);
24         g[v].push_back(u);
25     }
26
27     int Add_Query(int u, int v)
28     {
29         assert(1 <= u && u <= n && 1 <= v && v <= n);
30         int id = answer.size();
31         answer.push_back(0);
32         queries[u].push_back({v, id});
33         queries[v].push_back({u, id});
34         return id;
35     }
```

```

37     int Find(int u)
38     {
39         if ( parent[u] != u )
40             parent[u] = Find(parent[u]);
41         return parent[u];
42     }
43
44     void Merge(int u, int v)
45     {
46         u = Find(u);
47         v = Find(v);
48         if ( size[u] < size[v] )
49             swap(u, v);
50         parent[v] = u;
51         size[u] += size[v];
52     }
53
54     void Dfs(int u, int p)
55     {
56         for ( int v : g[u] )
57             if ( v != p )
58             {
59                 Dfs(v, u);
60                 Merge(u, v);
61                 ancestor[Find(u)] = u;
62             }
63         done[u] = 1;
64         for ( auto [v, id] : queries[u] )
65             if ( done[v] )
66                 answer[id] = ancestor[Find(v)];
67     }
68
69     // Input must be a nonempty tree. Query IDs are 0-based.
70     void Run(int root = 1)
71     {
72         assert(1 <= root && root <= n);
73         iota(parent, parent + n + 1, 0);
74         iota(ancestor, ancestor + n + 1, 0);
75         fill(size, size + n + 1, 1);
76         fill(done, done + n + 1, 0);
77         fill(answer.begin(), answer.end(), 0);
78         Dfs(root, 0);
79     }
80 };

```

第七十九章 在线最近公共祖先

79.1 欧拉序与 ST 表 LCA

算法约束见动态版本第 58 页。本节代码独立可抄；采用下列实际接口名。

声明 Euler_LCA<N,LOG>, Init(n) 清空树；Insert 加边，Build(root) 预处理，Lca 与 Distance 在线查询。LOG 层必须满足 2 的 LOG 次方大于 $2n-1$ ，默认 20 支持 50 万点；st 固定容量为 LOG 乘以 2N。大对象放全局或 static。

```
1 template <int N, int LOG = 20> struct Euler_LCA
2 {
3     static_assert(1 <= LOG && LOG <= 31);
4     int n;
5     bool ready = false;
6     vector<int> g[N + 1], euler;
7     int first[N + 1], depth[N + 1], lg[2 * N + 1];
8     int st[LOG][2 * N];
9
10    void Init(int vertices)
11    {
12        assert(0 < vertices && vertices <= N);
13        assert((1LL << LOG) > 2LL * vertices - 1);
14        n = vertices;
15        for ( int u = 1; u <= n; u++ )
16            g[u].clear();
17        euler.clear();
18        ready = false;
19    }
20
21    void Insert(int u, int v)
22    {
23        assert(1 <= u && u <= n && 1 <= v && v <= n);
24        g[u].push_back(v);
25        g[v].push_back(u);
26        ready = false;
27    }
28
29    int Shallower(int u, int v) const
30    {
31        return depth[u] <= depth[v] ? u : v;
32    }
```

```

34 void Dfs(int u, int parent)
35 {
36     first[u] = euler.size();
37     euler.push_back(u);
38     for ( int v : g[u] )
39         if ( v != parent )
40             {
41                 depth[v] = depth[u] + 1;
42                 Dfs(v, u);
43                 euler.push_back(u);
44             }
45 }
46
47 // Input must be a nonempty tree. Rebuilding may change the root.
48 void Build(int root = 1)
49 {
50     assert(1 <= root && root <= n);
51     euler.clear();
52     depth[root] = 0;
53     Dfs(root, 0);
54     int m = euler.size();
55     lg[0] = 0;
56     lg[1] = 0;
57     for ( int i = 2; i <= m; i++ )
58         lg[i] = lg[i / 2] + 1;
59     copy(euler.begin(), euler.end(), st[0]);
60     for ( int k = 1; k <= lg[m]; k++ )
61     {
62         for ( int i = 0; i + (1 << k) <= m; i++ )
63             st[k][i] = Shallower(st[k - 1][i], st[k - 1][i + (1 << (k - 1))]);
64     }
65     ready = true;
66 }
67
68 int Lca(int u, int v) const
69 {
70     assert(ready && 1 <= u && u <= n && 1 <= v && v <= n);
71     int l = first[u], r = first[v];
72     if ( l > r )
73         swap(l, r);
74     int k = lg[r - l + 1];
75     return Shallower(st[k][l], st[k][r - (1 << k) + 1]);
76 }
77
78 int Distance(int u, int v) const
79 {
80     int z = Lca(u, v);
81     return depth[u] + depth[v] - 2 * depth[z];
82 }
83 };

```

第八十章 倍增祖先与路径最值

80.1 倍增 LCA、祖先与最大边权

算法约束见动态版本第 60 页。本节代码独立可抄；采用下列实际接口名。

声明 `Lifting_LCA<N,LOG>`, `Init(n)` 清空树；`Insert`、`Build`、`Jump`、`Lca`、`Distance`、`Max_Edge` 对应动态接口。LOG 层要求 2 的 LOG 次方大于 n，默认 20；大对象放全局或 static。

```
1 template <int N, int LOG = 20> struct Lifting_LCA
2 {
3     using ll = long long;
4     static constexpr ll LOW = numeric_limits<ll>::lowest();
5     static_assert(1 <= LOG && LOG <= 31);
6     int n;
7     bool ready = false;
8     vector<pair<int, ll>> g[N + 1];
9     int up[LOG][N + 1], depth[N + 1];
10    ll mx[LOG][N + 1];
11
12    void Init(int vertices)
13    {
14        assert(0 < vertices && vertices <= N && (1LL << LOG) > vertices);
15        n = vertices;
16        for ( int u = 1; u <= n; u++ )
17            g[u].clear();
18        for ( int k = 0; k < LOG; k++ )
19        {
20            up[k][0] = 0;
21            mx[k][0] = LOW;
22        }
23        ready = false;
24    }
25
26    void Insert(int u, int v, ll w = 1)
27    {
28        assert(1 <= u && u <= n && 1 <= v && v <= n);
29        g[u].push_back({v, w});
30        g[v].push_back({u, w});
31        ready = false;
32    }
```

```

34 void Dfs(int u, int parent, ll weight)
35 {
36     up[0][u] = parent;
37     mx[0][u] = weight;
38     for ( int k = 1; k < LOG; k++ )
39     {
40         int p = up[k - 1][u];
41         up[k][u] = up[k - 1][p];
42         mx[k][u] = max(mx[k - 1][u], mx[k - 1][p]);
43     }
44     for ( auto [v, w] : g[u] )
45         if ( v != parent )
46         {
47             depth[v] = depth[u] + 1;
48             Dfs(v, u, w);
49         }
50 }
51
52 // Nonempty tree only. Node 0 means no ancestor.
53 void Build(int root = 1)
54 {
55     assert(1 <= root && root <= n);
56     depth[root] = 0;
57     Dfs(root, 0, LOW);
58     ready = true;
59 }
60
61 int Jump(int u, long long steps) const
62 {
63     assert(ready && 1 <= u && u <= n && steps >= 0);
64     if ( steps > depth[u] )
65         return 0;
66     for ( int k = 0; k < LOG; k++ )
67         if ( steps >> k & 1 )
68             u = up[k][u];
69     return u;
70 }

```



```

72  int Lca(int u, int v) const
73  {
74      assert(ready && 1 <= u && u <= n && 1 <= v && v <= n);
75      if ( depth[u] < depth[v] )
76          swap(u, v);
77      u = Jump(u, depth[u] - depth[v]);
78      if ( u == v )
79          return u;
80      for ( int k = LOG - 1; k >= 0; k-- )
81          if ( up[k][u] != up[k][v] )
82              {
83                  u = up[k][u];
84                  v = up[k][v];
85              }
86      return up[0][u];
87  }
88
89  int Distance(int u, int v) const
90  {
91      return depth[u] + depth[v] - 2 * depth[Lca(u, v)];
92  }
93
94  optional<ll> Max_Edge(int u, int v) const
95  {
96      assert(ready && 1 <= u && u <= n && 1 <= v && v <= n);
97      if ( u == v )
98          return nullopt;
99      if ( depth[u] < depth[v] )
100          swap(u, v);
101      ll answer = LOW;
102      int delta = depth[u] - depth[v];
103      for ( int k = 0; k < LOG; k++ )
104          if ( delta >> k & 1 )
105              {
106                  answer = max(answer, mx[k][u]);
107                  u = up[k][u];
108              }
109      if ( u == v )
110          return answer;
111      for ( int k = LOG - 1; k >= 0; k-- )
112          if ( up[k][u] != up[k][v] )
113              {
114                  answer = max(answer, max(mx[k][u], mx[k][v]));
115                  u = up[k][u];
116                  v = up[k][v];
117              }
118      return max(answer, max(mx[0][u], mx[0][v]));
119  }
120 };

```

第八十一章 树上路径交

81.1 两条树路径的交集

算法约束见动态版本第 63 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Path_Intersection_Result
2 {
3     int u, v, vertices;
4 };
5
6 // Same nonempty tree and root for depth[] and lca(). Empty result is {0,0,0}.
7 template <class Depth, class Lca>
8 Path_Intersection_Result
9 Path_Intersection(int a, int b, int c, int d, const Depth &depth, Lca lca)
10 {
11     int z = lca(a, b);
12     auto project = [&](int x)
13     {
14         int p = z;
15         for ( int v : {lca(a, x), lca(b, x)} )
16             if ( depth[v] > depth[p] )
17                 p = v;
18         return p;
19     };
20     auto distance = [&](int u, int v)
21     {
22         return 1LL * depth[u] + depth[v] - 2LL * depth[lca(u, v)];
23     };
24     int p = project(c), q = project(d);
25     if ( p == q && distance(c, p) + distance(p, d) != distance(c, d) )
26         return {0, 0, 0};
27     return {p, q, int(distance(p, q) + 1)};
28 }
```

第八十二章 点分治

82.1 点分治：树上距离点对计数

算法约束见动态版本第 64 页。本节代码独立可抄；采用下列实际接口名。

声明 Centroid_Pairs<N>, Init(n) 初始化, Insert 加边, Build 预处理; Count_Leq 和 Count_Exact 返回计数, parent 保存点分树父节点。大对象放在全局或 static。

```
1 template <int N> struct Centroid_Pairs
2 {
3     using ll = long long;
4     int n;
5     vector<pair<int, ll>> g[N + 1];
6     int siz[N + 1], removed[N + 1], parent[N + 1];
7     vector<ll> all[N + 1];
8     vector<vector<ll>> part[N + 1];
9
10    void Init(int vertices)
11    {
12        assert(0 <= vertices && vertices <= N);
13        n = vertices;
14        for ( int u = 1; u <= n; u++ )
15            g[u].clear();
16    }
17
18    void Insert(int u, int v, ll w)
19    {
20        assert(1 <= u && u <= n && 1 <= v && v <= n && u != v && w >= 0);
21        g[u].push_back({v, w});
22        g[v].push_back({u, w});
23    }
24
25    void Size_Dfs(int u, int p)
26    {
27        siz[u] = 1;
28        for ( auto [v, w] : g[u] )
29        {
30            if ( v == p || removed[v] )
31                continue;
32            Size_Dfs(v, u);
33            siz[u] += siz[v];
34        }
35    }
36
37    int Centroid(int u, int p, int total)
38    {
39        for ( auto [v, w] : g[u] )
40        {
41            if ( v != p && !removed[v] && siz[v] > total / 2 )
42                return Centroid(v, u, total);
43        }
44        return u;
```

```

45     }
46
47     void Collect(int u, int p, ll d, vector<ll> &out)
48     {
49         out.push_back(d);
50         for ( auto [v, w] : g[u] )
51         {
52             if ( v == p || removed[v] )
53                 continue;
54             assert(d <= numeric_limits<ll>::max() - w);
55             Collect(v, u, d + w, out);
56         }
57     }
58
59     void Decompose(int entry, int p)
60     {
61         Size_Dfs(entry, 0);
62         int c = Centroid(entry, 0, siz[entry]);
63         removed[c] = 1;
64         parent[c] = p;
65         all[c] = {0};
66         for ( auto [v, w] : g[c] )
67         {
68             if ( removed[v] )
69                 continue;
70             part[c].push_back({});
71             auto &d = part[c].back();
72             Collect(v, c, w, d);
73             sort(d.begin(), d.end());
74             all[c].insert(all[c].end(), d.begin(), d.end());
75         }
76         sort(all[c].begin(), all[c].end());
77         for ( auto [v, w] : g[c] )
78         {
79             if ( !removed[v] )
80                 Decompose(v, c);
81         }
82     }
83
84     // Input must be a forest. Rebuild after changing edges.
85     void Build()
86     {
87         fill(removed, removed + n + 1, 0);
88         for ( int u = 1; u <= n; u++ )
89         {
90             all[u].clear();
91             part[u].clear();
92         }
93         for ( int u = 1; u <= n; u++ )
94         {
95             if ( !removed[u] )
96                 Decompose(u, 0);
97         }
98     }
99
100     ll Pairs(const vector<ll> &d, ll k) const

```

```

101 {
102     int l = 0, r = (int)d.size() - 1;
103     ll answer = 0;
104     while ( l < r )
105     {
106         if ( d[l] <= k - d[r] )
107         {
108             answer += r - l;
109             l++;
110         }
111         else
112             r--;
113     }
114     return answer;
115 }
116
117 // Unordered Pairs of distinct connected vertices. Nonnegative edge weights.
118 ll Count_Leq(ll k) const
119 {
120     if ( k < 0 )
121         return 0;
122     ll answer = 0;
123     for ( int c = 1; c <= n; c++ )
124     {
125         answer += Pairs(all[c], k);
126         for ( const auto &d : part[c] )
127             answer -= Pairs(d, k);
128     }
129     return answer;
130 }
131
132 ll Count_Exact(ll k) const
133 {
134     if ( k < 0 )
135         return 0;
136     return Count_Leq(k) - (k ? Count_Leq(k - 1) : 0);
137 }
138 };

```

第八十三章 树上启发式合并

83.1 树上启发式合并：子树众数颜色和

算法约束见动态版本第 67 页。本节代码独立可抄；采用下列实际接口名。

声明 Subtree_Colors<N>, Init(n) 初始化, Insert 加边, Run(colors,root) 计算; answer 与 max_freq 为最终结果。Add_Color 为内部统计函数。大对象放在全局或 static。

```
1 template <int N> struct Subtree_Colors
2 {
3     int n, best = 0;
4     long long sum = 0;
5     vector<int> g[N + 1], order, value, freq;
6     int siz[N + 1], heavy[N + 1], tin[N + 1], tout[N + 1];
7     int color[N + 1], max_freq[N + 1];
8     long long answer[N + 1];
9
10    void Init(int vertices)
11    {
12        assert(0 <= vertices && vertices <= N);
13        n = vertices;
14        for ( int u = 1; u <= n; u++ )
15            g[u].clear();
16    }
17
18    void Insert(int u, int v)
19    {
20        assert(1 <= u && u <= n && 1 <= v && v <= n && u != v);
21        g[u].push_back(v);
22        g[v].push_back(u);
23    }
24
25    void Prepare(int u, int p)
26    {
27        siz[u] = 1;
28        tin[u] = order.size();
29        order.push_back(u);
30        for ( int v : g[u] )
31        {
32            if ( v == p )
33                continue;
34            Prepare(v, u);
35            siz[u] += siz[v];
36            if ( siz[v] > siz[heavy[u]] )
37                heavy[u] = v;
38        }
39        tout[u] = (int)order.size() - 1;
40    }
41
42    void Add_Color(int u)
43    {
44        int c = color[u];
```

```

45     freq[c]++;
46     if ( freq[c] > best )
47     {
48         best = freq[c];
49         sum = value[c];
50     }
51     else if ( freq[c] == best )
52         sum += value[c];
53 }
54
55 void Dfs(int u, int p, bool keep)
56 {
57     for ( int v : g[u] )
58     {
59         if ( v != p && v != heavy[u] )
60             Dfs(v, u, false);
61     }
62     if ( heavy[u] )
63         Dfs(heavy[u], u, true);
64     for ( int v : g[u] )
65     {
66         if ( v == p || v == heavy[u] )
67             continue;
68         for ( int i = tin[v]; i <= tout[v]; i++ )
69             Add_Color(order[i]);
70     }
71     Add_Color(u);
72     max_freq[u] = best;
73     answer[u] = sum;
74     if ( !keep )
75     {
76         for ( int i = tin[u]; i <= tout[u]; i++ )
77             freq[color[order[i]]]--;
78         best = 0;
79         sum = 0;
80     }
81 }
82
83 // A connected tree, colors[1..n]. Sum all colors tied for maximum frequency.
84 void Run(const vector<int> &colors, int root = 1)
85 {
86     assert((int)colors.size() == n + 1);
87     fill(siz, siz + n + 1, 0);
88     fill(heavy, heavy + n + 1, 0);
89     fill(tin, tin + n + 1, 0);
90     fill(tout, tout + n + 1, 0);
91     fill(color, color + n + 1, 0);
92     fill(max_freq, max_freq + n + 1, 0);
93     fill(answer, answer + n + 1, 0);
94     order.clear();
95     value.assign(colors.begin() + 1, colors.end());
96     sort(value.begin(), value.end());
97     value.erase(unique(value.begin(), value.end()), value.end());
98     freq.assign(value.size(), 0);
99     best = 0;
100    sum = 0;

```

```
101     if ( !n )
102         return;
103     assert(1 <= root && root <= n);
104     for ( int u = 1; u <= n; u++ )
105         color[u] =
106             lower_bound(value.begin(), value.end(), colors[u]) - value.begin();
107     Prepare(root, 0);
108     Dfs(root, 0, false);
109 }
110 };
```


第八十四章 虚树

84.1 虚树：关键点压缩与路径信息

算法约束见动态版本第 70 页。本节代码独立可抄；采用下列实际接口名。

声明 `Virtual_Tree<N,LOG>`，要求 2 的 LOG 次方大于实际点数，默认 LOG=20。Init(n)、Insert、Prepare、Build 对应初始化、加边、预处理和查询；Lca、Ancestor、Path 为对应辅助接口。大对象放在全局或 static。

```
1 template <int N, int LOG = 20> struct Virtual_Tree
2 {
3     using ll = long long;
4
5     struct Edge
6     {
7         int from, to;
8         ll distance, minimum;
9     };
10
11     struct Result
12     {
13         vector<int> vertices;
14         vector<Edge> edges;
15     };
16
17     int n, log = 1, timer = 0;
18     vector<pair<int, ll>> g[N + 1];
19     int up[LOG][N + 1], depth[N + 1], tin[N + 1], tout[N + 1];
20     ll mn[LOG][N + 1], dist[N + 1];
21
22     void Init(int vertices)
23     {
24         assert(0 < vertices && vertices <= N);
25         n = vertices;
26         for ( int u = 1; u <= n; u++ )
27             g[u].clear();
28     }
29
30     void Insert(int u, int v, ll w = 1)
31     {
32         assert(1 <= u && u <= n && 1 <= v && v <= n && u != v && w >= 0);
33         g[u].push_back({v, w});
34         g[v].push_back({u, w});
35     }
36
37     void Dfs(int u, int p)
38     {
39         tin[u] = ++timer;
40         up[0][u] = p;
41         for ( int j = 1; j < log; j++ )
42         {
43             up[j][u] = up[j - 1][up[j - 1][u]];
44             mn[j][u] = min(mn[j - 1][u], mn[j - 1][up[j - 1][u]]);
```

```

45     }
46     for ( auto [v, w] : g[u] )
47     {
48         if ( v == p )
49             continue;
50         assert(dist[u] <= numeric_limits<ll>::max() - w);
51         depth[v] = depth[u] + 1;
52         dist[v] = dist[u] + w;
53         mn[0][v] = w;
54         Dfs(v, u);
55     }
56     tout[u] = timer;
57 }
58
59 void Prepare(int root = 1)
60 {
61     assert(1 <= root && root <= n);
62     log = 1;
63     while ( (1LL << log) <= n )
64         log++;
65     assert(log <= LOG);
66     for ( int j = 0; j < log; j++ )
67     {
68         fill(up[j], up[j] + n + 1, 0);
69         fill(mn[j], mn[j] + n + 1, numeric_limits<ll>::max());
70     }
71     fill(depth, depth + n + 1, 0);
72     fill(tin, tin + n + 1, 0);
73     fill(tout, tout + n + 1, 0);
74     fill(dist, dist + n + 1, 0);
75     timer = 0;
76     Dfs(root, root);
77 }
78
79 bool Ancestor(int u, int v) const
80 {
81     return tin[u] <= tin[v] && tout[v] <= tout[u];
82 }
83
84 int Lca(int u, int v) const
85 {
86     if ( Ancestor(u, v) )
87         return u;
88     if ( Ancestor(v, u) )
89         return v;
90     for ( int j = log - 1; j >= 0; j-- )
91     {
92         if ( !Ancestor(up[j][u], v) )
93             u = up[j][u];
94     }
95     return up[0][u];
96 }
97
98 Edge Path(int u, int v) const
99 {
100     assert(u != v && Ancestor(u, v));

```

```

101     ll minimum = numeric_limits<ll>::max();
102     int x = v, steps = depth[v] - depth[u];
103     for ( int j = 0; j < log; j++ )
104     {
105         if ( steps >> j & 1 )
106         {
107             minimum = min(minimum, mn[j][x]);
108             x = up[j][x];
109         }
110     }
111     return {u, v, dist[v] - dist[u], minimum};
112 }
113
114 // Uses original vertex IDs. Empty input gives an empty result.
115 Result Build(vector<int> keys) const
116 {
117     if ( keys.empty() )
118         return {};
119     for ( int u : keys )
120         assert(1 <= u && u <= n);
121     auto cmp = [&](int u, int v)
122     {
123         return tin[u] < tin[v];
124     };
125     sort(keys.begin(), keys.end(), cmp);
126     keys.erase(unique(keys.begin(), keys.end()), keys.end());
127     int k = keys.size();
128     for ( int i = 1; i < k; i++ )
129         keys.push_back(Lca(keys[i - 1], keys[i]));
130     sort(keys.begin(), keys.end(), cmp);
131     keys.erase(unique(keys.begin(), keys.end()), keys.end());
132     Result result{keys, {}};
133     vector<int> st;
134     for ( int u : keys )
135     {
136         while ( !st.empty() && !Ancestor(st.back(), u) )
137             st.pop_back();
138         if ( !st.empty() )
139             result.edges.push_back(Path(st.back(), u));
140         st.push_back(u);
141     }
142     return result;
143 }
144 };

```

第八十五章 字符串

85.1 前缀函数

算法约束见动态版本第 73 页。本节代码独立可抄；采用下列实际接口名。

```
1 inline vector<int> Prefix_Function(const string &s)
2 {
3     int n = (int)s.size();
4     vector<int> p(n);
5     for ( int i = 1; i < n; i++ )
6     {
7         int j = p[i - 1];
8         while ( j && s[i] != s[j] )
9             j = p[j - 1];
10        p[i] = j + (s[i] == s[j]);
11    }
12    return p;
13 }
```

85.2 KMP 匹配

算法约束见动态版本第 73 页。本节代码独立可抄；采用下列实际接口名。

```
1 inline vector<int> Kmp_Match(const string &s, const string &t)
2 {
3     assert(!t.empty());
4     auto p = Prefix_Function(t);
5     vector<int> ans;
6     for ( int i = 0, j = 0; i < (int)s.size(); i++ )
7     {
8         while ( j && s[i] != t[j] )
9             j = p[j - 1];
10        j += (s[i] == t[j]);
11        if ( j == (int)t.size() )
12        {
13            ans.push_back(i - j + 1);
14            j = p[j - 1];
15        }
16    }
17    return ans;
18 }
```

85.3 Z 函数

算法约束见动态版本第 74 页。本节代码独立可抄；采用下列实际接口名。

```

1 inline vector<int> Z_Function(const string &s)
2 {
3     int n = (int)s.size();
4     vector<int> a(n);
5     if ( n )
6         a[0] = n;
7     for ( int i = 1, l = 0, r = 0; i < n; i++ )
8     {
9         if ( i < r )
10            a[i] = min(r - i, a[i - l]);
11        while ( i + a[i] < n && s[a[i]] == s[i + a[i]] )
12            ++a[i];
13        if ( i + a[i] > r )
14        {
15            l = i;
16            r = i + a[i];
17        }
18    }
19    return a;
20 }
```

85.4 Manacher 回文半径

算法约束见动态版本第 74 页。本节代码独立可抄；采用下列实际接口名。

```

1 // odd[i] includes the center; even[i] is centered before i.
2 inline pair<vector<int>, vector<int>> Manacher(const string &s)
3 {
4     int n = (int)s.size();
5     vector<int> odd(n), even(n);
6     for ( int i = 0, l = 0, r = -1; i < n; i++ )
7     {
8         int k = i > r ? 1 : min(odd[l + r - i], r - i + 1);
9         while ( i - k >= 0 && i + k < n && s[i - k] == s[i + k] )
10            ++k;
11        odd[i] = k--;
12        if ( i + k > r )
13        {
14            l = i - k;
15            r = i + k;
16        }
17    }
18    for ( int i = 0, l = 0, r = -1; i < n; i++ )
19    {
20        int k = i > r ? 0 : min(even[l + r - i + 1], r - i + 1);
21        while ( i - k - 1 >= 0 && i + k < n && s[i - k - 1] == s[i + k] )
22            ++k;
```

```

23     even[i] = k--;
24     if ( i + k > r )
25     {
26         l = i - k - 1;
27         r = i + k;
28     }
29 }
30 return {odd, even};
31 }

```

85.5 最小循环表示

算法约束见动态版本第 75 页。本节代码独立可抄；采用下列实际接口名。

```

1 inline int Minimum_Rotation(const string &s)
2 {
3     int n = (int)s.size(), i = 0, j = 1, k = 0;
4     if ( !n )
5         return 0;
6     while ( i < n && j < n && k < n )
7     {
8         unsigned char a = s[(i + k) % n], b = s[(j + k) % n];
9         if ( a == b )
10            {
11                ++k;
12                continue;
13            }
14         if ( a > b )
15            {
16                i += k + 1;
17                if ( i == j )
18                    ++i;
19            }
20         else
21            {
22                j += k + 1;
23                if ( i == j )
24                    ++j;
25            }
26         k = 0;
27     }
28     return min(i, j);
29 }

```

85.6 AC 自动机出现次数

算法约束见动态版本第 75 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct AC_Automaton
2 {
3     struct Node
4     {
5         array<int, 26> go{};
6         int fail = 0;

```

```

7     };
8
9     vector<Node> a{Node{}};
10    vector<int> order;
11    bool built = false;
12
13    int Insert(const string &s)
14    {
15        assert(!built && !s.empty());
16        int u = 0;
17        for ( char ch : s )
18        {
19            int c = ch - 'a';
20            assert(0 <= c && c < 26);
21            if ( !a[u].go[c] )
22            {
23                int v = (int)a.size();
24                a[u].go[c] = v;
25                a.push_back(Node{});
26            }
27            u = a[u].go[c];
28        }
29        return u;
30    }
31
32    void Build()
33    {
34        assert(!built);
35        built = true;
36        queue<int> q;
37        for ( int v : a[0].go )
38            if ( v )
39                q.push(v);
40        while ( !q.empty() )
41        {
42            int u = q.front();
43            q.pop();
44            order.push_back(u);
45            for ( int c = 0; c < 26; c++ )
46            {
47                int v = a[u].go[c];
48                if ( v )
49                {
50                    a[v].fail = a[a[u].fail].go[c];
51                    q.push(v);
52                }
53                else
54                    a[u].go[c] = a[a[u].fail].go[c];
55            }
56        }
57    }
58
59    vector<long long> count(const string &s) const
60    {
61        assert(built);
62        vector<long long> ans(a.size());

```

```
63     int u = 0;
64     for ( char ch : s )
65     {
66         u = a[u].go[ch - 'a'];
67         ++ans[u];
68     }
69     for ( int i = (int)order.size() - 1; i >= 0; i-- )
70         ans[a[order[i]].fail] += ans[order[i]];
71     return ans;
72 }
73 };
```


85.7 倍增后缀数组与 LCP

算法约束见动态版本第 78 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Suffix_Array
2 {
3     vector<int> sa, rk, lcp;
4
5     Suffix_Array(const string &s)
6     {
7         vector<int> a;
8         a.reserve(s.size());
9         for ( unsigned char c : s )
10             a.push_back(c);
11         Init(a, 256);
12     }
13
14     Suffix_Array(const vector<int> &s, int alphabet)
15     {
16         Init(s, alphabet);
17     }
18
19     void Init(const vector<int> &s, int alphabet)
20     {
21         assert(s.size() < INT_MAX);
22         assert(1 <= alphabet && alphabet < INT_MAX);
23         int n = (int)s.size();
24         sa.resize(n);
25         rk.resize(n);
26         lcp.assign(n, 0);
27         if ( !n )
28             return;
29         vector<int> order(n), old(n), cnt(max(n + 1, alphabet + 1));
30         int m = alphabet;
31         for ( int i = 0; i < n; i++ )
32         {
33             assert(0 <= s[i] && s[i] < alphabet);
34             rk[i] = s[i] + 1;
35             cnt[rk[i]]++;
36         }
37         for ( int i = 1; i <= m; i++ )
38             cnt[i] += cnt[i - 1];
39         for ( int i = n - 1; i >= 0; i-- )
40             sa[--cnt[rk[i]]] = i;

```

```

41     for ( int k = 1; k < n; k *= 2 )
42     {
43         int p = 0;
44         for ( int i = n - k; i < n; i++ )
45             order[p++] = i;
46         for ( int i : sa )
47             if ( i >= k )
48                 order[p++] = i - k;
49         fill(cnt.begin(), cnt.begin() + m + 1, 0);
50         for ( int x : rk )
51             cnt[x]++;
52         for ( int i = 1; i <= m; i++ )
53             cnt[i] += cnt[i - 1];
54         for ( int i = n - 1; i >= 0; i-- )
55             sa[--cnt[rk[order[i]]]] = order[i];
56         old.swap(rk);
57         auto second = [&](int i)
58         {
59             return i < n - k ? old[i + k] : 0;
60         };
61         m = 1;
62         rk[sa[0]] = m;
63         for ( int i = 1; i < n; i++ )
64         {
65             int a = sa[i - 1], b = sa[i];
66             if ( old[a] != old[b] || second(a) != second(b) )
67                 m++;
68             rk[b] = m;
69         }
70         if ( m == n )
71             break;
72     }
73     for ( int i = 0; i < n; i++ )
74         rk[sa[i]] = i;
75     for ( int i = 0, k = 0; i < n; i++ )
76     {
77         if ( !rk[i] )
78         {
79             k = 0;
80             continue;
81         }
82         int j = sa[rk[i] - 1];
83         while ( k < n - i && k < n - j && s[i + k] == s[j + k] )
84             ++k;
85         lcp[rk[i]] = k;
86         if ( k )
87             --k;
88     }
89 }
90 };

```

85.8 后缀自动机

算法约束见动态版本第 80 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Suffix_Automaton
2 {
3     struct Node
4     {
5         array<int, 26> go{};
6         int link = -1, len = 0;
7         long long occ = 0;
8     };
9
10    vector<Node> a{Node{}};
11    int last = 0;
12
13    void Extend(int c)
14    {
15        assert(0 <= c && c < 26);
16        int cur = (int)a.size();
17        a.push_back(Node{});
18        a[cur].len = a[last].len + 1;
19        a[cur].occ = 1;
20        int p = last;
21        while ( p != -1 && !a[p].go[c] )
22        {
23            a[p].go[c] = cur;
24            p = a[p].link;
25        }
26        if ( p == -1 )
27            a[cur].link = 0;
28        else
29        {
30            int q = a[p].go[c];
31            if ( a[p].len + 1 == a[q].len )
32                a[cur].link = q;
33            else
34            {
35                int clone = (int)a.size();
36                Node copy = a[q];
37                a.push_back(copy);
38                a[clone].len = a[p].len + 1;
39                a[clone].occ = 0;
40                while ( p != -1 && a[p].go[c] == q )
41                {
42                    a[p].go[c] = clone;
43                    p = a[p].link;
44                }
45                a[q].link = a[cur].link = clone;
46            }
47        }
48        last = cur;
49    }
50
51    vector<long long> Counts() const
52    {

```

```
53     vector<int> bucket(a[last].len + 1), order(a.size());
54     vector<long long> ans(a.size());
55     for ( auto &v : a )
56         ++bucket[v.len];
57     partial_sum(bucket.begin(), bucket.end(), bucket.begin());
58     for ( int i = 0; i < (int)a.size(); i++ )
59     {
60         order[--bucket[a[i].len]] = i;
61         ans[i] = a[i].occ;
62     }
63     for ( int i = (int)order.size() - 1; i > 0; i-- )
64         ans[a[order[i]].link] += ans[order[i]];
65     return ans;
66 }
67
68 long long Distinct() const
69 {
70     long long ans = 0;
71     for ( int i = 1; i < (int)a.size(); i++ )
72         ans += a[i].len - a[a[i].link].len;
73     return ans;
74 }
75 };
```

第八十六章 后缀查询与重复子串

86.1 后缀 LCP 与子串比较

算法约束见动态版本第 82 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Suffix_LCP
2 {
3     int n;
4     vector<int> rk, lg;
5     vector<vector<int>> st;
6
7     Suffix_LCP(const Suffix_Array &suffix)
8     {
9         Init(suffix);
10    }
11
12    void Init(const Suffix_Array &suffix)
13    {
14        n = suffix.sa.size();
15        rk = suffix.rk;
16        lg.assign(n + 1, 0);
17        for ( int i = 2; i <= n; i++ )
18            lg[i] = lg[i / 2] + 1;
19        st.assign(lg[n] + 1, {});
20        st[0] = suffix.lcp;
21        for ( int k = 1; k <= lg[n]; k++ )
22        {
23            int len = 1 << k;
24            st[k].resize(n - len + 1);
25            for ( int i = 0; i <= n - len; i++ )
26                st[k][i] = min(st[k - 1][i], st[k - 1][i + len / 2]);
27        }
28    }
```

```

30  int Query(int x, int y) const
31  {
32      assert(0 <= x && x <= n && 0 <= y && y <= n);
33      if ( x == y || x == n || y == n )
34          return n - max(x, y);
35      int a = rk[x], b = rk[y];
36      if ( a > b )
37          swap(a, b);
38      int k = lg[b - a];
39      return min(st[k][a + 1], st[k][b - (1 << k) + 1]);
40  }
41
42  // Compare [l1,r1) and [l2,r2); return -1, 0 or 1.
43  int Compare(int l1, int r1, int l2, int r2) const
44  {
45      assert(0 <= l1 && l1 <= r1 && r1 <= n);
46      assert(0 <= l2 && l2 <= r2 && r2 <= n);
47      int a = r1 - l1, b = r2 - l2;
48      if ( Query(l1, l2) >= min(a, b) )
49          return (a > b) - (a < b);
50      return rk[l1] < rk[l2] ? -1 : 1;
51  }
52 };

```

86.2 两个前缀的最长公共后缀

算法约束见动态版本第 83 页。本节代码独立可抄；采用下列实际接口名。

```

1  // reversed must be built on the reversed original sequence.
2  inline int Prefix_Lcs(const Suffix_LCP &reversed, int x, int y)
3  {
4      assert(0 <= x && x <= reversed.n && 0 <= y && y <= reversed.n);
5      return reversed.Query(reversed.n - x, reversed.n - y);
6  }

```

86.3 重复子串 AA 的起止位置计数

算法约束见动态版本第 84 页。本节代码独立可抄；采用下列实际接口名。

```

1 // Count nonempty squares AA by their starting and ending positions.
2 inline pair<vector<long long>, vector<long long>>
3 Square_Counts(const Suffix_LCP &forward, const Suffix_LCP &reversed)
4 {
5     int n = forward.n;
6     assert(reversed.n == n);
7     vector<long long> start(n + 1), finish(n + 1);
8     for ( int len = 1; len <= n / 2; len++ )
9         for ( int i = 0; i < n - len; i += len )
10            {
11                int j = i + len;
12                int left = min(len - 1, Prefix_Lcs(reversed, i, j));
13                int right = min(len, forward.Query(i, j));
14                if ( left + right < len )
15                    continue;
16                int l = i - left, r = i + right - len;
17                start[l]++;
18                start[r + 1]--;
19                finish[l + 2 * len - 1]++;
20                finish[r + 2 * len]--;
21            }
22     for ( int i = 1; i < n; i++ )
23     {
24         start[i] += start[i - 1];
25         finish[i] += finish[i - 1];
26     }
27     start.resize(n);
28     finish.resize(n);
29     return {start, finish};
30 }
```

第八十七章 回文树

87.1 回文树 PAM: 回文后缀与出现次数

算法约束见动态版本第 85 页。本节代码独立可抄；采用下列实际接口名。

声明 `Palindromic_Tree<N>` 后调用 `Init()`；`N` 为最大追加字符数，内部另留两个根节点。`Insert`、`Distinct`、`Occurrences` 对应动态版接口；`t`、`last`、`total` 可直接读。大对象放在全局或 `static`。

```
1 template <int N> struct Palindromic_Tree
2 {
3     struct Node
4     {
5         int next[26]{};
6         int len = 0, fail = 0, suffixes = 0, end = -1;
7         long long hits = 0;
8     };
9
10    Node t[N + 2];
11    int tot = 1;
12    vector<int> s;
13    int last = 0;
14    long long total = 0;
15
16    void Init()
17    {
18        tot = 1;
19        t[0] = Node{};
20        t[1] = Node{};
21        t[0].fail = 1;
22        t[1].len = -1;
23        s.assign(1, -1);
24        last = 0;
25        total = 0;
26    }
27
28    int Get_Fail(int u) const
29    {
30        int pos = s.size() - 1;
31        while ( s[pos - t[u].len - 1] != s[pos] )
32            u = t[u].fail;
33        return u;
34    }
35
36    // Append a lowercase letter; return the longest palindromic suffix node.
37    int Insert(char ch)
38    {
39        int c = ch - 'a';
40        assert(0 <= c && c < 26);
41        assert(s.size() <= N);
42        s.push_back(c);
43        int u = Get_Fail(last);
44        if ( !t[u].next[c] )
```



```

45     {
46         int v = ++tot;
47         t[v] = Node{};
48         t[v].len = t[u].len + 2;
49         t[v].fail = t[Get_Fail(t[u].fail)].next[c];
50         t[v].suffixes = t[t[v].fail].suffixes + 1;
51         t[v].end = (int)s.size() - 2;
52         t[u].next[c] = v;
53     }
54     last = t[u].next[c];
55     t[last].hits++;
56     total += t[last].suffixes;
57     return last;
58 }
59
60 int Distinct() const
61 {
62     return tot - 1;
63 }
64
65 // Copy-and-propagate: repeated calls and later appends remain valid.
66 vector<long long> Occurrences() const
67 {
68     vector<long long> count(tot + 1);
69     for ( int u = 2; u <= tot; u++ )
70         count[u] = t[u].hits;
71     for ( int u = tot; u >= 2; u-- )
72         count[t[u].fail] += count[u];
73     count[0] = 0;
74     count[1] = 0;
75     return count;
76 }
77 };

```

第八十八章 数论

88.1 模运算、素性、CRT 与类欧几里德

算法约束见动态版本第 87 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Number_Theory
2 {
3     using ll = long long;
4     using ull = unsigned long long;
5     using i128 = __int128_t;
6     using u128 = __uint128_t;
7
8     static ull Mul(ull a, ull b, ull m)
9     {
10         return u128(a) * b % m;
11     }
12
13     static ull Power(ull a, ull b, ull m)
14     {
15         assert(m);
16         ull r = 1 % m;
17         a %= m;
18         for ( ; b; b >>= 1, a = Mul(a, a, m) )
19             if ( b & 1 )
20                 r = Mul(r, a, m);
21         return r;
22     }
23
24     static bool Prime(ull n)
25     {
26         if ( n < 2 )
27             return false;
28         for ( ull p : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37} )
29             if ( n % p == 0 )
30                 return n == p;
31         ull d = n - 1;
32         int s = 0;
33         while ( !(d & 1) )
34         {
35             d >>= 1;
36             ++s;
37         }
38         for ( ull a :
39             {2ULL, 325ULL, 9375ULL, 28178ULL, 450775ULL, 9780504ULL, 1795265022ULL} )
40         {
41             if ( a % n == 0 )
42                 continue;
43             ull x = Power(a, d, n);
44             if ( x == 1 || x == n - 1 )
45                 continue;
46             bool ok = false;
```

```

47     for ( int j = 1; j < s; j++ )
48     {
49         x = Mul(x, x, n);
50         if ( x == n - 1 )
51         {
52             ok = true;
53             break;
54         }
55     }
56     if ( !ok )
57         return false;
58 }
59 return true;
60 }
61
62 static i128 Exgcd(i128 a, i128 b, i128 &x, i128 &y)
63 {
64     if ( !b )
65     {
66         x = 1;
67         y = 0;
68         return a;
69     }
70     i128 d = Exgcd(b, a % b, y, x);
71     y -= a / b * x;
72     return d;
73 }
74
75 static ll Inverse(ll a, ll m)
76 {
77     assert(m > 0);
78     a %= m;
79     if ( a < 0 )
80         a += m;
81     i128 x, y;
82     if ( Exgcd(a, m, x, y) != 1 )
83         return -1;
84     return (x % m + m) % m;
85 }
86
87 // x = r (mod m); false means inconsistent. Throws if lcm exceeds int64.
88 static bool Crt(ll &r, ll &m, ll b, ll n)
89 {
90     assert(m > 0 && n > 0);
91     r %= m;
92     if ( r < 0 )
93         r += m;
94     b %= n;
95     if ( b < 0 )
96         b += n;
97     i128 x, y;
98     ll g = (ll)Exgcd(m, n, x, y);
99     i128 diff = i128(b) - r;
100     if ( diff % g )
101         return false;
102     i128 q = n / g, k = (diff / g * x % q + q) % q, mod = i128(m) * q;

```

```

103     if ( mod > LLONG_MAX )
104         throw overflow_error("CRT modulus");
105     r = (r + i128(m) * k) % mod;
106     m = (ll)mod;
107     return true;
108 }
109
110 // sum_{i=0}^{n-1} floor((a*i+b)/m), signed a,b; 0<=n<=1e9, 1<=m<=1e9.
111 static i128 Floor_Sum(ll n, ll m, ll a, ll b)
112 {
113     assert(n >= 0 && m > 0);
114     i128 ans = 0;
115     auto norm = [&](ll &v)
116     {
117         ll q = v / m;
118         if ( v % m < 0 )
119             --q;
120         v = (ll)(i128(v) - i128(q) * m);
121         return q;
122     };
123     ans += i128(norm(a)) * n * (n - 1) / 2;
124     ans += i128(norm(b)) * n;
125     while ( true )
126     {
127         ans += i128(n) * (n - 1) / 2 * (a / m);
128         a %= m;
129         ans += i128(n) * (b / m);
130         b %= m;
131         i128 y = i128(a) * n + b;
132         if ( y < m )
133             break;
134         n = (ll)(y / m);
135         b = (ll)(y % m);
136         swap(a, m);
137     }
138     return ans;
139 }
140 };

```

88.2 Pollard–Rho 因数分解

算法约束见动态版本第 90 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Pollard_Rho
2 {
3     using ull = unsigned long long;
4     using u128 = __uint128_t;
5     mt19937_64 rng;
6
7     Pollard_Rho(ull seed = 712367821) : rng(seed)
8     {
9     }
10
11     ull Rho(ull n)
12     {
13         if ( n % 2 == 0 )
14             return 2;
15         for ( ;; )
16         {
17             ull c = rng() % (n - 1) + 1, x = rng() % n, y = x, d = 1;
18             auto f = [&](ull v)
19             {
20                 return (Number_Theory::Mul(v, v, n) + u128(c)) % n;
21             };
22             // Retry bounded attempts; Las Vegas: only exact divisors returned.
23             for ( int i = 0; i < 200000 && d == 1; i++ )
24             {
25                 x = f(x);
26                 y = f(f(y));
27                 d = gcd(x > y ? x - y : y - x, n);
28             }
29             if ( 1 < d && d < n )
30                 return d;
31         }
32     }
33
34     void Split(ull n, vector<ull> &a)
35     {
36         if ( n == 1 )
37             return;
38         if ( Number_Theory::Prime(n) )
39         {
40             a.push_back(n);
41             return;
42         }
43         ull d = Rho(n);
44         Split(d, a);
45         Split(n / d, a);
46     }

```

```

47
48     vector<ull> Factor(ull n)
49     {
50         assert(n >= 1);
51         vector<ull> a;
52         Split(n, a);
53         sort(a.begin(), a.end());
54         return a;
55     }
56 };

```

88.3 线性筛、欧拉函数与 Möbius 函数

算法约束见动态版本第 91 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Linear_Sieve
2 {
3     vector<int> Prime, lp, phi, mu;
4
5     Linear_Sieve(int n) : lp(n + 1), phi(n + 1), mu(n + 1)
6     {
7         if ( n )
8             phi[1] = mu[1] = 1;
9         for ( int i = 2; i <= n; i++ )
10            {
11                if ( !lp[i] )
12                {
13                    lp[i] = i;
14                    Prime.push_back(i);
15                    phi[i] = i - 1;
16                    mu[i] = -1;
17                }
18                for ( int p : Prime )
19                {
20                    if ( p > n / i )
21                        break;
22                    int j = i * p;
23                    lp[j] = p;
24                    if ( i % p == 0 )
25                    {
26                        phi[j] = phi[i] * p;
27                        mu[j] = 0;
28                        break;
29                    }
30                    phi[j] = phi[i] * (p - 1);
31                    mu[j] = -mu[i];
32                }
33            }
34    }
35 };

```

88.4 静态模整数

算法约束见动态版本第 92 页。本节代码独立可抄；采用下列实际接口名。

```

1 template <int mod> struct Mod_Int
2 {
3     int v;
4
5     Mod_Int(long long x = 0) : v((x % mod + mod) % mod)
6     {
7     }
8
9     Mod_Int operator+(Mod_Int b) const
10    {
11        return Mod_Int((long long)v + b.v);
12    }
13
14    Mod_Int operator-(Mod_Int b) const
15    {
16        return Mod_Int((long long)v - b.v);
17    }
18
19    Mod_Int operator*(Mod_Int b) const
20    {
21        return Mod_Int(1LL * v * b.v);
22    }
23
24    Mod_Int Pow(long long e) const
25    {
26        assert(e >= 0);
27        Mod_Int a = *this, r = 1;
28        for ( ; e; e >>= 1, a = a * a )
29            if ( e & 1 )
30                r = r * a;
31        return r;
32    }
33
34    Mod_Int Inv() const
35    {
36        assert(v);
37        return Pow(mod - 2);
38    } // prime modulus
39
40    Mod_Int operator/(Mod_Int b) const
41    {
42        return *this * b.Inv();
43    }
44 };

```

88.5 阶乘组合数

算法约束见动态版本第 93 页。本节代码独立可抄；采用下列实际接口名。

Combination<prime>(n=0) 构造，Init(n) 扩大上限；Choose、Permute 对应组合数和排列数，fac、ifac 可只读访问。

```

1  template <int mod> struct Combination
2  {
3      using Z = Mod_Int<mod>;
4      vector<Z> fac{Z(1)}, ifac{Z(1)};
5
6      Combination(int n = 0)
7      {
8          Init(n);
9      }
10
11     void Init(int n)
12     {
13         assert(0 <= n && n < mod);
14         int old = (int)fac.size() - 1;
15         if ( n <= old )
16             return;
17         fac.resize(n + 1);
18         ifac.resize(n + 1);
19         for ( int i = old + 1; i <= n; i++ )
20             fac[i] = fac[i - 1] * i;
21         ifac[n] = fac[n].Inv();
22         for ( int i = n; i > old; i-- )
23             ifac[i - 1] = ifac[i] * i;
24     }
25
26     Z Choose(int n, int k) const
27     {
28         if ( k < 0 || k > n )
29             return 0;
30         assert(n < (int)fac.size());
31         return fac[n] * ifac[k] * ifac[n - k];
32     }
33
34     Z Permute(int n, int k) const
35     {
36         if ( k < 0 || k > n )
37             return 0;
38         assert(n < (int)fac.size());
39         return fac[n] * ifac[n - k];
40     }
41 };

```


第八十九章 批量模逆元

89.1 任意数组的批量逆元

算法约束见动态版本第 94 页。本节代码独立可抄；采用下列实际接口名。

```
1 inline optional<vector<long long>> Batch_Inverse(vector<long long> a, long long mod)
2 {
3     assert(mod > 0);
4     int n = a.size();
5     vector<long long> prefix(n + 1, 1 % mod);
6     for ( int i = 0; i < n; i++ )
7     {
8         a[i] %= mod;
9         if ( a[i] < 0 )
10             a[i] += mod;
11         prefix[i + 1] = __int128_t(prefix[i]) * a[i] % mod;
12     }
13     long long inverse = Number_Theory::Inverse(prefix[n], mod);
14     if ( inverse == -1 )
15         return nullopt;
16     for ( int i = n - 1; i >= 0; i-- )
17     {
18         long long value = a[i];
19         a[i] = __int128_t(inverse) * prefix[i] % mod;
20         inverse = __int128_t(inverse) * value % mod;
21     }
22     return a;
23 }
```

第九十章 混合进制中国剩余定理

90.1 Garner：大周期 CRT 与混合进制

算法约束见动态版本第 95 页。本节代码独立可抄；采用下列实际接口名。

```
1 // Pairwise coprime positive moduli. Digits use the input order.
2 inline vector<long long> Garner_Digits(const vector<long long> &b,
3                                       const vector<long long> &moduli)
4 {
5     using ll = long long;
6     using i128 = __int128_t;
7     int n = b.size();
8     assert(moduli.size() == b.size());
9     vector<ll> value(n), coefficient(n, 1), digit(n);
10    for ( ll m : moduli )
11        assert(m > 0);
12    for ( int i = 0; i < n; i++ )
13    {
14        ll m = moduli[i];
15        ll inverse = Number_Theory::Inverse(coefficient[i], m);
16        if ( inverse == -1 )
17            throw invalid_argument("Garner moduli are not coprime");
18        i128 delta = (i128(b[i]) - value[i]) % m;
19        if ( delta < 0 )
20            delta += m;
21        digit[i] = delta * inverse % m;
22        for ( int j = i + 1; j < n; j++ )
23        {
24            value[j] = (value[j] + i128(coefficient[j]) * digit[i]) % moduli[j];
25            coefficient[j] = i128(coefficient[j]) * m % moduli[j];
26        }
27    }
28    return digit;
29 }
30
31 // Return the least nonnegative CRT solution modulo target, not all solutions.
32 inline long long
33 Garner(const vector<long long> &b, const vector<long long> &moduli, long long target)
34 {
35     assert(target > 0);
36     auto digit = Garner_Digits(b, moduli);
37     long long answer = 0;
38     for ( int i = (int)digit.size() - 1; i >= 0; i-- )
39         answer = (__int128_t(answer) * moduli[i] + digit[i]) % target;
40     return answer;
41 }
```

第九十一章 乘法阶与原根

91.1 乘法阶与原根

算法约束见动态版本第 97 页。本节代码独立可抄；采用下列实际接口名。

构造 `Primitive_Root(n)`, `Init`、`Order`、`Is_Root`、`Minimum`、`All` 对应动态接口；`Factors` 为 ϕ 的不同素因子，依赖 `Number_Theory::Power`。

```
1 struct Primitive_Root
2 {
3     int mod, phi;
4     bool exists;
5     vector<int> Factors;
6
7     static vector<int> Factor(int n)
8     {
9         vector<int> result;
10        for ( int p = 2; p <= n / p; p++ )
11            if ( n % p == 0 )
12            {
13                result.push_back(p);
14                while ( n % p == 0 )
15                    n /= p;
16            }
17        if ( n > 1 )
18            result.push_back(n);
19        return result;
20    }
21
22    Primitive_Root(int n)
23    {
24        Init(n);
25    }
```

```
27 void Init(int n)
28 {
29     assert(n >= 2);
30     mod = n;
31     phi = n;
32     for ( int p : Factor(n) )
33         phi -= phi / p;
34     int odd = n;
35     if ( odd % 2 == 0 )
36         odd /= 2;
37     exists = n == 2 || n == 4 || (odd % 2 && Factor(odd).size() == 1);
38     Factors = Factor(phi);
39 }
40
41 int Order(long long a) const
42 {
43     a %= mod;
44     if ( a < 0 )
45         a += mod;
46     if ( gcd(a, (long long)mod) != 1 )
47         return 0;
48     int result = phi;
49     for ( int p : Factors )
50         while ( result % p == 0 && Number_Theory::Power(a, result / p, mod) == 1 )
51             result /= p;
52     return result;
53 }
```

```

55     bool Is_Root(long long a) const
56     {
57         if ( !exists )
58             return false;
59         a %= mod;
60         if ( a < 0 )
61             a += mod;
62         if ( gcd(a, (long long)mod) != 1 )
63             return false;
64         for ( int p : Factors )
65             if ( Number_Theory::Power(a, phi / p, mod) == 1 )
66                 return false;
67         return true;
68     }
69
70     int Minimum() const
71     {
72         if ( !exists )
73             return -1;
74         for ( int g = 1; g < mod; g++ )
75             if ( Is_Root(g) )
76                 return g;
77         return -1;
78     }
79
80     vector<int> All() const
81     {
82         int g = Minimum();
83         vector<int> roots;
84         if ( g == -1 )
85             return roots;
86         long long value = 1;
87         for ( int k = 1; k <= phi; k++ )
88         {
89             value = value * g % mod;
90             if ( gcd(k, phi) == 1 )
91                 roots.push_back(value);
92         }
93         sort(roots.begin(), roots.end());
94         return roots;
95     }
96 };

```

第九十二章 Carmichael 函数

92.1 Carmichael: 模乘法群的公共指数

算法约束见动态版本第 100 页。本节代码独立可抄；采用下列实际接口名。

```
1 inline int Carmichael(int n)
2 {
3     assert(n >= 1);
4     long long answer = 1;
5     for ( int p = 2; p <= n / p; p++ )
6         if ( n % p == 0 )
7         {
8             int power = 1;
9             while ( n % p == 0 )
10             {
11                 n /= p;
12                 power *= p;
13             }
14             int value = power / p * (p - 1);
15             if ( p == 2 && power >= 8 )
16                 value /= 2;
17             answer = answer / gcd(answer, (long long)value) * value;
18         }
19     if ( n > 1 )
20         answer = answer / gcd(answer, (long long)(n - 1)) * (n - 1);
21     return answer;
22 }
```

第九十三章 素数模数组合数

93.1 Lucas：小素数模的大下标组合数

算法约束见动态版本第 101 页。本节代码独立可抄；采用下列实际接口名。

声明 Lucas_Theorem<N>, Init(p) 重建素数模数下的表, Choose(n,k) 查询。要求 p 不超过 N；大对象放全局或 static。传统版空间 $O(N)$ 。

```
1 template <int N> struct Lucas_Theorem
2 {
3     int mod;
4     int fac[N + 1], ifac[N + 1];
5
6     void Init(int p)
7     {
8         assert(2 <= p && p <= N);
9         mod = p;
10        fac[0] = 1;
11        for ( int i = 1; i < p; i++ )
12            fac[i] = 1LL * fac[i - 1] * i % p;
13        long long a = fac[p - 1], r = 1;
14        for ( int e = p - 2; e; e >>= 1 )
15        {
16            if ( e & 1 )
17                r = r * a % p;
18            a = a * a % p;
19        }
20        ifac[p - 1] = r;
21        for ( int i = p - 1; i; i-- )
22            ifac[i - 1] = 1LL * ifac[i] * i % p;
23    }
24
25    int Choose(unsigned long long n, unsigned long long k) const
26    {
27        if ( k > n )
28            return 0;
29        long long answer = 1;
30        while ( n || k )
31        {
32            int a = n % mod, b = k % mod;
33            if ( b > a )
34                return 0;
35            answer = answer * fac[a] % mod;
36            answer = answer * ifac[b] % mod;
37            answer = answer * ifac[a - b] % mod;
38            n /= mod;
39            k /= mod;
40        }
41        return answer;
42    }
43 };
```

第九十四章 合数模数组合数

94.1 合数模数组合数（扩展 Lucas）

算法约束见动态版本第 103 页。本节代码独立可抄；采用下列实际接口名。

声明 `Ex_Lucas<N>` 后 `Init(mod)`，要求 `mod` 不超过 `N`；`Choose` 为查询接口。`fac` 是共用静态前缀积数组，各素数幂通过 `start` 分段；各段总长度不超过 `mod+1`，所以 `N` 限制模数即可。`Init` 可切换模数并清空旧预处理，大对象放在全局或 `static`。

```
1 template <int N> struct Ex_Lucas
2 {
3     using ull = unsigned long long;
4     using ll = long long;
5
6     struct Factor
7     {
8         int p, q, exponent, start, crt;
9     };
10
11     int mod;
12     int fac[N + 1], used;
13     vector<Factor> factors;
14
15     static ll Power(ll a, ull b, int m)
16     {
17         ll result = 1;
18         while ( b )
19         {
20             if ( b & 1 )
21                 result = result * a % m;
22             a = a * a % m;
23             b >>= 1;
24         }
25         return result;
26     }
27
28     void Init(int m)
29     {
30         assert(1 <= m && m <= N);
31         mod = m;
32         used = 0;
33         factors.clear();
34         int x = m;
35         for ( int p = 2; p <= x / p; p++ )
36         {
37             if ( x % p )
38                 continue;
39             int q = 1, e = 0;
40             do
41             {
42                 x /= p;
43                 q *= p;
```



```

44         e++;
45     } while ( x % p == 0 );
46     Add_Factor(p, q, e);
47 }
48 if ( x > 1 )
49     Add_Factor(x, x, 1);
50 }
51
52 void Add_Factor(int p, int q, int e)
53 {
54     int start = used;
55     fac[used++] = 1;
56     for ( int i = 1; i < q; i++ )
57     {
58         fac[used] = ll(fac[used - 1]) * (i % p ? i : 1) % q;
59         used++;
60     }
61     fac[used] = fac[used - 1];
62     used++;
63     int phi = q - q / p;
64     int crt = ll(mod / q) * Power(mod / q % q, phi - 1, q) % mod;
65     factors.push_back({p, q, e, start, crt});
66 }
67
68 static ull Valuation(ull n, int p)
69 {
70     ull result = 0;
71     while ( n )
72     {
73         n /= p;
74         result += n;
75     }
76     return result;
77 }
78
79 ll Unit(ull n, const Factor &f) const
80 {
81     ll result = 1;
82     while ( n )
83     {
84         result = result * Power(fac[f.start + f.q], n / f.q, f.q) % f.q;
85         result = result * fac[f.start + n % f.q] % f.q;
86         n /= f.p;
87     }
88     return result;
89 }

```

```
91     int Choose(ull n, ull k) const
92     {
93         if ( k > n || mod == 1 )
94             return 0;
95         ll answer = 0;
96         for ( const auto &f : factors )
97         {
98             ull e = Valuation(n, f.p) - Valuation(k, f.p) - Valuation(n - k, f.p);
99             if ( e >= ull(f.exponent) )
100                 continue;
101             ll denominator = Unit(k, f) * Unit(n - k, f) % f.q;
102             ll residue =
103                 Unit(n, f) * Power(denominator, f.q - f.q / f.p - 1, f.q) % f.q;
104             residue = residue * Power(f.p, e, f.q) % f.q;
105             answer = (answer + residue * f.crt) % mod;
106         }
107         return answer;
108     }
109 };
```

第九十五章 二次剩余

95.1 素数模平方根 (Cipolla)

算法约束见动态版本第 106 页。本节代码独立可抄；采用下列实际接口名。

```
1 inline vector<long long> Mod_Sqrt(long long n, int p)
2 {
3     using ll = long long;
4     assert(p >= 2);
5     n = (n % p + p) % p;
6     if ( n == 0 || p == 2 )
7         return {n};
8     auto Power = [p](ll a, ll b)
9     {
10         ll result = 1;
11         while ( b )
12         {
13             if ( b & 1 )
14                 result = result * a % p;
15             a = a * a % p;
16             b >>= 1;
17         }
18         return result;
19     };
20     if ( Power(n, (p - 1) / 2) != 1 )
21         return {};
22     ll answer;
23     if ( p % 4 == 3 )
24         answer = Power(n, (ll(p) + 1) / 4);
25     else
26     {
27         static mt19937_64 rng(712367);
28         ll a, w;
29         do
30         {
31             a = rng() % p;
32             w = (a * a % p - n + p) % p;
33         } while ( Power(w, (p - 1) / 2) != p - 1 );
34         using P = pair<ll, ll>;
35         auto Mul = [p, w](P x, P y)
36         {
37             ll real = (x.first * y.first + x.second * y.second % p * w) % p;
38             ll imag = (x.first * y.second + x.second * y.first) % p;
39             return P{real, imag};
40         };
41         P result{1, 0}, base{a, 1};
42         ll exponent = (ll(p) + 1) / 2;
43         while ( exponent )
44         {
45             if ( exponent & 1 )
46                 result = Mul(result, base);
```

```
47         base = Mul(base, base);
48         exponent >>= 1;
49     }
50     answer = result.first;
51 }
52 ll other = p - answer;
53 if ( answer > other )
54     swap(answer, other);
55 return {answer, other};
56 }
```

第九十六章 多项式插值

96.1 拉格朗日插值与连续点优化

算法约束见动态版本第 108 页。本节代码独立可抄；采用下列实际接口名。

声明 Lagrange_Interpolation<N>; Init、Consecutive、Query 对应动态接口，N 为最大样本数。坐标、权值与工作数组均为静态数组，大对象放在全局或 static；Query 会改写工作数组。

```
1 template <int N> struct Lagrange_Interpolation
2 {
3     using ll = long long;
4     int n, mod;
5     ll x[N], w[N], invfac[N], pre[N + 1];
6
7     ll Norm(ll a) const
8     {
9         a %= mod;
10        if ( a < 0 )
11            a += mod;
12        return a;
13    }
14
15    ll Power(ll a, int b) const
16    {
17        ll result = 1;
18        while ( b )
19        {
20            if ( b & 1 )
21                result = result * a % mod;
22            a = a * a % mod;
23            b >>= 1;
24        }
25        return result;
26    }
27
28    void Init(const vector<ll> &xs, const vector<ll> &ys, int p)
29    {
30        n = ys.size();
31        mod = p;
32        assert(p >= 2 && n <= p && xs.size() == ys.size());
33        assert(n <= N);
34        for ( int i = 0; i < n; i++ )
35            x[i] = Norm(xs[i]);
36        for ( int i = 0; i < n; i++ )
37        {
38            ll denominator = 1;
39            for ( int j = 0; j < n; j++ )
40                if ( i != j )
41                    denominator = denominator * Norm(x[i] - x[j]) % mod;
42            assert(denominator != 0);
43            w[i] = Norm(ys[i]) * Power(denominator, mod - 2) % mod;
44        }
```

```

45     }
46
47     void Consecutive(const vector<ll> &ys, int p)
48     {
49         n = ys.size();
50         mod = p;
51         assert(p >= 2 && n <= p);
52         assert(n <= N);
53         if ( !n )
54             return;
55
56         ll factorial = 1;
57         for ( int i = 1; i < n; i++ )
58             factorial = factorial * i % mod;
59         invfac[n - 1] = Power(factorial, mod - 2);
60         for ( int i = n - 1; i > 0; i-- )
61             invfac[i - 1] = invfac[i] * i % mod;
62         for ( int i = 0; i < n; i++ )
63         {
64             x[i] = i;
65             w[i] = Norm(ys[i]) * invfac[i] % mod * invfac[n - 1 - i] % mod;
66             if ( (n - 1 - i) & 1 )
67                 w[i] = Norm(-w[i]);
68         }
69     }
70
71     ll Query(ll k)
72     {
73         k = Norm(k);
74         pre[0] = 1;
75         for ( int i = 0; i < n; i++ )
76             pre[i + 1] = pre[i] * Norm(k - x[i]) % mod;
77         ll answer = 0, suffix = 1;
78         for ( int i = n - 1; i >= 0; i-- )
79         {
80             answer = (answer + w[i] * pre[i] % mod * suffix) % mod;
81             suffix = suffix * Norm(k - x[i]) % mod;
82         }
83         return answer;
84     }
85 };

```

第九十七章 多项式

97.1 NTT、形式幂级数、FWT 与递推

算法约束见动态版本第 110 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Polynomial
2 {
3     static constexpr int mod = 998244353;
4     using Z = Mod_Int<mod>;
5     using Poly = vector<Z>;
6
7     static void Ntt(Poly &a, bool invert = false)
8     {
9         int n = (int)a.size();
10        assert(n > 0 && (n & (n - 1)) == 0 && n <= (1 << 23));
11        for ( int i = 1, j = 0; i < n; i++ )
12            {
13                int bit = n >> 1;
14                for ( ; j & bit; bit >>= 1 )
15                    j ^= bit;
16                j ^= bit;
17                if ( i < j )
18                    swap(a[i], a[j]);
19            }
20        for ( int len = 2; len <= n; len *= 2 )
21            {
22                Z step = Z(3).Pow((mod - 1) / len);
23                if ( invert )
24                    step = step.Inv();
25                for ( int i = 0; i < n; i += len )
26                    {
27                        Z w = 1;
28                        for ( int j = 0; j < len / 2; j++, w = w * step )
29                            {
30                                Z u = a[i + j], v = a[i + j + len / 2] * w;
31                                a[i + j] = u + v;
32                                a[i + j + len / 2] = u - v;
33                            }
34                    }
35            }
36        if ( invert )
37            {
38                Z Inv = Z(n).Inv();
39                for ( auto &x : a )
40                    x = x * Inv;
41            }
42    }
43
44    static Poly Multiply(Poly a, Poly b)
45    {
46        if ( a.empty() || b.empty() )
```

```

47     return {};
48     int size = (int)a.size() + (int)b.size() - 1, n = 1;
49     assert(size <= (1 << 23));
50     while ( n < size )
51         n *= 2;
52     a.resize(n);
53     b.resize(n);
54     Ntt(a);
55     Ntt(b);
56     for ( int i = 0; i < n; i++ )
57         a[i] = a[i] * b[i];
58     Ntt(a, true);
59     a.resize(size);
60     return a;
61 }
62
63 static Poly Inverse(const Poly &a, int n)
64 {
65     assert(n > 0 && !a.empty() && a[0].v);
66     Poly b{a[0].Inv()};
67     while ( (int)b.size() < n )
68     {
69         int m = min(n, 2 * (int)b.size());
70         Poly f(a.begin(), a.begin() + min(m, (int)a.size()));
71         auto t = Multiply(f, b);
72         t.resize(m);
73         for ( auto &x : t )
74             x = Z(0) - x;
75         t[0] = t[0] + 2;
76         b = Multiply(b, t);
77         b.resize(m);
78     }
79     return b;
80 }
81
82 static Poly Derivative(const Poly &a)
83 {
84     Poly b;
85     for ( int i = 1; i < (int)a.size(); i++ )
86         b.push_back(a[i] * i);
87     return b;
88 }
89
90 static Poly Integral(const Poly &a)
91 {
92     Poly b(a.size() + 1);
93     vector<Z> Inv(a.size() + 1);
94     if ( !a.empty() )
95         Inv[1] = 1;
96     for ( int i = 2; i <= (int)a.size(); i++ )
97         Inv[i] = Z(mod - mod / i) * Inv[mod % i];
98     for ( int i = 0; i < (int)a.size(); i++ )
99         b[i + 1] = a[i] * Inv[i + 1];
100     return b;
101 }
102

```



```

103 static Poly Log(const Poly &a, int n)
104 {
105     assert(n > 0 && !a.empty() && a[0].v == 1);
106     auto b = Multiply(Derivative(a), Inverse(a, n));
107     b.resize(n - 1);
108     return Integral(b);
109 }
110
111 static Poly Exp(const Poly &a, int n)
112 {
113     assert(n > 0 && (a.empty() || a[0].v == 0));
114     Poly b{1};
115     while ( (int)b.size() < n )
116     {
117         int m = min(n, 2 * (int)b.size());
118         auto t = Log(b, m);
119         for ( int i = 0; i < m; i++ )
120             t[i] = (i < (int)a.size() ? a[i] : Z(0)) - t[i];
121         t[0] = t[0] + 1;
122         b = Multiply(b, t);
123         b.resize(m);
124     }
125     return b;
126 }
127
128 static void Fwt(Poly &a, char op, bool invert = false)
129 {
130     int n = (int)a.size();
131     assert(n > 0 && (n & (n - 1)) == 0);
132     const Z inv2 = (mod + 1) / 2;
133     for ( int len = 1; len < n; len *= 2 )
134         for ( int i = 0; i < n; i += len * 2 )
135             for ( int j = 0; j < len; j++ )
136             {
137                 Z &u = a[i + j], &v = a[i + j + len], x = u, y = v;
138                 if ( op == '^' )
139                 {
140                     u = x + y;
141                     v = x - y;
142                     if ( invert )
143                     {
144                         u = u * inv2;
145                         v = v * inv2;
146                     }
147                 }
148                 else if ( op == '|' )
149                     v = invert ? y - x : y + x;
150                 else
151                 {
152                     assert(op == '&');
153                     u = invert ? x - y : x + y;
154                 }
155             }
156 }
157
158 // c: s[n]=sum c[i-1]*s[n-i], i=1..L.

```

```

159 static Poly Berlekamp_Massey(const Poly &s)
160 {
161     Poly c{1}, b{1};
162     int l = 0, m = 1;
163     Z last = 1;
164     for ( int n = 0; n < (int)s.size(); n++ )
165     {
166         Z d = s[n];
167         for ( int i = 1; i <= l; i++ )
168             d = d + c[i] * s[n - i];
169         if ( !d.v )
170         {
171             ++m;
172             continue;
173         }
174         auto old = c;
175         Z coef = d / last;
176         if ( c.size() < b.size() + m )
177             c.resize(b.size() + m);
178         for ( int j = 0; j < (int)b.size(); j++ )
179             c[j + m] = c[j + m] - coef * b[j];
180         if ( 2 * l <= n )
181         {
182             l = n + 1 - l;
183             b = old;
184             last = d;
185             m = 1;
186         }
187         else
188             ++m;
189     }
190     c.erase(c.begin());
191     for ( auto &x : c )
192         x = Z(0) - x;
193     return c;
194 }
195
196 static Z Recurrence(Poly init, const Poly &c, unsigned long long n)
197 {
198     int k = (int)c.size();
199     if ( n < init.size() )
200         return init[n];
201     if ( !k )
202         return 0;
203     assert((int)init.size() >= k);
204     auto combine = [&](const Poly &a, const Poly &b)
205     {
206         Poly t(2 * k - 1);
207         for ( int i = 0; i < k; i++ )
208             for ( int j = 0; j < k; j++ )
209                 t[i + j] = t[i + j] + a[i] * b[j];
210         for ( int i = 2 * k - 2; i >= k; i-- )
211             for ( int j = 1; j <= k; j++ )
212                 t[i - j] = t[i - j] + t[i] * c[j - 1];
213         t.resize(k);
214         return t;

```

```
215     };
216     Poly a(k), r(k);
217     r[0] = 1;
218     if ( k == 1 )
219         a[0] = c[0];
220     else
221         a[1] = 1;
222     for ( ; n; n >>= 1, a = combine(a, a) )
223         if ( n & 1 )
224             r = combine(r, a);
225     Z ans = 0;
226     for ( int i = 0; i < k; i++ )
227         ans = ans + r[i] * init[i];
228     return ans;
229 }
230 };
```

第九十八章 代数与数论进阶

98.1 模高斯消元、行列式、矩阵树定理

算法约束见动态版本第 115 页。本节代码独立可抄；采用下列实际接口名。

```
1 template <int mod> struct Linear_Algebra
2 {
3     using Z = Mod_Int<mod>;
4     using Matrix = vector<vector<Z>>>;
5
6     struct Solution
7     {
8         bool consistent;
9         int rank;
10        vector<Z> particular;
11        Matrix kernel;
12    };
13
14    // Augmented m x (n+1), prime modulus. Empty system needs explicit n.
15    static Solution Solve(Matrix a, int n)
16    {
17        int m = (int)a.size(), row = 0;
18        vector<int> where(n, -1);
19        for ( const auto &v : a )
20            assert((int)v.size() == n + 1);
21        for ( int col = 0; col < n && row < m; col++ )
22        {
23            int p = row;
24            while ( p < m && !a[p][col].v )
25                ++p;
26            if ( p == m )
27                continue;
28            swap(a[p], a[row]);
29            Z Inv = a[row][col].Inv();
30            for ( int j = col; j <= n; j++ )
31                a[row][j] = a[row][j] * Inv;
32            for ( int i = 0; i < m; i++ )
33                if ( i != row && a[i][col].v )
34                {
35                    Z f = a[i][col];
36                    for ( int j = col; j <= n; j++ )
37                        a[i][j] = a[i][j] - f * a[row][j];
38                }
39            where[col] = row++;
40        }
41        for ( int i = row; i < m; i++ )
42            if ( a[i][n].v )
43                return {false, row, {}, {}};
44        Solution ans{true, row, vector<Z>(n), {}};
45        for ( int j = 0; j < n; j++ )
46            if ( where[j] != -1 )
```

```

47         ans.particular[j] = a[where[j]][n];
48     for ( int j = 0; j < n; j++ )
49         if ( where[j] == -1 )
50         {
51             vector<Z> v(n);
52             v[j] = 1;
53             for ( int k = 0; k < n; k++ )
54                 if ( where[k] != -1 )
55                     v[k] = Z(0) - a[where[k]][j];
56             ans.kernel.push_back(v);
57         }
58     return ans;
59 }
60
61 static Z Determinant(Matrix a)
62 {
63     int n = (int)a.size();
64     Z ans = 1;
65     for ( int i = 0; i < n; i++ )
66     {
67         assert((int)a[i].size() == n);
68         int p = i;
69         while ( p < n && !a[p][i].v )
70             ++p;
71         if ( p == n )
72             return 0;
73         if ( p != i )
74         {
75             swap(a[p], a[i]);
76             ans = Z(0) - ans;
77         }
78         ans = ans * a[i][i];
79         Z Inv = a[i][i].Inv();
80         for ( int j = i + 1; j < n; j++ )
81         {
82             Z f = a[j][i] * Inv;
83             for ( int k = i; k < n; k++ )
84                 a[j][k] = a[j][k] - f * a[i][k];
85         }
86     }
87     return ans;
88 }
89
90 static Matrix Multiply(const Matrix &a, const Matrix &b)
91 {
92     assert(!a.empty() && !b.empty());
93     int n = a.size(), m = b[0].size(), k = b.size();
94     assert((int)a[0].size() == k);
95     Matrix c(n, vector<Z>(m));
96     for ( int i = 0; i < n; i++ )
97         for ( int t = 0; t < k; t++ )
98             for ( int j = 0; j < m; j++ )
99                 c[i][j] = c[i][j] + a[i][t] * b[t][j];
100     return c;
101 }
102

```

```

103 static Matrix Power(Matrix a, unsigned long long e)
104 {
105     int n = a.size();
106     assert(n > 0 && (int)a[0].size() == n);
107     Matrix r(n, vector<Z>(n));
108     for ( int i = 0; i < n; i++ )
109         r[i][i] = 1;
110     for ( ; e; e >>= 1, a = Multiply(a, a) )
111         if ( e & 1 )
112             r = Multiply(r, a);
113     return r;
114 }
115
116 // Undirected multigraph, vertices 0..n-1; loops ignored.
117 static Z Spanning_Trees(int n, const vector<pair<int, int>> &edges)
118 {
119     assert(n > 0);
120     Matrix lap(n, vector<Z>(n));
121     for ( auto [u, v] : edges )
122         if ( u != v )
123         {
124             lap[u][u] = lap[u][u] + 1;
125             lap[v][v] = lap[v][v] + 1;
126             lap[u][v] = lap[u][v] - 1;
127             lap[v][u] = lap[v][u] - 1;
128         }
129     lap.pop_back();
130     for ( auto &row : lap )
131         row.pop_back();
132     return Determinant(lap);
133 }
134 };

```

98.2 杜教筛

算法约束见动态版本第 118 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Du_Jiao
2 {
3     using ll = long long;
4     using I = __int128_t;
5     int limit;
6     vector<ll> pmu, pphi;
7     unordered_map<ll, ll> mmu;
8     unordered_map<ll, I> mphi;
9
10    Du_Jiao(int limit) : limit(limit), pmu(limit + 1), pphi(limit + 1)
11    {
12        assert(limit >= 1);
13        Linear_Sieve s(limit);
14        for ( int i = 1; i <= limit; i++ )
15        {
16            pmu[i] = pmu[i - 1] + s.mu[i];
17            pphi[i] = pphi[i - 1] + s.phi[i];
18        }
19    }
20
21    ll Mertens(ll n)
22    {
23        if ( n <= limit )
24            return pmu[n];
25        if ( mmu.count(n) )
26            return mmu[n];
27        ll ans = 1;
28        for ( ll l = 2, r; l <= n; l = r + 1 )
29        {
30            r = n / (n / l);
31            ans -= (r - l + 1) * Mertens(n / l);
32        }
33        return mmu[n] = ans;
34    }
35
36    I Totient_Sum(ll n)
37    {
38        if ( n <= limit )
39            return pphi[n];
40        if ( mphi.count(n) )
41            return mphi[n];
42        I ans = I(n) * (n + 1) / 2;
43        for ( ll l = 2, r; l <= n; l = r + 1 )
44        {
45            r = n / (n / l);
46            ans -= I(r - l + 1) * Totient_Sum(n / l);
47        }
48        return mphi[n] = ans;
49    }
50 };

```

98.3 扩展 BSGS

算法约束见动态版本第 119 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Discrete_Log
2 {
3     using ll = long long;
4
5     // 1<=m<=1e12; Expected O(sqrt(m)) time, O(sqrt(m)) storage. Smallest x>=0, or -1.
6     static ll Solve(ll a, ll b, ll m)
7     {
8         assert(m >= 1 && m <= 1000000000000LL);
9         a = (a % m + m) % m;
10        b = (b % m + m) % m;
11        if ( m == 1 || b == 1 )
12            return 0;
13        if ( a == 1 )
14            return -1;
15        ll offset = 0, k = 1;
16        for ( ll g; (g = gcd(a, m)) > 1; )
17        {
18            if ( b == k )
19                return offset;
20            if ( b % g )
21                return -1;
22            b /= g;
23            m /= g;
24            k = (__int128)k * (a / g) % m;
25            ++offset;
26        }
27        ll target = (__int128)b * Number_Theory::Inverse(k, m) % m;
28        ll step = sqrtl(m) + 1;
29        unordered_map<ll, ll> baby;
30        baby.reserve(step);
31        ll cur = 1 % m;
32        for ( ll j = 0; j < step; j++ )
33        {
34            if ( !baby.count(cur) )
35                baby[cur] = j;
36            cur = (__int128)cur * a % m;
37        }
38        ll Inv = Number_Theory::Inverse(Number_Theory::Power(a, step, m), m);
39        cur = target;
40        for ( ll i = 0; i <= step; i++ )
41        {
42            auto it = baby.find(cur);
43            if ( it != baby.end() )
44                return offset + i * step + it->second;
45            cur = (__int128)cur * Inv % m;
46        }
47        return -1;
48    }
49 };

```


第九十九章 计算几何

99.1 整数精确二维几何

算法约束见动态版本第 121 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Integer_Geometry
2 {
3     using ll = long long;
4     using I = __int128_t;
5
6     // Contract: |coordinate|<=1e12. Differences/products widened before arithmetic.
7     struct Point
8     {
9         ll x = 0, y = 0;
10
11         bool operator<(Point b) const
12         {
13             return tie(x, y) < tie(b.x, b.y);
14         }
15
16         bool operator==(Point b) const
17         {
18             return x == b.x && y == b.y;
19         }
20     };
21
22     static I Cross(Point a, Point b, Point c)
23     {
24         return (I(b.x) - a.x) * (I(c.y) - a.y) - (I(b.y) - a.y) * (I(c.x) - a.x);
25     }
26
27     static I Dot(Point a, Point b, Point c)
28     {
29         return (I(b.x) - a.x) * (I(c.x) - a.x) + (I(b.y) - a.y) * (I(c.y) - a.y);
30     }
31
32     static I Distance_Squared(Point a, Point b)
33     {
34         return Dot(a, b, b);
35     }
36
37     static int Sign(I x)
38     {
39         return (x > 0) - (x < 0);
40     }
41
42     static bool On_Segment(Point p, Point a, Point b)
43     {
44         return Cross(a, b, p) == 0 && Dot(p, a, b) <= 0;
45     }
46 }
```

```

47 static bool Intersect(Point a, Point b, Point c, Point d)
48 {
49     int x = Sign(Cross(a, b, c)), y = Sign(Cross(a, b, d));
50     int Z_Function = Sign(Cross(c, d, a)), w = Sign(Cross(c, d, b));
51     if ( !x && On_Segment(c, a, b) )
52         return true;
53     if ( !y && On_Segment(d, a, b) )
54         return true;
55     if ( !Z_Function && On_Segment(a, c, d) )
56         return true;
57     if ( !w && On_Segment(b, c, d) )
58         return true;
59     return x * y < 0 && Z_Function * w < 0;
60 }
61
62 static vector<Point> Convex_Hull(vector<Point> p)
63 {
64     sort(p.begin(), p.end());
65     p.erase(unique(p.begin(), p.end()), p.end());
66     int n = (int)p.size();
67     if ( n < 3 )
68         return p;
69     vector<Point> h;
70     for ( Point v : p )
71     {
72         while ( h.size() > 1 && Cross(h[h.size() - 2], h.back(), v) <= 0 )
73             h.pop_back();
74         h.push_back(v);
75     }
76     int low = (int)h.size();
77     for ( int i = n - 2; i >= 0; i-- )
78     {
79         while ( (int)h.size() > low && Cross(h[h.size() - 2], h.back(), p[i]) <= 0 )
80             h.pop_back();
81         h.push_back(p[i]);
82     }
83     h.pop_back();
84     return h; // CCW, no repeated endpoint, no collinear interior points.
85 }
86
87 static I Area_Twice(const vector<Point> &p)
88 {
89     I ans = 0;
90     int n = (int)p.size();
91     for ( int i = 0; i < n; i++ )
92         ans += I(p[i].x) * p[(i + 1) % n].y - I(p[i].y) * p[(i + 1) % n].x;
93     return ans;
94 }
95
96 // 0 outside, 1 boundary, 2 inside. Simple polygon; either orientation.
97 static int Contains(const vector<Point> &p, Point q)
98 {
99     int winding = 0, n = (int)p.size();
100     for ( int i = 0; i < n; i++ )
101     {
102         Point a = p[i], b = p[(i + 1) % n];

```

```

103         if ( On_Segment(q, a, b) )
104             return 1;
105         I c = Cross(a, b, q);
106         if ( a.y <= q.y && b.y > q.y && c > 0 )
107             ++winding;
108         if ( a.y > q.y && b.y <= q.y && c < 0 )
109             --winding;
110     }
111     return winding ? 2 : 0;
112 }
113
114 // Input must be the strict CCW hull above. O(log n).
115 static int Convex_Contains(const vector<Point> &p, Point q)
116 {
117     int n = (int)p.size();
118     if ( n < 3 )
119         return Contains(p, q);
120     I a = Cross(p[0], p[1], q), b = Cross(p[0], p[n - 1], q);
121     if ( a < 0 || b > 0 )
122         return 0;
123     if ( !a )
124         return On_Segment(q, p[0], p[1]) ? 1 : 0;
125     if ( !b )
126         return On_Segment(q, p[0], p[n - 1]) ? 1 : 0;
127     int l = 1, r = n - 1;
128     while ( l + 1 < r )
129     {
130         int m = (l + r) / 2;
131         if ( Cross(p[0], p[m], q) >= 0 )
132             l = m;
133         else
134             r = m;
135     }
136     I c = Cross(p[l], p[r], q);
137     return c < 0 ? 0 : c == 0 ? 1 : 2;
138 }
139
140 static I Diameter_Squared(const vector<Point> &p)
141 {
142     int n = (int)p.size();
143     if ( n < 2 )
144         return 0;
145     if ( n == 2 )
146         return Distance_Squared(p[0], p[1]);
147     I ans = 0;
148     int j = 1;
149     for ( int i = 0; i < n; i++ )
150     {
151         int k = (i + 1) % n;
152         while ( Cross(p[i], p[k], p[(j + 1) % n]) > Cross(p[i], p[k], p[j]) )
153             j = (j + 1) % n;
154         ans =
155             max({ans, Distance_Squared(p[i], p[j]), Distance_Squared(p[k], p[j])});
156     }
157     return ans;
158 }

```

```

159
160 struct PolarLess
161 {
162     static int half(Point p)
163     {
164         return p.y < 0 || (p.y == 0 && p.x < 0);
165     }
166
167     bool operator()(Point a, Point b) const
168     {
169         if ( half(a) != half(b) )
170             return half(a) < half(b);
171         I c = Cross({0, 0}, a, b);
172         if ( c )
173             return c > 0;
174         return Distance_Squared({0, 0}, a) < Distance_Squared({0, 0}, b);
175     }
176 };
177 };

```

99.2 浮点点线圆构造

算法约束见动态版本第 124 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Real_Geometry
2 {
3     using R = long double;
4     // Tolerance is explicit, not used in sorting. Requires finite, well-scaled input.
5     R eps;
6
7     Real_Geometry(R eps = 1e-12L) : eps(eps)
8     {
9         assert(eps > 0);
10    }
11
12    struct Point
13    {
14        R x = 0, y = 0;
15
16        Point operator+(Point b) const
17        {
18            return {x + b.x, y + b.y};
19        }
20
21        Point operator-(Point b) const
22        {
23            return {x - b.x, y - b.y};
24        }
25
26        Point operator*(R k) const
27        {
28            return {x * k, y * k};
29        }
30
31        Point operator/(R k) const

```

```

32     {
33         return {x / k, y / k};
34     }
35 };
36
37 struct Circle
38 {
39     Point o;
40     R r;
41 };
42 enum class Kind
43 {
44     none,
45     one,
46     two,
47     infinite,
48     degenerate
49 };
50
51 struct Result
52 {
53     Kind kind;
54     vector<Point> p;
55 };
56
57 static R Dot(Point a, Point b)
58 {
59     return a.x * b.x + a.y * b.y;
60 }
61
62 static R Cross(Point a, Point b)
63 {
64     return a.x * b.y - a.y * b.x;
65 }
66
67 static R norm(Point a)
68 {
69     return hypotl(a.x, a.y);
70 }
71
72 static Point perp(Point a)
73 {
74     return {-a.y, a.x};
75 }
76
77 Point Projection(Point p, Point a, Point b) const
78 {
79     Point v = b - a;
80     R d = Dot(v, v);
81     if ( d == 0 )
82         return a;
83     return a + v * (Dot(p - a, v) / d);
84 }
85
86 R Segment_Distance(Point p, Point a, Point b) const
87 {

```

```

88     Point v = b - a;
89     R d = Dot(v, v);
90     if ( d == 0 )
91         return norm(p - a);
92     R t = clamp(Dot(p - a, v) / d, R(0), R(1));
93     return norm(p - (a + v * t));
94 }
95
96 Result Line_Intersection(Point a, Point b, Point c, Point d) const
97 {
98     Point u = b - a, v = d - c;
99     R nu = norm(u), nv = norm(v);
100    if ( nu == 0 || nv == 0 )
101        return {Kind::degenerate, {}};
102    R det = Cross(u, v);
103    if ( fabsl(det) <= eps * nu * nv )
104        return {fabsl(Cross(c - a, u)) <= eps * nu * max(R(1), norm(c - a))
105                ? Kind::infinite
106                : Kind::none,
107                {}};
108    return {Kind::one, {a + u * (Cross(c - a, v) / det)}};
109 }
110
111 Result Line_Circle(Point a, Point b, Circle c) const
112 {
113     assert(c.r >= 0);
114     Point v = b - a;
115     R len = norm(v);
116     if ( len == 0 )
117         return {Kind::degenerate, {}};
118     Point h = Projection(c.o, a, b);
119     R d = norm(h - c.o), tol = eps * max(d, c.r);
120     if ( d > c.r + tol )
121         return {Kind::none, {}};
122     if ( fabsl(d - c.r) <= tol )
123         return {Kind::one, {h}};
124     R t = sqrtl(max(R(0), (c.r - d) * (c.r + d)));
125     Point w = v * (t / len);
126     return {Kind::two, {h - w, h + w}};
127 }
128
129 Result Circles(Circle a, Circle b) const
130 {
131     assert(a.r >= 0 && b.r >= 0);
132     Point v = b.o - a.o;
133     R d = norm(v);
134     R tol = eps * max({d, a.r, b.r});
135     // Infinite intersections require identical positive-radius circles.
136     if ( d == 0 )
137     {
138         if ( a.r != b.r )
139             return {Kind::none, {}};
140         if ( a.r == 0 )
141             return {Kind::one, {a.o}};
142         return {Kind::infinite, {}};
143     }

```

```

144     if ( a.r == 0 && b.r == 0 )
145         return {Kind::none, {}};
146     if ( a.r == 0 || b.r == 0 )
147     {
148         R radius = max(a.r, b.r);
149         if ( fabsl(d - radius) > tol )
150             return {Kind::none, {}};
151         return {Kind::one, {a.r == 0 ? a.o : b.o}};
152     }
153     if ( d > a.r + b.r + tol || d < fabsl(a.r - b.r) - tol )
154         return {Kind::none, {}};
155     R x = (d * d + (a.r - b.r) * (a.r + b.r)) / (2 * d);
156     Point h = a.o + v * (x / d);
157     // Near-concentric circles must not be mistaken for internal tangency.
158     if ( fabsl(x) > a.r + tol )
159         return {Kind::none, {}};
160     if ( fabsl(fabsl(x) - a.r) <= tol )
161         return {Kind::one, {h}};
162     R y = sqrtl(max(R(0), (a.r - x) * (a.r + x)));
163     Point w = perp(v) * (y / d);
164     return {Kind::two, {h - w, h + w}};
165 }
166
167 R Overlap(Circle a, Circle b) const
168 {
169     assert(a.r >= 0 && b.r >= 0);
170     R d = norm(a.o - b.o), pi = acosl(-1.L);
171     if ( d >= a.r + b.r )
172         return 0;
173     if ( d <= fabsl(a.r - b.r) )
174         return pi * min(a.r, b.r) * min(a.r, b.r);
175     R difference = (a.r - b.r) * (a.r + b.r);
176     R x = acosl(clamp((d * d + difference) / (2 * d * a.r), R(-1), R(1)));
177     R y = acosl(clamp((d * d - difference) / (2 * d * b.r), R(-1), R(1)));
178     auto segment = [](R angle)
179     {
180         if ( angle < 1e-3L )
181         {
182             R z = angle * angle;
183             return angle * z * (R(2) / 3 + z * (-R(2) / 15 + z * R(4) / 315));
184         }
185         return angle - sinl(2 * angle) / 2;
186     };
187     return a.r * a.r * segment(x) + b.r * b.r * segment(y);
188 }
189 };

```

第一百章 优化与可持久化

100.1 Max-plus 矩阵幂

算法约束见动态版本第 128 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Max_Plus_Matrix
2 {
3     using I = __int128_t;
4     static constexpr I neg = -(I(1) << 120);
5     int n;
6     vector<vector<I>> a;
7
8     Max_Plus_Matrix(int n, bool identity = false) : n(n), a(n, vector<I>(n, neg))
9     {
10         if ( identity )
11             for ( int i = 0; i < n; i++ )
12                 a[i][i] = 0;
13     }
14
15     Max_Plus_Matrix operator*(const Max_Plus_Matrix &b) const
16     {
17         assert(n == b.n);
18         Max_Plus_Matrix c(n);
19         for ( int i = 0; i < n; i++ )
20             for ( int k = 0; k < n; k++ )
21                 if ( a[i][k] != neg )
22                     for ( int j = 0; j < n; j++ )
23                         if ( b.a[k][j] != neg )
24                             c.a[i][j] = max(c.a[i][j], a[i][k] + b.a[k][j]);
25         return c;
26     }
27
28     Max_Plus_Matrix Power(unsigned long long e) const
29     {
30         Max_Plus_Matrix b = *this, r(n, true);
31         for ( ; e >>= 1, b = b * b )
32             if ( e & 1 )
33                 r = r * b;
34         return r;
35     }
36 };
```

100.2 离散李超树与最优直线编号

算法约束见动态版本第 128 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Li_Chao_Tree
2 {
3     using ll = long long;
4     using I = __int128_t;
5 }
```



```

6   struct Line
7   {
8       ll k = 0, b = 0;
9       int id = -1;
10
11       I value(ll x) const
12       {
13           return I(k) * x + b;
14       }
15   };
16
17   struct Node
18   {
19       Line line;
20       int left = -1, right = -1;
21   };
22
23   vector<ll> xs;
24   vector<Node> a{Node{}};
25
26   Li_Chao_Tree(vector<ll> x) : xs(x)
27   {
28       sort(xs.begin(), xs.end());
29       xs.erase(unique(xs.begin(), xs.end()), xs.end());
30       assert(!xs.empty());
31   }
32
33   bool better(Line u, Line v, ll x) const
34   {
35       return v.id == -1 ||
36              (u.id != -1 && pair{u.value(x), u.id} < pair{v.value(x), v.id});
37   }
38
39   void Insert(Line line)
40   {
41       assert(line.id >= 0);
42       Insert(0, 0, (int)xs.size() - 1, line);
43   }
44
45   void Insert(int p, int l, int r, Line line)
46   {
47       int m = (l + r) / 2;
48       if ( better(line, a[p].line, xs[m]) )
49           swap(line, a[p].line);
50       if ( l == r || line.id == -1 )
51           return;
52       bool left = better(line, a[p].line, xs[l]);
53       bool right = better(line, a[p].line, xs[r]);
54       if ( !left && !right )
55           return;
56       int next = left ? a[p].left : a[p].right;
57       if ( next == -1 )
58       {
59           next = (int)a.size();
60           a.push_back(Node{});
61           if ( left )

```

```

62         a[p].left = next;
63     else
64         a[p].right = next;
65 }
66 if ( left )
67     Insert(next, l, m, line);
68 else
69     Insert(next, m + 1, r, line);
70 }
71
72 pair<I, int> Query(ll x) const
73 {
74     int idx = lower_bound(xs.begin(), xs.end(), x) - xs.begin();
75     assert(idx < (int)xs.size() && xs[idx] == x);
76     int p = 0, l = 0, r = (int)xs.size() - 1;
77     Line best;
78     while ( p != -1 )
79     {
80         if ( better(a[p].line, best, x) )
81             best = a[p].line;
82         if ( l == r )
83             break;
84         int m = (l + r) / 2;
85         if ( idx <= m )
86         {
87             p = a[p].left;
88             r = m;
89         }
90         else
91         {
92             p = a[p].right;
93             l = m + 1;
94         }
95     }
96     assert(best.id != -1);
97     return {best.value(x), best.id};
98 }
99 };

```

100.3 主席树：静态区间第 k 小

算法约束见动态版本第 130 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Persistent_Kth
2 {
3     struct Node
4     {
5         int l = 0, r = 0, sum = 0;
6     };
7
8     vector<Node> t{Node{}};
9     vector<int> root{0};
10    vector<long long> vals;
11
12    Persistent_Kth(const vector<long long> &a) : vals(a)

```

```

13 {
14     sort(vals.begin(), vals.end());
15     vals.erase(unique(vals.begin(), vals.end()), vals.end());
16     for ( long long x : a )
17     {
18         int p = lower_bound(vals.begin(), vals.end(), x) - vals.begin();
19         root.push_back(Insert(root.back(), 0, (int)vals.size() - 1, p));
20     }
21 }
22
23 int Insert(int old, int l, int r, int x)
24 {
25     int p = t.size();
26     Node copy = t[old];
27     t.push_back(copy);
28     ++t[p].sum;
29     if ( l < r )
30     {
31         int m = (l + r) / 2;
32         if ( x <= m )
33         {
34             int v = Insert(t[old].l, l, m, x);
35             t[p].l = v;
36         }
37         else
38         {
39             int v = Insert(t[old].r, m + 1, r, x);
40             t[p].r = v;
41         }
42     }
43     return p;
44 }
45
46 long long Kth(int l, int r, int k) const
47 {
48     assert(1 <= l && l <= r && r < (int)root.size() && 1 <= k && k <= r - l + 1);
49     int u = root[l - 1], v = root[r], L = 0, R = (int)vals.size() - 1;
50     while ( L < R )
51     {
52         int m = (L + R) / 2, c = t[t[v].l].sum - t[t[u].l].sum;
53         if ( k <= c )
54         {
55             u = t[u].l;
56             v = t[v].l;
57             R = m;
58         }
59         else

```

```
60         {
61             k -= c;
62             u = t[u].r;
63             v = t[v].r;
64             L = m + 1;
65         }
66     }
67     return vals[L];
68 }
69 };
```

第一百零一章 可持久化数组

101.1 可持久化数组：历史版本分叉与单点修改

算法约束见动态版本第 133 页。本节代码独立可抄；采用下列实际接口名。

声明 Persistent_Array<NODES>, Init(a) 初始化；NODES 限制节点总数，不是数组长度。建议按 $2n-1$ 加上修改数乘以 $(\text{ceil}(\log_2(n))+1)$ 分配；Copy 不分配节点。Set、Copy、Query 对应动态接口。大对象放在全局或 static，Init 会丢弃全部旧版本。

```
1  template <int NODES> struct Persistent_Array
2  {
3      using ll = long long;
4
5      struct Node
6      {
7          int l = 0, r = 0;
8          ll value = 0;
9      };
10
11     int n;
12     Node t[NODES + 1];
13     int tot = 0;
14     vector<int> root;
15
16     void Init(const vector<ll> &a)
17     {
18         n = a.size();
19         assert(n > 0);
20         tot = 0;
21         root.clear();
22         root.push_back(Build(1, n, a));
23     }
24
25     int Build(int l, int r, const vector<ll> &a)
26     {
27         assert(tot < NODES);
28         int p = ++tot;
29         t[p] = Node{};
30         if ( l == r )
31             t[p].value = a[l - 1];
32         else
33         {
34             int m = (l + r) / 2;
35             int left = Build(l, m, a);
36             int right = Build(m + 1, r, a);
37             t[p].l = left;
38             t[p].r = right;
39         }
40         return p;
41     }
42
43     int Modify(int old, int l, int r, int pos, ll value)
```

```

44     {
45         assert(tot < NODES);
46         int p = ++tot;
47         t[p] = t[old];
48         if ( l == r )
49             t[p].value = value;
50         else
51         {
52             int m = (l + r) / 2;
53             if ( pos <= m )
54             {
55                 int child = Modify(t[old].l, l, m, pos, value);
56                 t[p].l = child;
57             }
58             else
59             {
60                 int child = Modify(t[old].r, m + 1, r, pos, value);
61                 t[p].r = child;
62             }
63         }
64         return p;
65     }
66
67     // Create a new version from any existing version. Positions are 1-based.
68     int Set(int version, int pos, ll value)
69     {
70         assert(0 <= version && version < (int)root.size());
71         assert(1 <= pos && pos <= n);
72         int p = Modify(root[version], 1, n, pos, value);
73         root.push_back(p);
74         return (int)root.size() - 1;
75     }
76
77     int Copy(int version)
78     {
79         assert(0 <= version && version < (int)root.size());
80         root.push_back(root[version]);
81         return (int)root.size() - 1;
82     }
83
84     ll Query(int version, int pos) const
85     {
86         assert(0 <= version && version < (int)root.size());
87         assert(1 <= pos && pos <= n);
88         int p = root[version], l = 1, r = n;
89         while ( l < r )
90         {
91             int m = (l + r) / 2;
92             if ( pos <= m )
93             {
94                 p = t[p].l;
95                 r = m;

```

```
96         }
97         else
98         {
99             p = t[p].r;
100             l = m + 1;
101         }
102     }
103     return t[p].value;
104 }
105 };
```

第一百零二章 树上路径第 k 小

102.1 树上路径点权第 k 小

算法约束见动态版本第 136 页。本节代码独立可抄；采用下列实际接口名。

声明 `Tree_Path_Kth<N,NODES,LOG>`, `Init(a)` 清空图并保存权值; `Insert`、`Build`、`Kth`、`Lca` 对应动态接口。

要求 2 的 LOG 次方大于实际点数，默认 LOG=20。若不同权值数为 D，NODES 至少取 n 乘以 $(\text{ceil}(\log_2(D))+1)$ ，另有编号 0 空节点。大对象放在全局或 static；`Init` 与 `Build` 分别清空图和重建版本。

```
1 template <int N, int NODES, int LOG = 20> struct Tree_Path_Kth
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, sum = 0;
8     };
9
10    static_assert(1 <= LOG && LOG <= 31);
11    int n, tot;
12    vector<int> g[N + 1];
13    int up[LOG][N + 1], root[N + 1], dep[N + 1];
14    ll value[N + 1];
15    vector<ll> vals;
16    Node t[NODES + 1];
17
18    void Init(const vector<ll> &a)
19    {
20        n = a.size();
21        assert(0 < n && n <= N && (1LL << LOG) > n);
22        for (int u = 1; u <= n; u++)
23        {
24            value[u] = a[u - 1];
25            g[u].clear();
26        }
27        root[0] = 0;
28        for (int j = 0; j < LOG; j++)
29            up[j][0] = 0;
30    }
31
32    void Insert(int u, int v)
33    {
34        g[u].push_back(v);
35        g[v].push_back(u);
36    }
37
38    int Update(int old, int l, int r, int x)
39    {
40        assert(tot < NODES);
41        int p = ++tot;
42        t[p] = t[old];
43        t[p].sum++;
```



```

44     if ( l == r )
45         return p;
46     int m = (l + r) / 2;
47     if ( x <= m )
48     {
49         int child = Update(t[old].l, l, m, x);
50         t[p].l = child;
51     }
52     else
53     {
54         int child = Update(t[old].r, m + 1, r, x);
55         t[p].r = child;
56     }
57     return p;
58 }
59
60 void Dfs(int u, int p)
61 {
62     up[0][u] = p;
63     for ( int j = 1; j < LOG; j++ )
64         up[j][u] = up[j - 1][up[j - 1][u]];
65     int x = lower_bound(vals.begin(), vals.end(), value[u]) - vals.begin();
66     root[u] = Update(root[p], 0, (int)vals.size() - 1, x);
67     for ( int v : g[u] )
68     {
69         if ( v == p )
70             continue;
71         dep[v] = dep[u] + 1;
72         Dfs(v, u);
73     }
74 }
75
76 void Build(int r = 1)
77 {
78     assert(1 <= r && r <= n);
79     vals.assign(value + 1, value + n + 1);
80     sort(vals.begin(), vals.end());
81     vals.erase(unique(vals.begin(), vals.end()), vals.end());
82     tot = 0;
83     t[0] = Node{};
84     dep[r] = 0;
85     Dfs(r, 0);
86 }
87
88 int Lca(int u, int v) const
89 {
90     if ( dep[u] < dep[v] )
91         swap(u, v);
92     int diff = dep[u] - dep[v];
93     for ( int j = 0; j < LOG; j++ )
94         if ( diff >> j & 1 )
95             u = up[j][u];
96     if ( u == v )
97         return u;
98     for ( int j = LOG - 1; j >= 0; j-- )
99         if ( up[j][u] != up[j][v] )

```

```

100         {
101             u = up[j][u];
102             v = up[j][v];
103         }
104         return up[0][u];
105     }
106
107 ll Kth(int u, int v, int k) const
108 {
109     int z = Lca(u, v);
110     assert(1 <= k && k <= dep[u] + dep[v] - 2 * dep[z] + 1);
111     int a = root[u], b = root[v], c = root[z], d = root[up[0][z]];
112     int l = 0, r = (int)vals.size() - 1;
113     while ( l < r )
114     {
115         int count = t[t[a].l].sum + t[t[b].l].sum - t[t[c].l].sum - t[t[d].l].sum;
116         int m = (l + r) / 2;
117         if ( k <= count )
118         {
119             a = t[a].l;
120             b = t[b].l;
121             c = t[c].l;
122             d = t[d].l;
123             r = m;
124         }
125         else
126         {
127             k -= count;
128             a = t[a].r;
129             b = t[b].r;
130             c = t[c].r;
131             d = t[d].r;
132             l = m + 1;
133         }
134     }
135     return vals[l];
136 }
137 };

```

第一百零三章 动态区间顺序统计

103.1 动态区间第 k 小 (树状数组套权值线段树)

算法约束见动态版本第 139 页。本节代码独立可抄；采用下列实际接口名。

声明 `Dynamic_Kth<N,NODES>` 后 `Init(a,future)`, `Set` 与 `Kth` 为公开操作。N 限制数组长度, NODES 限制分配节点的最大编号, 另有编号 0 空节点。按照上面的 B 分配一定足够; 例如 n 不超过 100000 且 D 不超过 200000 时, B 不超过 16682000。节点回收使此界与修改次数无关。大对象放在全局或 static, Init 清空旧状态。

```
1  template <int N, int NODES> struct Dynamic_Kth
2  {
3      using ll = long long;
4
5      struct Node
6      {
7          int l = 0, r = 0, sum = 0;
8      };
9
10     int n, tot, free_head;
11     int bit[N + 1], rank[N + 1];
12     vector<ll> vals;
13     Node t[NODES + 1];
14
15     void Init(const vector<ll> &a, const vector<ll> &future = {})
16     {
17         n = a.size();
18         assert(0 < n && n <= N);
19         tot = 0;
20         free_head = 0;
21         t[0] = Node{};
22         fill(bit, bit + n + 1, 0);
23         vals = future;
24         vals.insert(vals.end(), a.begin(), a.end());
25         sort(vals.begin(), vals.end());
26         vals.erase(unique(vals.begin(), vals.end()), vals.end());
27         for (int i = 1; i <= n; i++)
28         {
29             rank[i] = Id(a[i - 1]);
30             Modify(i, rank[i], 1);
31         }
32     }
33
34     int Id(ll value) const
35     {
36         int p = lower_bound(vals.begin(), vals.end(), value) - vals.begin();
37         assert(p < (int)vals.size() && vals[p] == value);
38         return p;
39     }
40
41     int New_Node()
42     {
43         if (!free_head)
```

```

44     {
45         assert(tot < NODES);
46         int p = ++tot;
47         t[p] = Node{};
48         return p;
49     }
50     int p = free_head;
51     free_head = t[p].l;
52     t[p] = Node{};
53     return p;
54 }
55
56 int Update(int p, int l, int r, int x, int delta)
57 {
58     if ( !p )
59         p = New_Node();
60     t[p].sum += delta;
61     if ( l < r )
62     {
63         int m = (l + r) / 2;
64         if ( x <= m )
65         {
66             int child = Update(t[p].l, l, m, x, delta);
67             t[p].l = child;
68         }
69         else
70         {
71             int child = Update(t[p].r, m + 1, r, x, delta);
72             t[p].r = child;
73         }
74     }
75     if ( !t[p].sum )
76     {
77         t[p].l = free_head;
78         free_head = p;
79         return 0;
80     }
81     return p;
82 }
83
84 void Modify(int pos, int x, int delta)
85 {
86     for ( int i = pos; i <= n; i += i & -i )
87         bit[i] = Update(bit[i], 0, (int)vals.size() - 1, x, delta);
88 }

```

```

90 void Set(int pos, ll value)
91 {
92     assert(1 <= pos && pos <= n);
93     int x = Id(value);
94     if ( rank[pos] == x )
95         return;
96     Modify(pos, rank[pos], -1);
97     Modify(pos, x, 1);
98     rank[pos] = x;
99 }
100
101 ll Kth(int l, int r, int k) const
102 {
103     assert(1 <= l && l <= r && r <= n && 1 <= k && k <= r - l + 1);
104     vector<int> a, b;
105     for ( int i = r; i; i -= i & -i )
106         a.push_back(bit[i]);
107     for ( int i = l - 1; i; i -= i & -i )
108         b.push_back(bit[i]);
109     int L = 0, R = (int)vals.size() - 1;
110     while ( L < R )
111     {
112         int count = 0;
113         for ( int p : a )
114             count += t[t[p].l].sum;
115         for ( int p : b )
116             count -= t[t[p].l].sum;
117         bool left = k <= count;
118         if ( !left )
119             k -= count;
120         for ( int &p : a )
121             p = left ? t[p].l : t[p].r;
122         for ( int &p : b )
123             p = left ? t[p].l : t[p].r;
124         int m = (L + R) / 2;
125         if ( left )
126             R = m;
127         else
128             L = m + 1;
129     }
130     return vals[L];
131 }
132 };

```

第一百零四章 区间不同数统计

104.1 静态区间不同数计数（前驱位置主席树）

算法约束见动态版本第 142 页。本节代码独立可抄；采用下列实际接口名。

声明 `Persistent_Distinct<N,NODES>` 后 `Init(a)`, `Query(l,r)` 为查询。`N` 限制数组长度；`NODES` 至少取 n 乘以 $(\text{ceil}(\log_2(n))+1)$ ，另有编号 0 空节点。 n 不超过 1000000 时可取 `NODES=21000000`。大对象放在全局或 `static`，`Init` 可重复调用并清空全部旧版本。

```
1 template <int N, int NODES> struct Persistent_Distinct
2 {
3     struct Node
4     {
5         int l = 0, r = 0, sum = 0;
6     };
7
8     int n, tot;
9     Node t[NODES + 1];
10    int root[N + 1];
11
12    void Init(const vector<long long> &a)
13    {
14        n = a.size();
15        assert(0 < n && n <= N);
16        tot = 0;
17        t[0] = Node{};
18        root[0] = 0;
19        vector<long long> vals = a;
20        sort(vals.begin(), vals.end());
21        vals.erase(unique(vals.begin(), vals.end()), vals.end());
22        vector<int> last(vals.size());
23        for ( int i = 1; i <= n; i++ )
24        {
25            int x = lower_bound(vals.begin(), vals.end(), a[i - 1]) - vals.begin();
26            root[i] = Insert(root[i - 1], 0, n - 1, last[x]);
27            last[x] = i;
28        }
29    }
30
31    int Insert(int old, int l, int r, int x)
32    {
33        assert(tot < NODES);
34        int p = ++tot;
35        t[p] = t[old];
36        t[p].sum++;
37        if ( l < r )
38        {
39            int m = (l + r) / 2;
40            if ( x <= m )
41            {
42                int child = Insert(t[old].l, l, m, x);
43                t[p].l = child;
```

```
44         }
45         else
46         {
47             int child = Insert(t[old].r, m + 1, r, x);
48             t[p].r = child;
49         }
50     }
51     return p;
52 }
53
54 int Query(int l, int r) const
55 {
56     assert(1 <= l && l <= r && r <= n);
57     int u = root[l - 1], v = root[r];
58     int L = 0, R = n - 1, answer = 0;
59     while ( L < R )
60     {
61         int m = (L + R) / 2;
62         if ( l - 1 <= m )
63         {
64             u = t[u].l;
65             v = t[v].l;
66             R = m;
67         }
68         else
69         {
70             answer += t[t[v].l].sum - t[t[u].l].sum;
71             u = t[u].r;
72             v = t[v].r;
73             L = m + 1;
74         }
75     }
76     return answer + t[v].sum - t[u].sum;
77 }
78 };
```

第一百零五章 图论进阶

105.1 矩形二分图最小权匹配

算法约束见动态版本第 144 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Kuhn_Munkres
2 {
3     using I = __int128_t;
4
5     // Minimum cost injection from n rows to m columns, n<=m. Zero-based matrix.
6     static pair<I, vector<int>> Solve(const vector<vector<long long>> &cost)
7     {
8         int n = cost.size();
9         if ( !n )
10             return {0, {}};
11         int m = cost[0].size();
12         assert(n <= m);
13         vector<I> u(n + 1), v(m + 1);
14         vector<int> p(m + 1), way(m + 1);
15         const I inf = I(1) << 120;
16         for ( int i = 1; i <= n; i++ )
17         {
18             assert((int)cost[i - 1].size() == m);
19             p[0] = i;
20             int j0 = 0;
21             vector<I> dis(m + 1, inf);
22             vector<bool> Used(m + 1);
23             do
24             {
25                 Used[j0] = true;
26                 int i0 = p[j0], j1 = 0;
27                 I delta = inf;
28                 for ( int j = 1; j <= m; j++ )
29                     if ( !Used[j] )
30                     {
31                         I cur = I(cost[i0 - 1][j - 1]) - u[i0] - v[j];
32                         if ( cur < dis[j] )
33                         {
34                             dis[j] = cur;
35                             way[j] = j0;
36                         }
37                         if ( dis[j] < delta )
38                         {
39                             delta = dis[j];
40                             j1 = j;
41                         }
42                     }
43                 for ( int j = 0; j <= m; j++ )
44                     if ( Used[j] )
45                     {
46                         u[p[j]] += delta;
```



```

47         v[j] -= delta;
48     }
49     else
50         dis[j] -= delta;
51     j0 = j1;
52 } while ( p[j0] );
53 do
54 {
55     int j1 = way[j0];
56     p[j0] = p[j1];
57     j0 = j1;
58 } while ( j0 );
59 }
60 vector<int> ans(n);
61 for ( int j = 1; j <= m; j++ )
62     if ( p[j] )
63         ans[p[j] - 1] = j - 1;
64 return {-v[0], ans};
65 }
66 };

```

105.2 有向最小树形图

算法约束见动态版本第 145 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Directed_MST
2 {
3     using I = __int128_t;
4
5     struct Edge
6     {
7         int u, v;
8         I w;
9     };
10
11     // Directed minimum rooted OUT-arborescence cost, 0-based, or nullopt.
12     static optional<I> Solve(int n, int root, vector<Edge> e)
13     {
14         const I inf = I(1) << 120;
15         I ans = 0;
16         while ( true )
17         {
18             vector<I> in(n, inf);
19             vector<int> pre(n, -1), id(n, -1), vis(n, -1);
20             for ( auto a : e )
21                 if ( a.u != a.v && a.w < in[a.v] )
22                 {
23                     in[a.v] = a.w;
24                     pre[a.v] = a.u;
25                 }
26             in[root] = 0;
27             for ( I x : in )
28             {
29                 if ( x == inf )
30                     return nullopt;

```

```

31         ans += x;
32     }
33     int cnt = 0;
34     for ( int i = 0; i < n; i++ )
35     {
36         int u = i;
37         while ( vis[u] != i && id[u] == -1 && u != root )
38         {
39             vis[u] = i;
40             u = pre[u];
41         }
42         if ( u != root && id[u] == -1 )
43         {
44             id[u] = cnt;
45             for ( int v = pre[u]; v != u; v = pre[v] )
46                 id[v] = cnt;
47             ++cnt;
48         }
49     }
50     if ( !cnt )
51         return ans;
52     for ( int i = 0; i < n; i++ )
53         if ( id[i] == -1 )
54             id[i] = cnt++;
55     for ( auto &a : e )
56     {
57         int v = a.v;
58         a.u = id[a.u];
59         a.v = id[a.v];
60         if ( a.u != a.v )
61             a.w -= in[v];
62     }
63     root = id[root];
64     n = cnt;
65 }
66 }
67 };

```

105.3 无向全局最小割

算法约束见动态版本第 147 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Global_Min_Cut
2 {
3     using I = __int128_t;
4
5     // Symmetric nonnegative adjacency matrix; returns weight and one cut side.
6     static pair<I, vector<int>> Solve(vector<vector<I>> w)
7     {
8         int n = w.size();
9         assert(n >= 2);
10        vector<int> active(n);
11        iota(active.begin(), active.end(), 0);
12        vector<vector<int>> group(n);
13        for ( int i = 0; i < n; i++ )
14            group[i] = {i};
15        I best = I(1) << 120;
16        vector<int> side;
17        while ( active.size() > 1 )
18        {
19            vector<I> dis(n);
20            vector<bool> Used(n);
21            int prev = -1;
22            for ( int step = 0; step < (int)active.size(); step++ )
23            {
24                int sel = -1;
25                for ( int v : active )
26                    if ( !Used[v] && (sel == -1 || dis[v] > dis[sel]) )
27                        sel = v;
28                if ( step + 1 == (int)active.size() )
29                {
30                    if ( dis[sel] < best )
31                    {
32                        best = dis[sel];
33                        side = group[sel];
34                    }
35                    for ( int v : active )
36                        if ( v != prev && v != sel )
37                            w[prev][v] = w[v][prev] = w[prev][v] + w[sel][v];
38                    group[prev].insert(
39                        group[prev].end(), group[sel].begin(), group[sel].end());
40                    active.erase(find(active.begin(), active.end(), sel));
41                    break;
42                }
43                Used[sel] = true;
44                prev = sel;
45                for ( int v : active )
46                    if ( !Used[v] )
47                        dis[v] += w[sel][v];
48            }
49        }
50        return {best, side};
51    }
52 };

```

第一百零六章 带权二分图匹配

106.1 二分图最大权匹配（缺边与可选匹配）

算法约束见动态版本第 149 页。本节代码独立可抄；采用下列实际接口名。

声明 `Weighted_Matching<N,M>` 后 `Init(n,m)`，`N`、`M` 分别限制左右点数。`Insert` 加边，`Solve` 求解；`l`、`r` 保存方案，`Init` 清空原图。权值矩阵是静态数组，内部增广辅助数组按实际规模分配，大对象放在全局或 `static`。

```
1 template <int N, int M> struct Weighted_Matching
2 {
3     using I = __int128_t;
4     static constexpr I inf = I(1) << 120;
5     int n, m;
6     I w[N + 1][M + 1];
7     int l[N + 1], r[M + 1];
8
9     void Init(int left_size, int right_size)
10    {
11        assert(0 <= left_size && left_size <= N);
12        assert(0 <= right_size && right_size <= M);
13        n = left_size;
14        m = right_size;
15        for ( int i = 0; i <= n; i++ )
16            fill(w[i], w[i] + m + 1, -inf);
17        fill(l, l + n + 1, 0);
18        fill(r, r + m + 1, 0);
19    }
20
21    void Insert(int x, int y, long long weight)
22    {
23        assert(1 <= x && x <= n && 1 <= y && y <= m);
24        w[x][y] = max(w[x][y], I(weight));
25    }
26
27    optional<I> Solve(bool allow_unmatched = false)
28    {
29        fill(l, l + n + 1, 0);
30        fill(r, r + m + 1, 0);
31        if ( !allow_unmatched && n > m )
32            return nullopt;
33        int cols = m + (allow_unmatched ? n : 0);
34        vector<I> u(n + 1), v(cols + 1);
35        vector<int> p(cols + 1), way(cols + 1);
36        for ( int i = 1; i <= n; i++ )
37        {
38            p[0] = i;
39            int j0 = 0;
40            vector<I> dis(cols + 1, inf);
41            vector<bool> used(cols + 1);
42            do
43            {
44                used[j0] = true;
```

```

45     int x = p[j0], j1 = 0;
46     I delta = inf;
47     for ( int j = 1; j <= cols; j++ )
48     {
49         if ( used[j] )
50             continue;
51         I weight = j <= m ? w[x][j] : 0;
52         if ( weight != -inf )
53         {
54             I cur = -weight - u[x] - v[j];
55             if ( cur < dis[j] )
56             {
57                 dis[j] = cur;
58                 way[j] = j0;
59             }
60         }
61         if ( dis[j] < delta )
62         {
63             delta = dis[j];
64             j1 = j;
65         }
66     }
67     if ( delta == inf )
68         return nullopt;
69     for ( int j = 0; j <= cols; j++ )
70     {
71         if ( used[j] )
72         {
73             u[p[j]] += delta;
74             v[j] -= delta;
75         }
76         else if ( dis[j] != inf )
77             dis[j] -= delta;
78     }
79     j0 = j1;
80     } while ( p[j0] );
81     do
82     {
83         int j1 = way[j0];
84         p[j0] = p[j1];
85         j0 = j1;
86     } while ( j0 );
87 }
88 I answer = 0;
89 for ( int j = 1; j <= m; j++ )
90 {
91     if ( !p[j] )
92         continue;
93     l[p[j]] = j;
94     r[j] = p[j];
95     answer += w[p[j]][j];
96 }
97 return answer;
98 }
99 };

```

第一百零七章 精确几何进阶

107.1 整数最近点对与 Minkowski 和

算法约束见动态版本第 151 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Geometry_Extra
2 {
3     using G = Integer_Geometry;
4     using P = G::Point;
5     using I = G::I;
6
7     // No pair exists for n<2. Exact squared distance, O(n log n).
8     static optional<I> Closest_Pair(vector<P> p)
9     {
10         int n = p.size();
11         if ( n < 2 )
12             return nullopt;
13         sort(p.begin(), p.end());
14         for ( int i = 1; i < n; i++ )
15             if ( p[i] == p[i - 1] )
16                 return I(0);
17         vector<P> tmp(n);
18         auto by_y = [](P a, P b)
19         {
20             return tie(a.y, a.x) < tie(b.y, b.x);
21         };
22         function<I(int, int)> Solve = [&](int l, int r) -> I
23         {
24             if ( r - l <= 3 )
25             {
26                 I best = I(1) << 120;
27                 for ( int i = l; i < r; i++ )
28                     for ( int j = i + 1; j < r; j++ )
29                         best = min(best, G::Distance_Squared(p[i], p[j]));
30                 sort(p.begin() + l, p.begin() + r, by_y);
31                 return best;
32             }
33             int m = (l + r) / 2;
34             long long x = p[m].x;
35             I best = min(Solve(l, m), Solve(m, r));
36             merge(p.begin() + l,
37                 p.begin() + m,
38                 p.begin() + m,
39                 p.begin() + r,
40                 tmp.begin() + l,
41                 by_y);
42             copy(tmp.begin() + l, tmp.begin() + r, p.begin() + l);
43             vector<P> strip;
44             for ( int i = l; i < r; i++ )
45                 if ( (I(p[i].x) - x) * (I(p[i].x) - x) < best )
46                     {
```

```

47         for ( int j = (int)strip.size() - 1; j >= 0; j-- )
48         {
49             I dy = I(p[i].y) - strip[j].y;
50             if ( dy * dy >= best )
51                 break;
52             best = min(best, G::Distance_Squared(p[i], strip[j]));
53         }
54         strip.push_back(p[i]);
55     }
56     return best;
57 };
58 return Solve(0, n);
59 }
60
61 // Inputs strict CCW hulls. Coordinate sums must obey IntegerGeometry's bound.
62 static vector<P> Minkowski(vector<P> a, vector<P> b)
63 {
64     if ( a.empty() || b.empty() )
65         return {};
66     auto plus = [](P u, P v)
67     {
68         return P{u.x + v.x, u.y + v.y};
69     };
70     if ( a.size() < 3 || b.size() < 3 )
71     {
72         vector<P> p;
73         for ( P u : a )
74             for ( P v : b )
75                 p.push_back(plus(u, v));
76         return G::Convex_Hull(p);
77     }
78     auto start = [](vector<P> &p)
79     {
80         auto it = min_element(p.begin(),
81                               p.end(),
82                               [](P u, P v)
83                               {
84                                   return tie(u.y, u.x) < tie(v.y, v.x);
85                               });
86         rotate(p.begin(), it, p.end());
87     };
88     start(a);
89     start(b);
90     int n = a.size(), m = b.size(), i = 0, j = 0;
91     vector<P> result;
92     while ( i < n || j < m )
93     {
94         result.push_back(plus(a[i % n], b[j % m]));
95         if ( i == n )
96         {
97             ++j;
98             continue;
99         }
100         if ( j == m )
101         {
102             ++i;

```

```
103         continue;
104     }
105     P u{a[(i + 1) % n].x - a[i].x, a[(i + 1) % n].y - a[i].y};
106     P v{b[(j + 1) % m].x - b[j].x, b[(j + 1) % m].y - b[j].y};
107     I c = G::Cross({0, 0}, u, v);
108     if ( c >= 0 )
109         ++i;
110     if ( c <= 0 )
111         ++j;
112 }
113 return result;
114 }
115 };
```


107.2 整数三维共线、共面与方向

算法约束见动态版本第 154 页。本节代码独立可抄；采用下列实际接口名。

```

1 struct Integer_Geometry_3D
2 {
3     using I = __int128_t;
4
5     // |coordinates|<=1e9: triple products fit signed int128.
6     struct Point
7     {
8         long long x, y, Z_Function;
9     };
10
11     struct Vector
12     {
13         I x, y, Z_Function;
14     };
15
16     static Vector diff(Point a, Point b)
17     {
18         return {I(a.x) - b.x, I(a.y) - b.y, I(a.Z_Function) - b.Z_Function};
19     }
20
21     static Vector Cross(Vector a, Vector b)
22     {
23         return {a.y * b.Z_Function - a.Z_Function * b.y,
24                 a.Z_Function * b.x - a.x * b.Z_Function,
25                 a.x * b.y - a.y * b.x};
26     }
27
28     static I Dot(Vector a, Vector b)
29     {
30         return a.x * b.x + a.y * b.y + a.Z_Function * b.Z_Function;
31     }
32
33     static I orient(Point a, Point b, Point c, Point d)
34     {
35         return Dot(Cross(diff(b, a), diff(c, a)), diff(d, a));
36     }
37
38     static bool collinear(Point a, Point b, Point c)
39     {
40         auto v = Cross(diff(b, a), diff(c, a));
41         return !v.x && !v.y && !v.Z_Function;
42     }
43
44     static bool On_Segment(Point p, Point a, Point b)
45     {
46         return collinear(p, a, b) && Dot(diff(a, p), diff(b, p)) <= 0;
47     }
48 };

```

第一百零八章 凸包支撑点查询

108.1 支撑点查询恢复下凸包

算法约束见动态版本第 155 页。本节代码独立可抄；采用下列实际接口名。

```
1 struct Support_Hull
2 {
3     using ll = long long;
4     using I = __int128_t;
5
6     struct Point
7     {
8         ll x, y;
9     };
10
11     static I Cross(Point a, Point b, Point c)
12     {
13         return (I(b.x) - a.x) * (I(c.y) - a.y) - (I(b.y) - a.y) * (I(c.x) - a.x);
14     }
```

```

16 // oracle(a,b) maximizes a*x+b*y exactly; |coordinates| <= 1e18.
17 // left/right: lowest points at the minimum/maximum x.
18 template <class Oracle>
19 static vector<Point> Build(Point left, Point right, Oracle &&oracle)
20 {
21     assert(left.x <= right.x);
22     vector<Point> hull{left};
23     auto dfs = [&](auto &&self, Point a, Point b) -> void
24     {
25         Point c = oracle(b.y - a.y, a.x - b.x);
26         I turn = Cross(a, b, c);
27         assert(turn <= 0);
28         if ( turn < 0 )
29         {
30             assert(a.x < c.x && c.x < b.x);
31             self(self, a, c);
32             self(self, c, b);
33         }
34         else
35         {
36             while ( hull.size() >= 2 &&
37                     Cross(hull[hull.size() - 2], hull.back(), b) == 0 )
38                 hull.pop_back();
39             hull.push_back(b);
40         }
41     };
42     if ( left.x < right.x )
43         dfs(dfs, left, right);
44     else
45         assert(left.y == right.y);
46     return hull;
47 }
48 };

```

第一百零九章 动态树

109.1 Link-cut tree: 动态森林与路径异或

算法约束见动态版本第 157 页。本节代码独立可抄；采用下列实际接口名。

声明 `Link_Cut_Tree<N>`，然后 `Init(n)` 初始化；`N` 为最大点号。`Set`、`Link`、`Cut`、`Connected`、`Query` 分别对应动态版的小写接口。大对象放在全局或 `static`。

```
1 template <int N> struct Link_Cut_Tree
2 {
3     using U = unsigned long long;
4
5     struct Node
6     {
7         int ch[2]{}, fa = 0;
8         U val = 0, sum = 0;
9         bool rev = false;
10    };
11
12    Node tree[N + 1];
13    int st[N + 1];
14
15    void Init(int n)
16    {
17        assert(0 <= n && n <= N);
18        fill(tree, tree + n + 1, Node{});
19    }
20
21    bool Is_Root(int x) const
22    {
23        int f = tree[x].fa;
24        return tree[f].ch[0] != x && tree[f].ch[1] != x;
25    }
26
27    void Push_Up(int x)
28    {
29        tree[x].sum = tree[tree[x].ch[0]].sum ^ tree[x].val ^ tree[tree[x].ch[1]].sum;
30    }
31
32    void Reverse(int x)
33    {
34        if ( !x )
35            return;
36        swap(tree[x].ch[0], tree[x].ch[1]);
37        tree[x].rev = !tree[x].rev;
38    }
39
40    void Push_Down(int x)
41    {
42        if ( !tree[x].rev )
43            return;
44        Reverse(tree[x].ch[0]);
```

```

45     Reverse(tree[x].ch[1]);
46     tree[x].rev = false;
47 }
48
49 void Rotate(int x)
50 {
51     int y = tree[x].fa, z = tree[y].fa;
52     int k = tree[y].ch[1] == x;
53     int w = tree[x].ch[k ^ 1];
54     if ( !Is_Root(y) )
55         tree[z].ch[tree[z].ch[1] == y] = x;
56     tree[x].fa = z;
57     tree[y].ch[k] = w;
58     if ( w )
59         tree[w].fa = y;
60     tree[x].ch[k ^ 1] = y;
61     tree[y].fa = x;
62     Push_Up(y);
63     Push_Up(x);
64 }
65
66 void Splay(int x)
67 {
68     int top = 0, y = x;
69     st[top++] = y;
70     while ( !Is_Root(y) )
71     {
72         y = tree[y].fa;
73         st[top++] = y;
74     }
75     while ( top )
76         Push_Down(st[--top]);
77     while ( !Is_Root(x) )
78     {
79         int y = tree[x].fa, z = tree[y].fa;
80         if ( !Is_Root(y) )
81         {
82             bool same = (tree[y].ch[1] == x) == (tree[z].ch[1] == y);
83             Rotate(same ? y : x);
84         }
85         Rotate(x);
86     }
87 }
88
89 void Access(int x)
90 {
91     int last = 0;
92     for ( int y = x; y; y = tree[y].fa )
93     {
94         Splay(y);
95         // The old right child's fa remains y: it becomes a path-parent.
96         tree[y].ch[1] = last;
97         if ( last )
98             tree[last].fa = y;
99         Push_Up(y);
100        last = y;

```

```

101     }
102     Splay(x);
103 }
104
105 void Make_Root(int x)
106 {
107     Access(x);
108     Reverse(x);
109 }
110
111 int Find_Root(int x)
112 {
113     Access(x);
114     Push_Down(x);
115     while ( tree[x].ch[0] )
116     {
117         x = tree[x].ch[0];
118         Push_Down(x);
119     }
120     Splay(x);
121     return x;
122 }
123
124 bool Connected(int x, int y)
125 {
126     return x == y || Find_Root(x) == Find_Root(y);
127 }
128
129 bool Link(int x, int y)
130 {
131     Make_Root(x);
132     if ( Find_Root(y) == x )
133         return false;
134     tree[x].fa = y;
135     return true;
136 }
137
138 bool Cut(int x, int y)
139 {
140     Make_Root(x);
141     Access(y);
142     if ( tree[y].ch[0] != x || tree[x].ch[1] )
143         return false;
144     tree[y].ch[0] = 0;
145     tree[x].fa = 0;
146     Push_Up(y);
147     return true;
148 }
149
150 void Set(int x, U value)
151 {
152     Access(x);
153     tree[x].val = value;
154     Push_Up(x);

```

```
155     }
156
157     // x and y must be connected. This is a path aggregate, not a subtree sum.
158     U Query(int x, int y)
159     {
160         Make_Root(x);
161         Access(y);
162         return tree[y].sum;
163     }
164 };
```

第一百一十章 随机平衡树

110.1 FHQ Treap: 可重集合与顺序统计

算法约束见动态版本第 161 页。本节代码独立可抄；采用下列实际接口名。

声明容量模板参数 N，然后 Init() 或 Init(seed)。N 限制累计插入次数，不是当前元素个数；大对象放在全局或 static。接口采用 Insert、Erase、Kth/Query 等命名。

```
1 template <int N> struct Ordered_Treap
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, siz = 0;
8         ll val = 0;
9         unsigned long long pri = 0;
10    };
11
12    Node tree[N + 1];
13    int tot = 0;
14    mt19937_64 rng;
15    int root = 0;
16
17    void Init(unsigned long long seed = 712367821)
18    {
19        root = 0;
20        tot = 0;
21        tree[0] = Node{};
22        rng.seed(seed);
23    }
24
25    void Push_Up(int p)
26    {
27        tree[p].siz = tree[tree[p].l].siz + tree[tree[p].r].siz + 1;
28    }
29
30    int New_Node(ll x)
31    {
32        assert(tot < N);
33        int p = ++tot;
34        tree[p] = {0, 0, 1, x, rng()};
35        return p;
36    }
37
38    // left: val < x, or val <= x when equal is true. No x-1 overflow.
39    void Split(int p, ll x, int &l, int &r, bool equal = false)
40    {
41        if ( !p )
42        {
43            l = 0;
44            r = 0;
```



```

45         return;
46     }
47     if ( tree[p].val < x || (equal && tree[p].val == x) )
48     {
49         l = p;
50         Split(tree[p].r, x, tree[p].r, r, equal);
51     }
52     else
53     {
54         r = p;
55         Split(tree[p].l, x, l, tree[p].l, equal);
56     }
57     Push_Up(p);
58 }
59
60 int Merge(int l, int r)
61 {
62     if ( !l || !r )
63         return l ? l : r;
64     if ( tree[l].pri > tree[r].pri )
65     {
66         tree[l].r = Merge(tree[l].r, r);
67         Push_Up(l);
68         return l;
69     }
70     tree[r].l = Merge(l, tree[r].l);
71     Push_Up(r);
72     return r;
73 }
74
75 void Insert(ll x)
76 {
77     int p = New_Node(x), l, r;
78     Split(root, x, l, r);
79     root = Merge(Merge(l, p), r);
80 }
81
82 bool Erase(ll x)
83 {
84     int l, mid, r;
85     Split(root, x, l, mid);
86     Split(mid, x, mid, r, true);
87     bool found = mid != 0;
88     if ( mid )
89         mid = Merge(tree[mid].l, tree[mid].r);
90     root = Merge(l, Merge(mid, r));
91     return found;
92 }
93
94 int Size() const
95 {
96     return tree[root].siz;
97 }
98
99 int Count_Less(ll x, bool equal = false) const
100 {

```

```

101     int p = root, ans = 0;
102     while ( p )
103     {
104         if ( tree[p].val < x || (equal && tree[p].val == x) )
105         {
106             ans += tree[tree[p].l].siz + 1;
107             p = tree[p].r;
108         }
109         else
110             p = tree[p].l;
111     }
112     return ans;
113 }
114
115 int Rank(ll x) const
116 {
117     return Count_Less(x) + 1;
118 }
119
120 ll Kth(int k) const
121 {
122     assert(1 <= k && k <= Size());
123     int p = root;
124     while ( true )
125     {
126         int left = tree[tree[p].l].siz;
127         if ( k == left + 1 )
128             return tree[p].val;
129         if ( k <= left )
130             p = tree[p].l;
131         else
132         {
133             k -= left + 1;
134             p = tree[p].r;
135         }
136     }
137 }
138
139 optional<ll> Predecessor(ll x) const
140 {
141     int k = Count_Less(x);
142     if ( !k )
143         return nullopt;
144     return Kth(k);
145 }
146
147 optional<ll> Successor(ll x) const
148 {
149     int k = Count_Less(x, true);
150     if ( k == Size() )
151         return nullopt;
152     return Kth(k + 1);
153 }
154 };

```

110.2 隐式 Treap：序列翻转、加值与区间和

算法约束见动态版本第 163 页。本节代码独立可抄；采用下列实际接口名。

声明容量模板参数 N，然后 Init() 或 Init(seed)。N 限制累计插入次数，不是当前元素个数；大对象放在全局或 static。接口采用 Insert、Erase、Kth/Query 等命名。

```

1 template <int N> struct Sequence_Treap
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, siz = 0;
8         ll val = 0, sum = 0, tag = 0;
9         unsigned long long pri = 0;
10        bool rev = false;
11    };
12
13    Node tree[N + 1];
14    int tot = 0;
15    mt19937_64 rng;
16    int root = 0;
17
18    void Init(unsigned long long seed = 712367821)
19    {
20        root = 0;
21        tot = 0;
22        tree[0] = Node{};
23        rng.seed(seed);
24    }
25
26    int Size() const
27    {
28        return tree[root].siz;
29    }
30
31    void Push_Up(int p)
32    {
33        tree[p].siz = tree[tree[p].l].siz + tree[tree[p].r].siz + 1;
34        tree[p].sum = tree[tree[p].l].sum + tree[p].val + tree[tree[p].r].sum;
35    }
36
37    void Apply_Add(int p, ll x)
38    {
39        if ( !p )
40            return;
41        tree[p].val += x;
42        tree[p].sum += x * tree[p].siz;
43        tree[p].tag += x;
44    }
45
46    void Apply_Reverse(int p)
47    {
48        if ( !p )
49            return;
50        swap(tree[p].l, tree[p].r);

```

```

51     tree[p].rev = !tree[p].rev;
52 }
53
54 void Push_Down(int p)
55 {
56     if ( tree[p].rev )
57     {
58         Apply_Reverse(tree[p].l);
59         Apply_Reverse(tree[p].r);
60         tree[p].rev = false;
61     }
62     if ( tree[p].tag )
63     {
64         Apply_Add(tree[p].l, tree[p].tag);
65         Apply_Add(tree[p].r, tree[p].tag);
66         tree[p].tag = 0;
67     }
68 }
69
70 // Split into the first k elements and the rest.
71 void Split(int p, int k, int &l, int &r)
72 {
73     if ( !p )
74     {
75         l = 0;
76         r = 0;
77         return;
78     }
79     Push_Down(p);
80     int left = tree[tree[p].l].siz;
81     if ( k > left )
82     {
83         l = p;
84         Split(tree[p].r, k - left - 1, tree[p].r, r);
85     }
86     else
87     {
88         r = p;
89         Split(tree[p].l, k, l, tree[p].l);
90     }
91     Push_Up(p);
92 }
93
94 int Merge(int l, int r)
95 {
96     if ( !l || !r )
97         return l ? l : r;
98     if ( tree[l].pri > tree[r].pri )
99     {
100         Push_Down(l);
101         tree[l].r = Merge(tree[l].r, r);
102         Push_Up(l);
103         return l;
104     }
105     Push_Down(r);
106     tree[r].l = Merge(l, tree[r].l);

```

```

107     Push_Up(r);
108     return r;
109 }
110
111 // Insert after the first k elements, 0 <= k <= size().
112 void Insert(int k, ll value)
113 {
114     assert(0 <= k && k <= Size());
115     assert(tot < N);
116     int p = ++tot, l, r;
117     tree[p] = {0, 0, 1, value, value, 0, rng(), false};
118     Split(root, k, l, r);
119     root = Merge(Merge(l, p), r);
120 }
121
122 // All range operations use 1-based inclusive intervals.
123 void Erase(int l, int r)
124 {
125     assert(1 <= l && l <= r && r <= Size());
126     int x, y, z;
127     Split(root, r, y, z);
128     Split(y, l - 1, x, y);
129     root = Merge(x, z);
130 }
131
132 void Reverse(int l, int r)
133 {
134     assert(1 <= l && l <= r && r <= Size());
135     int x, y, z;
136     Split(root, r, y, z);
137     Split(y, l - 1, x, y);
138     Apply_Reverse(y);
139     root = Merge(Merge(x, y), z);
140 }
141
142 void Add(int l, int r, ll value)
143 {
144     assert(1 <= l && l <= r && r <= Size());
145     int x, y, z;
146     Split(root, r, y, z);
147     Split(y, l - 1, x, y);
148     Apply_Add(y, value);
149     root = Merge(Merge(x, y), z);
150 }
151
152 ll Query(int l, int r)
153 {
154     assert(1 <= l && l <= r && r <= Size());
155     int x, y, z;
156     Split(root, r, y, z);
157     Split(y, l - 1, x, y);
158     ll ans = tree[y].sum;
159     root = Merge(Merge(x, y), z);
160     return ans;
161 }
162

```

```
163 void Collect(int p, vector<ll> &out)
164 {
165     if ( !p )
166         return;
167     Push_Down(p);
168     Collect(tree[p].l, out);
169     out.push_back(tree[p].val);
170     Collect(tree[p].r, out);
171 }
172
173 vector<ll> Values()
174 {
175     vector<ll> answer;
176     Collect(root, answer);
177     return answer;
178 }
179 };
```

第一百一十一章 伸展树

111.1 Splay：可重集合与顺序统计

算法约束见动态版本第 168 页。本节代码独立可抄；采用下列实际接口名。

声明 `Ordered_Splay<N>` 后调用 `Init()`；`N` 为累计新建节点上限。接口采用 `Insert`、`Erase`、`Rank`、`Kth`、`Prev`、`Next`。大对象放在全局或 `static`。

```
1 template <int N> struct Ordered_Splay
2 {
3     using ll = long long;
4
5     struct Node
6     {
7         int ch[2]{}, fa = 0, cnt = 0, siz = 0;
8         ll val = 0;
9     };
10
11     Node tree[N + 1];
12     int tot = 0;
13     int root = 0;
14
15     void Init()
16     {
17         root = 0;
18         tot = 0;
19         tree[0] = Node{};
20     }
21
22     int Size() const
23     {
24         return tree[root].siz;
25     }
26
27     void Push_Up(int x)
28     {
29         tree[x].siz = tree[tree[x].ch[0]].siz + tree[x].cnt + tree[tree[x].ch[1]].siz;
30     }
31
32     int New_Node(ll value, int parent)
33     {
34         assert(tot < N);
35         int x = ++tot;
36         tree[x] = {0, 0}, parent, 1, 1, value;
37         return x;
38     }
39
40     void Rotate(int x)
41     {
42         int y = tree[x].fa, z = tree[y].fa;
43         int k = tree[y].ch[1] == x;
44         int w = tree[x].ch[k ^ 1];
```

```

45     if ( z )
46         tree[z].ch[tree[z].ch[1] == y] = x;
47     else
48         root = x;
49     tree[x].fa = z;
50     tree[y].ch[k] = w;
51     if ( w )
52         tree[w].fa = y;
53     tree[x].ch[k ^ 1] = y;
54     tree[y].fa = x;
55     Push_Up(y);
56     Push_Up(x);
57 }
58
59 void Splay(int x)
60 {
61     while ( tree[x].fa )
62     {
63         int y = tree[x].fa, z = tree[y].fa;
64         if ( z )
65         {
66             bool same = (tree[y].ch[1] == x) == (tree[z].ch[1] == y);
67             Rotate(same ? y : x);
68         }
69         Rotate(x);
70     }
71 }
72
73 // Splay the last visited node even when the value does not exist.
74 int Find(ll value)
75 {
76     int x = root, last = 0;
77     while ( x && tree[x].val != value )
78     {
79         last = x;
80         x = tree[x].ch[value > tree[x].val];
81     }
82     if ( x )
83         Splay(x);
84     else if ( last )
85         Splay(last);
86     return x;
87 }
88
89 void Insert(ll value)
90 {
91     int x = root, parent = 0;
92     while ( x && tree[x].val != value )
93     {
94         parent = x;
95         x = tree[x].ch[value > tree[x].val];
96     }
97     if ( x )
98     {
99         ++tree[x].cnt;
100         Push_Up(x);

```



```

101     }
102     else
103     {
104         x = New_Node(value, parent);
105         if ( parent )
106             tree[parent].ch[value > tree[parent].val] = x;
107         else
108             root = x;
109     }
110     Splay(x);
111 }
112
113 bool Erase(ll value)
114 {
115     int x = Find(value);
116     if ( !x )
117         return false;
118     if ( tree[x].cnt > 1 )
119     {
120         --tree[x].cnt;
121         Push_Up(x);
122         return true;
123     }
124     int l = tree[x].ch[0], r = tree[x].ch[1];
125     if ( l )
126         tree[l].fa = 0;
127     if ( r )
128         tree[r].fa = 0;
129     if ( !l )
130     {
131         root = r;
132         return true;
133     }
134     root = l;
135     x = l;
136     while ( tree[x].ch[1] )
137         x = tree[x].ch[1];
138     Splay(x);
139     tree[x].ch[1] = r;
140     if ( r )
141         tree[r].fa = x;
142     Push_Up(x);
143     return true;
144 }
145
146 int Rank(ll value)
147 {
148     int x = root, last = 0, ans = 1;
149     while ( x )
150     {
151         last = x;
152         if ( value <= tree[x].val )
153             x = tree[x].ch[0];
154         else
155         {
156             ans += tree[tree[x].ch[0]].siz + tree[x].cnt;

```

```

157         x = tree[x].ch[1];
158     }
159 }
160 if ( last )
161     Splay(last);
162 return ans;
163 }
164
165 ll Kth(int k)
166 {
167     assert(1 <= k && k <= Size());
168     int x = root;
169     while ( true )
170     {
171         int left = tree[tree[x].ch[0]].siz;
172         if ( k <= left )
173             x = tree[x].ch[0];
174         else if ( k <= left + tree[x].cnt )
175         {
176             Splay(x);
177             return tree[x].val;
178         }
179         else
180         {
181             k -= left + tree[x].cnt;
182             x = tree[x].ch[1];
183         }
184     }
185 }
186
187 optional<ll> Predecessor(ll value)
188 {
189     int x = root, last = 0, answer = 0;
190     while ( x )
191     {
192         last = x;
193         if ( tree[x].val < value )
194         {
195             answer = x;
196             x = tree[x].ch[1];
197         }
198         else
199             x = tree[x].ch[0];
200     }
201     if ( last )
202         Splay(last);
203     if ( !answer )
204         return nullopt;
205     Splay(answer);
206     return tree[answer].val;
207 }
208
209 optional<ll> Successor(ll value)
210 {
211     int x = root, last = 0, answer = 0;
212     while ( x )

```

```
213     {
214         last = x;
215         if ( tree[x].val > value )
216         {
217             answer = x;
218             x = tree[x].ch[0];
219         }
220         else
221             x = tree[x].ch[1];
222     }
223     if ( last )
224         Splay(last);
225     if ( !answer )
226         return nullopt;
227     Splay(answer);
228     return tree[answer].val;
229 }
230 };
```

第一百一十二章 状态序列维护

112.1 状态序列 Treap：分组区间 GCD

算法约束见动态版本第 173 页。本节代码独立可抄；采用下列实际接口名。

声明 `Gcd_Sequence_Treap<N>` 后调用 `Init(seed)`；`N` 为峰值元素数。`Build` 重建序列，`Insert`、`Erase`、`Set`、`Toggle`、`Query` 对应动态版接口。大对象放在全局或 `static`。

```
1 template <int N> struct Gcd_Sequence_Treap
2 {
3     using U = unsigned long long;
4
5     struct Node
6     {
7         int l = 0, r = 0, siz = 0, tag = 0;
8         int cnt[2]{};
9         U val = 0, g[2]{}, pri = 0;
10    };
11
12    Node tree[N + 1];
13    int tot = 0;
14    vector<int> recycled;
15    mt19937_64 rng;
16    int root = 0;
17
18    void Init(U seed = 712367821)
19    {
20        root = 0;
21        tot = 0;
22        tree[0] = Node{};
23        recycled.clear();
24        rng.seed(seed);
25    }
26
27    int Size() const
28    {
29        return tree[root].siz;
30    }
31
32    int New_Node(U value, int tag)
33    {
34        assert(tag == 0 || tag == 1);
35        int p;
36        if ( recycled.empty() )
37        {
38            assert(tot < N);
39            p = ++tot;
40            tree[p] = Node{};
41        }
42        else
43        {
44            p = recycled.back();
```

```

45         recycled.pop_back();
46         tree[p] = Node{};
47     }
48     tree[p].siz = 1;
49     tree[p].val = value;
50     tree[p].tag = tag;
51     tree[p].cnt[tag] = 1;
52     tree[p].g[tag] = value;
53     tree[p].pri = rng();
54     return p;
55 }
56
57 void Push_Up(int p)
58 {
59     int l = tree[p].l, r = tree[p].r;
60     tree[p].siz = tree[l].siz + tree[r].siz + 1;
61     for ( int t = 0; t < 2; t++ )
62     {
63         tree[p].cnt[t] = tree[l].cnt[t] + tree[r].cnt[t] + (tree[p].tag == t);
64         tree[p].g[t] = gcd(tree[l].g[t], tree[r].g[t]);
65         if ( tree[p].tag == t )
66             tree[p].g[t] = gcd(tree[p].g[t], tree[p].val);
67     }
68 }
69
70 void Split(int p, int k, int &l, int &r)
71 {
72     if ( !p )
73     {
74         l = 0;
75         r = 0;
76         return;
77     }
78     int left = tree[tree[p].l].siz;
79     if ( k > left )
80     {
81         l = p;
82         Split(tree[p].r, k - left - 1, tree[p].r, r);
83     }
84     else
85     {
86         r = p;
87         Split(tree[p].l, k, l, tree[p].l);
88     }
89     Push_Up(p);
90 }
91
92 int Merge(int l, int r)
93 {
94     if ( !l || !r )
95         return l ? l : r;
96     if ( tree[l].pri > tree[r].pri )
97     {
98         tree[l].r = Merge(tree[l].r, r);
99         Push_Up(l);
100        return l;

```

```

101     }
102     tree[r].l = Merge(l, tree[r].l);
103     Push_Up(r);
104     return r;
105 }
106
107 void Rebuild(int p)
108 {
109     if ( !p )
110         return;
111     Rebuild(tree[p].l);
112     Rebuild(tree[p].r);
113     Push_Up(p);
114 }
115
116 void Release(int p)
117 {
118     if ( !p )
119         return;
120     Release(tree[p].l);
121     Release(tree[p].r);
122     recycled.push_back(p);
123 }
124
125 // O(n) GCD calls; replaces the current sequence.
126 void Build(const vector<pair<U, int>> &v)
127 {
128     assert(v.size() <= N);
129     tot = 0;
130     tree[0] = Node{};
131     recycled.clear();
132     root = 0;
133     vector<int> st;
134     for ( auto [value, tag] : v )
135     {
136         int p = New_Node(value, tag), last = 0;
137         while ( !st.empty() && tree[st.back()].pri < tree[p].pri )
138         {
139             last = st.back();
140             st.pop_back();
141         }
142         tree[p].l = last;
143         if ( !st.empty() )
144             tree[st.back()].r = p;
145         st.push_back(p);
146     }
147     if ( st.empty() )
148         return;
149     root = st.front();
150     Rebuild(root);
151 }
152
153 // Insert after the first k entries. Range operations below are 1-based.
154 void Insert(int k, U value, int tag)
155 {
156     assert(0 <= k && k <= Size());

```

```

157     int p = New_Node(value, tag), l, r;
158     Split(root, k, l, r);
159     root = Merge(Merge(l, p), r);
160 }
161
162 void Erase(int l, int r)
163 {
164     assert(1 <= l && l <= r && r <= Size());
165     int x, y, z;
166     Split(root, r, y, z);
167     Split(y, l - 1, x, y);
168     root = Merge(x, z);
169     Release(y);
170 }
171
172 void Set(int k, U value)
173 {
174     assert(1 <= k && k <= Size());
175     int x, y, z;
176     Split(root, k, y, z);
177     Split(y, k - 1, x, y);
178     tree[y].val = value;
179     Push_Up(y);
180     root = Merge(Merge(x, y), z);
181 }
182
183 void Toggle(int k)
184 {
185     assert(1 <= k && k <= Size());
186     int x, y, z;
187     Split(root, k, y, z);
188     Split(y, k - 1, x, y);
189     tree[y].tag ^= 1;
190     Push_Up(y);
191     root = Merge(Merge(x, y), z);
192 }
193
194 optional<U> Query(int l, int r, int tag)
195 {
196     assert(1 <= l && l <= r && r <= Size());
197     assert(tag == 0 || tag == 1);
198     int x, y, z;
199     Split(root, r, y, z);
200     Split(y, l - 1, x, y);
201     optional<U> answer;
202     if ( tree[y].cnt[tag] )
203         answer = tree[y].g[tag];
204     root = Merge(Merge(x, y), z);
205     return answer;
206 }
207 };

```

第一百一十三章 一般图匹配

113.1 一般图最大基数匹配：带花树

算法约束见动态版本第 177 页。本节代码独立可抄；采用下列实际接口名。

声明 `General_Matching<N>` 后调用 `Init(n)`，`N` 为点数上限。`Insert` 加边，`Solve` 返回匹配边数，`mate` 保存配对点，点号仍为 0-based。大对象放在全局或 `static`。

```
1 template <int N> struct General_Matching
2 {
3     int n;
4     vector<int> g[N];
5     int mate[N], pre[N], base[N], q[N];
6     bool outer[N], flower[N], seen[N];
7
8     void Init(int vertices)
9     {
10         assert(0 <= vertices && vertices <= N);
11         n = vertices;
12         for ( int u = 0; u < n; u++ )
13             g[u].clear();
14         fill(mate, mate + n, -1);
15     }
16
17     void Insert(int u, int v)
18     {
19         assert(0 <= u && u < n && 0 <= v && v < n);
20         if ( u == v )
21             return;
22         g[u].push_back(v);
23         g[v].push_back(u);
24     }
25
26     int Lca(int u, int v)
27     {
28         fill(seen, seen + n, false);
29         while ( true )
30         {
31             u = base[u];
32             seen[u] = true;
33             if ( mate[u] == -1 )
34                 break;
35             u = pre[mate[u]];
36         }
37         while ( !seen[base[v]] )
38             v = pre[mate[base[v]]];
39         return base[v];
40     }
41
42     void Mark(int u, int b, int v)
43     {
44         while ( base[u] != b )
```



```

45     {
46         flower[base[u]] = true;
47         flower[base[mate[u]]] = true;
48         pre[u] = v;
49         v = mate[u];
50         u = pre[v];
51     }
52 }
53
54 bool Augment(int root)
55 {
56     fill(outer, outer + n, false);
57     fill(pre, pre + n, -1);
58     iota(base, base + n, 0);
59     int tail = 0;
60     q[tail++] = root;
61     outer[root] = true;
62     for ( int head = 0; head < tail; head++ )
63     {
64         int u = q[head];
65         for ( int v : g[u] )
66         {
67             if ( base[u] == base[v] || mate[u] == v )
68                 continue;
69             if ( v == root || (mate[v] != -1 && pre[mate[v]] != -1) )
70             {
71                 int b = Lca(u, v);
72                 fill(flower, flower + n, false);
73                 Mark(u, b, v);
74                 Mark(v, b, u);
75                 for ( int x = 0; x < n; x++ )
76                 {
77                     if ( !flower[base[x]] )
78                         continue;
79                     base[x] = b;
80                     if ( !outer[x] )
81                     {
82                         outer[x] = true;
83                         q[tail++] = x;
84                     }
85                 }
86             }
87             else if ( pre[v] == -1 )
88             {
89                 pre[v] = u;
90                 if ( mate[v] == -1 )
91                 {
92                     while ( v != -1 )
93                     {
94                         int p = pre[v];
95                         int next = mate[p];
96                         mate[v] = p;
97                         mate[p] = v;
98                         v = next;
99                     }
100                     return true;

```

```
101         }
102         int x = mate[v];
103         outer[x] = true;
104         q[tail++] = x;
105     }
106 }
107 }
108 return false;
109 }
110
111 // Rebuild the maximum matching; mate[u] is -1 for an unmatched vertex.
112 int Solve()
113 {
114     fill(mate, mate + n, -1);
115     int answer = 0;
116     for ( int u = 0; u < n; u++ )
117     {
118         if ( mate[u] == -1 && Augment(u) )
119             answer++;
120     }
121     return answer;
122 }
123 };
```

第一百一十四章 线性递推

114.1 Berlekamp–Massey：最短线性递推

算法约束见动态版本第 180 页。本节代码独立可抄；采用下列实际接口名。

```
1 // Prime p; normalized input. s[i] = sum(c[j-1] * s[i-j]), i >= c.size().
2 vector<int> Berlekamp_Massey(const vector<int> &s, int p = 998244353)
3 {
4     assert(p >= 2);
5     vector<int> c{1}, b{1};
6     int len = 0, shift = 1, last = 1;
7     for ( int i = 0; i < (int)s.size(); i++ )
8     {
9         assert(0 <= s[i] && s[i] < p);
10        long long d = s[i];
11        for ( int j = 1; j <= len; j++ )
12            d = (d + 1LL * c[j] * s[i - j]) % p;
13        if ( d == 0 )
14        {
15            shift++;
16            continue;
17        }
18        long long inv = 1, x = last;
19        for ( int e = p - 2; e; e >>= 1 )
20        {
21            if ( e & 1 )
22                inv = inv * x % p;
23            x = x * x % p;
24        }
25        long long factor = d * inv % p;
26        auto old = c;
27        if ( c.size() < b.size() + shift )
28            c.resize(b.size() + shift);
29        for ( int j = 0; j < (int)b.size(); j++ )
30            c[j + shift] = (c[j + shift] - factor * b[j] % p + p) % p;
31        if ( 2 * len <= i )
32        {
33            len = i + 1 - len;
34            b = old;
35            last = d;
36            shift = 1;
37        }
38        else
39            shift++;
40    }
41    c.resize(len + 1);
42    c.erase(c.begin());
43    for ( int &x : c )
44        x = x == 0 ? 0 : p - x;
45    return c;
46 }
```

114.2 特征多项式求线性递推第 k 项

算法约束见动态版本第 182 页。本节代码独立可抄；采用下列实际接口名。

```

1 // init has exactly k entries; c[j-1] multiplies a[n-j]. Modulus need not be prime.
2 int Recurrence_Nth(const vector<int> &init,
3                   const vector<int> &c,
4                   unsigned long long n,
5                   int p = 998244353)
6 {
7     assert(p >= 2 && init.size() == c.size());
8     int k = c.size();
9     if ( !k )
10         return 0;
11     for ( int x : init )
12         assert(0 <= x && x < p);
13     for ( int x : c )
14         assert(0 <= x && x < p);
15     if ( n < (unsigned)k )
16         return init[n];
17     auto multiply = [&](const vector<int> &a, const vector<int> &b)
18     {
19         vector<int> t(2 * k - 1);
20         for ( int i = 0; i < k; i++ )
21         {
22             for ( int j = 0; j < k; j++ )
23                 t[i + j] = (t[i + j] + 1LL * a[i] * b[j]) % p;
24         }
25         for ( int i = 2 * k - 2; i >= k; i-- )
26         {
27             for ( int j = 1; j <= k; j++ )
28                 t[i - j] = (t[i - j] + 1LL * t[i] * c[j - 1]) % p;
29         }
30         t.resize(k);
31         return t;
32     };
33     vector<int> answer(k), x(k);
34     answer[0] = 1;
35     if ( k == 1 )
36         x[0] = c[0];
37     else
38         x[1] = 1;
39     while ( n )
40     {
41         if ( n & 1 )
42             answer = multiply(answer, x);
43         x = multiply(x, x);
44         n >>= 1;
45     }
46     long long value = 0;
47     for ( int i = 0; i < k; i++ )
48         value = (value + 1LL * answer[i] * init[i]) % p;
49     return value;
50 }

```

第三部分

数学速查与维护

第一百一十五章 数学速查：条件先于公式

115.1 整除、卷积与筛法

对正整数定义狄利克雷卷积

$$(f * g)(n) = \sum_{d|n} f(d)g(n/d).$$

单位元为 $\varepsilon(n) = [n = 1]$ ，常函数 $1(n) = 1$ ，恒等函数 $\text{id}(n) = n$ 。关键恒等式是 $\mu * 1 = \varepsilon$ 、 $\varphi * 1 = \text{id}$ 。若 $F(n) = \sum_{d|n} f(d)$ ，则 $f(n) = \sum_{d|n} \mu(d)F(n/d)$ 。

$$\sum_{i=1}^a \sum_{j=1}^b [\gcd(i, j) = 1] = \sum_{d=1}^{\min(a, b)} \mu(d) \lfloor a/d \rfloor \lfloor b/d \rfloor.$$

整除分块：当左端点为 l 、商为 $q = \lfloor n/l \rfloor$ ，右端点为 $r = \lfloor n/q \rfloor$ 。同时包含多个商时，取各自右端点的最小值。杜教筛由卷积恒等式获得

$$M(n) = 1 - \sum_{l=2}^n M(\lfloor n/l \rfloor), \quad \Phi(n) = \frac{n(n+1)}{2} - \sum_{l=2}^n \Phi(\lfloor n/l \rfloor),$$

求和按等商区间合并；第二式求和中没有额外乘以 l 。

115.2 模运算的适用条件

模逆元存在当且仅当 $\gcd(a, m) = 1$ 。费马逆元 a^{p-2} 仅适用于素数模数及 $a \neq 0$ 。对正模数 m, n ，同余式 $x \equiv a \pmod{m}$ 、 $x \equiv b \pmod{n}$ 可合并当且仅当 $\gcd(m, n) \mid (b - a)$ ；新模数为 $\text{lcm}(m, n)$ 。

$$\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod{p} \quad (p \text{ 为素数}, n_i, k_i \text{ 是 } p \text{ 进制位}).$$

Lucas 不能直接套在合数模数。阶乘与逆阶乘预处理要求上限小于模数；超过模数会出现阶乘为零。扩展欧拉降幂中，非互质情况不能直接将指数对 $\varphi(m)$ 取模；需要判断原指数是否达到足够阈值。大整数指数可以在读入时同时维护余数与是否超过阈值。

115.3 任意数组的批量逆元

设 $s_i = \prod_{j < i} a_j \pmod{m}$ ， $s_0 = 1$ 。仅对总乘积 s_n 求一次逆元，随后从右向左处理。若当前 $t = s_{i+1}^{-1}$ ，则

$$a_i^{-1} = t s_i \pmod{m}, \quad s_i^{-1} = t a_i \pmod{m}.$$

先保存原 a_i 再覆盖输出，避免误用已写入的逆元更新 t 。总乘积与 m 互素当且仅当每一项均与 m 互素，因此一次扩展欧几里得也能判断整个批次是否可行；模数无需为素数。实现见第 94 页。

接口返回可选向量：无解是空的可选值，空输入是有效的空向量，两者不同。模数为 1 时按接口约定返回全零。调用者可以移动不再使用的输入数组，避免额外复制。P5431 的驱动程序计算 $\sum_{i=1}^n k^i / a_i$ ，注意权重从 k^1 开始。五百万项规模采用缓冲的字符读取，并移动输入以控制峰值内存。双版本已做乘法证书和非互素拒绝的穷举检查，以及 64 位极值、空数组、模数 1 的检查；完整驱动另用逐项逆元和几何级数进行对拍，在线 AC 待补。

该方法可处理任意可逆数组。连续整数 $1, 2, \dots, N$ 在素数模数下的专用线性递推仍另列为待补。来源：<https://oi-wiki.org/math/number-theory/inverse/>。

115.4 扩展 CRT：逐式合并

实现见第 87 页的 `crt`，传统版对应 `Crt`。初始取 $r = 0, m = 1$ ，依次合并 $x \equiv b \pmod{n}$ 。先把余数归一化。设 $g = \gcd(m, n)$ ，由扩展欧几里得得到 $mu + nv = g$ 。写 $x = r + mt$ ，原问题化为 $mt \equiv b - r \pmod{n}$ 。若 $g \nmid (b - r)$ 则无解，否则

$$q = n/g, \quad t \equiv \frac{b-r}{g}u \pmod{q}, \quad 0 \leq t < q.$$

新周期为 mq ，新余数为 $(r + mt) \bmod (mq)$ 。所有可行 t 相差 q ，故新余数加新周期的任意整数倍恰好是全部解；逐式合并时这一不变量保持成立。模数不必互素，模数 1 也可直接参与。

代码要求正模数且最终周期不超过有符号 64 位上限；中间乘法与扩展欧几里得系数使用 128 位整数。无解返回假；有解而周期越界抛出异常。失败时旧周期不变，旧余数已归一化。 k 条方程的时间为 $O(k \log M)$ ，其中 M 是输入及中间模数的上界；递归欧几里得使用 $O(\log M)$ 栈空间。P4777 的完整程序按“模数、余数”读入，题目保证有解且周期不超过 10^{18} 。本地验证包含双版本小模数穷举、任意精度整数对拍、有符号极值、无解与越界，以及十万条同余式的完整程序测试。双版本均已通过 P4777 全部 15 个测试点；动态版记录 297521102，传统版记录 297521173。

kuangbin 的“解同余方程组”是模 7 的线性方程组消元应用，见后面的工厂工期一节。来源：<https://oi-wiki.org/math/number-theory/crt/>；WIDA 数论模板的 `excrt` 节。

115.5 Garner：总周期可以超过整数范围

设正模数 m_i 两两互素， $P = \prod_i m_i$ 。经典构造为 $x = \sum_i b_i P_i (P_i^{-1} \bmod m_i) \bmod P$ ，其中 $P_i = P/m_i$ 。模 m_i 时其他项都为零，本项为 b_i ；两解之差被所有 m_i 整除，因而被 P 整除。但直接写此式需要存储整个 P ，并防止乘法溢出。

Garner 改用混合进制

$$x = d_0 + d_1 m_0 + d_2 m_0 m_1 + \cdots, \quad 0 \leq d_i < m_i.$$

在确定第 i 位前，对每个后续模数维护已确定部分的余数 v_j ，以及当前位权 $c_j = \prod_{t < i} m_t \bmod m_j$ 。于是

$$d_i = (b_i - v_i) c_i^{-1} \bmod m_i.$$

把 $c_j d_i$ 加到 v_j ，再将 c_j 乘 m_i 即可进入下一位。互素性保证逆元存在；模数 1 的位固定为 0，可直接保留。位数范围保证重建的 x 恰在 $[0, P)$ ，因此就是最小非负解。实现见第 95 页。

若只需 $x \bmod T$ ，从高位向低位做 Horner 计算：每步将当前值乘对应模数，加上该位，再对 T 取模。 T 无需与 m_i 互素，因为这里不对 T 求逆。这返回的是规范解 x 的余数；一般解 $x + kP$ 对 T 的余数可能不同。例如模 3 余 2、模 5 余 3，规范解是 8；对 7 取模为 1，但另一个解 23 的余数是 2。若用多模 NTT 恢复整数系数，仍须证明真实系数位于所选规范区间，不能在乘积不够大时直接把 CRT 的代表元当成原整数。

双版本本地验证使用任意精度整数的经典 CRT 求和作为独立对拍，检查混合进制的完整重建、负余数、有符号极值、非互素输入拒绝和 1000 个模数。P1495 完整程序按题目保证的 $P \leq 10^{18}$ 输出规范解，双版本均通过全部 11 个测试点；动态版记录 297521594，传统版记录 297521658。超出该题范围的能力以上述本地测试验证。来源：<https://oi-wiki.org/math/number-theory/crt/>。

115.6 素数模平方根

奇素数 p 下，对 $n \not\equiv 0 \pmod{p}$ ，Euler 判别式 $n^{(p-1)/2} \equiv 1 \pmod{p}$ 当且仅当存在平方根，否则为 -1 。当 $p \equiv 3 \pmod{4}$ 且有解时， $r = n^{(p+1)/4}$ 为一根，另一根是 $p - r$ 。

Cipolla 选择 a 使 $w = a^2 - n$ 是非平方，令 $\omega^2 = w$ 。在扩域中取 $\alpha = a + \omega$ ，有 $\alpha^p = a - \omega$ ，故 $\alpha^{p+1} = n$ 。令 $\beta = \alpha^{(p+1)/2}$ ，则 $\beta^2 = n$ 且 $\beta^{p-1} = n^{(p-1)/2} = 1$ ，所以 β 落在原素数域中。实现见第 106 页；零和模数 2 单独处理。模数为合数时不能照搬此判别和扩域构造。

115.7 乘法阶与原根

对 $\gcd(a, n) = 1$, 阶是最小的正整数 d 使 $a^d \equiv 1 \pmod{n}$, 且 $d \mid \varphi(n)$ 。从 $d = \varphi(n)$ 开始, 对每个素因子 q , 只要 $a^{d/q} \equiv 1$ 就继续除去 q , 所得即为实际阶。对于不可逆的 a , 接口返回 0, 表示此处没有定义乘法阶。

原根要求阶达到 $\varphi(n)$ 。只需对其每个不同素因子 q 检查 $a^{\varphi(n)/q} \not\equiv 1$: 若实际阶是真因子, 商中必含某个这样的 q 。必须先检查 $\gcd(a, n) = 1$; 例如模 9 的候选 3 虽通过上述非 1 判定, 却不是原根。 $n \geq 2$ 时, 原根恰存在于 $2, 4, p^k, 2p^k$, 其中 p 是奇素数。原根存在且取一个原根 g 后, g^t 的阶是 $\varphi(n)/\gcd(t, \varphi(n))$, 所以枚举与 $\varphi(n)$ 互素的指数即可得到全部原根, 共 $\varphi(\varphi(n))$ 个; 这些数按大小输出时还需排序。

实现见第 97 页。本地验证穷举小模数的幂循环, 与双版本的阶、最小值和完整集合逐项比较; 另验证大素数、奇素数幂、无原根合数和有符号输入边界。P6091 的完整程序输出排序后第 $d, 2d, \dots$ 个原根, 零个输出时仍保留空行。双版本均通过全部 11 个测试点; 动态版记录 297558740, 传统版记录 297559082。求阶、最小原根的最小性及更大输入范围仍以上述本地证据验证。若 $d = \text{ord}_n(a)$, 则 $\text{ord}_n(a^k) = d/\gcd(d, k)$ 。两个可逆元素乘积的阶整除各自阶的最小公倍数; 两阶互素时, 乘积的阶等于两阶之积。来源: <https://oi-wiki.org/math/number-theory/primitive-root/>; kuangbin 2.17。

115.8 Carmichael: 最小公共指数

$\lambda(n)$ 是模 n 的所有可逆元素之阶的最小公倍数, 约定 $\lambda(1) = 1$ 。它整除 $\varphi(n)$, 但可以严格更小, 例如 $\lambda(8) = 2$ 而 $\varphi(8) = 4$ 。按互素的素数幂分解 $n = \prod_i p_i^{e_i}$ 后, 由 CRT, 对每个分量同时成立的幂等式也对 n 成立, 因此

$$\lambda(n) = \text{lcm}_i \lambda(p_i^{e_i}), \quad \lambda(p^e) = \begin{cases} 2^{e-2}, & p = 2, e \geq 3, \\ p^{e-1}(p-1), & \text{其他情形} \end{cases}$$

奇素数幂的单位群有原根, 公共指数即其大小。模 2^e 的奇数可表示为 $\pm 5^k$, 当 $e \geq 3$ 时 5 的阶为 2^{e-2} , 这给出二幂情形的上界和达到上界的元素。实现见第 100 页, 试除版本适用于有符号 32 位正模数。

对可逆底数可把指数对 $\lambda(n)$ 取模; 非互素底数仍须另行处理。例如模 8 下 $2^2 = 4$, 但 $2^{2 \bmod \lambda(8)} = 1$, 直接降幂会错误。另有 $\lambda(\text{lcm}(a, b)) = \text{lcm}(\lambda(a), \lambda(b))$, 这也提供了独立的组合检查条件。

合数 n 是 Carmichael 数当且仅当 $\lambda(n) \mid (n-1)$ 。等价的 Korselt 判别是: n 无平方因子, 且每个素因子 p 都满足 $(p-1) \mid (n-1)$ 。素数也满足前一整除式, 判定时必须排除素数。这些合数对所有互素底数都通过 Fermat 判定, 不能用单纯的 Fermat 检验代替素性判定。无平方因子再配合 CRT, 也使所有整数 a 满足 $a^n \equiv a \pmod{n}$ 。

双版本以小模数的全部实际阶之最小公倍数对拍, 另检验大二幂、奇素数幂、32 位边界及公共指数证书。

UVa10006 完整程序对全部合法输入进行本地 Korselt 对拍; 在线 AC 待补。

来源: <https://oi-wiki.org/math/number-theory/primitive-root/>;

应用题: <https://onlinejudge.org/external/100/10006.pdf>。

115.9 Lucas: 按素数进制逐位计算

在特征为素数 p 的域中, $(1+x)^p = 1+x^p$ 。若 $n = \sum_i n_i p^i$, 则

$$(1+x)^n = \prod_i (1+x^{p^i})^{n_i}.$$

每个因子选择的次数在 $0, \dots, n_i$ 之间, 严格小于 p , 因此位数表示唯一。比较 x^k 的系数即得 $\binom{n}{k} = \prod_i \binom{n_i}{k_i} \pmod{p}$; 若某位 $k_i > n_i$, 答案立即为零。仅预处理 $0!, \dots, (p-1)!$ 及其逆元, 这些数均非零。 $p!$ 为零, 不能把阶乘表延长后继续用除法公式。实现见第 101 页。

预处理后每位只做常数乘法, 适合固定小素数模数下的多次查询。接口接收 n, k , 不接收 “ n, m 表示选 n 个” 的隐含约定; P3807 的驱动程序显式调用 `choose(n+m, n)`。题目范围内求和安全, 完整 64 位输入自行求和时仍要检查溢出。本地验证包含 Pascal 三角、精确大整数、小模数重建和最大无符号 64 位下标; P3807 双版本均已通过全部 25 个测试点; 动态版记录 297556209, 传统版记录 297556596。完整 64 位参数及额外边界以上述本地对拍验证。来源: <https://oi-wiki.org/math/number-theory/lucas/>。

115.10 合数模数组合数与插值

对素数幂 $q = p^e$, 记 $v_p(n!) = \sum_{j \geq 1} \lfloor n/p^j \rfloor$, $U_p(n) = n!/p^{v_p(n!)}$ 。设 $t = v_p(n!) - v_p(k!) - v_p((n-k)!)$ 。当 $t \geq e$ 时组合数模 q 为零; 否则

$$\binom{n}{k} \equiv U_p(n) U_p(k)^{-1} U_p(n-k)^{-1} p^t \pmod{q}.$$

这里仅对与 p 互素的单位部分求逆。各素数幂上的结果通过 CRT 合并, 对应第 103 页的实现。

在素数域中, n 个模 p 两两不同的横坐标确定唯一的次数小于 n 的多项式。定义 $w_i = y_i / \prod_{j \neq i} (x_i - x_j)$, 则

$$f(k) = \sum_{i=0}^{n-1} w_i \prod_{j < i} (k - x_j) \prod_{j > i} (k - x_j).$$

该式不除以 $k - x_i$, 查询点命中样本点时仍然成立。当 $x_i = i$ 时, 分母为 $i!(n-1-i)!(-1)^{n-1-i}$, 要求 $n \leq p$ 。实现见第 108 页。对整数幂前缀和 $S_d(k) = \sum_{i=1}^k i^d$, 当 $0 \leq d \leq p-2$ 时, 它在模 p 意义下可用次数至多 $d+1$ 的多项式表示; 取 $0, 1, \dots, d+1$ 的样本即可。不能只因某序列前几项能插值, 就断言后续仍由该多项式给出。

115.11 阶乘表: 组合数与排列数

当素数模数为 p 且 $0 \leq k \leq n < p$ 时, 阶乘均非零, 因此

$$C(n, k) = n! (k!)^{-1} ((n-k)!)^{-1}, \quad P(n, k) = n! ((n-k)!)^{-1} \pmod{p}.$$

将表从旧上限扩展到 N , 先正向补齐阶乘, 只对 $N!$ 求一次逆元, 再用 $(i-1)!^{-1} = i(i!)^{-1}$ 反向补齐新逆阶乘项即可。不需要每一项分别快速幂求逆。较小上限的初始化请求保留已有表。查询前显式扩表, 之后查询为常数时间; 内部两表重分配时旧元素引用不能继续使用。实现见第 93 页。

WIDA 的“组合数 (Lucas)”文件实际提供的是普通阶乘、逆阶乘、组合数和排列数, 没有逐位 Lucas 递推。本书以此节覆盖该文件的数学功能, 用显式扩表取代查询时自动扩表, 适用范围仍须限制在模数以内。验证使用独立 Pascal 三角和下降阶乘, 另检查百万项表及扩表前后的原查询结果。

115.12 组合计数

容斥: $|\bigcup A_i| = \sum_{\emptyset \neq S} (-1)^{|S|+1} |\bigcap_{i \in S} A_i|$ 。二项式反演: 若 $b_n = \sum_{k=0}^n \binom{n}{k} a_k$, 则 $a_n = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} b_k$ 。

$$\sum_k \binom{a}{k} \binom{b}{n-k} = \binom{a+b}{n}, \quad C_n = \binom{2n}{n} - \binom{2n}{n+1}.$$

差分形式的 Catalan 公式避免直接除以 $n+1$, 在模数下尤其有用, 但组合数本身仍须正确求模。第二类 Stirling 数满足

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \left\{ \begin{matrix} n-1 \\ k-1 \end{matrix} \right\} + k \left\{ \begin{matrix} n-1 \\ k \end{matrix} \right\}, \quad S(0, 0) = 1.$$

第一类无符号 Stirling 数把系数 k 换为 $n-1$ 。

球与盒	空盒允许	空盒禁止
不同球, 不同盒	k^n	$k!S(n, k)$
相同球, 不同盒	$\binom{n+k-1}{k-1}$	$\binom{n-1}{k-1}$
不同球, 相同盒	$\sum_{j=0}^k S(n, j)$	$S(n, k)$

上表默认 $k \geq 1$; $n = 0, k = 0$ 等边界应按原问题约定单独处理。

115.13 群作用计数

有限群 G 作用在有限集合 X 上时, 轨道数为

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |\text{Fix}(g)|.$$

对位置置换的 q 色染色, $|\text{Fix}(g)| = q^{c(g)}$, 其中 $c(g)$ 为循环数。环形项链只商去旋转; 手链还商去反射, 二者群不同, 不能共用分母。模数整除 $|G|$ 时不能使用模逆元, 须先在合适的整数模数下求和再做精确除法。

115.14 生成函数与递推

普通生成函数 $A(x) = \sum a_n x^n$ 的乘法对应普通卷积。指数生成函数 $A(x) = \sum a_n x^n / n!$ 的乘法对应 $c_n = \sum_k \binom{n}{k} a_k b_{n-k}$ 。形式幂级数求逆要求常数项可逆; $\log A$ 取 $A(0) = 1$, $\exp A$ 取 $A(0) = 0$ 。有限域中积分涉及 $1/i$, 必须避免 i 被特征整除。

$$B_{\text{new}} = B(2 - AB) \pmod{x^{2m}}, \quad (\log A)' = A'/A.$$

BM 从样本得到最短线性递推。要对未来项使用, 必须另外证明序列确实具有不超过相应阶数的线性递推; 少量数值吻合不是证明。矩阵转移、有限状态计数常可提供阶数上界。

115.15 线性代数与生成树

域上的线性方程组一致时, 解集为一个特解加零空间, 零空间维数为未知数个数减秩。浮点消元中的秩依赖误差阈值; 模素数消元是精确域运算; 合数模数下不可随意除以非零元。无向图 Laplacian 的对角元为度数, 非对角元为负的边重数。删去同一行列后的行列式为生成树数。自环不参与生成树; 重边需要累加。只有一个顶点时空余子式行列式约定为 1。有向图需要重新明确入度/出度 Laplacian 与根向内/向外树的方向, 不能直接复用无向封装。

115.16 模线性方程组：工厂工期

POJ2947 (问题 UVa1564) 给出每位工人制造的产品类型和起止星期。设第 j 种产品需 x_j 天, 第 i 条记录中出现 a_{ij} 次, 以星期一为 0、星期日为 6, 则

$$\sum_j a_{ij} x_j \equiv \text{finish}_i - \text{start}_i + 1 \pmod{7}.$$

起止两天均计入; 同一种产品重复出现必须累加系数。跨星期无需特判, 模数类型负责归一化负数。每个余数在 $[3, 9]$ 中恰有一个代表, 因此模方程的解数就是原问题的解数。

直接使用第 115 页的模高斯消元, 模数取 7。增广矩阵有 m 行、 $n+1$ 列, 最后一列是右端项。先判断 consistent: 假时输出无解; 为真且秩小于 n 时输出多解; 否则读取特解, 余数小于 3 的分量加 7。不能先以秩不足判断多解, 因为零系数行仍可能有非零右端项。本题所有非零系数均可逆; 合数模数不具备这一性质, 不能直接套用此消元。

完整双版本程序位于 `verify/poj/2947.*.cpp`。验证包含所有起止星期组合、产品重复 7 次导致系数归零、矛盾的零行、独立枚举所有 3~9 天赋值, 以及用可逆行变换构造的 300 阶稠密系统。该应用验证覆盖分类及唯一解输出, 不代替零空间基等其他接口的验证。来源: <https://onlinejudge.org/external/15/1564.pdf>; kuangbin 2.10.2。

115.17 概率、期望与博弈

期望线性性不要求独立: $\mathbb{E}[\sum X_i] = \sum \mathbb{E}[X_i]$ 。方差可加需要协方差为零; 独立是充分条件。非负整数变量 $\mathbb{E}[X] = \sum_{k \geq 1} \Pr(X \geq k)$ 。吸收过程的期望方程 $E_i = 1 + \sum_j p_{ij} E_j$ 需要检查是否几乎必然吸收及期望是否有限;

不能对闭合非吸收分量直接求唯一有限解。有限无环公平组合游戏在正常玩法下

$$\text{SG}(u) = \text{mex}\{\text{SG}(v) : u \rightarrow v\}.$$

独立子游戏的 SG 值异或为零当且仅当先手必败。反常玩法、含环游戏、双方可走步不同的游戏不能直接套这个结论。

115.18 数值算法与几何

排序必须使用严格弱序；不能以“差小于 eps 则相等”定义通用排序比较器。叉积与行列式容易发生抵消；整数题优先精确整数判定，浮点构造单独处理误差。对于简单格点多边形（无孔），Pick 定理 $A = I + B/2 - 1$ ，边界格点数 $B = \sum \gcd(|\Delta x|, |\Delta y|)$ ；不要重复计入边端点。自适应积分的局部误差估计需要函数足够规则；奇点、间断与强振荡必须先分段处理。三分搜索要求单峰性，整数域最后枚举剩余区间；二分答案要求判定单调性。

来源索引 本章为重新组织的公式速查，扩展范围依据 <https://oi-wiki.org/math/>。小模数 exLucas 与有限域拉格朗日插值已有双风格实现和本地验证；素数模平方根已有 Cipolla 实现；仍需补齐二次剩余的其他方法与定理、Min25、Newton 插值、子集卷积、行列式进阶、数值积分等内容及验证。本章公式不能代替这些待完成代码。

115.19 线性递推：系数方向与外推条件

约定 $a_i = \sum_{j=1}^k c_{j-1} a_{i-j}$ ，初值是 $a_0, \dots, a_{k-1}$ 。对应特征多项式为

$$F(x) = x^k - c_0 x^{k-1} - c_1 x^{k-2} - \dots - c_{k-1}.$$

若 $x^n \bmod F(x) = \sum_{i=0}^{k-1} r_i x^i$ ，则 $a_n = \sum_{i=0}^{k-1} r_i a_i$ 。代码由高次向低次消去多项式项，第 j 个系数写入原次数减 j 的位置，避免倒置系数。例如 Fibonacci 初值为 $[0, 1]$ ，递推系数为 $[1, 1]$ 。

BM 在域上求有限前缀的最短递推。本书实现要求素数模数；一般的模合数环不能保证非零偏差存在逆元。仅凭若干样本无法证明真实序列以后一直遵循同一个规律。若事先知道序列满足阶数至多 d 的齐次线性递推，至少给出前 $2d$ 项，再用恢复出的系数求后续项。系数向量末尾的零不能擅自删除：它可能对应非平稳的初始段，改变阶数会改变递推开始适用的位置。

第一百一十六章 来源与覆盖维护

116.1 参考基线

- kuangbin, ACM 常用算法模板, 2018 年版本: <https://kuangbin.github.io/2018/08/01/ACM-template/>。
- WIDA XCPC: <https://github.com/hh2048/XCPC>, 参考快照 484e9e0。
- OI Wiki 数学: <https://oi-wiki.org/math/>。

本项目重新实现的代码与公式速查不能冒用原作者的验证记录。完整覆盖审计应为每个来源条目记录实现文件、替代关系与验证证据, 保留遗漏项直至完成。

116.2 后续覆盖项

尚需完成的范围以覆盖表为准, 包括动态树、平衡树、一般图匹配、最小树形图、上下界流、高阶几何构造、半平面交、三维凸包、数值积分、复杂数论及多项式、搜索与动态规划等。不能用通用模块的存在推断其覆盖了所有原书例题。

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